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Reformulated Vortex Particle Method and Meshless Large Eddy Simulation of Multirotor Aircraft

Eduardo J. Alvarez

A dissertation submitted to the faculty of
Brigham Young University
in partial fulfillment of the requirements for the degree of

Doctor of Philosophy

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and Meshless Large Eddy Simulation of Multirotor Aircraft

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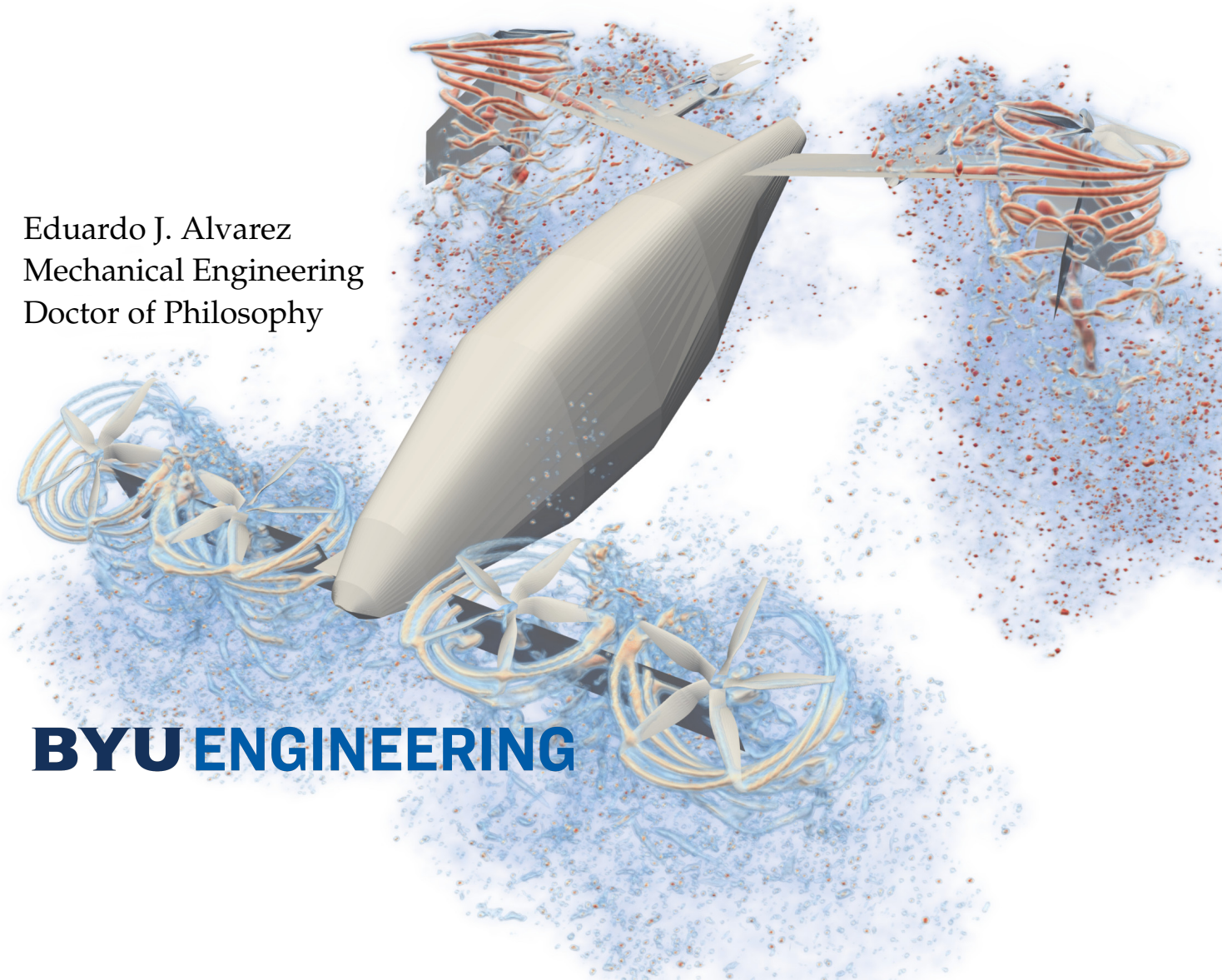

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Abstract

The vortex particle method (VPM) is a mesh-free approach to computational fluid dynamics (CFD) solving the Navier-Stokes equations in their velocity-vorticity form. The VPM uses a Lagrangian scheme, which not only avoids the hurdles of mesh generation, but it also conserves vortical structures over long distances with minimal numerical dissipation while being orders of magnitude faster than conventional mesh-based CFD. However, VPM is known to be numerically unstable when vortical structures break down close to the turbulent regime. In this study, we reformulate the VPM as a large eddy simulation (LES) in a scheme that is numerically stable, without increasing its computational cost. A new set of VPM governing equations are derived from the LES-filtered Navier-Stokes equations. The new equations reinforce conservation of mass and angular momentum by reshaping the vortex elements subject to vortex stretching. In addition to the VPM reformulation, a new anisotropic dynamic model of subfilter-scale (SFS) vortex stretching is developed. This SFS model is well suited for turbulent flows with coherent vortical structures where the predominant cascade mechanism is vortex stretching. Extensive validation is presented, asserting the scheme comprised of the reformulated VPM and SFS model as a meshless LES that accurately resolves large-scale features of turbulent flow. Advection, viscous diffusion, and vortex stretching are validated through simulation of isolated and leapfrogging vortex rings. Mean and fluctuating components of turbulent flow are validated through simulation of a turbulent round jet, in which Reynolds stresses are resolved directly and compared to experimental measurements. Finally, the computational efficiency of the scheme is showcased in the simulation of an aircraft rotor in hover, showing our meshless LES to be 100x faster than a mesh-based LES with similar fidelity.

The ability to accurately and rapidly assess unsteady interactional aerodynamics is a shortcoming and bottleneck in the design of various next-generation aerospace systems: from electric vertical takeoff and landing (eVTOL) aircraft to airborne wind energy and wind farms. For instance, current models used in preliminary design fail to predict and assess configurations that may lead to the wake of a rotor impinging on another rotor or a wing during an eVTOL transition maneuver. In the second part of this dissertation, we address this shortcoming as we present a variable-fidelity CFD framework based on the reformulated

VPM for simulating complex interactional aerodynamics. We further develop our meshless LES scheme to include rotors and wings in the computational domain through actuator models. A novel, vorticity-based, actuator surface model (ASM) is developed for wings, which is suitable for rotor-wing interactions when a wake impinges on the surface of a wing. This ASM imposes the no-flow-through condition at the airfoil centerline by calculating the circulation that meets this condition and by immersing the associated vorticity following a pressure-like distribution. Extensive validation of rotor-rotor and rotor-wing interactions predicted with our LES is presented, simulating two side-by-side rotors in hover, a tailplane with tip-mounted propellers, and a wing with propellers mounted mid-span. To conclude, the capabilities of the framework are showcased through the simulation of a multirotor tiltwing vehicle. The vehicle is simulated mid maneuver as it transitions from powered lift to wing-borne flight, featuring rotors with variable RPM and variable pitch, tilting of wings and rotors, and significant rotor-rotor and rotor-wing interactions from hover to cruise. Thus, the reformulated VPM provides aircraft designers with a high-fidelity LES tool that is orders of magnitude faster than mesh-based CFD, while also featuring variable-fidelity capabilities.

Keywords: vortex particle, interactional aerodynamics, multirotor, eVTOL, aircraft, large eddy simulation, meshless, mesh-free, CFD, LES

Acknowledgments

"And it came to pass that the Lord commanded them that they should go forth into the wilderness, yea, into that quarter where there never had man been. And it came to pass that the Lord did go before them, and did talk with them as he stood in a cloud, and gave directions whither they should travel."

– Ether 2:5, The Book of Mormon

I have been blessed throughout my life with extraordinary mentors: from loving parents and enlightened professors, to God himself, who has frequently opened my mind to the workings of both the natural world and the spirit. This work has been enabled by all these mentors.

I was fortunate to have Professor Andrew Ning as my PhD advisor. The intellectual freedom that he constantly entrusted me with was a key enabler, allowing me to pursue the ideas that lead to the discoveries of this doctoral work. It is remarkable that a man as productive as him would consistently make himself available to meet when I was in need of advise, and would invariably reply to messages within the hour. While being a prolific researcher, it is clear that his priority is to build and lift up his students. I am grateful for his guidance, mentorship, and friendship.

I am also immensely grateful for Professor Scott Thomson. His passion for fluid dynamics, teaching, and research ended up converting me into a "fluids guy." His mentorship during my undergrad gave me confidence to pursue a career as a scientist and researcher. Few people have shaped the course of my profession as he has.

The fellowship that Brigham Young University granted me through the High Impact Doctoral Research Award funded the first three years of my doctorate, which allowed me to start research in a topic that few other institutions were researching at the time. This work was then continued with the support of National Science Foundation under Grant No. 2006219 and Department of Defense through AFWERX contract FA8649-21-P-0095, the latter in collaboration with Whisper Aero. I am grateful to Mark Moore, Ian Villa, and the Whisper team for their association, mentorship, and friendship as we have worked together to bring forth the future of electric aviation.

I am thankful for the good company and engaging conversations with my colleagues of the Flight, Optimization, and Wind Laboratory (FLOW Lab). The need of reformulating the vortex particle method became clear to me as I was answering a question from my colleague Ryan Anderson. Thanks, Ryan, for asking just the right question. I am also grateful for the association, collaboration, and conversations with colleagues in other institutions, especially Racheal Erhard at Stanford University and Professor Adrin Gharakhani at Claremont Graduate University.

I am grateful for my parents who provided a solid foundation at home of love and peace, which allowed me to appreciate the beauty of the natural and spiritual world. Any achievement in my life is a product of my parents' characters and teachings. Also, thanks to my parents-in-law for their continuous support, love, and encouragement, and for the countless Sunday dinners together.

Foremost, I am grateful for my wife, best friend, and eternal companion. This dissertation was enabled by her support through the countless nights that I was up working. Your divine character gives me the strength to be my best self.

Ed Alvarez
Provo, June 2022

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Nomenclature

Vector fields are represented with bold variables, while scalar fields and values are represented with non-bold variables. Tensor notation is indicated with subscripted variables.

Physical Variables and Properties

α, θ	Angle of attack	deg
$\boldsymbol{\omega}$	Vorticity field, $\boldsymbol{\omega}(\mathbf{x}, t)$	$1/\mathrm{s}$
$\boldsymbol{\omega}^*$	Non-dimensional vorticity, $\boldsymbol{\omega}^* \equiv {}^{\omega b}/_{U_\infty}$ or $\boldsymbol{\omega}^* \equiv {}^{\omega D}/_{V_{\text{disk}}}$	
δ_e	Elevator deflection	deg
η	Propulsive efficiency, $\eta = \frac{TU_\infty}{2\pi n Q}$	
Γ	Circulation	m^2/s
$\mathbf{u}$	Velocity field, $\mathbf{u}(\mathbf{x}, t)$	m/s
$\mathbf{u}_\infty$	Freestream velocity	m/s
$\mathbf{x}$	Position	m
Re	Reynolds number	
ν	Kinematic viscosity	m^2/s
ρ	Density	kg/m^3
ξ	Local or global enstrophy	$1/\mathrm{s}^2$
a	Ring cross-sectional radius	m
b	Wing span	m
c	Chord length	m
c_ℓ	Sectional lift coefficient, $c_\ell = \frac{\ell}{\frac{1}{2}\rho U_\infty^2 c}$	
C_D	drag coefficient, $C_D = \frac{D}{\frac{1}{2}\rho U_\infty^2 bc}$	
c_d	Sectional drag coefficient, $c_d = \frac{d}{\frac{1}{2}\rho U_\infty^2 c}$	
C_L	lift coefficient, $C_L = \frac{L}{\frac{1}{2}\rho U_\infty^2 bc}$	
c_n	Normal force coefficient, $c_n = \frac{n}{\frac{1}{2}\rho U_\infty^2 c}$	
C_Q	Torque coefficient, $C_Q = \frac{Q}{\rho n^2 d^5}$	
C_T	Thrust coefficient, $C_T = \frac{T}{\rho n^2 d^4}$	
c_t	Tangential force coefficient, $c_t = \frac{t}{\frac{1}{2}\rho U_\infty^2 c}$	
d, D	Distance or diameter	m

I	Moment of inertia	$\mathrm{kg\,m^2}$
J	Advance ratio, U_∞/nD	
m	Mass	kg
n	Revolutions per second	1/s
p	Pressure field, $p(\mathbf{x}, t)$	Pa
Q	Torque	N m
R	Radius	m
r	Radial position	m
T	Thrust	N
t	Time	s
t_{ref}	Reference time	s
u'_z, u'_r	Streamwise and radial fluctuating components	m/s
U_∞, V_∞	Magnitude of freestream velocity	m/s
U_c	Centerline velocity	m/s
x, y, z	Coordinates	m
Z	Position of ring centroid	m

Mathematical Variables and Functions

α	Relaxation parameter	
α_τ	Tunning parameter associated to scaling factor $3\alpha_\tau - 2$	
$\boldsymbol{\Gamma}_p$	Vortex strength of p-th particle, $\boldsymbol{\Gamma}_p(t)$	$\mathrm{m^3/s}$
$\boldsymbol{\omega}_\sigma$	RBF-approximated vorticity field, $\boldsymbol{\omega}_\sigma(\mathbf{x}) \equiv \sum_p \boldsymbol{\Gamma}_p \zeta_{\sigma_p}(\mathbf{x} - \mathbf{x}_p)$	
$\boldsymbol{\psi}$	Vector potential of velocity, $\mathbf{u} = \nabla \times \boldsymbol{\psi}$	
Δt	Time step size	s
Δx	Discretization length (distance between particles)	m
δ	Dirac delta	
δ_{ij}	Kronecker delta	
f, g	Formulation parameters	
λ	Particle overlap $\lambda \equiv \sigma/\Delta x$	
$\mathbf{E}_{\text{adv}}$	SFS vorticity advection, $(\mathbf{E}_{\text{adv}})_i \equiv \frac{\partial T'_{ij}}{\partial x_j}$	$1/\mathrm{s^2}$
$\mathbf{E}_{\text{str}}$	SFS vortex stretching, $(\mathbf{E}_{\text{str}})_i \equiv -\frac{\partial T_{ij}}{\partial x_j}$	$1/\mathrm{s^2}$
$\mathbf{K}$	Newtonian kernel, $\mathbf{K}(\mathbf{x}) \equiv -\frac{\mathbf{x}}{4\pi\|\mathbf{x}\|^3}$	
$\mathbf{K}_\sigma$	Regularized Newtonian kernel, $\mathbf{K}_\sigma \equiv g_\sigma \mathbf{K}$	
$\mathbf{x}_p$	Position of p-th particle, $\mathbf{x}_p(t)$	m
σ	Filter width (smoothing radius or core size)	m
σ_p	Filter width at position of p-th particle, $\sigma_p(t) \equiv \sigma(\mathbf{x}_p, t)$	m
$\tilde{\boldsymbol{\omega}}$	Divergence-free vorticity field, $\tilde{\boldsymbol{\omega}} \equiv \nabla \times \overline{\mathbf{u}}$	
ζ	Radial basis function	
ζ_σ	Filter kernel, $\zeta_\sigma(\mathbf{x}) \equiv \frac{1}{\sigma^3}\zeta(\frac{\|\mathbf{x}\|}{\sigma})$	$1/\mathrm{m^3}$
C_d	SFS model coefficient, $C_d(\mathbf{x}, t)$	

f_σ Particle smoothing factor, $f_\sigma \equiv \sigma / R$

G_σ Green's function

g_σ Regularizing function, $g_\sigma(\mathbf{x}) \equiv 4\pi \int_0^{\|\mathbf{x}\|/\sigma} \zeta(t) t^2 \, \mathrm{d}t$

m Common multiple between N_{steps} and N_{sheds}

n_{blade} Number of blade elements per blade

N_{sheds} Number of particle sheds per revolution

N_{steps} Number of time steps per revolution

n_{wing} Number of wing elements

T_{ij} SFS vorticity stress tensor, $T_{ij} \equiv \overline{u_i \omega_j} - \overline{u_i}\,\overline{\omega_j}$ $\mathrm{m/s^2}$

Symbols and Operators

$\langle\rangle$ Average operator

$\overline{()}$ Filter operator, or primary filter

$\frac{\mathrm{D}}{\mathrm{D}t}$ Total or material derivative, $\frac{\mathrm{D}}{\mathrm{D}t}() \equiv \frac{\partial}{\partial t}() + (\mathbf{u} \cdot \nabla)()$

$\frac{\mathrm{d}}{\mathrm{d}t}$ Linearized filtered material derivative, $\frac{\mathrm{d}}{\mathrm{d}t}() \equiv \frac{\partial}{\partial t}() + (\overline{\mathbf{u}} \cdot \nabla)()$

$\mathbb{C}$ Complex set

$\mathbb{R}$ Real set

∇ Del operator

∇^2 Laplace operator

Im Imaginary part

$\widehat{()}$, $\breve{()}$, $\breve{\breve{()}}$ Test filters

$\|\|$ Euclidean norm

$\widehat{()}$ Unit vector, for instance $\widehat{\boldsymbol{\omega}} \equiv \omega / \|\omega\|$

Acronyms

ALM Actuator line model

AOA Angle of attack

ASM Actuator surface model

CFD Computational fluid dynamics

cVPM Classic VPM

DES Detached eddy simulation

DNS Direct numerical simulation

FMM Fast multipole method

LBM Lattice Boltzmann method

LES Large eddy simulation

PIV Particle image velocimetry

RBF Radial basis function

rVPM Reformulated VPM

SFS Subfilter-scale

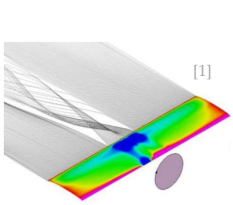
URANS Unsteady Reynolds-average Navier-Stokes

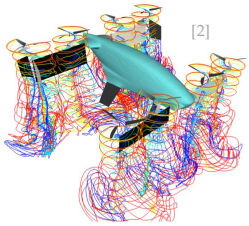
VPM Vortex particle method

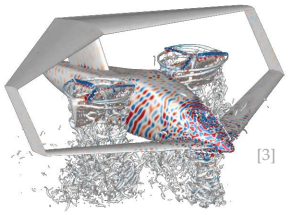
Preamble and Outline

The ability to accurately and rapidly assess unsteady interactional aerodynamics is a shortcoming and bottleneck in the design of various next-generation aerospace systems: from electric vertical takeoff and landing (eVTOL) aircraft to airborne wind energy (AWE) and wind farms. For instance, current models used in preliminary design fail to predict and assess eVTOL or AWE multirotor configurations that may lead to the wake of a rotor impinging on another rotor or a wing during a given maneuver. These unsteady interactions can be analyzed with computational fluid dynamics (CFD) methods. However, resolving wake dynamics in conventional CFD tools requires high-order numerical schemes and mesh resolutions with associated computational costs that make them prohibitive for design space exploration.

In the year 2017, when this doctoral work commenced, it became evident that there was a gap in the range of analysis tools that were available to aircraft designers, shown in Table A. Since then, academia







Low fidelity
Low computation
High fidelity
High computation

	Steady-State Methods	Free-wake Methods	Mesh-Based CFD
	BEMT, lifting line, VLM, panel methods	filament wake, UVLM	URANS, DES, LES, LBM
Unsteady interactions?	✗	✓	✓
Wake mixing?	✗	✗	✓
Fast enough for design?	conceptual	conceptual / preliminary	detailed

[1] Sheridan et al. (2021). *Evaluation of VSPAERO Analysis Capabilities for Conceptual Design of Aircraft with Propeller-Blown Wings*. AIAA AVIATION Forum.
[2] Droandi et al. (2018). *Tiltwing Multi-Rotor Aerodynamic Modeling in Hover, Transition and Cruise Flight Conditions*. AHS International 74th Annual Forum.
[3] Ventura Diaz, P., & Yoon, S. (2018). *High-Fidelity Computational Aerodynamics of Multi-Rotor Unmanned Aerial Vehicles*. AIAA Aerospace Sciences Meeting.

Table A: Gap between low and high-fidelity analysis tools for multirotor aircraft design.

and industry have turned their attention to a promising candidate that could fill the gap: the vortex particle method (VPM). In spite of its growing popularity, VPM is known to be numerically unstable when vortical structures break down close to the turbulent regime. This has limited its range of applications to mostly benign cases with well-behaved numerics (*e.g.,* coarse simulations of rotors with an axial inflow). Furthermore, little validation has been provided, since the lack of numerical stability typically embroils the efforts to show that simulations are convergent. In short, the classic VPM suffers from the following shortcomings due to its poor numerical stability:

Issue 1 Applicable to only low and mid-fidelity simulations (simulations become numerically unstable as spatial resolution is increased).

Issue 2 Applicable to only numerically well-behaved cases (*e.g.,* propellers in forward flight and wakes before breakdown/mixing).

Issue 3 Lack of numerical convergence.

Issue 4 Scarcity of validation studies in the literature.

In this doctoral work, the VPM is overhauled by reformulating the method, and each of the aforementioned issues are respectively addressed through the following outcomes:

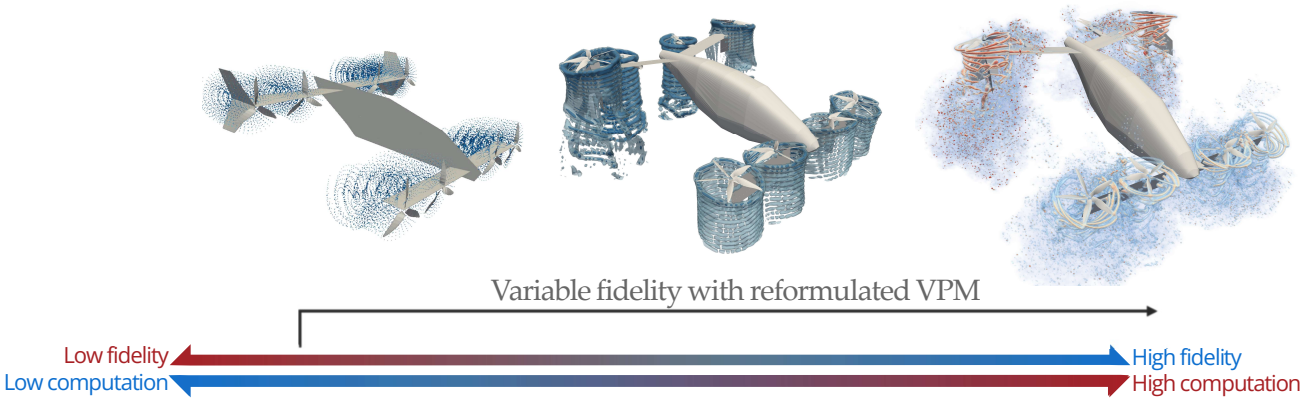
Outcome 1 The method is reformulated as a large eddy simulation (LES) that is meshless and capable of resolving turbulent phenomena in a high-fidelity manner.

Outcome 2 The reformulated VPM is applied in a variety of turbulent scenarios showing remarkable numerical stability: from a turbulent round jet to the complex aerodynamic interactions of a propeller blowing on a wing.

Outcome 3 Numerical convergence is characterized through studies on spatial and temporal discretization of vortex ring, rotor, and wing simulations.

Outcome 4 Extensive validation is presented by comparison to experimental and numerical results reported in the literature on the following cases: Isolated and leapfrogging vortex rings, turbulent round jet, rotor in hover, propeller in forward flight, two side-by-side rotors, isolated wing, propeller wake, and rotor-wing interactions.

Furthermore, the LES obtained with the reformulated VPM is compared to other state-of-art conventional CFD approaches, showing this approach to be 100x faster than conventional LES with comparable fidelity. The overall outcome of this doctoral research is a novel CFD method capable of resolving mean and fluctuating large-scale features of turbulent



	Low Fidelity	Mid Fidelity	High Fidelity
Resolution	coarse (10k – 100k particles)	mid (100k – 1M particles)	fine (1M – 10M particles)
Wall-clock time (on laptop computer)	3 minutes – 30 minutes	1 hour – 12 hours	1 day – 4 days
Design stage	conceptual	preliminary	detailed

Table B: Variable fidelity achieved with the reformulated VPM, filling the gap in analysis tools.

flow with minimal computational effort, providing aircraft designers with a robust variable-fidelity tool that fills the gap in analysis tools, as shown in Table B.

Outline of Part I

The first part of this dissertation presents the reformulated VPM, deriving its governing equations from the LES-filtered Navier-Stokes equations and developing a meshless LES scheme for the simulation of turbulent flow with coherent vortical structures. The content and novel contributions of each chapter in Part I are summarized as follows.

Chapter 1 – Fundamentals of the Vortex Particle Method

Summary: The vorticity-velocity form of the Navier-Stokes (NS) equations is introduced, along with all the fundamental concepts necessary to bring the reader up to speed with LES and vortex methods. The vorticity NS equation is discretized in a Lagrangian scheme and used to derive governing equations.

Contributions: A new set of VPM governing equations are derived from the LES-filtered Navier-Stokes equations. The classic VPM is shown to be only one out of several possible formulations that arise from these governing equations.

Chapter 2 – Reformulated Vortex Particle Method

Summary: Multiple formulations of the governing equations are considered using different element shapes and conserva-

tion laws. One formulation is selected, referred to as the *reformulated VPM*, which uses the particle size to reinforce local conservation of both mass and angular momentum.

Contributions: The classic VPM formulation is shown to locally violate conservation of mass and angular momentum under vortex stretching, explaining why the classic VPM tends to be numerically unstable. The reformulated VPM is proposed, reinforcing local conservation of both mass and angular momentum. The proposed rVPM equations do not require more computation than the classic VPM.

Chapter 3 – Anisotropic Dynamic SFS Model

Summary: Existing subfilter scale (SFS) models for LES based on the vorticity-velocity NS equations are briefly reviewed. A novel low-dissipation anisotropic SFS model that uses vortex stretching as the physical mechanism for turbulence is proposed. This low-dissipation SFS model is well suited for turbulent flows with coherent vortical structures where the predominant cascade mechanism is vortex stretching.

Contributions: An anisotropic structural model of SFS vortex stretching is developed. A dynamic procedure for computing the model coefficient is proposed, which is based on a simultaneous balance of enstrophy-production and derivatives between true and modeled SFS contributions. It is described how the model can be implemented in conventional mesh-based CFD with a pressure-velocity solver.

Chapter 4 – Numerical Schemes

Summary: All numerical schemes that are implemented along the reformulated VPM are described. This includes the fast multipole method (FMM), relaxation of $\boldsymbol{\omega}$ divergence, viscous diffusion, time integration, etc.

Contributions: A novel approach to computing vortex stretching and derivatives of $\overline{\mathbf{u}}$ through a complex-step derivative approximation implemented in the FMM is proposed. A correction to a popular relaxation scheme is proposed.

Chapter 5 – Validation of Meshless LES Scheme

Summary: The scheme comprised of the reformulated VPM and SFS model is extensively tested, validating the scheme as an LES method. Simulations of vortex rings, a turbulent round jet, and a rotor in hover are compared to experimental and numerical results reported in the literature.

Contributions: Extensive validation is presented, incrementally testing advection, viscous diffusion, and vortex stretching. Mean and fluctuating components of a turbulent jet are validated, resolving Reynolds stresses directly. Initial and boundary conditions for a meshless scheme are demonstrated in vortex ring, turbulent jet, and rotor cases. Using the rotor case, it is shown that the meshless LES is 100x faster than a mesh-based LES with similar fidelity, while being 10x faster than a low-fidelity URANS simulation and 1000x faster than a high-fidelity DES.

In the spirit of reproducibility and scientific cooperation, the meshless LES used in this study is implemented and hereby released as an open-source software called FLOWVPM.

Outline of Part II

The second part of this dissertation further develops the meshless LES scheme to include rotors and wings in the computational domain through actuator models. Extensive validation is presented on the rotor-rotor and rotor-wing interactions predicted by the meshless LES, culminating with the simulation of a full multirotor vehicle. The content and novel contributions of each chapter in Part II are summarized as follows.

Chapter 6 – Interactional Aerodynamics Solver

Summary: Actuator models for wings and rotating blades are formulated, immersing their vorticity in the Navier-Stokes equations. An aeroacoustics solver is coupled to the meshless LES to predict aeroacoustic noise radiated by rotors.

Contributions: A novel, vorticity-based, actuator surface model (ASM) is developed for wings, which is suitable for rotor-wing interactions when a wake impinges on the surface of a wing.

Chapter 7 – Validation of Rotor-Rotor Interactions

Summary: Validation of rotor-rotor interactions is presented, comparing the meshless LES simulation to experimental results reported in the literature.

Contributions: A detailed convergence study of the rotor simulation is presented. Simulations of isolated propellers and two side-by-side rotors in hover are validated.

Chapter 8 – Validation of Rotor-Wing Interactions

> **Summary:** Validation of rotor-wing interactions is presented, comparing the meshless LES simulation to experimental and numerical results reported in the literature.
>
> **Contributions:** A detailed convergence study of the propeller wake is presented. The ASM and predicted rotor-wing interactions are validated through the simulation of a swept-back wing, a tailplane with tip-mounted propellers, and a wing with propellers mounted mid-span.

Chapter 9 – Maneuvering eVTOL Aircraft

> **Summary:** The capabilities of the meshless LES are showcased through the simulation of a multirotor tiltwing vehicle. The vehicle is simulated mid maneuver as it transitions from powered lift to wing-borne flight.
>
> **Contributions:** This simulation features rotors with variable RPM and variable blade pitch, tilting of wings and rotors, and significant rotor-rotor and rotor-wing interactions from hover to cruise. Thrust, powered lift, and wing-borne lift are characterized throughout the maneuver.

In the spirit of reproducibility and scientific cooperation, the framework developed for this study is hereby released as an open-source software called FLOWUnsteady.

This dissertation assumes that the reader is familiar with multivariate calculus, tensor notation, numerical methods, fundamental fluid dynamics, and notions of aerodynamic principles. This pre-requisite knowledge is commonly attained through undergraduate studies of aerospace/mechanical engineering.

Publications

This dissertation reproduces partial or total content of the following peer-reviewed journal publications led by the author:

- E. J. Alvarez & A. Ning, "Meshless Large Eddy Simulation of Rotor-Wing Interactions Through the Reformulated Vortex Particle Method," 2022, (in preparation).
- E. J. Alvarez & A. Ning, "Reviving the Vortex Particle Method: A Stable Formulation for Meshless Large Eddy Simulation," 2022, (in review).
- E. J. Alvarez & A. Ning, "High-Fidelity Modeling of Multirotor Aerodynamic Interactions for Aircraft Design," 2020, AIAA Journal.

and the following publications in conference proceedings:

- E. J. Alvarez & A. Ning, "FLOWUnsteady: An Interactional Aerodynamics Solver for Multirotor Aircraft and Wind Energy," 2022, AIAA AVIATION Forum (in progress).
- E. J. Alvarez, A. Schenk, T. Critchfield, & A. Ning, "Rotor-on-Rotor Aeroacoustic Interactions of Multirotor in Hover," 2020, VFS 76th Forum.
- E. J. Alvarez & A. Ning, "Modeling Multirotor Aerodynamic Interactions Through the Vortex Particle Method," 2019, AIAA AVIATION Forum.
- E. J. Alvarez & A. Ning, "Development of a Vortex Particle Code for the Modeling of Wake Interaction in Distributed Propulsion," 2018, AIAA AVIATION Forum.

The author hereby confirms that the use of these articles is compliant with all publishing agreements.

The codes developed in this doctoral work are publicly available in the following repositories:

- github.com/byuflowlab/FLOWVPM.jl
- github.com/byuflowlab/FLOWUnsteady

Part I

Vortex Particle Method Reformulation for Meshless Large Eddy Simulation

Introduction

Large eddy simulation (LES) is a class of computational fluid dynamics (CFD) that filters the Navier-Stokes equations to decompose small-scale fluctuations of the flow from large ones. The large scales are then resolved directly, while the effects of the smaller scales are modeled. The most common LES methods use a mesh or grid to discretize the space and calculate fluxes and derivatives, classified as finite volume/element/difference methods. Significant user effort is spent in the generation and manipulation of the mesh, with studies showing that about 67% of engineering time in mesh-based CFD is spent in these efforts.[1]

Vortex methods[2] are a class of meshless CFD solving the Navier-Stokes equations in their velocity-vorticity form. This form is solved in a Lagrangian scheme, which not only avoids the hurdles of mesh generation, but also conserves vortical structures over long distances with minimal numerical dissipation.

The vortex particle method[3] (VPM) is a vortex method that uses particles to discretize the Navier-Stokes equations, with the particles representing radial basis functions that construct a continuous vorticity field. This meshless CFD has several advantages over conventional mesh-based CFD. In the absence of a mesh, the VPM (1) does not suffer from the conventional Courant–Friedrichs–Lewy (CFL) condition, (2) does not suffer from the numerical dissipation introduced by a mesh, (3) derivatives are calculated exactly rather than approximated through a stencil, and (4) it has been shown to be 100x to 1000x faster than mesh-based CFD with comparable fidelity.[4] Furthermore, the VPM is spatially second-order accurate,[5] it is highly efficient since elements are placed only where vorticity exists, the spatial discretization can be automatically adapted[6] in the fashion of an adaptive mesh refinement, and simulations are highly-parallelizable in heterogeneous CPU and GPU architectures.[7]

VPM has gained popularity in recent years due to a growing need to predict complex aerodynamic interactions in modern electric aircraft at a computational cost fit for conceptual design.[8] For example, VPM has been used for high-fidelity simulation of rotorcraft forward flight[9] and multirotor interactions,[10] and as a mid and low-fidelity tool for ground effect,[11] distributed electric propulsion,[12] aeromechanics of unconventional rotorcraft,[13] rotor-rotor interactions,[14] and wind energy.[15] Furthermore, a couple of open-source VPM codes have been recently

1 M. F. Hardwick et al. (2005). *DART System analysis*.

2 G. S. Winckelmans (2004). "Vortex methods"; C. Mimeau and I. Mortazavi (2021). "A Review of Vortex Methods and Their Applications: From Creation to Recent Advances".

3 G. S. Winckelmans et al. (1993). "Contributions to Vortex Particle Methods for the Computation of Three-Dimensional Incompressible Unsteady Flows".

4 E. J. Alvarez, A. Schenk, et al. (2020). "Rotor-on-Rotor Aeroacoustic Interactions of Multirotor in Hover".

5 R. Yokota and L. A. Barba (2013). "FMM-based vortex method for simulation of isotropic turbulence on GPUs , compared with a spectral method".

6 L. A. Barba et al. (2005b). "Vortex method with meshless spatial adaption for accurate simulation of viscous, unsteady vortical flows"; M. J. Stock and A. Gharakhani (2021). "Solution-Responsive Particle Size Adaptivity in Lagrangian Vortex Particle Methods".

7 R. Yokota, L. A. Barba, et al. (2013). "Petascale turbulence simulation using a highly parallel fast multipole method on GPUs"; Q. Hu et al. (2014). "Scalable Fast Multipole Accelerated Vortex Methods".

8 C. Silva, W. R. Johnson, et al. (2018). "VTOL Urban Air Mobility Concept Vehicles for Technology Development"; A. Kasliwal et al. (2019). "Role of flying cars in sustainable mobility"; R. A. McDonald et al. (2021). "Future aircraft concepts and design methods".

9 M. Stock, A. Gharakhani, and C. Stone (2010). "Modeling Rotor Wakes with a Hybrid OVERFLOW-Vortex Method on a GPU Cluster".

10 E. J. Alvarez and A. Ning (2020). "High-Fidelity Modeling of Multirotor Aerodynamic Interactions for Aircraft Design"; Alvarez, Schenk, et al., "Rotor-on-Rotor Aeroacoustic Interactions of Multirotor in Hover".

11 J. F. Tan, J. G. Cai, et al. (2020). "Computational Study on the Aerodynamic Interference Between Tandem Rotors and Nearby Obstacles"; J. F. Tan, J. er Gao, et al. (2021). "Novel approach to helicopter brownout based on vortex and discrete element methods".

released.[16] Fig. I.1 shows the simulation of an eVTOL aircraft using VPM.

In spite of its growing popularity, VPM is known to be numerically unstable when vortical structures break down close to the turbulent regime. This has limited its range of applications in the aforementioned studies to mostly benign cases with well-behaved numerics (*e.g.*, coarse simulations of rotors with axial inflow). The instability is triggered when vortex stretching gives raise to a rapid increase of local vorticity, which the low numerical dissipation of the method fails to damp out. Such numerical instability is believed to be caused by a combination of Lagrangian distortion, a vorticity field that is not divergence-free, and the absence of subfilter-scale (SFS) turbulent diffusion, among other reasons.

Multiple meshless schemes have been developed over the years to address Lagrangian distortion,[17] divergence of the vorticity field,[18] and SFS effects.[19] While some of these schemes are effective in two-dimensional cases, they have not succeeded at making the three-dimensional VPM numerically stable. The need to circumvent this challenge motivated further development of the vortex-in-cell (VIC) method,[20] also known as vortex particle-mesh method. In this method, the particles are projected onto a background mesh at every time step, and vortex stretching, viscous diffusion, and the Biot-Savart law are computed in a mesh-based scheme. This approach has shown to be numerically stable, recently enabling the study of wake dynamics with unprecedented fidelity.[21]However, the introduction of this mesh also forfeits a few of the aforementioned benefits of a purely Lagrangian (meshless) scheme.

In this study, we propose an LES formulation of the VPM that is both numerically stable and meshless. A new set of VPM governing equations

12 P. C. Teixeira et al. (2019). "Propeller Effects on the Response of High-Altitude Long-Endurance Aircraft".

13 P. Singh et al. (2021). "Aeromechanics and Aeroelastic Stability of Coaxial Rotors"; G. Jacobellis et al. (2021). "Experimental and computational investigation of stacked rotor performance in hover"; E. Corle et al. (2021). "On the Influence of Inflow Model Selection for Time-Domain Tiltrotor Aeroelastic Analysis".

14 H. Lee and D. J. Lee (2020). "Rotor interactional effects on aerodynamic and noise characteristics of a small multirotor unmanned aerial vehicle".

15 H. Lee and D.-J. Lee (2019). "Effects of platform motions on aerodynamic performance and unsteady wake evolution of a floating offshore wind turbine".

16 K. S. Kuzmina et al. (2019). "The VM2D Open Source Code for Incompressible Flow Simulation by Using Meshless Lagrangian Vortex Methods on CPU and GPU"; M. J. Stock and A. Gharakhani (2020). "Open-Source Accelerated Vortex Particle Methods For Unsteady Flow Simulation"; M. Tugnoli et al. (2021). "Mid-fidelity approach to aerodynamic simulations of unconventional VTOL aircraft configurations".

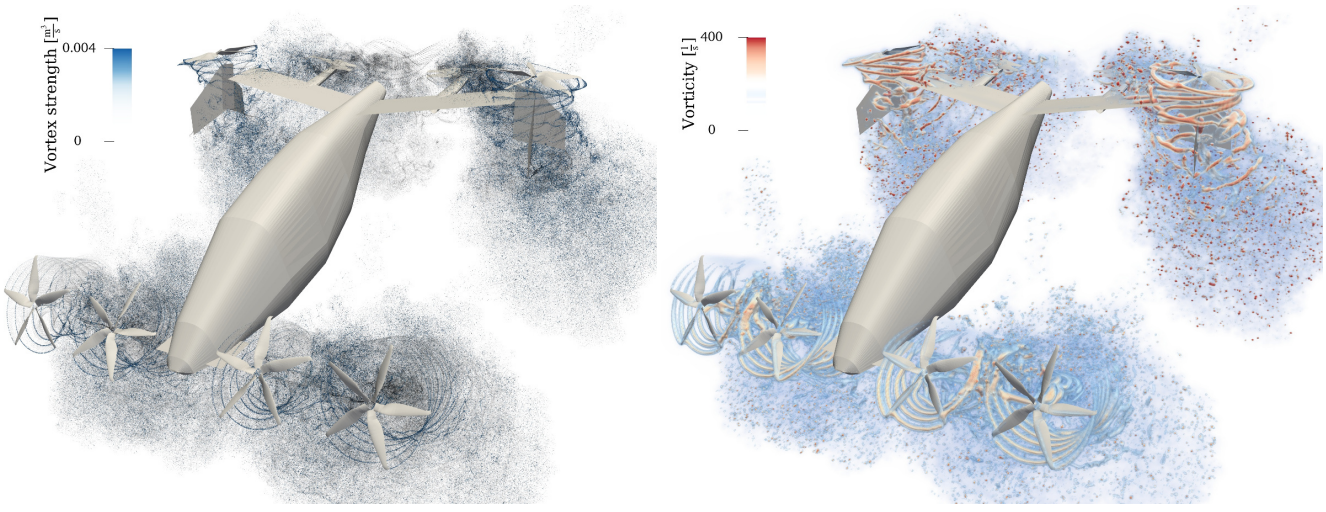


Figure I.1: Meshless LES of eVTOL aircraft using reformulated VPM: (left) computational elements (vortex particles and strength), and (right) volume rendering of vorticity field.

are derived from the LES-filtered Navier-Stokes equations in Chapter 1. In Section 1.6, the classic VPM is shown to be only one out of multiple possible formulations of these governing equations. Such formulation is shown to locally violate conservation of mass and angular momentum under vortex stretching, explaining why the classic VPM tends to be numerically unstable. In Chapter 2, we consider multiple possible formulations using different element shapes and conservation laws. One formulation is selected, which uses the particle size to reinforce local conservation of both mass and angular momentum. This formulation, referred to as the *reformulated VPM*, is implemented and hereby released as the open-source software FLOWVPM.[22]

In addition to the reformulated VPM, a novel anisotropic structural model of subfilter-scale (SFS) vortex stretching is developed in Chapter 3. The means for backscatter control are also provided, along with a dynamic procedure for the automatic computation of the model coefficient. This SFS model is apt for both meshless and mesh-based CFD, and is well suited for turbulent flows with coherent vortical structures where the predominant cascade mechanism is vortex stretching.

The scheme comprised of the reformulated VPM and SFS model is extensively tested in Chapter 5, validating the scheme as an LES method. Advection, viscous diffusion, and vortex stretching are validated through simulation of isolated and leapfrogging vortex rings. Large-scale turbulent dynamics are validated through simulation of a turbulent round jet, where predicted fluctuations and Reynolds stress are compared to experimental measurements. Finally, the computational efficiency of our meshless LES is showcased in an engineering application through the simulation of an aircraft rotor in hover.

17 L. A. Barba et al. (2005a). "Advances in viscous vortex methods - Meshless spatial adaption based on radial basis function interpolation"; I. Lakkis et al. (2009). "A high resolution spatially adaptive vortex method for separating flows. Part I: Two-dimensional domains"; M. Kirchhart et al. (2021). "Discrete Projections: A Step Towards Particle Methods on Bounded Domains without Remeshing".

18 G. Pedrizzetti (1992). "Insight into singular vortex flows"; A. Gharakhani (1997). *A Survey of Grid-Free Methods for the Simulation of 3-D Incompressible Flows in Bounded Domains*.

19 A. Gharakhani (2002). "A Grid-Free Method for LES of Incompressible Flow"; A. Gharakhani (2005). "A Lagrangian Vortex Method for Grid-Free Dynamic LES".

20 J. Christiansen (1997). "Numerical Simulation of Hydrodynamics by the Method of Point Vortices"; G.-H. Cottet and P. Poncet (2004). "Advances in direct numerical simulations of 3D wall-bounded flows by Vortex-in-Cell methods"; P. Koumoutsakos (2005). "Multi-scale flow simulations using particles".

21 P. Chatelain, A. Curioni, et al. (2008). "Billion vortex particle direct numerical simulations of aircraft wakes"; P. Chatelain, M. Duponcheel, et al. (2017). "Vortex particle-mesh simulations of vertical axis wind turbine flows: from the airfoil performance to the very far wake"; D.-G. Caprace, P. Chatelain, et al. (2020). "Wakes of rotorcraft in advancing flight: A large-eddy simulation study".

22 The code is available at github.com/byuflowlab/FLOWVPM.jl

1 Fundamentals of the Vortex Particle Method

A new set of VPM governing equations are hereby derived from the LES-filtered Navier-Stokes equations in their vorticity form. Along the way, the derivation here presented introduces all the fundamental concepts necessary to bring the reader up to speed with LES and vortex methods.

1.1 Vorticity Form of the Navier-Stokes Equations

In a Newtonian, incompressible fluid with constant viscosity, the linear momentum of a differential fluid element is governed by the Navier-Stokes equation

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u}\cdot\nabla)\mathbf{u} = -\frac{1}{\rho}\nabla p + \nu\nabla^2\mathbf{u}, \tag{1.1}$$

where $\mathbf{u}(\mathbf{x},t)$ is the velocity field, $p(\mathbf{x},t)$ is the pressure field, and ρ and ν are the density and kinematic viscosity of the fluid, respectively.

Taking the curl over Eq. (1.1), the pressure dependence disappears obtaining

$$\frac{\mathrm{D}}{\mathrm{D}t}\boldsymbol{\omega} = (\boldsymbol{\omega}\cdot\nabla)\mathbf{u} + \nu\nabla^2\boldsymbol{\omega}, \tag{1.2}$$

where $\frac{\mathrm{D}}{\mathrm{D}t}$ denotes the material derivative operator, $\frac{\mathrm{D}}{\mathrm{D}t}() \equiv \frac{\partial}{\partial t}() + (\mathbf{u}\cdot\nabla)()$, and $\boldsymbol{\omega}(\mathbf{x},t) = \nabla\times\mathbf{u}(\mathbf{x},t)$ is the vorticity field.

Eq. (1.2) is the vorticity form of the Navier-Stokes linear-momentum equation. This equation depends on $\boldsymbol{\omega}$ alone since $\mathbf{u}$ can be calculated from $\boldsymbol{\omega}$ through the Biot-Savart law. While Eq. (1.1) stems from conservation of linear momentum, Eq. (1.2) can be interpreted as some form of conservation of angular momentum,[23] which is further discussed in Section 2.1. From the right-hand side, we see that the evolution of the vorticity field is governed by vortex stretching (first term) and viscous diffusion (second term).

[23] P. C. Chatwin (1973). "The vorticity equation as an angular momentum equation".

1.2 Large Eddy Simulation Equation

Large eddy simulation (LES) is a class of computational fluid dynamics that filters the Navier-Stokes equations to decompose small-scale fluctuations of the flow from large ones. The large scales are then resolved directly, while the smaller scales (called *subfilter scales*, or SFS) remain

unresolved. In the turbulent cascade, however, all eddies break into progressively smaller eddies, transferring the kinetic energy of large eddies down to the molecular level. At the same time, small eddies can locally coalesce to transfer energy to the large scales. This poses the problem of determining the energy that is constantly transferred between both resolved and unresolved domains.

We now derive the filtered vorticity Navier-Stokes equation—here referred to as *large-eddy simulation equation*—and define the stress tensor that transfers energy between resolved scales and subfilter scales.

Let ϕ be a field and ζ_σ a filter kernel, the filtered field is denoted by a bar and defined as

$$\overline{\phi}\,(\mathbf{x}) \equiv \int_{-\infty}^{\infty} \phi(\mathbf{y})\zeta_\sigma(\mathbf{x}-\mathbf{y})\,\mathrm{d}\mathbf{y},$$

where the filter ζ_σ is associated to a certain cutoff length scale σ. In this study, ζ_σ is defined as $\zeta_\sigma(\mathbf{x}) \equiv \frac{1}{\sigma^3}\zeta(\frac{\|\mathbf{x}\|}{\sigma})$ where ζ is a radial basis function, and is required to have a volume integral of unity, $\int_{-\infty}^{\infty} \zeta_\sigma(\mathbf{y})\,\mathrm{d}\mathbf{y} = 1$.

In order to derive the LES version of the vorticity equation, Eq. (1.2) is written in tensor notation and filtered as

$$\overline{\frac{\partial \omega_i}{\partial t}} + \overline{u_j \frac{\partial \omega_i}{\partial x_j}} = \overline{\omega_j \frac{\partial u_i}{\partial x_j}} + \nu \overline{\nabla^2 \omega_i}. \tag{1.3}$$

In this equation, $\overline{u_j \frac{\partial \omega_i}{\partial x_j}}$ and $\overline{\omega_j \frac{\partial u_i}{\partial x_j}}$ are non-linear terms that cannot be calculated from resolved quantities, but are rather approximated through a tensor T_{ij} that encapsulates the error between $\overline{u_i \omega_j}$ and $\overline{u_i}\,\overline{\omega_j}$ as

$$\overline{u_i \omega_j} = \overline{u_i}\,\overline{\omega_j} + T_{ij}.$$

The gradient of T_{ij} and its transpose result in

$$\frac{\partial T'_{ij}}{\partial x_j} = \overline{u_j \frac{\partial \omega_i}{\partial x_j}} - \overline{u_j} \frac{\partial \overline{\omega_i}}{\partial x_j}$$
$$\frac{\partial T_{ij}}{\partial x_j} = \overline{\omega_j \frac{\partial u_i}{\partial x_j}} - \overline{\omega_j} \frac{\partial \overline{u_i}}{\partial x_j}$$

Replacing this into Eq. (1.3) and using the derivative-filter commutation property, we obtain the LES vorticity equation:

$$\frac{\partial \overline{\omega_i}}{\partial t} + \overline{u_j} \frac{\partial \overline{\omega_i}}{\partial x_j} = \overline{\omega_j} \frac{\partial \overline{u_i}}{\partial x_j} + \nu \nabla^2 \overline{\omega_i} - \frac{\partial T'_{ij}}{\partial x_j} + \frac{\partial T_{ij}}{\partial x_j}. \tag{1.4}$$

The term $\frac{\partial T'_{ij}}{\partial x_j}$ represents the subfilter-scale (SFS) contributions arising from the advective term (vorticity advection), while $\frac{\partial T_{ij}}{\partial x_j}$ represents the contributions arising from vortex stretching. T_{ij} is associated to the

SFS vorticity stress that encapsulates the interactions between large scale dynamics and SFS dynamics and has to be modeled in terms of resolved quantities. The accuracy of LES hinges on the modeling of this tensor. Its divergence represents the rate at which enstrophy—a measure of rotational kinetic energy—is transferred from resolved scales to subfilter scales (diffusion) and from subfilter scales to resolved scales (backscatter). In Chapter 3 we will develop a model that approximates the vortex-stretching component of this tensor, and provide the means for backscatter control.

For convenience, we write Eq. (1.4) in vector notation as

$$\frac{\mathrm{d}}{\mathrm{d}t}\overline{\boldsymbol{\omega}} = (\overline{\boldsymbol{\omega}}\cdot\nabla)\,\overline{\mathbf{u}} + \nu\nabla^2\overline{\boldsymbol{\omega}} - \mathbf{E}_{\text{adv}} - \mathbf{E}_{\text{str}}, \tag{1.5}$$

where $(\mathbf{E}_{\text{adv}})_i \equiv \frac{\partial T'_{ij}}{\partial x_j}$ is the SFS vorticity advection, $(\mathbf{E}_{\text{str}})_i \equiv -\frac{\partial T_{ij}}{\partial x_j}$ is the SFS vortex stretching, and the $\frac{\mathrm{d}}{\mathrm{d}t}$ operator is a linearized version of the filtered material derivative, $\frac{\mathrm{d}}{\mathrm{d}t}() \equiv \frac{\partial}{\partial t}() + (\overline{\mathbf{u}}\cdot\nabla)()$.

1.3 Viscous Diffusion

One common practice for solving non-linear partial differential equations (PDE), like Eq. (1.5), is that of splitting the PDE operator into a linear sum of its non-linear pieces. This permits discretizing and solving each non-linear piece in a separate numerical scheme. In this study we will split the PDE in Eq. (1.5) into inviscid and viscous pieces as

$$\frac{\mathrm{d}}{\mathrm{d}t}\overline{\boldsymbol{\omega}} = \left(\frac{\mathrm{d}}{\mathrm{d}t}\overline{\boldsymbol{\omega}}\right)_{\text{inviscid}} + \left(\frac{\mathrm{d}}{\mathrm{d}t}\overline{\boldsymbol{\omega}}\right)_{\text{viscous}},$$

where

$$\left(\frac{\mathrm{d}}{\mathrm{d}t}\overline{\boldsymbol{\omega}}\right)_{\text{inviscid}} = (\overline{\boldsymbol{\omega}}\cdot\nabla)\,\overline{\mathbf{u}} - \mathbf{E}_{\text{adv}} - \mathbf{E}_{\text{str}}$$
$$\left(\frac{\mathrm{d}}{\mathrm{d}t}\overline{\boldsymbol{\omega}}\right)_{\text{viscous}} = \nu\nabla^2\overline{\boldsymbol{\omega}}.$$

Over the years, multiple Lagrangian schemes have been developed that accurately resolve the viscous component, like the vortex redistribution method,[24] particle strength exchange,[25] and core spreading,[26] to name a few. In this study, viscous diffusion will be solved through the core spreading method coupled with the radial basis function interpolation approach for spatial adaptation developed by Barba,[27] discussed in more detail in Section 4.6. This viscous scheme diffuses the vorticity by thickening each particle's core size σ over time, while using an RBF interpolation to reset core sizes when they have overgrown. As shown by Rossi,[28] the core spreading method has second-order spatial convergence, while showing linear convergence when coupled with spatial adaptation.

In the following sections we will focus on the inviscid part of the PDE, developing a scheme for its numerical solution.

24 S. Shankar et al. (1996). "A New Diffusion Procedure for Vortex Methods"; A. Gharakhani (2001). "Grid-free simulation of 3-D vorticity diffusion by a high-order vorticity redistribution method".

25 P. Degond et al. (1989). "The Weighted Particle Method for Convection-Diffusion Equations. Part 1: The Case of an Isotropic Viscosity".

26 L. F. Rossi (1996). "Resurrecting Core Spreading Vortex Methods: A New Scheme that is Both Deterministic and Convergent".

27 Barba et al., "Advances in viscous vortex methods - Meshless spatial adaption based on radial basis function interpolation"; L. A. Barba (2004). "Vortex Method for computing high-Reynolds number Flows: Increased accuracy with a fully mesh-less formulation"; C. E. Torres et al. (2009). "Fast radial basis function interpolation with Gaussians by localization and iteration".

28 Rossi, "Resurrecting Core Spreading Vortex Methods: A New Scheme that is Both Deterministic and Convergent".

1.4 Lagrangian Discretization: The Vortex Particle

The material derivative in Eq. (1.2) and the material-conservative nature of the vorticity makes the $\boldsymbol{\omega}$ field especially well fit for a Lagrangian description. We now discretize the vorticity equation with Lagrangian elements, termed *vortex particles*. Each particle represents a volume of fluid that is convected by the velocity field carrying an integral quantity of vorticity.

The unfiltered $\boldsymbol{\omega}$ field is discretized with singular vortex particles of positions $\mathbf{x}_p$ and coefficients $\boldsymbol{\Gamma}_p$, approximating $\boldsymbol{\omega}$ as

$$\boldsymbol{\omega}(\mathbf{x}, t) \approx \sum_p \boldsymbol{\Gamma}_p(t)\delta(\mathbf{x} - \mathbf{x}_p(t)), \tag{1.6}$$

where δ is the Dirac delta. Each particle travels with the local velocity as

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathbf{x}_p = \mathbf{u}(\mathbf{x}_p),$$

where $\mathbf{x}_p$ is the position of the p-th particle. Thus, each coefficient $\boldsymbol{\Gamma}_p$ (termed *vortex strength*) represents the average vorticity that is carried in the volume of each particle since

$$\int_{-\infty}^{\infty} \boldsymbol{\omega}(\mathbf{x}, t)\, \mathrm{d}\mathbf{x} \approx \sum_p \boldsymbol{\Gamma}_p(t).$$

Using the singular particle approximation in the filtered vorticity $\overline{\boldsymbol{\omega}}$,

$$\begin{aligned}\overline{\boldsymbol{\omega}}\,(\mathbf{x}) &= \int_{-\infty}^{\infty} \boldsymbol{\omega}\,(\mathbf{y})\, \zeta_\sigma(\mathbf{x} - \mathbf{y})\, \mathrm{d}\mathbf{y} \\ &\approx \int_{-\infty}^{\infty} \left(\sum_p \boldsymbol{\Gamma}_p \delta(\mathbf{y} - \mathbf{x}_p) \right) \zeta_\sigma(\mathbf{x} - \mathbf{y})\, \mathrm{d}\mathbf{y},\end{aligned}$$

we obtain an approximation of the filtered vorticity field,

$$\overline{\boldsymbol{\omega}}\,(\mathbf{x}) \approx \sum_p \boldsymbol{\Gamma}_p \zeta_\sigma(\mathbf{x} - \mathbf{x}_p). \tag{1.7}$$

As seen in Eq. (1.7), the filter operator has the effect of spreading the vortex strength $\boldsymbol{\Gamma}_p$ in space, regularizing the singularity originally introduced by the Dirac delta in Eq. (1.6). Thus, the filter kernel takes the role of a basis function that is used to discretize and approximate the filtered vorticity field through particles. We let the filter width σ change in time and space according to the evolution of each individual particle. Here on, the filter width is referred to as *smoothing radius* or *core size*, denoted σ_p, and defined as $\sigma_p(t) \equiv \sigma(\mathbf{x}_p, t)$. We approximate the filtered vorticity as $\overline{\boldsymbol{\omega}} \approx \boldsymbol{\omega}_\sigma$, with

$$\boldsymbol{\omega}_\sigma\,(\mathbf{x},\, t) \equiv \sum_p \boldsymbol{\Gamma}_p(t) \zeta_{\sigma_p}(\mathbf{x} - \mathbf{x}_p(t))$$

and

$$\zeta_{\sigma_p}(\mathbf{x}-\mathbf{x}_p(t)) = \frac{1}{\sigma_p^3(t)}\zeta\left(\frac{\|\mathbf{x}-\mathbf{x}_p(t)\|}{\sigma_p(t)}\right).$$

Replacing the original filter ζ_σ with the variable-width filter ζ_{σ_p} introduces an error from commuting filter and differentiation operators in the LES equations. However, this error is second order in the filter width,[29] hence it is assumed to be negligible or part of the discretization error of the VPM. Also, Eq. (1.7) introduces numerical issues by approximating $\overline{\boldsymbol{\omega}}$ through a field that is in general not divergence-free. Cottet[30] suggested that the divergence of $\overline{\boldsymbol{\omega}}$ arises from unphysical small scales produced by the computation, which could be properly addressed with subfilter-scale diffusion. The divergence of $\overline{\boldsymbol{\omega}}$ will be treated using a modified version of the relaxation proposed by Pedrizzetti[31] developed in Section 4.4, which, as shown in Chapter 5, will be sufficient to attain both numerical stability and physical accuracy with the LES formulation and subfilter-scale model developed in this study.

The rotational part of the velocity field is calculated from the particles using the Helmholtz decomposition $\overline{\mathbf{u}} = \nabla \times \boldsymbol{\psi}$ where $\boldsymbol{\psi}$ is some vector potential. This is done by analytically solving the Poisson equation $\nabla^2\boldsymbol{\psi} = -\overline{\boldsymbol{\omega}}$, which solution is a regularized Biot-Savart law, as discussed in Section 4.1. Fig. 1.1 shows a vortex particle with unit size and strength, the spreading of the vortex strength by the filter kernel, and the resulting velocity field.

29 S. Ghosal et al. (1995). "The Basic Equations for the Large Eddy Simulation of Turbulent Flows in Complex Geometry".

30 G.-H. Cottet and P. D. Koumoutsakos (2000a). *Vortex Methods Theory and Practice*.

31 Pedrizzetti, "Insight into singular vortex flows".

1.5 General VPM Governing Equations

We will now use vortex particles to discretize the LES vorticity equation, Eq. (1.5), and derive the equations governing the evolution of the Lagrangian elements. For ease of notation, here on we denote the filtered velocity field $\overline{\mathbf{u}}$ simply $\mathbf{u}$ and use $\frac{\mathrm{D}}{\mathrm{D}t}$ and $\frac{\mathrm{d}}{\mathrm{d}t}$ interchangeably. Also, time dependence is no longer explicitly indicated, but $\mathbf{x}_p$, $\boldsymbol{\Gamma}_p$, and σ_p are time-dependent variables.

Starting from the inviscid part of the LES-filtered vorticity equation,

$$\frac{\mathrm{d}}{\mathrm{d}t}\overline{\boldsymbol{\omega}} = (\overline{\boldsymbol{\omega}}\cdot\nabla)\,\mathbf{u} - \mathbf{E}_{\text{adv}} - \mathbf{E}_{\text{str}},$$

we write the filter operator explicitly,

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\int_{-\infty}^{\infty}\boldsymbol{\omega}(\mathbf{y})\,\zeta_\sigma(\mathbf{x}-\mathbf{y})\,\mathrm{d}\mathbf{y}\right) = \left[\left(\int_{-\infty}^{\infty}\boldsymbol{\omega}(\mathbf{y})\,\zeta_\sigma(\mathbf{x}-\mathbf{y})\,\mathrm{d}\mathbf{y}\right)\cdot\nabla\right]\mathbf{u}(\mathbf{x}) - \mathbf{E}_{\text{adv}}(\mathbf{x}) - \mathbf{E}_{\text{str}}(\mathbf{x}).$$

Using the singular particle approximation,

$$\boldsymbol{\omega}(\mathbf{y}) \approx \sum_q \boldsymbol{\Gamma}_q \delta(\mathbf{y}-\mathbf{x}_q),$$

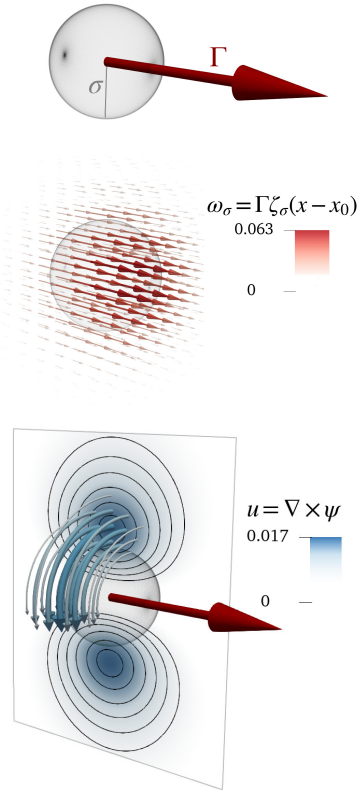


Figure 1.1: Vortex particle with unit size and strength: (top) core size σ and vortex strength Γ, (middle) vorticity field, and (bottom) contours of velocity field and streamlines (represented as curved arrows).

both integrals collapse resulting in

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\sum_q \boldsymbol{\Gamma}_q \zeta_{\sigma_q}(\mathbf{x}-\mathbf{x}_q)\right) = \left[\left(\sum_q \boldsymbol{\Gamma}_q \zeta_{\sigma_q}(\mathbf{x}-\mathbf{x}_q)\right)\cdot\nabla\right]\mathbf{u}(\mathbf{x}) - \mathbf{E}_{\mathrm{adv}}(\mathbf{x}) - \mathbf{E}_{\mathrm{str}}(\mathbf{x}). \tag{1.8}$$

With the help of Appendix A.1, the left-hand side unfolds as

$$\begin{aligned}
&\frac{\mathrm{d}}{\mathrm{d}t}\left(\sum_q \boldsymbol{\Gamma}_q \zeta_{\sigma_q}(\mathbf{x}-\mathbf{x}_q)\right) \\
&\qquad = \sum_q \left[\frac{\mathrm{d}\boldsymbol{\Gamma}_q}{\mathrm{d}t}\zeta_{\sigma_q}(\mathbf{x}-\mathbf{x}_q) + \boldsymbol{\Gamma}_q \frac{\mathrm{d}}{\mathrm{d}t}\left(\zeta_{\sigma_q}(\mathbf{x}-\mathbf{x}_q)\right)\right] \\
&\qquad = \sum_q \left\{\frac{\mathrm{d}\boldsymbol{\Gamma}_q}{\mathrm{d}t}\zeta_{\sigma_q}(\mathbf{x}-\mathbf{x}_q) + \boldsymbol{\Gamma}_q \frac{\partial\zeta_{\sigma_q}}{\partial t}(\mathbf{x}-\mathbf{x}_q)\right. \\
&\qquad\qquad \left. + \boldsymbol{\Gamma}_q\left[\left(\mathbf{u}(\mathbf{x})-\mathbf{u}(\mathbf{x}_q)\right)\cdot\nabla\right]\zeta_{\sigma_q}(\mathbf{x}-\mathbf{x}_q)\right\}
\end{aligned}$$

while vortex stretching is rearranged as

$$\left[\left(\sum_q \boldsymbol{\Gamma}_q \zeta_{\sigma_q}(\mathbf{x}-\mathbf{x}_q)\right)\cdot\nabla\right]\mathbf{u}(\mathbf{x}) = \sum_q \zeta_{\sigma_q}(\mathbf{x}-\mathbf{x}_q)\left(\boldsymbol{\Gamma}_q\cdot\nabla\right)\mathbf{u}(\mathbf{x}).$$

Evaluating at the position of the p-th particle, $\mathbf{x} = \mathbf{x}_p$, and pulling the p-th term out of each sum,

$$\begin{aligned}
&\frac{\mathrm{d}}{\mathrm{d}t}\left(\sum_q \boldsymbol{\Gamma}_q \zeta_{\sigma_q}(\mathbf{x}_p-\mathbf{x}_q)\right) = \\
&\quad \frac{\mathrm{d}\boldsymbol{\Gamma}_p}{\mathrm{d}t}\zeta_{\sigma_p}(\mathbf{0}) + \boldsymbol{\Gamma}_p\frac{\partial\zeta_{\sigma_p}}{\partial t}(\mathbf{0}) \\
&\qquad + \sum_{q\neq p}\left\{\frac{\mathrm{d}\boldsymbol{\Gamma}_q}{\mathrm{d}t}\zeta_{\sigma_q}(\mathbf{x}_p-\mathbf{x}_q) + \boldsymbol{\Gamma}_q\frac{\partial\zeta_{\sigma_q}}{\partial t}(\mathbf{x}_p-\mathbf{x}_q)\right. \\
&\qquad\qquad \left. + \boldsymbol{\Gamma}_q\left[\left(\mathbf{u}(\mathbf{x}_p)-\mathbf{u}(\mathbf{x}_q)\right)\cdot\nabla\right]\zeta_{\sigma_q}(\mathbf{x}_p-\mathbf{x}_q)\right\}
\end{aligned}$$

and

$$\begin{aligned}
&\left[\left(\sum_q \boldsymbol{\Gamma}_q \zeta_{\sigma_q}(\mathbf{x}_p-\mathbf{x}_q)\right)\cdot\nabla\right]\mathbf{u}(\mathbf{x}_p) = \\
&\qquad \zeta_{\sigma_p}(\mathbf{0})\left(\boldsymbol{\Gamma}_p\cdot\nabla\right)\mathbf{u}(\mathbf{x}_p) + \sum_{q\neq p}\zeta_{\sigma_q}(\mathbf{x}_p-\mathbf{x}_q)\left(\boldsymbol{\Gamma}_q\cdot\nabla\right)\mathbf{u}(\mathbf{x}_p).
\end{aligned}$$

Thus, Eq. (1.8) evaluated at $\mathbf{x} = \mathbf{x}_p$ becomes

$$\frac{\mathrm{d}\boldsymbol{\Gamma}_p}{\mathrm{d}t}\zeta_{\sigma_p}(\mathbf{0}) + \boldsymbol{\Gamma}_p\frac{\partial\zeta_{\sigma_p}}{\partial t}(\mathbf{0}) + \mathbf{M}_p^0 = \zeta_{\sigma_p}(\mathbf{0})\left(\boldsymbol{\Gamma}_p\cdot\nabla\right)\mathbf{u}(\mathbf{x}_p) + \left(\mathbf{M}_p^1 + \mathbf{M}_p^2\right) - \left(\mathbf{E}_{\mathrm{adv}}(\mathbf{x}_p) + \mathbf{E}_{\mathrm{str}}(\mathbf{x}_p)\right), \tag{1.9}$$

where

- $\mathbf{M}_p^0 \equiv \sum_{q\neq p}\left(\frac{\mathrm{d}\boldsymbol{\Gamma}_q}{\mathrm{d}t}\zeta_{\sigma_q}(\mathbf{x}_p-\mathbf{x}_q)+\boldsymbol{\Gamma}_q\frac{\partial\zeta_{\sigma_q}}{\partial t}(\mathbf{x}_p-\mathbf{x}_q)\right)$ (1.10)
- $\mathbf{M}_p^1 \equiv -\sum_{q\neq p}\boldsymbol{\Gamma}_q\left(\mathbf{u}(\mathbf{x}_p)-\mathbf{u}(\mathbf{x}_q)\right)\cdot\nabla\zeta_{\sigma_q}(\mathbf{x}_p-\mathbf{x}_q)$ (1.11)
- $\mathbf{M}_p^2 \equiv \sum_{q\neq p}\zeta_{\sigma_q}(\mathbf{x}_p-\mathbf{x}_q)\left(\boldsymbol{\Gamma}_q\cdot\nabla\right)\mathbf{u}(\mathbf{x}_p)$ (1.12)

These **M**-terms pose an interdependence between the p-th particle and neighboring particles, which arises from filtering the vorticity equation. In this study we will neglect this interdependence as explained in Section 1.6. However, we will see in Chapter 3 that $\mathbf{M}_p^1$ and $\mathbf{M}_p^2$ are closely related to advection and vortex stretching at the subfilter scale.

Recalling that ζ_σ is defined as $\zeta_\sigma(\mathbf{x}) = \frac{1}{\sigma^3}\zeta(\frac{\|\mathbf{x}\|}{\sigma})$, its time derivative is calculated as

$$\begin{aligned}\frac{\partial\zeta_\sigma}{\partial t}(\mathbf{x}) &= -3\frac{1}{\sigma^4}\frac{\partial\sigma}{\partial t}\zeta\left(\frac{\|\mathbf{x}\|}{\sigma}\right)+\frac{1}{\sigma^3}\frac{\partial}{\partial t}\left(\zeta\left(\frac{\|\mathbf{x}\|}{\sigma}\right)\right)\\ &= -3\frac{1}{\sigma}\frac{\partial\sigma}{\partial t}\zeta_\sigma(\mathbf{x})-\frac{1}{\sigma^3}\frac{\partial\zeta}{\partial r}\left(\frac{\|\mathbf{x}\|}{\sigma}\right)\frac{\|\mathbf{x}\|}{\sigma^2}\frac{\partial\sigma}{\partial t}.\end{aligned}$$

Assuming that $\zeta(r)$ reaches a maximum at $r=0$ (*i.e.*, $\frac{\partial\zeta}{\partial r}(0)=0$) and evaluating at $\mathbf{x}=\mathbf{0}$, we get

$$\frac{\partial\zeta_\sigma}{\partial t}(\mathbf{0}) = -3\frac{1}{\sigma}\frac{\partial\sigma}{\partial t}\zeta_\sigma(\mathbf{0}). \tag{1.13}$$

Since viscous effects have been set aside through operator splitting, $\frac{\partial\sigma}{\partial t}$ in this derivation only accounts for inviscid effects. For clarity, this means that core spreading due to viscous diffusion must not be included in Eq. (1.13).

Finally, substituting Eq. (1.13) into Eq. (1.9) and assuming $\zeta_{\sigma_p}(\mathbf{0})\neq 0$, we arrive to the equation governing the evolution of vortex strength,

$$\begin{aligned}\frac{\mathrm{d}\boldsymbol{\Gamma}_p}{\mathrm{d}t} = \left(\boldsymbol{\Gamma}_p\cdot\nabla\right)\mathbf{u}(\mathbf{x}_p)&+3\boldsymbol{\Gamma}_p\frac{1}{\sigma_p}\frac{\partial\sigma_p}{\partial t}\\ &+\frac{1}{\zeta_{\sigma_p}(\mathbf{0})}(-\mathbf{M}_p^0+\mathbf{M}_p^1+\mathbf{M}_p^2)\\ &-\frac{1}{\zeta_{\sigma_p}(\mathbf{0})}\left(\mathbf{E}_{\mathrm{adv}}(\mathbf{x}_p)+\mathbf{E}_{\mathrm{str}}(\mathbf{x}_p)\right).\end{aligned} \tag{1.14}$$

Thus, particle convection as in

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathbf{x}_p = \mathbf{u}(\mathbf{x}_p), \tag{1.15}$$

strength evolution as in Eq. (1.14), some expression for $\frac{\partial\sigma_p}{\partial t}$, and the viscous diffusion equation make up the general governing equations

of the vortex particle method that solve the LES-filtered Navier-Stokes vorticity equation.

We see in Eq. (1.14) that the evolution of the vortex strength is dictated by vortex stretching (first term), inviscid expansion/contraction of the particle size σ_p (second term), a dependence on other particles through the **M** terms, and SFS contributions through the **E** terms. One obtains different formulations of the VPM depending on how $\frac{\partial \sigma}{\partial t}$, the **M** terms, and the SFS **E** terms are handled. In particular, we will see in Section 1.6 that the classic VPM is equivalent to assuming[32] $\frac{\partial \sigma}{\partial t} = 0$ while neglecting all **M** terms. In Chapter 2 we propose a new formulation that uses $\frac{\partial \sigma}{\partial t} \neq 0$ to reinforce conservations laws in spherical elements. In Chapter 3 we discuss existing SFS models for meshless schemes and develop a new anisotropic dynamic model of SFS vortex stretching, $\mathbf{E}_{\text{str}}$.

32 The only instance that the classic VPM uses $\frac{\partial \sigma}{\partial t} \neq 0$ is in solving the viscous diffusion equation through the core spreading scheme, but even in this case the effects of $\frac{\partial \sigma}{\partial t}$ on vortex strength are neglected.

1.6 Classic VPM Governing Equations

Before further exploring the general equation governing vortex stretching, Eq. (1.14), we pause to analyze the equation that has been used extensively in the literature throughout the years:

$$\frac{\mathrm{d}\boldsymbol{\Gamma}_p}{\mathrm{d}t} = \left(\boldsymbol{\Gamma}_p \cdot \nabla\right) \mathbf{u}(\mathbf{x}_p). \tag{1.16}$$

We denote this equation as the *Classic VPM*.

Multiple variations of Eq. (1.16) have been used over the years. For instance, Gharakhani[33] introduced a new term accounting for the divergence of the approximated vorticity field, Winckelmans and Leonard[34] replaced the differential operator in vortex stretching with its transpose, and Mansfield et al.[35] and Cottet[36] reintroduced the SFS contributions. However, all these variations and the classic equation have in common that they all neglect $\frac{\partial \sigma_p}{\partial t}$ and are free of the interdependence posed by the **M**-terms in Eq. (1.14).

We now delineate the assumption underlying the classic VPM that will justify neglecting the **M**-terms, termed *localized-vorticity assumption*, as follows.

Given a vorticity field $\boldsymbol{\omega}$ that is compact in a small volume Vol, the average vorticity $\langle \boldsymbol{\omega} \rangle$ is calculated as

$$\langle \boldsymbol{\omega} \rangle = \frac{\int_{\text{Vol}} \boldsymbol{\omega}(\mathbf{y})\, \mathrm{d}\mathbf{y}}{\text{Vol}}.$$

The field can be approximated through a radially-symmetric field $\tilde{\boldsymbol{\omega}}$ defined as

$$\tilde{\boldsymbol{\omega}}(\mathbf{x}) \equiv \langle \boldsymbol{\omega} \rangle \, \text{Vol} \zeta_\sigma(\mathbf{x} - \mathbf{x}_0),$$

where ζ_σ is a radial basis function of spread σ and center $\mathbf{x}_0$ approximating the vorticity distribution of the original field $\boldsymbol{\omega}$. If ζ_σ is normalized

33 Gharakhani, *A Survey of Grid-Free Methods for the Simulation of 3-D Incompressible Flows in Bounded Domains*.

34 Winckelmans et al., "Contributions to Vortex Particle Methods for the Computation of Three-Dimensional Incompressible Unsteady Flows".

35 J. R. Mansfield et al. (1998). "A Dynamic LES Scheme for the Vorticity Transport Equation: Formulation and a Priori Tests"; J. R. Mansfield et al. (1999). "Dynamic LES of Colliding Vortex Rings Using a 3D Vortex Method".

36 G.-H. Cottet (1996). "Artificial Viscosity Models for Vortex and Particle Methods".

such as to have a volume integral of unity, $\tilde{\omega}$ approximates ω in an average sense since

$$\int_{-\infty}^{\infty} \tilde{\omega}(\mathbf{y})\,\mathrm{d}\mathbf{y} = \int_{\mathrm{Vol}} \omega(\mathbf{y})\,\mathrm{d}\mathbf{y}.$$

Defining $\boldsymbol{\Gamma} = \langle\omega\rangle\,\mathrm{Vol}$, the localized-vorticity field ω can be approximated with a single particle as

$$\omega(\mathbf{x}) \approx \boldsymbol{\Gamma}\zeta_\sigma(\mathbf{x}-\mathbf{x}_0).$$

Replacing this in the inviscid part of the vorticity equation,

$$\frac{\mathrm{d}}{\mathrm{d}t}\omega = (\omega\cdot\nabla)\mathbf{u},$$

evaluating at $\mathbf{x} = \mathbf{x}_0$, and following similar steps as in Section 1.5, we get

$$\frac{\mathrm{d}\boldsymbol{\Gamma}}{\mathrm{d}t} = (\boldsymbol{\Gamma}\cdot\nabla)\,\mathbf{u}(\mathbf{x}_0) + 3\boldsymbol{\Gamma}\frac{1}{\sigma}\frac{\partial\sigma}{\partial t}. \tag{1.17}$$

This is the evolution equation of the vortex strength approximating the field of localized vorticity, discretized with only one particle. In contrast, in Section 1.5 we showed that a more general vorticity field that is discretized using multiple particles leads to a governing equation that is slightly different:

$$\begin{aligned}\frac{\mathrm{d}\boldsymbol{\Gamma}_p}{\mathrm{d}t} = (\boldsymbol{\Gamma}_p\cdot\nabla)\,\mathbf{u}(\mathbf{x}_p) + 3\boldsymbol{\Gamma}_p\frac{1}{\sigma_p}\frac{\partial\sigma_p}{\partial t} \\ + \frac{1}{\zeta_{\sigma_p}(\mathbf{0})}(-\mathbf{M}_p^0 + \mathbf{M}_p^1 + \mathbf{M}_p^2),\end{aligned} \tag{1.18}$$

after ignoring the SFS contributions. Here, the **M**-terms, as defined in Eqs. (1.10) to (1.12), pose a dependence on neighboring particles that arises from having filtered the vorticity equation.

Discarding these **M**-terms is equivalent to assuming that the vorticity field can be approximated by the superposition of blobs of fluid with compact vorticity that evolve somewhat independently from each other. [37] This assumption, which we call *localized-vorticity assumption*, reduces Eq. (1.18) into Eq. (1.17) and is valid as long as there is no significant particle overlap. In both derivations of the classic VPM by Winckelmans and Leonard[38] and Cottet and Koumutsakos,[39] these **M**-terms are not present due to the construction of the method: In the classic derivation, the unfiltered vorticity equation (Eq. (1.2)) is discretized with singular particles and only the velocity field is filtered to obtain a regularized field, while in our derivation we discretized the LES-filtered vorticity equation (Eq. (1.5)) which leads to Eq. (1.18). In order to bring both approaches into agreement, in Chapter 2 we will use the localized-vorticity assumption in the derivation of our reformulated VPM to neglect these **M**-terms.

37 The interactions neglected here are only the ones pertaining to the evolution of vortex strength, while other interactions are still accounted for in the convection and viscous diffusion of the particles.

38 Winckelmans et al., "Contributions to Vortex Particle Methods for the Computation of Three-Dimensional Incompressible Unsteady Flows".

39 G.-H. Cottet and P. D. Koumoutsakos (2000b). *Vortex Methods Theory and Practice*.

Interestingly, discarding the **M**-terms resembles the LES decomposition approach of truncated basis functions.[40] In such an approach, the flow field is expanded using orthonormal basis functions. The summation of bases is then truncated to define the large-scale field, and the discarded modes represent the range of subfilter scales. In the localized-vorticity assumption, the sum over all the particles is truncated after the leading term, p, while neglecting the contributions of neighboring particles. Hence, the localized-vorticity assumption can be regarded as a secondary LES filter with the neglected **M**-terms becoming part of the subfilter-scale contributions.

While it is justifiable to neglect the **M**-terms through the localized-vorticity assumption, it is unclear to us what the basis is for the classic VPM to assume $\frac{\partial \sigma_p}{\partial t} = 0$. In fact, we hypothesize that this last assumption is the cause of the numerical instabilities that pervade the classic VPM.

The classic VPM simply regards σ as a numerical parameter with no physical significance. However, Leonard[41] suggested that σ should change according to conservation of mass, and Nakanishi, Ojima, and Maremoto[42] suggested that σ should change according to Kelvin's theorem, which was more recently implemented by Kornev *et al.*[43] Even though these authors let σ evolve in time, they did not include these effects back in the equation that governs $\mathbf{\Gamma}$, effectively assuming $\frac{\partial \sigma_p}{\partial t} = 0$ in the evolution of $\mathbf{\Gamma}$. In the following chapter we propose a formulation that uses $\frac{\partial \sigma_p}{\partial t} \neq 0$ in the governing equation of $\mathbf{\Gamma}$, while letting σ change as to reinforce conservation of both mass and angular momentum, which will be shown to lead to remarkable numerical stability.

41 A. Leonard (1980). "Vortex methods for flow simulation".

42 Y. Nakanishi et al. (1993). "Numerical Simulation of Flow around a Sphere with Vortex Blobs"; A. Ojima et al. (2000). "Numerical Simulation of Unsteady Flow around Three Dimensional Bluff Bodies by an Advanced Vortex Method."

43 N. Kornev et al. (2020). "Theoretical Background of the Hybrid VpiLES Method for Flows with Variable Transport Properties".

2 *Reformulated Vortex Particle Method*

The general governing equations derived in Section 1.5 call for an expression for $\frac{\partial \sigma_p}{\partial t}$. While the classic VPM simply assumes $\frac{\partial \sigma_p}{\partial t} = 0$, we will now go back to first principles to find some plausible expressions for $\frac{\partial \sigma_p}{\partial t}$. As a preamble, in Section 2.1 we put forth two physical implications that distill from the vorticity equation, which will then guide our search for possible candidate formulations of $\frac{\partial \sigma_p}{\partial t}$ in Section 2.2. The process for constructing candidate formulation is then generalized in Section 2.3, and the entire space of possible formulation is analyzed in Section 2.4. This will then lead us to one formulation, termed *reformulated VPM* and summarized in Section 2.5, which reinforces conservation of both mass and angular momentum by reshaping the vortex elements as they are subject to vortex stretching.

2.1 Physical Implications of the Vorticity Equation

In Section 1.1 we showed that the linear-momentum Navier-Stokes equation can be transformed into an expression that only depends on vorticity[44], namely Eq. (1.2), here repeated:

$$\frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{\omega} = (\boldsymbol{\omega}\cdot\nabla)\mathbf{u} + \nu\nabla^2\boldsymbol{\omega}. \tag{1.2}$$

44 It only depends on vorticity since the velocity field can be calculated from the vorticity field by inverting the relation $\boldsymbol{\omega} = \nabla \times \mathbf{u}$ as shown in Section 4.1.

We now point our attention to two laws that follow from Eq. (1.2): conservation of mass and angular momentum. The discussion that follows is inspired by the writing of P. A. Davidson.[45]

45 P. A. Davidson (2001). *An Introduction to Magnetohydrodynamics*; P. A. Davidson (2019). *An Introduction to Electrodynamics*.

2.1.1 Conservation of Angular Momentum

Consider a spherical differential fluid element carrying a mean vorticity $\boldsymbol{\omega}$ with moment of inertia I. Due to $\boldsymbol{\omega}$, the element is then rotating at an angular velocity of $\boldsymbol{\omega}/2$ and its angular momentum $\mathbf{L}$ is calculated as[46]

$$\mathbf{L} = \frac{I\boldsymbol{\omega}}{2}. \tag{2.1}$$

46 G. K. Batchelor (1967). *An Introduction to Fluid Dynamics*, p. 82.

Given that the element is spherical (and before strain distorts the element), the pressure field exerts no torque on the element and the only torque-producing forces are due to viscous effects, namely $\boldsymbol{\tau}_{\text{viscous}}$. The change of angular momentum is then calculated as

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathbf{L} = \boldsymbol{\tau}_{\text{viscous}}. \tag{2.2}$$

Replacing Eq. (2.1) in Eq. (2.2), we arrive to the following expression of the total derivative of vorticity[47],

$$\frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{\omega} = -\frac{\boldsymbol{\omega}}{I}\frac{\mathrm{d}}{\mathrm{d}t}I + \boldsymbol{\tau}^*_{\text{viscous}}, \tag{2.3}$$

where $\boldsymbol{\tau}^*_{\text{viscous}} = 2\boldsymbol{\tau}_{\text{viscous}}/I$. When viscous effects are ignored, Eq. (2.3) implies that, in order to conserve angular momentum, the angular velocity (or vorticity) must decrease whenever the moment of inertia increases, and vice versa.

47 More generally, this equation should be expressed with an inertia tensor, but for a spherical body this tensor reduces to a scalar due to the symmetry of the body.

Eqs. (1.2) and (2.3) suggest that the vorticity Navier-Stokes equation is simply an expression of the conservation of angular momentum in a spherical fluid element.[48] Furthermore, the first term in the right hand side of Eq. (1.2) accounts for the change of moment of inertia as

48 Chatwin, "The vorticity equation as an angular momentum equation".

$$(\boldsymbol{\omega}\cdot\nabla)\mathbf{u} = -\frac{\boldsymbol{\omega}}{I}\frac{\mathrm{d}}{\mathrm{d}t}I, \tag{2.4}$$

which leads to an increase/decrease of vorticity to conserve momentum as the moment of inertia decreases/increases. Hence, $(\boldsymbol{\omega}\cdot\nabla)\mathbf{u}$ is referred to as vortex stretching as it accounts for the deformation exerted by the velocity field on the fluid element, intensifying the vorticity in the direction that the element is stretched, as illustrated in Fig. 2.1.

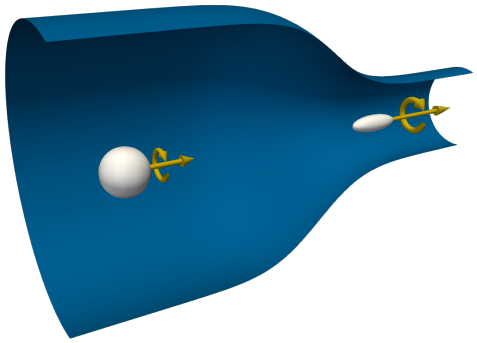


Figure 2.1: Stretching of a spherical fluid element and increase in vorticity (or angular velocity).

2.1.2 Conservation of Mass

Another physical implication of Eq. (1.2) is the existence of vortex lines and vortex tubes, described as follows. We define a short material line $\boldsymbol{\ell}$ that moves with the fluid, with start and end points $\mathbf{x}$ and $\mathbf{x}+\boldsymbol{\ell}$, respectively. If $\boldsymbol{\ell}$ is infinitesimally small, the evolution of $\boldsymbol{\ell}$ is given by

$$\frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{\ell} = \mathbf{u}(\mathbf{x}+\boldsymbol{\ell}) - \mathbf{u}(\mathbf{x}) = (\boldsymbol{\ell}\cdot\nabla)\,\mathbf{u}(\mathbf{x}). \tag{2.5}$$

Defining $\boldsymbol{\ell}$ as being tangent to the local vorticity, and comparing Eq. (2.5) to the inviscid part of Eq. (1.2), we see that vorticity evolves as the material line, being identically stretched and reoriented by the velocity field. This led Helmholtz to conclude the existence of lines of vorticity, termed *vortex lines*, that move with the fluid. We then define a *vortex tube*

as the surface formed by all the vortex lines passing through a closed curve, depicted in Fig. 2.2.

The fact that vortex lines move with the fluid implies that the volume of a vortex tube segment must be conserved in incompressible flow. For instance, let us discretize and approximate a vortex tube through Lagrangian cylindrical elements of length ℓ and cross section r. As the length of the element is stretched, the cross section r must shrink to conserve the same volume $\text{Vol} = \pi r^2 \ell$. This is,

$$\frac{\mathrm{d}}{\mathrm{d}t}\text{Vol} = 0,$$

leading to

$$\frac{\mathrm{d}}{\mathrm{d}t}r = -\frac{r}{2\ell}\frac{\mathrm{d}}{\mathrm{d}t}\ell. \tag{2.6}$$

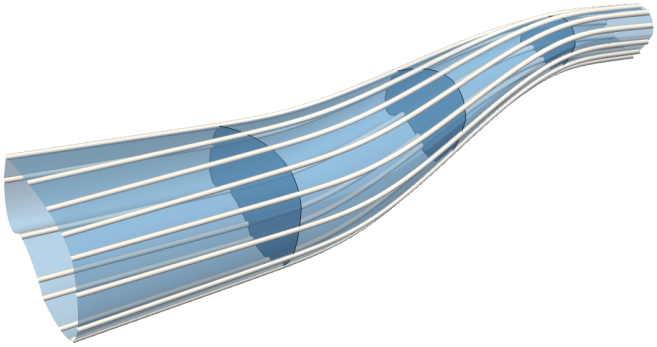


Figure 2.2: Vortex tube formed by the vortex lines passing through a closed curve.

2.1.3 Implications for the VPM

The aforementioned laws of conservation point our attention to two major pitfalls of the classic VPM, if particles are considered as material elements. First, Eq. (2.4) establishes that vortex stretching must lead to a change in the shape of the fluid element. Otherwise, if the vorticity (angular velocity) has changed due to vortex stretching while the shape of the element (moment of inertia) is kept constant, the conservation of angular momentum has been locally violated. This is exactly the case when the classic VPM assumes a constant core size, $\frac{\mathrm{d}}{\mathrm{d}t}\sigma = 0$.

Second, in an incompressible tube of vorticity, Eqs. (1.2) and (2.5) establish that an increase in vorticity due to vortex stretching is associated to a lengthening of the tube, which must cause the cross section of the tube to shrink according to Eq. (2.6). Otherwise, the tube has kept a constant cross section while being stretched, thus locally violating conservation of mass. Again, this is exactly the case when the classic VPM assumes $\frac{\mathrm{d}}{\mathrm{d}t}\sigma = 0$.

In summary, we believe that the classic VPM violates conservation of both angular momentum and mass when it assumes $\frac{\mathrm{d}}{\mathrm{d}t}\sigma = 0$. We also hypothesize that this is the reason for its poor numerical stability. In the following sections we will consider some possible formulations that use $\frac{\mathrm{d}}{\mathrm{d}t}\sigma$ as a degree of freedom to locally reinforce these conservation laws.

2.2 Candidate Formulations

Starting from the general equation governing vortex strength (Eq. (1.14)), neglecting SFS terms, and using the localized-vorticity assumption, we obtain

$$\frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{\Gamma}_p = \left(\boldsymbol{\Gamma}_p \cdot \nabla\right)\mathbf{u}(\mathbf{x}_p) + 3\boldsymbol{\Gamma}_p \frac{1}{\sigma_p}\frac{\partial \sigma_p}{\partial t}. \tag{2.7}$$

We will now explore multiple combinations of element shapes and conservation laws to derive possible expressions for $\frac{\partial \sigma_p}{\partial t}$.

2.2.1 Sphere with Conservation of Momentum

Let us consider a spherical fluid element with angular velocity $\boldsymbol{\omega}/2$. Given that the moment of inertia of a solid sphere is $I = \frac{2}{5}mr^2$, where $m = \frac{4}{3}\rho\pi r^3$ is the mass of the sphere and r its radius, the inertial term in Eq. (2.4) becomes

$$\frac{1}{I}\frac{\mathrm{d}}{\mathrm{d}t}I = \frac{1}{r^5}\frac{\mathrm{d}}{\mathrm{d}t}\left(r^5\right) = \frac{5}{r}\frac{\mathrm{d}}{\mathrm{d}t}r.$$

Replacing this into Eq. (2.4) and taking the dot product with $\boldsymbol{\omega}$,

$$-\left(\boldsymbol{\omega}\cdot\boldsymbol{\omega}\right)\frac{5}{r}\frac{\mathrm{d}}{\mathrm{d}t}r = \left[(\boldsymbol{\omega}\cdot\nabla)\mathbf{u}\right]\cdot\boldsymbol{\omega},$$

results in

$$\begin{aligned}\frac{1}{r}\frac{\mathrm{d}}{\mathrm{d}t}r &= -\frac{1}{5}\frac{1}{\|\boldsymbol{\omega}\|^2}\left[(\boldsymbol{\omega}\cdot\nabla)\mathbf{u}\right]\cdot\boldsymbol{\omega} \\ &= -\frac{1}{5}\left[(\hat{\boldsymbol{\omega}}\cdot\nabla)\mathbf{u}\right]\cdot\hat{\boldsymbol{\omega}}.\end{aligned}$$

When $\boldsymbol{\omega}$ is filtered and discretized with vortex particles, and the localized-vorticity assumption is applied, the spherical element becomes the vortex particle itself. Vortex strengths are roughly aligned with the vorticity field[49], meaning $\hat{\boldsymbol{\Gamma}}_p \approx \hat{\boldsymbol{\omega}}\left(\mathbf{x}_p\right)$, and the size of the particle can be expressed as $\sigma_p = \alpha r$ with α some scaling factor, obtaining

$$\frac{1}{\sigma_p}\frac{\mathrm{d}}{\mathrm{d}t}\sigma_p = -\frac{1}{5}\left[(\hat{\boldsymbol{\Gamma}}_p\cdot\nabla)\mathbf{u}\left(\mathbf{x}_p\right)\right]\cdot\hat{\boldsymbol{\Gamma}}_p,$$

[49] This is especially true when using Pedrizzetti's relaxation scheme, described in Section 4.4.

or, equivalently

$$\frac{\mathrm{d}}{\mathrm{d}t}\sigma_p = -\frac{1}{5}\frac{\sigma_p}{\|\boldsymbol{\Gamma}_p\|}\left[(\boldsymbol{\Gamma}_p\cdot\nabla)\mathbf{u}\left(\mathbf{x}_p\right)\right]\cdot\hat{\boldsymbol{\Gamma}}_p. \tag{2.8}$$

Noticing that, in a Lagrangian scheme, σ_p is only a function of time (*i.e.*, $\frac{\mathrm{d}}{\mathrm{d}t}\sigma_p = \frac{\partial\sigma_p}{\partial t}$), we replace Eq. (2.8) into Eq. (2.7) to obtain the equation governing $\boldsymbol{\Gamma}_p$:

$$\frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{\Gamma}_p = \left(\boldsymbol{\Gamma}_p\cdot\nabla\right)\mathbf{u}(\mathbf{x}_p) - \frac{3}{5}\left\{\left[(\boldsymbol{\Gamma}_p\cdot\nabla)\mathbf{u}\left(\mathbf{x}_p\right)\right]\cdot\hat{\boldsymbol{\Gamma}}_p\right\}\hat{\boldsymbol{\Gamma}}_p. \tag{2.9}$$

Thus, Eqs. (2.8) and (2.9) are the governing equations required for the particles to preserve angular momentum.

2.2.2 Tube with Conservation of Mass

Assuming that the field $\boldsymbol{\omega}$ encompasses a tube of vorticity of radius r, we now use conservation of mass to derive an expression for $\frac{\mathrm{d}}{\mathrm{d}t}\sigma$. When $\boldsymbol{\omega}$ is filtered and discretized through a vortex tube method, as described in Appendix B, the tubular segments become vortex tube elements of length ℓ and radius $\sigma_t = \alpha_t r$ with α_t some scaling factor. Then, Eq. (2.6) becomes

$$\frac{1}{\sigma_t}\frac{\mathrm{d}}{\mathrm{d}t}\sigma_t = -\frac{1}{2\ell}\frac{\mathrm{d}}{\mathrm{d}t}\ell. \tag{2.10}$$

The vortex particle method and the vortex tube method are closely related, where the latter can be thought of as a line integral of the former. In Appendix B.5, we show that $\frac{\mathrm{d}\ell}{\mathrm{d}t}$ in the vortex tube scheme relates to $\frac{\mathrm{d}\boldsymbol{\Gamma}}{\mathrm{d}t}$ in the vortex particle scheme through

$$\frac{1}{\ell}\frac{\mathrm{d}}{\mathrm{d}t}\ell = \frac{1}{\|\boldsymbol{\Gamma}\|}\frac{\mathrm{d}\boldsymbol{\Gamma}}{\mathrm{d}t}\cdot\hat{\boldsymbol{\Gamma}}.$$

Hence, the analogous to Eq. (2.10) in the vortex particle scheme is

$$\frac{1}{\sigma_p}\frac{\mathrm{d}}{\mathrm{d}t}\sigma_p = -\frac{1}{2\|\boldsymbol{\Gamma}_p\|}\frac{\mathrm{d}\boldsymbol{\Gamma}_p}{\mathrm{d}t}\cdot\hat{\boldsymbol{\Gamma}}_p. \tag{2.11}$$

Using Eq. (2.11) in the vortex strength equation, Eq. (2.7), we obtain

$$\frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{\Gamma}_p = \left(\boldsymbol{\Gamma}_p\cdot\nabla\right)\mathbf{u}(\mathbf{x}_p) - \frac{3}{2}\left(\frac{\mathrm{d}\boldsymbol{\Gamma}_p}{\mathrm{d}t}\cdot\hat{\boldsymbol{\Gamma}}_p\right)\hat{\boldsymbol{\Gamma}}_p, \tag{2.12}$$

and from its dot product with $\hat{\boldsymbol{\Gamma}}_p$ we get

$$\frac{\mathrm{d}\boldsymbol{\Gamma}_p}{\mathrm{d}t}\cdot\hat{\boldsymbol{\Gamma}}_p = \frac{2}{5}\left[\left(\boldsymbol{\Gamma}_p\cdot\nabla\right)\mathbf{u}(\mathbf{x}_p)\right]\cdot\hat{\boldsymbol{\Gamma}}_p.$$

Finally, substituting this back into Eqs. (2.11) and (2.12), we obtain the governing equations required for the particles to preserve mass in a tube of vorticity:

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}t}\sigma_p &= -\frac{1}{5}\frac{\sigma_p}{\|\boldsymbol{\Gamma}_p\|}\left[\left(\boldsymbol{\Gamma}_p\cdot\nabla\right)\mathbf{u}(\mathbf{x}_p)\right]\cdot\hat{\boldsymbol{\Gamma}}_p \\
\frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{\Gamma}_p &= \left(\boldsymbol{\Gamma}_p\cdot\nabla\right)\mathbf{u}(\mathbf{x}_p) - \frac{3}{5}\left\{\left[\left(\boldsymbol{\Gamma}_p\cdot\nabla\right)\mathbf{u}(\mathbf{x}_p)\right]\cdot\hat{\boldsymbol{\Gamma}}_p\right\}\hat{\boldsymbol{\Gamma}}_p
\end{aligned}$$

Surprisingly, we have arrived to the same governing equations of the momentum-conserving sphere formulation[50], Eqs. (2.8) and (2.9). This stems from the fact that both conservation laws are physical consequences of the same equation: the vorticity Navier-Stokes equation. These new governing equations, Eqs. (2.8) and (2.9), ensure local conservation of both angular momentum and mass, thus overcoming the pitfalls of the classic VPM.

50 With the caveat that this is so only because we have neglected the SFS term. For this reason, in Section 2.3 we will revert back to Eq. (2.11).

2.2.3 Other Formulations

Combining the two conservations laws with the two element shapes, one can devise up to six different formulations: Each permutation between sphere/tube elements, momentum/mass conservation, and the assumption that mass conservation has or has not been already ensured by the convection of the elements. Two of them have already been discussed and turn out to be the same. For completeness, we now derive and briefly discuss the four remaining formulations.

Mass-Conserving Sphere

The mass of a spherical fluid element is conserved in incompressible flow as long as its volume remains constant. This translates into vortex particles of constant radii,

$$\frac{\mathrm{d}}{\mathrm{d}t}\sigma_p = 0, \tag{2.13}$$

which reduces the vortex strength equation to

$$\frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{\Gamma}_p = \left(\boldsymbol{\Gamma}_p \cdot \nabla\right)\mathbf{u}(\mathbf{x}_p).$$

Notice that these are the equations of the classic VPM. Even though this formulation preserves mass of spherical chunks of the flow, it is deemed unphysical as it violates local conservation of mass when the vorticity field encompasses a tube of vorticity that is being stretched, as discussed in Section 2.1.

Momentum-Conserving Tube

Consider a cylindrical element that rotates with angular velocity $\boldsymbol{\Omega} = \boldsymbol{\omega}/2$. Let $\mathbb{I}$ denote its inertia tensor and I_z the moment of inertia about the centerline axis. If we assume that $\boldsymbol{\Omega}$ is always aligned with its centerline axis, the inertia tensor product $\mathbb{I}\boldsymbol{\Omega}$ becomes $\mathbb{I}\boldsymbol{\Omega} = I_z\boldsymbol{\Omega}$ since $\boldsymbol{\Omega}$ is an eigenvector of $\mathbb{I}$, with I_z the corresponding eigenvalue. In order to conserve angular momentum, neglecting pressure torque and following the discussion in Section 2.1.1, the moment of inertia must change according to

$$-\frac{\boldsymbol{\omega}}{I_z}\frac{\mathrm{d}}{\mathrm{d}t}I_z = (\boldsymbol{\omega}\cdot\nabla)\mathbf{u}. \tag{2.14}$$

Using $I_z = \frac{1}{2}mr^2$ and without assuming that the mass $m = \rho\pi r^2\ell$ is automatically conserved, the dot product of Eq. (2.14) with $\boldsymbol{\omega}$ leads to

$$\frac{\mathrm{d}}{\mathrm{d}t}r = -\frac{r}{4\ell}\frac{\mathrm{d}}{\mathrm{d}t}\ell - \frac{r}{4\|\boldsymbol{\omega}\|^2}\left[(\boldsymbol{\omega}\cdot\nabla)\mathbf{u}\right]\cdot\boldsymbol{\omega}.$$

In the vortex particle scheme this becomes

$$\frac{\mathrm{d}}{\mathrm{d}t}\sigma_p = -\frac{1}{4}\frac{\sigma_p}{\|\boldsymbol{\Gamma}_p\|}\frac{\mathrm{d}\boldsymbol{\Gamma}_p}{\mathrm{d}t}\cdot\hat{\boldsymbol{\Gamma}}_p - \frac{1}{4}\frac{\sigma_p}{\|\boldsymbol{\Gamma}_p\|}\left[(\boldsymbol{\Gamma}_p\cdot\nabla)\mathbf{u}\right]\cdot\hat{\boldsymbol{\Gamma}}_p, \tag{2.15}$$

which in conjunction to the vortex strength equation, Eq. (2.7), we obtain the following governing equations:

$$\frac{\mathrm{d}}{\mathrm{d}t}\sigma_p = -\frac{2}{7}\frac{\sigma_p}{\|\boldsymbol{\Gamma}_p\|}\left[(\boldsymbol{\Gamma}_p\cdot\nabla)\mathbf{u}\right]\cdot\hat{\boldsymbol{\Gamma}}_p$$
$$\frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{\Gamma}_p = \left(\boldsymbol{\Gamma}_p\cdot\nabla\right)\mathbf{u}(\mathbf{x}_p) - \frac{6}{7}\left\{\left[(\boldsymbol{\Gamma}_p\cdot\nabla)\mathbf{u}\right]\cdot\hat{\boldsymbol{\Gamma}}_p\right\}\hat{\boldsymbol{\Gamma}}_p.$$

Assumed Mass Conservation

Vortex methods typically assume that the continuity equation is implicitly solved by convecting the elements with the velocity field, automatically satisfying mass conservation. Repeating the momentum-conserving tube derivation, but this time assuming that the convection of the particles automatically ensures $\frac{\mathrm{d}}{\mathrm{d}t}m = 0$, we get the following governing equations:

$$\frac{\mathrm{d}}{\mathrm{d}t}\sigma_p = -\frac{1}{2}\frac{\sigma_p}{\|\boldsymbol{\Gamma}_p\|}\left[(\boldsymbol{\Gamma}_p\cdot\nabla)\mathbf{u}\right]\cdot\hat{\boldsymbol{\Gamma}}_p \tag{2.16}$$
$$\frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{\Gamma}_p = \left(\boldsymbol{\Gamma}_p\cdot\nabla\right)\mathbf{u}(\mathbf{x}_p) - \frac{3}{2}\left\{\left[(\boldsymbol{\Gamma}_p\cdot\nabla)\mathbf{u}\right]\cdot\hat{\boldsymbol{\Gamma}}_p\right\}\hat{\boldsymbol{\Gamma}}_p.$$

Taking the dot product of $\hat{\boldsymbol{\Gamma}}_p$ over the vortex strength equation, we get

$$\frac{\mathrm{d}\boldsymbol{\Gamma}_p}{\mathrm{d}t}\cdot\hat{\boldsymbol{\Gamma}}_p = -\frac{1}{2}\left[(\boldsymbol{\Gamma}_p\cdot\nabla)\mathbf{u}\right]\cdot\hat{\boldsymbol{\Gamma}}_p.$$

In this last equation, notice that the derivative of the strength projected on its own direction is negative when vortex stretching is positive. This means that the magnitude of vortex strength (and associated vorticity) decreases when the particle is stretched, which is an unphysical result.

Along the same lines, we can repeat the momentum-conserving sphere derivation from Section 2.2.1, but this time assuming automatic mass conservation, $\frac{\mathrm{d}}{\mathrm{d}t}m = 0$. This turns out to lead to the same governing equations in Eq. (2.16). Hence, we conclude that it is not possible to impose momentum conservation with either tube or sphere elements while assuming automatic mass conservation.

2.3 Generalized Formulation

In the preceding section, the conservation laws were imposed in six different ways to derive various expressions of $\frac{\mathrm{d}}{\mathrm{d}t}\sigma_p$, namely Eqs. (2.8), (2.11), (2.13), (2.15) and (2.16). In turn, each version of $\frac{\mathrm{d}}{\mathrm{d}t}\sigma_p$ was used with the vortex strength equation, Eq. (2.7), to obtain a distinct VPM formulation. We can generalize this formulation procedure noticing that all versions of $\frac{\mathrm{d}}{\mathrm{d}t}\sigma_p$ take the functional form

$$\frac{\mathrm{d}}{\mathrm{d}t}\sigma_p = -f\sigma_p\frac{1}{\|\boldsymbol{\Gamma}_p\|}\frac{\mathrm{d}\boldsymbol{\Gamma}_p}{\mathrm{d}t}\cdot\hat{\boldsymbol{\Gamma}}_p - g\sigma_p\frac{1}{\|\boldsymbol{\Gamma}_p\|}\left[\left(\boldsymbol{\Gamma}_p\cdot\nabla\right)\mathbf{u}(\mathbf{x}_p)\right]\cdot\hat{\boldsymbol{\Gamma}}_p, \tag{2.17}$$

where the parameters $f, g \in \mathbb{R}$ are derived applying the conservation laws.

We will now use the f–g form of $\frac{\mathrm{d}}{\mathrm{d}t}\sigma_p$, Eq. (2.17), to derive a new set of governing equation of the VPM, also in the f–g form. We start from the general equation governing vortex strength, Eq. (1.14), here repeated

$$\frac{\mathrm{d}\boldsymbol{\Gamma}_p}{\mathrm{d}t} = \left(\boldsymbol{\Gamma}_p \cdot \nabla\right)\mathbf{u}(\mathbf{x}_p) + 3\boldsymbol{\Gamma}_p \frac{1}{\sigma_p}\frac{\partial \sigma_p}{\partial t} + \mathbf{M}_p, \tag{1.14}$$

where

$$\begin{aligned}\mathbf{M}_p \equiv & \frac{1}{\zeta_{\sigma_p}(\mathbf{0})}(-\mathbf{M}_p^0 + \mathbf{M}_p^1 + \mathbf{M}_p^2) \\ & - \frac{1}{\zeta_{\sigma_p}(\mathbf{0})}\left(\mathbf{E}_{\text{adv}}(\mathbf{x}_p) + \mathbf{E}_{\text{str}}(\mathbf{x}_p)\right).\end{aligned}$$

Recalling that $\frac{\mathrm{d}\sigma_p}{\mathrm{d}t} = \frac{\partial \sigma_p}{\partial t}$ since σ_p is a Lagrangian quantity, we substitute Eq. (2.17) into Eq. (1.14) and take its dot product with $\hat{\boldsymbol{\Gamma}}_p$ to get

$$\frac{\mathrm{d}\boldsymbol{\Gamma}_p}{\mathrm{d}t}\cdot\hat{\boldsymbol{\Gamma}}_p = \frac{1-3g}{1+3f}\left[\left(\boldsymbol{\Gamma}_p\cdot\nabla\right)\mathbf{u}(\mathbf{x}_p)\right]\cdot\hat{\boldsymbol{\Gamma}}_p + \frac{1}{1+3f}\mathbf{M}_p\cdot\hat{\boldsymbol{\Gamma}}_p. \tag{2.18}$$

Substituting Eq. (2.18) into Eq. (2.17), we arrive to the equation governing the evolution of particle size,

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}t}\sigma_p = & -\left(\frac{g+f}{1+3f}\right)\frac{\sigma_p}{\|\boldsymbol{\Gamma}_p\|}\left[\left(\boldsymbol{\Gamma}_p\cdot\nabla\right)\mathbf{u}(\mathbf{x}_p)\right]\cdot\hat{\boldsymbol{\Gamma}}_p \\ & -\left(\frac{f}{1+3f}\right)\frac{\sigma_p}{\|\boldsymbol{\Gamma}_p\|}\mathbf{M}_p\cdot\hat{\boldsymbol{\Gamma}}_p,\end{aligned} \tag{2.19}$$

and we substitute this into Eq. (1.14) to get the equation governing the evolution of vortex strength,

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{\Gamma}_p = & \left(\boldsymbol{\Gamma}_p\cdot\nabla\right)\mathbf{u}(\mathbf{x}_p) - \frac{g+f}{1/3+f}\left\{\left[\left(\boldsymbol{\Gamma}_p\cdot\nabla\right)\mathbf{u}(\mathbf{x}_p)\right]\cdot\hat{\boldsymbol{\Gamma}}_p\right\}\hat{\boldsymbol{\Gamma}}_p \\ & + \mathbf{M}_p - \frac{f}{1/3+f}\left(\mathbf{M}_p\cdot\hat{\boldsymbol{\Gamma}}_p\right)\hat{\boldsymbol{\Gamma}}_p.\end{aligned} \tag{2.20}$$

Eqs. (2.19) and (2.20) constitute the governing equations of the reformulated VPM, where the parameters f, g are derived from a specific implementation of the conservations laws. In particular when $f = g = 0$, Eqs. (2.19) and (2.20) collapse back to the classic VPM equations, making the reformulated VPM a generalization of the classic method. Furthermore, notice that the reformulated equations do not require more computation than the classic method: When SFS and non-localized vorticity effects are neglected ($\mathbf{M}_p = 0$), both $\frac{\mathrm{d}\sigma_p}{\mathrm{d}t}$ and $\frac{\mathrm{d}\boldsymbol{\Gamma}_p}{\mathrm{d}t}$ are calculated directly from vortex stretching, $\left(\boldsymbol{\Gamma}_p\cdot\nabla\right)\mathbf{u}(\mathbf{x}_p)$.

2.4 Formulation Analysis

In Section 2.2 we have shown several plausible formulations that can be derived from the conservation laws. We now cast each formulation into its f–g form, as summarized in Table 2.1, and contrast the physical implications of each formulation to help us choose one formulation to further explore.

We wish to determine how each formulation may hinder or augment the effects of vortex stretching on both vortex strength and vortex re-orientation. First, we focus on vortex re-orientation. The rate of vortex re-orientation, denoted $\frac{\mathrm{d}\hat{\boldsymbol{\Gamma}}_p}{\mathrm{d}t}$, is constructed as

$$\frac{\mathrm{d}\hat{\boldsymbol{\Gamma}}_p}{\mathrm{d}t} = \frac{1}{\|\boldsymbol{\Gamma}_p\|}\frac{\mathrm{d}\boldsymbol{\Gamma}_p}{\mathrm{d}t} - \frac{1}{\|\boldsymbol{\Gamma}_p\|}\left(\frac{\mathrm{d}\boldsymbol{\Gamma}_p}{\mathrm{d}t}\cdot\hat{\boldsymbol{\Gamma}}_p\right)\hat{\boldsymbol{\Gamma}}_p,$$

which is the change in direction of vortex strength over time, as shown in Appendix A.2. Using Eq. (2.18) and Eq. (2.20) and neglecting $\mathbf{M}$, this becomes

$$\frac{\mathrm{d}\hat{\boldsymbol{\Gamma}}_p}{\mathrm{d}t} = \left(\hat{\boldsymbol{\Gamma}}_p\cdot\nabla\right)\mathbf{u}(\mathbf{x}_p) - \left\{\left[\left(\hat{\boldsymbol{\Gamma}}_p\cdot\nabla\right)\mathbf{u}(\mathbf{x}_p)\right]\cdot\hat{\boldsymbol{\Gamma}}_p\right\}\hat{\boldsymbol{\Gamma}}_p,$$

which is independent of f and g. Hence, vortex re-orientation is not affected by the formulation, staying the same between the classic VPM and any f–g formulation.

Now, we shift our attention to the effects of vortex stretching on vortex strength. Neglecting SFS and non-localized vorticity effects ($\mathbf{M}_p = 0$), from Eq. (2.18) we have

$$\frac{\mathrm{d}\boldsymbol{\Gamma}_p}{\mathrm{d}t}\cdot\hat{\boldsymbol{\Gamma}}_p = \frac{1-3g}{1+3f}\left[\left(\boldsymbol{\Gamma}_p\cdot\nabla\right)\mathbf{u}(\mathbf{x}_p)\right]\cdot\hat{\boldsymbol{\Gamma}}_p, \tag{2.21}$$

Formulation	f	g	$h_\Gamma = \frac{1-3g}{1+3f}$	$h_\sigma = \frac{g+f}{1+3f}$
Mass-conserving sphere (Classic VPM)	0	0	1	0
Mass-conserving tube	1/2	0	0.4	0.2
Momentum-conserving sphere (Reformulated VPM)	0	1/5	0.4	0.2
Momentum-conserving tube	1/4	1/4	0.143	0.286
Momentum-conservation w/ assumed mass conservation	0	1/2	-0.5	0.5

Table 2.1: f–g form of formulations derived in Section 2.2.

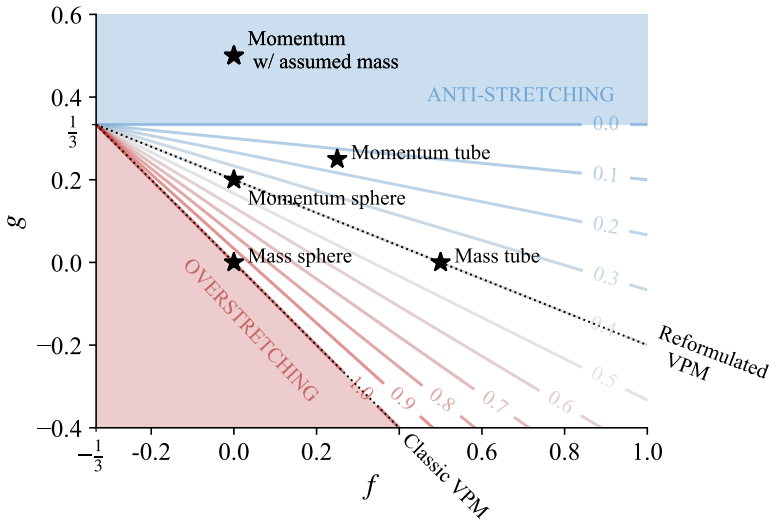


Figure 2.3: Space of all possible f-g formulations, with contour levels of $h_\Gamma = \frac{1-3g}{1+3f}$ that are solutions to the vorticity Navier-Stokes equation. Markers show each formulation in Table 2.1.

where $\frac{\mathrm{d}\boldsymbol{\Gamma}_p}{\mathrm{d}t} \cdot \hat{\boldsymbol{\Gamma}}_p$ is the change in the magnitude of vortex strength. We introduce the parameters

$$h_\Gamma \equiv \frac{1-3g}{1+3f}, \quad h_\sigma \equiv \frac{g+f}{1+3f}$$

and rewrite this as

$$\frac{\mathrm{d}\boldsymbol{\Gamma}_p}{\mathrm{d}t} \cdot \hat{\boldsymbol{\Gamma}}_p = h_\Gamma \left[\left(\boldsymbol{\Gamma}_p \cdot \nabla \right) \mathbf{u}(\mathbf{x}_p) \right] \cdot \hat{\boldsymbol{\Gamma}}_p,$$

while Eq. (2.19) becomes

$$\frac{\|\boldsymbol{\Gamma}_p\|}{\sigma_p} \frac{\mathrm{d}}{\mathrm{d}t} \sigma_p = -h_\sigma \left[\left(\boldsymbol{\Gamma}_p \cdot \nabla \right) \mathbf{u}(\mathbf{x}_p) \right] \cdot \hat{\boldsymbol{\Gamma}}_p.$$

It can be shown that h_Γ and h_σ must satisfy $h_\Gamma + 3h_\sigma = 1$ in order for a formulation to be a solution to the vorticity Navier-Stokes equation. Hence, any f–g formulation simply differs in how vortex stretching is distributed between the lengthening of vortex strength $\boldsymbol{\Gamma}_p$, and shrinking of particle size σ_p. The classic VPM results in $h_\Gamma = 1$ and $h_\sigma = 0$, meaning that vortex strength takes the full amount of vortex stretching.

The space of all possible f–g formulations is shown in Fig. 2.3, along with the contours of h_Γ that satisfy $h_\Gamma + 3h_\sigma = 1$. Any formulation with $h_\Gamma > 1$ will result in a negative h_σ, meaning that the lengthening of vortex strength under stretching is so aggressive that σ_p has to expand instead of shrink, which is unphysical. This is referred to as *overstretching*. On the other extreme, $h_\sigma > 1/3$ results in a negative h_Γ, meaning that the shrinking of the particle is so aggressive that $\boldsymbol{\Gamma}$ has to compensate

by shrinking when it ought to have lengthened with vortex stretching, which is also unphysical. This is referred to as *anti-stretching*. The region between overstretching and anti-stretching is the space of VPM formulations that are physically plausible.

There are four formulations that lay in the space of physically plausible formulations, shown in Fig. 2.3. First, notice that the classic VPM (mass-conserving sphere) lies at the threshold of overstretching, which explains its tendency to be numerically unstable. We are then left with three possible formulations, two of which lay on the same h_Γ contour line: the momentum-conserving sphere and the mass-conserving tube. Even though these two formulations stem from different physical principles resulting in different f–g values, they share the same h_Γ and h_σ values, leading to the same governing equations[51], as mentioned in Section 2.2.2. Thus, by choosing either set of f–g values, we end up in a formulation that preserves angular momentum in spherical chunks of the flow, while at the same time conserving mass in tubes of vorticity. Here on, the momentum-conserving sphere formulation is referred to as the *reformulated vortex particle method*, or rVPM.

51 This is only true after neglecting $\mathbf{M}$. When SFS and non-localized vorticity effects are re-incorporated, the $\mathbf{M}$-term ends up being multiplied by a factor $h_M \equiv \frac{1}{1+3f}$ as shown in Eq. (2.18). Hence, SFS and non-localized vorticity effects are stronger in the momentum-conserving sphere formulation ($h_M = 1$) than the mass-conserving tube formulation ($h_M = 0.4$).

2.5 Reformulated Governing Equations

Summarizing the chapter, the governing equations of the reformulated VPM are

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathbf{x}_p = \mathbf{u}(\mathbf{x}_p) \tag{2.22}$$

$$\frac{\mathrm{d}}{\mathrm{d}t}\sigma_p = -\left(\frac{g+f}{1+3f}\right)\frac{\sigma_p}{\|\boldsymbol{\Gamma}_p\|}\left[\left(\boldsymbol{\Gamma}_p\cdot\nabla\right)\mathbf{u}(\mathbf{x}_p)\right]\cdot\hat{\boldsymbol{\Gamma}}_p \tag{2.23}$$

$$\frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{\Gamma}_p = \left(\boldsymbol{\Gamma}_p\cdot\nabla\right)\mathbf{u}(\mathbf{x}_p) - \frac{g+f}{1/3+f}\left\{\left[\left(\boldsymbol{\Gamma}_p\cdot\nabla\right)\mathbf{u}(\mathbf{x}_p)\right]\cdot\hat{\boldsymbol{\Gamma}}_p\right\}\hat{\boldsymbol{\Gamma}}_p \tag{2.24}$$

$$\left(\frac{\mathrm{d}}{\mathrm{d}t}\overline{\boldsymbol{\omega}}\right)_{\text{viscous}} = \nu\nabla^2\overline{\boldsymbol{\omega}}, \tag{2.25}$$

where Eq. (2.22) resolves vorticity advection by convecting the particles, Eq. (2.23) governs the evolution of particle size, and Eq. (2.24) governs the evolution of vortex strength. Eq. (2.24) in conjunction with Eqs. (2.22) and (2.23) resolve the inviscid part of the LES-filtered vorticity Navier-Stokes equation, while the viscous part in Eq. (2.25) is resolved by one of the schemes mentioned in Section 1.3. For simplicity, Eqs. (2.23) and (2.24) omit the SFS contributions, but they are readily re-incorporated as shown in Eqs. (2.19) and (2.20).

The formulation used for the rest of this study implements $f = 0$ and $g = 1/5$, which ensures conservation of angular momentum in spherical chunks of the flow, while at the same time conserving mass in tubes of vorticity. As previously mentioned, the rVPM equations do not require more computation than the classic VPM since $\frac{\mathrm{d}\sigma_p}{\mathrm{d}t}$ and $\frac{\mathrm{d}\boldsymbol{\Gamma}_p}{\mathrm{d}t}$ are calculated directly and solely from vortex stretching, $\left(\boldsymbol{\Gamma}_p\cdot\nabla\right)\mathbf{u}(\mathbf{x}_p)$.

3 Anisotropic Dynamic SFS Model

In the preceding chapters we have formulated a numerical scheme for solving the large scales of the LES-filtered vorticity equation,

$$\frac{\mathrm{d}}{\mathrm{d}t}\overline{\boldsymbol{\omega}} = \left(\overline{\boldsymbol{\omega}}\cdot\nabla\right)\overline{\mathbf{u}} + \nu\nabla^2\overline{\boldsymbol{\omega}} - \mathbf{E}_{\text{adv}} - \mathbf{E}_{\text{str}}. \tag{1.5}$$

We will now focus on the subfilter-scale (SFS) stresses associated with advection and vortex stretching, $\mathbf{E}_{\text{adv}}$ and $\mathbf{E}_{\text{str}}$, respectively.

$\mathbf{E}_{\text{adv}}$ and $\mathbf{E}_{\text{str}}$ arise from the SFS vorticity stress tensor, $T_{ij} \equiv \overline{u_i\omega_j} - \overline{u_i}\,\overline{\omega_j}$, introduced in Section 1.2. T_{ij} encapsulates the interactions between large scale dynamics and SFS dynamics, which have to be modeled in terms of resolved quantities. Its divergence represents the rate at which enstrophy—a measure of rotational kinetic energy—is transferred from resolved scales to subfilter scales (turbulent diffusion) and from subfilter scales to resolved scales (backscatter). The accuracy of LES hinges on the modeling of this enstrophy transfer. As previously discussed in Section 1.2, $\mathbf{E}_{\text{adv}}$ is defined as

$$E_i^{\text{adv}} \equiv \frac{\partial T'_{ij}}{\partial x_j} = \overline{u_j\frac{\partial\omega_i}{\partial x_j}} - \overline{u_j}\frac{\partial\overline{\omega_i}}{\partial x_j} \tag{3.1}$$

and it represents the enstrophy transfer arising from SFS vorticity advection, while $\mathbf{E}_{\text{str}}$ is defined as

$$E_i^{\text{str}} \equiv -\frac{\partial T_{ij}}{\partial x_j} = -\left(\overline{\omega_j\frac{\partial u_i}{\partial x_j}} - \overline{\omega_j}\frac{\partial\overline{u_i}}{\partial x_j}\right), \tag{3.2}$$

and it represents the enstrophy transfer arising from SFS vortex stretching.

In Section 3.1 we briefly discuss existing SFS models relevant to meshless vortex methods. In Section 3.2 we introduce a novel anisotropic structural model of SFS vortex stretching that is suitable for turbulent flows where the predominant cascade mechanism is vortex stretching. In Sections 3.3 and 3.4 we develop a dynamic procedure for computing the model coefficient while also providing the means for backscatter control. Finally, in Section 3.5 we show how our model can be implemented in conventional mesh-based CFD with a pressure-velocity solver.

3.1 Existing Models

Over the years, only a few models have been proposed that are suitable for meshless vortex methods. The most popular one is a variant of the Smagorinsky eddy-viscosity model formulated for the vorticity stress,[52]

$$\mathbf{E} = -\nabla \times (\nu_{\text{SFS}} \nabla \times \overline{\boldsymbol{\omega}}),$$

where $\mathbf{E} \equiv \mathbf{E}_{\text{adv}} + \mathbf{E}_{\text{str}}$, $\nu_{\text{SFS}} = C_d^2 \sigma^2 \sqrt{2 S_{mn} S_{mn}}$, S_{mn} is the strain-rate tensor, σ is the filter width, and C_d is a model coefficient which is either prescribed or computed dynamically. This functional model and others alike were developed on the basis of homogeneous isotropic turbulence, which makes them overly diffusive in simulations with coherent vortical structures. In the latest developments of the vortex particle-mesh scheme,[53] this drawback has been avoided with the variational multiscale method,[54] however its applicability to a meshless scheme is not clear.

In a different approach, Cottet[55] developed an anisotropic structural model of the advective SFS term $\mathbf{E}_{\text{adv}}$ as

$$\mathbf{E}_{\text{adv}}(\mathbf{x}_p) = C_d \sum_q \text{Vol}_q \left(\boldsymbol{\omega}_p - \boldsymbol{\omega}_q\right) \left[\left(\mathbf{u}(\mathbf{x}_p) - \mathbf{u}(\mathbf{x}_q)\right) \cdot \nabla \zeta_{\sigma_q}(\mathbf{x}_p - \mathbf{x}_q)\right],$$

where the model coefficient C_d is usually prescribed with *ad hoc* calibration.[56] This model is reportedly[57] less dissipative than Smagorinsky-type models, however, it was developed on the basis of 2D flow which is absent of vortex stretching. Thus, this model neglects $\mathbf{E}_{\text{str}}$ even though vortex stretching is known to be one of the main mechanisms for enstrophy production in the energy cascade in three dimensions.

To address the need for a low-dissipation SFS model that captures vortex stretching as the physical mechanism for turbulence, in the following sections we develop an anisotropic structural model of SFS vortex stretching with a dynamic model coefficient.

3.2 SFS Vortex Stretching Model

Starting from the definition of SFS vortex stretching, here repeated

$$E_i^{\text{str}} = -\left(\overline{\omega_j \frac{\partial u_i}{\partial x_j}} - \overline{\omega_j} \frac{\partial \overline{u_i}}{\partial x_j}\right), \tag{3.2}$$

we expand the filter operator

$$E_i^{\text{str}}(\mathbf{x}) = -\left(\int \omega_j(\mathbf{y}) \frac{\partial u_i}{\partial x_j}(\mathbf{y}) \zeta_\sigma(\mathbf{x}-\mathbf{y})\, d\mathbf{y} - \frac{\partial \overline{u_i}}{\partial x_j}(\mathbf{x}) \int \omega_j(\mathbf{y}) \zeta_\sigma(\mathbf{x}-\mathbf{y})\, d\mathbf{y}\right),$$

and group terms as

$$E_i^{\text{str}}(\mathbf{x}) = -\int \omega_j(\mathbf{y}) \left(\frac{\partial u_i}{\partial x_j}(\mathbf{y}) - \frac{\partial \overline{u_i}}{\partial x_j}(\mathbf{x})\right) \zeta_\sigma(\mathbf{x}-\mathbf{y})\, d\mathbf{y}.$$

52 G. S. Winckelmans (1995). "Some progress in large-eddy simulation using the 3D vortex particle method"; Mansfield et al., "A Dynamic LES Scheme for the Vorticity Transport Equation: Formulation and a Priori Tests"; J. R. Mansfield et al. (1996). "Towards lagrangian large vortex simulation". Ed. by Y. Gagnon et al.; Mansfield et al., "Dynamic LES of Colliding Vortex Rings Using a 3D Vortex Method".

53 D.-G. Caprace et al. (2020). "An immersed lifting and dragging line model for the vortex particle-mesh method"; D.-G. Caprace et al. (2021). "Assessment of the Vortex Particle-Mesh Method for Efficient LES of Hovering Rotors and their Wakes".

54 T. J. Hughes et al. (2000). "Large Eddy Simulation and the variational multiscale method"; A. W. Vreman (2003). "The filtering analog of the variational multiscale method in large-eddy simulation"; H. Jeanmart et al. (2007). "Investigation of eddy-viscosity models modified using discrete filters: A simplified "regularized variational multiscale model" and an "enhanced field model""; R. Cocle et al. (2009). "Scale dependence and asymptotic very high Reynolds number spectral behavior of multiscale subgrid models".

55 Cottet, "Artificial Viscosity Models for Vortex and Particle Methods"; G.-H. Cottet (1998). *Anisotropic sub-grid scale numerical schemes for Large Eddy Simulations of turbulent flows*; G.-H. Cottet, D. Jiroveanu, et al. (2003). "Vorticity dynamics and turbulence models for Large-Eddy Simulations".

56 C. Mimeau et al. (2019). "A Hybrid Vortex Method for the Simulation of 3D Incompressible Flows".

57 Stock, Gharakhani, and Stone, "Modeling Rotor Wakes with a Hybrid OVERFLOW-Vortex Method on a GPU Cluster"; C. Mimeau et al. (2018). "An artificial viscosity model for 3D simulations with Vortex Methods".

Using the singular particle approximation, $\boldsymbol{\omega}(\mathbf{x}) \approx \sum_q \boldsymbol{\Gamma}_q \delta(\mathbf{x} - \mathbf{x}_q)$, the integral collapses to

$$E_i^{\text{str}}(\mathbf{x}) \approx -\sum_q \Gamma_j^q \left(\frac{\partial u_i}{\partial x_j}(\mathbf{x}_q) - \frac{\partial \overline{u_i}}{\partial x_j}(\mathbf{x}) \right) \zeta_\sigma(\mathbf{x} - \mathbf{x}_q).$$

Assuming $\frac{\partial u_i}{\partial x_j} \approx \frac{\partial \overline{u_i}}{\partial x_j}$ and writing in vector notation, the model then becomes

$$\mathbf{E}_{\text{str}}(\mathbf{x}) \approx \sum_q \zeta_\sigma(\mathbf{x} - \mathbf{x}_q) \left(\boldsymbol{\Gamma}_q \cdot \nabla \right) \left(\overline{\mathbf{u}}(\mathbf{x}) - \overline{\mathbf{u}}(\mathbf{x}_q) \right).$$

Interestingly, both Cottet's model and our model are strikingly similar to the non-localized vorticity terms $\mathbf{M}_p^1$ and $\mathbf{M}_p^2$, respectively, defined in Section 1.5 as

- $\mathbf{M}_p^1 \equiv -\sum_{q \neq p} \boldsymbol{\Gamma}_q \left(\mathbf{u}(\mathbf{x}_p) - \mathbf{u}(\mathbf{x}_q) \right) \cdot \nabla \zeta_{\sigma_q}(\mathbf{x}_p - \mathbf{x}_q)$ (1.11)
- $\mathbf{M}_p^2 \equiv \sum_{q \neq p} \zeta_{\sigma_q}(\mathbf{x}_p - \mathbf{x}_q) \left(\boldsymbol{\Gamma}_q \cdot \nabla \right) \mathbf{u}(\mathbf{x}_p)$, (1.12)

which are neglected (or filtered out) by the localized-vorticity assumption. Thus, these models can be thought of as soft deconvolutions of the localized-vorticity assumption. Furthermore, introducing a model coefficient, C_d, they can approximate full deconvolutions encompassing the entire spectrum of subfilter scales. This model coefficient is computed dynamically as follows.

3.3 Dynamic Procedure

Since SFS vortex stretching needs to be modeled from resolved quantities, the initial equality given in Eq. (3.2), written here in vector notation

$$\mathbf{E}_{\text{str}} = -\left[\overline{(\boldsymbol{\omega} \cdot \nabla)\mathbf{u}} - (\overline{\boldsymbol{\omega}} \cdot \nabla)\overline{\mathbf{u}} \right],$$

becomes only an approximation,

$$\mathbf{E}_{\text{str}} \approx -\left[\overline{(\boldsymbol{\omega} \cdot \nabla)\mathbf{u}} - (\overline{\boldsymbol{\omega}} \cdot \nabla)\overline{\mathbf{u}} \right]$$

once the $\mathbf{E}_{\text{str}}$ model is introduced. To recover the equality (or at least improve the approximation), we introduce a dynamic model coefficient $C_d(\mathbf{x}, t)$ satisfying

$$C_d \mathbf{E}_{\text{str}} = -\left[\overline{(\boldsymbol{\omega} \cdot \nabla)\mathbf{u}} - (\overline{\boldsymbol{\omega}} \cdot \nabla)\overline{\mathbf{u}} \right]. \tag{3.3}$$

We now propose a procedure for calculating the model coefficient C_d based on a simultaneous balance of enstrophy-production and derivatives between true and modeled SFS vortex stretching.

3.3.1 Derivative Balance

The relation given in Eq. (3.3) is not useful *per se* since, in order to determine C_d, Eq. (3.3) requires knowing the SFS quantity $\overline{(\boldsymbol{\omega}\cdot\nabla)\mathbf{u}} - (\overline{\boldsymbol{\omega}}\cdot\nabla)\overline{\mathbf{u}}$, which is exactly what we are trying to model with $\mathbf{E}_{\text{str}}$. However, assuming scale similarity (meaning that C_d is independent of the filter width), we differentiate this equation with respect to the filter width σ to obtain a more useful relation:

$$C_d\frac{\partial\mathbf{E}_{\text{str}}}{\partial\sigma} = -\frac{\partial}{\partial\sigma}\left[\overline{(\boldsymbol{\omega}\cdot\nabla)\mathbf{u}} - (\overline{\boldsymbol{\omega}}\cdot\nabla)\overline{\mathbf{u}}\right]. \tag{3.4}$$

Using the singular particle approximation, the right-hand side of Eq. (3.4) before differentiation becomes

$$\left[\overline{(\boldsymbol{\omega}\cdot\nabla)\mathbf{u}} - (\overline{\boldsymbol{\omega}}\cdot\nabla)\overline{\mathbf{u}}\right]\Big|_{\mathbf{x}} \approx \sum_q \zeta_\sigma(\mathbf{x}-\mathbf{x}_q)\left(\boldsymbol{\Gamma}_q\cdot\nabla\right)\left(\mathbf{u}\left(\mathbf{x}_q\right) - \overline{\mathbf{u}}\left(\mathbf{x}\right)\right),$$

and after differentiation,

$$\frac{\partial}{\partial\sigma}\left[\overline{(\boldsymbol{\omega}\cdot\nabla)\mathbf{u}} - (\overline{\boldsymbol{\omega}}\cdot\nabla)\overline{\mathbf{u}}\right]\Big|_{\mathbf{x}} \approx \sum_q \frac{\partial\zeta_\sigma}{\partial\sigma}(\mathbf{x}-\mathbf{x}_q)\left(\boldsymbol{\Gamma}_q\cdot\nabla\right)\left(\mathbf{u}\left(\mathbf{x}_q\right) - \overline{\mathbf{u}}\left(\mathbf{x}\right)\right) - \sum_q \zeta_\sigma(\mathbf{x}-\mathbf{x}_q)\left(\boldsymbol{\Gamma}_q\cdot\nabla\right)\frac{\partial\overline{\mathbf{u}}}{\partial\sigma}\left(\mathbf{x}\right).$$

Recalling that ζ_σ is defined as $\zeta_\sigma(\mathbf{x}) = \frac{1}{\sigma^3}\zeta\left(\frac{\|\mathbf{x}\|}{\sigma}\right)$, its width-derivative is

$$\frac{\partial\zeta_\sigma}{\partial\sigma}(\mathbf{x}) = -\frac{3}{\sigma}\zeta_\sigma\left(\mathbf{x}\right) - \frac{\|\mathbf{x}\|}{\sigma^5}\frac{\partial\zeta}{\partial r}\left(\frac{\|\mathbf{x}\|}{\sigma}\right),$$

and assuming that $\zeta_\sigma(\mathbf{x})$ reaches a maximum at $\mathbf{x}=\mathbf{0}$ (meaning, $\frac{\partial\zeta}{\partial r}(0)=0$),

$$\frac{\partial\zeta_\sigma}{\partial\sigma}(\mathbf{0}) = -\frac{3}{\sigma}\zeta_\sigma\left(\mathbf{0}\right).$$

Then, evaluating at $\mathbf{x}=\mathbf{x}_p$, assuming $\zeta_\sigma\left(\mathbf{0}\right)\neq 0$, and using the localized-vorticity assumption to neglect all terms $q\neq p$, Eq. (3.4) becomes

$$C_d(\mathbf{x}_p)\frac{1}{\zeta_\sigma\left(\mathbf{0}\right)}\frac{\partial\mathbf{E}_{\text{str}}}{\partial\sigma}(\mathbf{x}_p) = \frac{3}{\sigma}\left(\boldsymbol{\Gamma}_p\cdot\nabla\right)\left(\mathbf{u}(\mathbf{x}_p) - \overline{\mathbf{u}}(\mathbf{x}_p)\right) + \left(\boldsymbol{\Gamma}_p\cdot\nabla\right)\frac{\partial\overline{\mathbf{u}}}{\partial\sigma}(\mathbf{x}_p),$$

or

$$C_d\mathbf{m} = \mathbf{L}, \tag{3.5}$$

with

$$\mathbf{m} \equiv \frac{\sigma^3}{\zeta(0)}\frac{\partial\mathbf{E}_{\text{str}}}{\partial\sigma}(\mathbf{x}_p)$$
$$\mathbf{L} \equiv \frac{3}{\sigma}\left(\boldsymbol{\Gamma}_p\cdot\nabla\right)\left(\mathbf{u}(\mathbf{x}_p) - \overline{\mathbf{u}}(\mathbf{x}_p)\right) + \left(\boldsymbol{\Gamma}_p\cdot\nabla\right)\frac{\partial\overline{\mathbf{u}}}{\partial\sigma}(\mathbf{x}_p).$$

The relation in Eq. (3.5) will be the basis for our dynamic procedure. This procedure differs from the classic dynamic procedure developed by Germano *et al.*[58] in that the former is motivated by the balance of derivatives between true and modeled SFS contributions given in Eq. (3.4), while the latter is based on the Germano identity.

3.3.2 Enstrophy-Production Balance

The procedure aims at calculating C_d such as to impose the relation given in Eq. (3.5), however, this is an overdetermined system as there are three equations (one for each spatial dimension) and only one unknown, C_d. Thus, the relation is now contracted by also imposing a balance of enstrophy production between true and modeled SFS contributions as follows.

Enstrophy, a measure of the rotational kinetic energy of the flow, is defined locally as $\xi \equiv \frac{1}{2}\boldsymbol{\omega}\cdot\boldsymbol{\omega}$. The rate of local enstrophy production is then calculated as

$$\frac{\mathrm{d}}{\mathrm{d}t}\xi = \boldsymbol{\omega}\cdot\left(\frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{\omega}\right),$$

which can be decomposed between resolved and unresolved domains as

$$\frac{\mathrm{d}}{\mathrm{d}t}\xi = \boldsymbol{\omega}\cdot\left(\frac{\mathrm{d}}{\mathrm{d}t}\overline{\boldsymbol{\omega}}\right) + \boldsymbol{\omega}\cdot\left[\frac{\mathrm{d}}{\mathrm{d}t}\left(\boldsymbol{\omega}-\overline{\boldsymbol{\omega}}\right)\right],$$

The local enstrophy production in the resolved domain is then defined as

$$\frac{\mathrm{d}}{\mathrm{d}t}\xi_r \equiv \boldsymbol{\omega}\cdot\left(\frac{\mathrm{d}}{\mathrm{d}t}\overline{\boldsymbol{\omega}}\right),$$

and the global enstrophy production in the resolved domain is then calculated by integration,

$$\int_{-\infty}^{\infty}\frac{\mathrm{d}}{\mathrm{d}t}\xi_r\,\mathrm{d}\mathbf{y} = \int_{-\infty}^{\infty}\boldsymbol{\omega}\cdot\left(\frac{\mathrm{d}}{\mathrm{d}t}\overline{\boldsymbol{\omega}}\right)\mathrm{d}\mathbf{y}.$$

The SFS contribution to the rate of enstrophy production is isolated through operator splitting as

$$\left(\int_{-\infty}^{\infty}\frac{\mathrm{d}}{\mathrm{d}t}\xi_r\,\mathrm{d}\mathbf{y}\right)_{\mathrm{SFS}} \equiv \int_{-\infty}^{\infty}\boldsymbol{\omega}\cdot\left(\frac{\mathrm{d}}{\mathrm{d}t}\overline{\boldsymbol{\omega}}\right)_{\mathrm{SFS}}\mathrm{d}\mathbf{y}.$$

Now, we constrain the SFS model to match the enstrophy production of the true SFS contribution,

$$\int_{-\infty}^{\infty}\boldsymbol{\omega}\cdot\left(\frac{\mathrm{d}}{\mathrm{d}t}\overline{\boldsymbol{\omega}}\right)_{\mathrm{SFS\,model}}\mathrm{d}\mathbf{y} = \int_{-\infty}^{\infty}\boldsymbol{\omega}\cdot\left(\frac{\mathrm{d}}{\mathrm{d}t}\overline{\boldsymbol{\omega}}\right)_{\mathrm{true\,SFS}}\mathrm{d}\mathbf{y}.$$

Using the singular particle approximation, $\boldsymbol{\omega}(\mathbf{x}) \approx \sum_p \boldsymbol{\Gamma}_p\delta(\mathbf{x}-\mathbf{x}_p)$, the integrals collapse into

$$\sum_p \boldsymbol{\Gamma}_p\cdot\left(\frac{\mathrm{d}}{\mathrm{d}t}\overline{\boldsymbol{\omega}}\right)^{\mathrm{SFS\,model}}_{(\mathbf{x}_p)} = \sum_p \boldsymbol{\Gamma}_p\cdot\left(\frac{\mathrm{d}}{\mathrm{d}t}\overline{\boldsymbol{\omega}}\right)^{\mathrm{true\,SFS}}_{(\mathbf{x}_p)}.$$

One instance that this equality is satisfied is when each term in the sum satisfies

$$\boldsymbol{\Gamma}_p \cdot \left(\frac{\mathrm{d}}{\mathrm{d}t}\overline{\boldsymbol{\omega}}\right)^{\text{SFS model}}_{(\mathbf{x}_p)} = \boldsymbol{\Gamma}_p \cdot \left(\frac{\mathrm{d}}{\mathrm{d}t}\overline{\boldsymbol{\omega}}\right)^{\text{true SFS}}_{(\mathbf{x}_p)}.$$

Differentiating with respect to the filter width, we arrive to

$$\boldsymbol{\Gamma}_p \cdot \frac{\partial}{\partial\sigma}\left(\frac{\mathrm{d}}{\mathrm{d}t}\overline{\boldsymbol{\omega}}\right)^{\text{SFS model}}_{(\mathbf{x}_p)} = \boldsymbol{\Gamma}_p \cdot \frac{\partial}{\partial\sigma}\left(\frac{\mathrm{d}}{\mathrm{d}t}\overline{\boldsymbol{\omega}}\right)^{\text{true SFS}}_{(\mathbf{x}_p)}. \tag{3.6}$$

When SFS advection is neglected, the SFS model and the true SFS become

$$\left(\frac{\mathrm{d}}{\mathrm{d}t}\overline{\boldsymbol{\omega}}\right)^{\text{SFS model}} = -C_d\mathbf{E}_{\text{str}}$$
$$\left(\frac{\mathrm{d}}{\mathrm{d}t}\overline{\boldsymbol{\omega}}\right)^{\text{true SFS}} = \overline{(\boldsymbol{\omega}\cdot\nabla)\mathbf{u}} - (\overline{\boldsymbol{\omega}}\cdot\nabla)\overline{\mathbf{u}},$$

thus Eq. (3.6) is simply the dot-product of Eq. (3.4) with $\boldsymbol{\Gamma}_p$,

$$C_d\boldsymbol{\Gamma}\cdot\frac{\partial\mathbf{E}_{\text{str}}}{\partial\sigma} = -\boldsymbol{\Gamma}\cdot\frac{\partial}{\partial\sigma}\left[\overline{(\boldsymbol{\omega}\cdot\nabla)\mathbf{u}} - (\overline{\boldsymbol{\omega}}\cdot\nabla)\overline{\mathbf{u}}\right].$$

Finally, following the same steps that led from Eq. (3.4) to Eq. (3.5), the enstrophy balance in Eq. (3.6) becomes

$$C_d\boldsymbol{\Gamma}_p\cdot\mathbf{m} = \boldsymbol{\Gamma}_p\cdot\mathbf{L},$$

and C_d is then calculated dynamically as

$$C_d = \frac{\boldsymbol{\Gamma}_p\cdot\mathbf{L}}{\boldsymbol{\Gamma}_p\cdot\mathbf{m}}. \tag{3.7}$$

Thus, this C_d calculated at the position of every particle is the coefficient that approximates the derivative balance while satisfying a local balance of enstrophy production between the model and the true SFS contribution.

3.3.3 Lagrangian Average

Similar to the classic procedure based on the Germano identity,[59] Eq. (3.7) poses numerical issues when the denominator is close to zero, leading to large fluctuations. In the classic procedure, Meneveau *et al.*[60] addressed this issue by integrating both numerator and denominator along Lagrangian trajectories (pathlines), effectively building ensemble averages. Applying this technique, our dynamic procedure becomes

$$C_d = \frac{\langle\boldsymbol{\Gamma}_p\cdot\mathbf{L}\rangle}{\langle\boldsymbol{\Gamma}_p\cdot\mathbf{m}\rangle}, \tag{3.8}$$

where $\langle\cdot\rangle$ denotes the Lagrangian integration. As suggested by Meneveau *et al.*, the integration is performed as a relaxation process at every time step of the form

$$\langle\phi\rangle_{\text{new}} = (1-\alpha)\langle\phi\rangle_{\text{old}} + \alpha\phi,$$

where α is calculated as $\alpha = \Delta t/T \leq 1$, Δt is the time step of the simulation, and T is the time length of the ensemble average.

59 Germano et al., "A dynamic subgrid-scale eddy viscosity model".

60 C. Meneveau, T. S. Lund, et al. (1996). "A Lagrangian dynamic subgrid-scale model of turbulence".

3.4 Backscatter Control

For stability reasons, it is common in LES to control the amount of energy being backscattered from the unresolved scales into the resolved scales. The SFS term is purely-dissipative if its contributions to the enstrophy budget decrease the total enstrophy. This is,

$$\left(\int_{-\infty}^{\infty} \frac{\mathrm{d}}{\mathrm{d}t} \xi_r \, \mathrm{d}\mathbf{y} \right)_{\mathrm{SFS}} \leq 0.$$

Following the derivation in Section 3.3.2, this condition is satisfied if

$$\boldsymbol{\Gamma}_p \cdot \left(\frac{\mathrm{d}}{\mathrm{d}t} \overline{\boldsymbol{\omega}} \right)_{(\mathbf{x}_p)}^{\text{SFS model}} \leq 0$$

for each particle, or equivalently,

$$C_d \boldsymbol{\Gamma}_p \cdot \mathbf{E}_{\mathrm{str}}(\mathbf{x}_p) \geq 0$$

Therefore, enstrophy backscatter can be filtered out at each particle by clipping the model coefficient as $C_d = 0$ whenever the condition shown above is not satisfied.

3.5 Usage in Conventional Mesh-Based CFD

Even though our SFS model is tailored for vortex methods as it only uses the primitive variables of the VPM, the model can also be readily applied to conventional mesh-based CFD as follows. The vortex strength is expressed from mesh quantities as $\boldsymbol{\Gamma}_q = \langle \boldsymbol{\omega} \rangle_q \mathrm{Vol}_q$, where $\langle \boldsymbol{\omega} \rangle_q$ and Vol_q are the average vorticity and volume associated to each element in the grid (*i.e.*, cells in a finite volume method or nodes in a finite difference method). The model in the vorticity transport equation then becomes

$$\frac{\partial T_{ij}}{\partial x_j}(\mathbf{x}) = C_d \Delta^{-3} \sum_q \langle \omega \rangle_j^q \, \mathrm{Vol}_q \left(\frac{\partial \overline{u_i}}{\partial x_j}(\mathbf{x}_q) - \frac{\partial \overline{u_i}}{\partial x_j}(\mathbf{x}) \right) \zeta \left(\frac{\|\mathbf{x} - \mathbf{x}_q\|}{\Delta} \right),$$

where the index q iterates over each element in the grid, Δ is the width of the grid filter, and ζ is the filter kernel. Furthermore, the model can be used in the pressure-velocity form of the momentum equation noticing that the SFS terms in the vorticity transport equation are simply the curl of the SFS term in the pressure-velocity equation, $\frac{\partial \tau_{ij}}{\partial x_j}$, as

$$\epsilon_{ijk} \frac{\partial}{\partial x_j} \frac{\partial \tau_{kl}}{\partial x_l} = \frac{\partial T'_{ij}}{\partial x_j} - \frac{\partial T_{ij}}{\partial x_j},$$

where ϵ_{ijk} is the Levi-Civita tensor. Hence, the SFS models of the vorticity transport equation can be used to model $\frac{\partial \tau_{ij}}{\partial x_j}$ by "un-curling" the equation above.

3.6 Summary

Recapitulating Chapters 2 and 3, the LES-filtered vorticity equation

$$\frac{\mathrm{d}}{\mathrm{d}t}\overline{\boldsymbol{\omega}} = \left(\overline{\boldsymbol{\omega}}\cdot\nabla\right)\overline{\mathbf{u}} + \nu\nabla^2\overline{\boldsymbol{\omega}} - \mathbf{E}_{\text{adv}} - \mathbf{E}_{\text{str}} \tag{1.5}$$

is discretized and numerically solved in the reformulated VPM through the following governing equations

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}t}\mathbf{x}_p &= \overline{\mathbf{u}}(\mathbf{x}_p)\\
\frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{\Gamma}_p &= \left(\boldsymbol{\Gamma}_p\cdot\nabla\right)\overline{\mathbf{u}}(\mathbf{x}_p) - \frac{g+f}{1/3+f}\left\{\left[\left(\boldsymbol{\Gamma}_p\cdot\nabla\right)\overline{\mathbf{u}}(\mathbf{x}_p)\right]\cdot\hat{\boldsymbol{\Gamma}}_p\right\}\hat{\boldsymbol{\Gamma}}_p\\
&\qquad - \frac{C_d}{\zeta_{\sigma_p}(\mathbf{0})}\left[\mathbf{E}(\mathbf{x}_p) - \frac{f}{1/3+f}\left(\mathbf{E}(\mathbf{x}_p)\cdot\hat{\boldsymbol{\Gamma}}_p\right)\hat{\boldsymbol{\Gamma}}_p\right]\\
\frac{\mathrm{d}}{\mathrm{d}t}\sigma_p &= -\left(\frac{g+f}{1+3f}\right)\frac{\sigma_p}{\|\boldsymbol{\Gamma}_p\|}\left[\left(\boldsymbol{\Gamma}_p\cdot\nabla\right)\overline{\mathbf{u}}(\mathbf{x}_p)\right]\cdot\hat{\boldsymbol{\Gamma}}_p\\
&\qquad + \left(\frac{f}{1+3f}\right)\frac{\sigma_p}{\|\boldsymbol{\Gamma}_p\|}\frac{C_d}{\zeta_{\sigma_p}(\mathbf{0})}\mathbf{E}(\mathbf{x}_p)\cdot\hat{\boldsymbol{\Gamma}}_p\\
\left(\frac{\mathrm{d}}{\mathrm{d}t}\overline{\boldsymbol{\omega}}\right)_{\text{viscous}} &= \nu\nabla^2\overline{\boldsymbol{\omega}},
\end{aligned}$$

where $\mathbf{E} = \mathbf{E}_{\text{adv}} + \mathbf{E}_{\text{str}}$. In the derivation of the governing equations we have introduced only two assumptions: incompressible flow and localized vorticity. From here on, we also assume $\mathbf{E}_{\text{adv}} = 0$ since our interest is in testing and validating our SFS vortex-stretching model, but $\mathbf{E}_{\text{adv}}$ can be readily reincorporated using, for instance, Cottet's model. As discussed in Chapters 2 and 3, $f = 0$ and $g = 1/5$ reinforce conservation of both mass and angular momentum, and the governing equations then become

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}t}\mathbf{x}_p &= \overline{\mathbf{u}}(\mathbf{x}_p)\\
\frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{\Gamma}_p &= \left(\boldsymbol{\Gamma}_p\cdot\nabla\right)\overline{\mathbf{u}}(\mathbf{x}_p) - \frac{3}{5}\left\{\left[\left(\boldsymbol{\Gamma}_p\cdot\nabla\right)\overline{\mathbf{u}}(\mathbf{x}_p)\right]\cdot\hat{\boldsymbol{\Gamma}}_p\right\}\hat{\boldsymbol{\Gamma}}_p - \frac{C_d}{\zeta_{\sigma_p}(\mathbf{0})}\mathbf{E}_{\text{str}}(\mathbf{x}_p)\\
\frac{\mathrm{d}}{\mathrm{d}t}\sigma_p &= -\frac{1}{5}\frac{\sigma_p}{\|\boldsymbol{\Gamma}_p\|}\left[\left(\boldsymbol{\Gamma}_p\cdot\nabla\right)\overline{\mathbf{u}}(\mathbf{x}_p)\right]\cdot\hat{\boldsymbol{\Gamma}}_p\\
\left(\frac{\mathrm{d}}{\mathrm{d}t}\overline{\boldsymbol{\omega}}\right)_{\text{viscous}} &= \nu\nabla^2\overline{\boldsymbol{\omega}}.
\end{aligned}$$

Subfilter-scale vortex stretching is then modeled as

$$\mathbf{E}_{\text{str}}(\mathbf{x}) = \sum_q \zeta_\sigma(\mathbf{x}-\mathbf{x}_q)\left(\boldsymbol{\Gamma}_q\cdot\nabla\right)\left(\overline{\mathbf{u}}(\mathbf{x}) - \overline{\mathbf{u}}(\mathbf{x}_q)\right),$$

and the model coefficient C_d is calculated dynamically at the position of every particle as

$$C_d = \frac{\left\langle\boldsymbol{\Gamma}_p\cdot\mathbf{L}\right\rangle}{\left\langle\boldsymbol{\Gamma}_p\cdot\mathbf{m}\right\rangle}, \tag{3.9}$$

where $\langle\cdot\rangle$ denotes an integration along Lagrangian trajectories, and

$$\mathbf{m} = \frac{\sigma^3}{\zeta(0)} \frac{\partial \mathbf{E}_{\text{str}}}{\partial \sigma}(\mathbf{x}_p)$$
$$\mathbf{L} = \frac{3}{\sigma} \left(\mathbf{\Gamma}_p \cdot \nabla\right) \left(\mathbf{u}(\mathbf{x}_p) - \overline{\mathbf{u}}(\mathbf{x}_p)\right) + \left(\mathbf{\Gamma}_p \cdot \nabla\right) \frac{\partial \overline{\mathbf{u}}}{\partial \sigma}(\mathbf{x}_p).$$

Backscatter is controlled by clipping the model coefficient to $C_d = 0$ whenever the condition $C_d \mathbf{\Gamma}_p \cdot \mathbf{E}_{\text{str}}(\mathbf{x}_p) \geq 0$ is not satisfied.

In the next chapter we will describe the various numerical schemes needed to evaluate and integrate the governing equations. In particular, we will show how the filtered velocity field $\overline{\mathbf{u}}$ is calculated by inverting the relation $\boldsymbol{\omega} = \nabla \times \mathbf{u}$, how particle interactions are efficiently computed with complexity $\mathcal{O}(N)$ through the fast multipole method, and how to compute the filter derivatives $\frac{\partial}{\partial \sigma}$ needed to calculate $\mathbf{m}$ and $\mathbf{L}$.

4 Numerical Schemes

In the preceding chapters, we introduced the LES-filtered vorticity equation and discretized it with vortex particles to derive the governing equations that solve the LES equation numerically. While the particle discretization is the main numerical tool of the VPM, there are other minor numerical schemes at play. In this chapter, we describe each numerical scheme that is implemented along with the reformulated VPM.

We do not intend this chapter to be a literature review on the numerics of VPM, rather we limit our comments to only what is relevant to our meshless LES scheme. For instance, a few numerical schemes have been developed over the years to address Lagrangian distortion,[61] and the reader may benefit from recent work on this topic by Kirchhart et al.[62] and Salloum and Lakkis.[63] However, unlike the classic VPM, Lagrangian distortion does not significantly affect the wide range of cases studied in this doctoral work with our novel meshless LES scheme, hence we make no further comments on that topic.

In Section 4.1, we show how the filtered velocity field $\overline{\mathbf{u}}$ is calculated by inverting the relation $\boldsymbol{\omega} = \nabla \times \mathbf{u}$. In Section 4.2, we review how the computation of $\overline{\mathbf{u}}$ can be efficiently performed with complexity $\mathcal{O}(N)$ through the fast multipole method (FMM). In Section 4.3, we introduce a novel approach to computing vortex stretching and derivatives of $\overline{\mathbf{u}}$ through a complex-step derivative approximation implemented in the FMM. In Section 4.4, we address the divergence of the $\boldsymbol{\omega}$ field and develop a correction to a popular relaxation scheme. In Sections 4.5 and 4.6, we review the transposed vortex stretching and viscous diffusion schemes. Finally, in Sections 4.7 to 4.9, we describe the numerical implementation of the dynamic procedure, the governing equations, and the time integration scheme.

61 Barba et al., "Advances in viscous vortex methods - Meshless spatial adaption based on radial basis function interpolation"; Lakkis et al., "A high resolution spatially adaptive vortex method for separating flows. Part I: Two-dimensional domains".

62 Kirchhart et al., "Discrete Projections: A Step Towards Particle Methods on Bounded Domains without Remeshing".

63 S. Salloum and I. Lakkis (2020). "Proper evaluation of spherical harmonics-based expressions for the velocity and vortex stretching vectors in three-dimensional grid-free vortex methods".

4.1 Velocity Field Evaluation

The vorticity equation is solved in the vortex particle method as particles are convected and stretched by the velocity field. The velocity field is not known *a priori*; however, we will now show that $\mathbf{u}$ can be computed directly from the vorticity field by inverting the relationship $\boldsymbol{\omega} = \nabla \times \mathbf{u}$. The derivations that follow come from the work of Winckelmans et al.[64]

By the Helmholtz decomposition theorem, $\mathbf{u}$ can be decomposed into a uniform component, an irrotational component, and a solenoidal

64 Winckelmans et al., "Contributions to Vortex Particle Methods for the Computation of Three-Dimensional Incompressible Unsteady Flows".

component as

$$\mathbf{u} = \underbrace{\mathbf{u}_\infty}_{\text{uniform}} + \underbrace{\nabla\phi}_{\text{irrotational}} + \underbrace{\nabla\times\boldsymbol{\psi}}_{\text{solenoidal}},$$

where $\mathbf{u}_\infty(t)$ is the freestream, $\phi(\mathbf{x},t)$ is a scalar-potential field, and $\boldsymbol{\psi}(\mathbf{x},t)$ is a vector-potential field. From the definition of the vorticity field, $\boldsymbol{\omega} = \nabla\times\mathbf{u}$, we have

$$\begin{aligned}\boldsymbol{\omega} &= \nabla\times(\nabla\times\boldsymbol{\psi}) &(4.1)\\ &= \nabla(\nabla\cdot\boldsymbol{\psi}) - \nabla^2\boldsymbol{\psi}. &(4.2)\end{aligned}$$

The first term in the right side is nullified after applying a gauge transform,[65] leading to a three-dimensional unbounded Poisson equation,

$$\nabla^2\boldsymbol{\psi} = -\boldsymbol{\omega}.$$

65 E. M. Purcell et al. (2013). *Electricity and Magnetism*, Sec. 6.3.

Filtering the Poisson equation,

$$\nabla^2\overline{\boldsymbol{\psi}}(\mathbf{x}) = -\int \boldsymbol{\omega}(\mathbf{y})\zeta_\sigma(\mathbf{x}-\mathbf{y})\,\mathrm{d}\mathbf{y},$$

and using the singular particle approximation $\boldsymbol{\omega}(\mathbf{y}) \approx \sum_p \boldsymbol{\Gamma}_p\delta(\mathbf{y}-\mathbf{x}_p)$, the Poisson equation then becomes

$$\nabla^2\overline{\boldsymbol{\psi}}(\mathbf{x}) \approx -\sum_p \boldsymbol{\Gamma}_p\zeta_{\sigma_p}(\mathbf{x}-\mathbf{x}_p).$$

Recalling that ζ_σ is defined as $\zeta_\sigma(\mathbf{x}) = \frac{1}{\sigma^3}\zeta\left(\frac{\|\mathbf{x}\|}{\sigma}\right)$, the Poisson equation can solved defining a Green's function $G_\sigma(\mathbf{x}) = \frac{1}{\sigma}G\left(\frac{\|\mathbf{x}\|}{\sigma}\right)$ such that $\nabla^2 G(r) = -\zeta(r)$, resulting in

$$\overline{\boldsymbol{\psi}}(\mathbf{x}) \approx \sum_p \boldsymbol{\Gamma}_p G_{\sigma_p}(\mathbf{x}-\mathbf{x}_p).$$

The velocity field $\overline{\mathbf{u}}$ is then derived from the vector-potential field as[66]

$$\begin{aligned}\overline{\mathbf{u}}(\mathbf{x}) &= \nabla\times\overline{\boldsymbol{\psi}}(\mathbf{x})\\ &= \sum_p \nabla\times\left(\boldsymbol{\Gamma}_p G_{\sigma_p}(\mathbf{x}-\mathbf{x}_p)\right)\\ &= \sum_p \nabla\left(G_{\sigma_p}(\mathbf{x}-\mathbf{x}_p)\right)\times\boldsymbol{\Gamma}_p\\ &= \sum_p \frac{1}{\sigma_p^2}\frac{\partial G}{\partial r}\left(\frac{\|\mathbf{x}-\mathbf{x}_p\|}{\sigma_p}\right)\frac{\mathbf{x}-\mathbf{x}_p}{\|\mathbf{x}-\mathbf{x}_p\|}\times\boldsymbol{\Gamma}_p\\ &= \sum_p \frac{1}{\sigma_p^2}\left(-\frac{\sigma_p^2}{\|\mathbf{x}-\mathbf{x}_p\|^2}\int_0^{\frac{\|\mathbf{x}-\mathbf{x}_p\|}{\sigma_p}}\zeta(t)t^2\,\mathrm{d}t\right)\frac{\mathbf{x}-\mathbf{x}_p}{\|\mathbf{x}-\mathbf{x}_p\|}\times\boldsymbol{\Gamma}_p\\ &= -\sum_p\left(\int_0^{\frac{\|\mathbf{x}-\mathbf{x}_p\|}{\sigma_p}}\zeta(t)t^2\,\mathrm{d}t\right)\frac{\mathbf{x}-\mathbf{x}_p}{\|\mathbf{x}-\mathbf{x}_p\|^3}\times\boldsymbol{\Gamma}_p.\end{aligned}$$

66 In the third line we use the relationship

$$\frac{\partial G}{\partial r}(r) = -\frac{1}{r^2}\int_0^r \zeta(t)t^2\,\mathrm{d}t,$$

which is derived as follows:

$$\begin{aligned}\int_0^r \zeta(t)t^2\,\mathrm{d}t &= -\int_0^r\left[\nabla^2 G(t)\right]t^2\,\mathrm{d}t\\ &= -\int_0^r\left[\frac{1}{t^2}\frac{\partial}{\partial t}\left(t^2\frac{\partial G}{\partial t}(t)\right)\right]t^2\,\mathrm{d}t\\ &= -\int_0^r\frac{\partial}{\partial t}\left(t^2\frac{\partial G}{\partial t}(t)\right)\mathrm{d}t\\ &= -r^2\frac{\partial G}{\partial r}(r).\end{aligned}$$

Thus, the velocity induced by the field of vortex particles is calculated analytically as

$$\overline{\mathbf{u}}\left(\mathbf{x}\right) = \sum_p g_{\sigma_p}\left(\mathbf{x}-\mathbf{x}_p\right)\mathbf{K}\left(\mathbf{x}-\mathbf{x}_p\right)\times\boldsymbol{\Gamma}_p, \tag{4.3}$$

where g_σ is a normalized regularizing function calculated from the chosen basis function ζ as

$$g_\sigma\left(\mathbf{x}\right) \equiv 4\pi\int_0^{\frac{\|\mathbf{x}\|}{\sigma}}\zeta(t)t^2\,\mathrm{d}t,$$

and

$$\mathbf{K}\left(\mathbf{x}\right) \equiv -\frac{\mathbf{x}}{4\pi\|\mathbf{x}\|^3}$$

is the singular Newtonian kernel resulting from the three-dimensional Green's function of the singular unbounded Poisson's equation.

The resulting velocity field is a regularized version of the Newtonian kernel, $\mathbf{K}_\sigma \equiv g_\sigma\mathbf{K}$, and is dependent on the chosen radial basis function ζ (or filter). To illustrate this, Fig. 4.1 compares ζ_σ, g_σ, and $\mathbf{K}_\sigma$ obtained from three radial basis functions commonly used in VPM. Following the same derivation that led to Eq. (4.15), it can be shown that the unfiltered velocity field $\mathbf{u}$ is calculated as

$$\mathbf{u}\left(\mathbf{x}\right) = \sum_p \mathbf{K}\left(\mathbf{x}-\mathbf{x}_p\right)\times\boldsymbol{\Gamma}_p, \tag{4.4}$$

which poses a singularity close to the center of each particle.

In order to evaluate the governing equations, not only we need to compute $\overline{\mathbf{u}}$, but, due to vortex stretching, we also need the spatial derivatives of $\overline{\mathbf{u}}$. These derivatives are calculated as

$$\frac{\partial\overline{\mathbf{u}}}{\partial x_j}\left(\mathbf{x}\right) = \sum_p\left[\frac{\partial g_{\sigma_p}}{\partial x_j}\left(\mathbf{x}-\mathbf{x}_p\right)\mathbf{K}\left(\mathbf{x}-\mathbf{x}_p\right)\times\boldsymbol{\Gamma}_p \tag{4.5}$$

$$+ g_{\sigma_p}\left(\mathbf{x}-\mathbf{x}_p\right)\frac{\partial\mathbf{K}}{\partial x_j}\left(\mathbf{x}-\mathbf{x}_p\right)\times\boldsymbol{\Gamma}_p\Bigg], \tag{4.6}$$

with $\frac{\partial K_i}{\partial x_j}\left(\mathbf{x}\right) = \frac{3x_ix_j}{4\pi\|\mathbf{x}\|^5} - \frac{1}{4\pi\|\mathbf{x}\|^3}\delta_{ij}$, where δ_{ij} is the Kronecker delta.

All simulations presented in this study implement the Gaussian basis function $\zeta\left(r\right) = \frac{1}{(2\pi)^{3/2}}\exp\left(-\frac{r^2}{2}\right)$, which leads to

$$g_\sigma\left(\mathbf{x}\right) = \mathrm{erf}\left(\sqrt{\frac{\|\mathbf{x}\|^2}{2\sigma^2}}\right) - \sqrt{\frac{4}{\pi}}\sqrt{\frac{\|\mathbf{x}\|^2}{2\sigma^2}}\exp\left(-\frac{\|\mathbf{x}\|^2}{2\sigma^2}\right),$$

and

$$\frac{\partial g_\sigma}{\partial x_j}\left(\mathbf{x}\right) = \frac{x_j}{\sigma^2}\sqrt{\frac{4}{\pi}}\sqrt{\frac{\|\mathbf{x}\|^2}{2\sigma^2}}\exp\left(-\frac{\|\mathbf{x}\|^2}{2\sigma^2}\right).$$

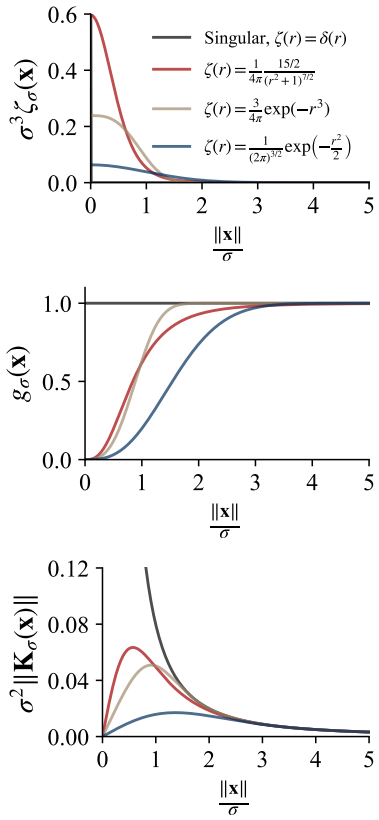


Figure 4.1: Characteristic functions of common VPM kernels: (top) filter or radial basis function, (middle) regularizing function, and (bottom) velocity function.

4.2 Fast Multipole Method

The governing equations require evaluating the velocity $\overline{\mathbf{u}}$ and its derivatives $\partial\overline{\mathbf{u}}/\partial x_j$ at the position of each particle. Eqs. (4.5) and (4.15) iterate over all particles to compute $\overline{\mathbf{u}}$ and its derivatives, which poses an N-body problem with computational complexity $\mathcal{O}(N^2)$, where N is the number of particles. This problem illustrated in Fig. 4.2 as all *particles-to-particle* (P2P) interactions are computed directly. The resulting complexity is prohibitively expensive as simulations typically use millions of particles.

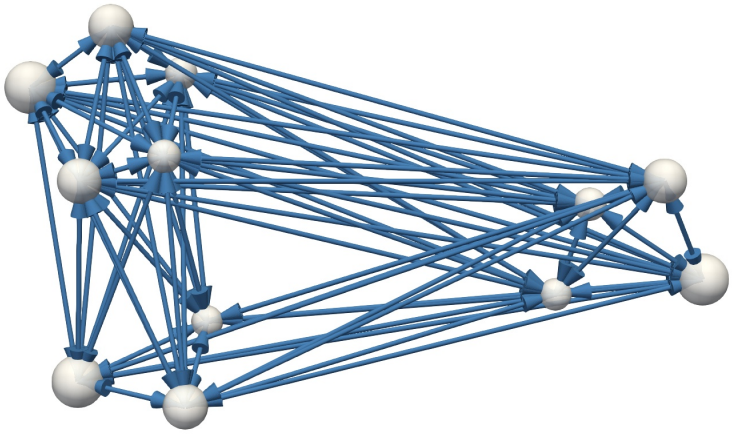


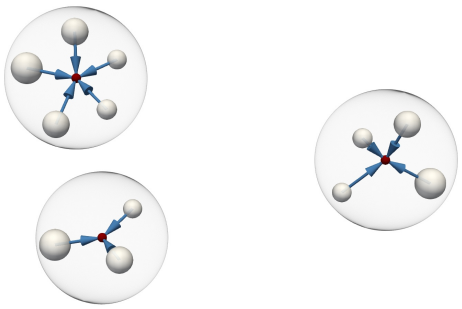
Figure 4.2: Direct evaluation of particle-to-particle (P2P) interactions in complexity $\mathcal{O}(N^2)$.

As an alternative to directly computing all P2P interactions, particles can be lumped into clusters and direct P2P interactions between well-separated particles are approximated with their interactions with the clusters. Each cluster is represented as a multipole expansion. This process is illustrated in Fig. 4.3. First, cluster multipoles are computed (*particle-to-multipole* step, or P2M), and clusters are recursively lumped into clusters of clusters (*multipole-to-multipole* step, or M2M). Then, the influence of each cluster on far particles is computed (*multipole-to-particle* step, or M2P), and P2P is performed only among nearby particles. This algorithm, known as the Barnes-Hut treecode method,[67] reduces the computational effort from $\mathcal{O}(N^2)$ to $\mathcal{O}(N\log(N))$.

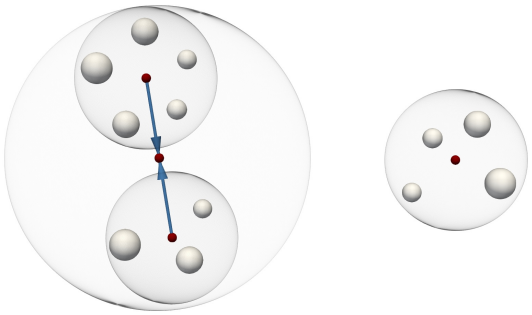
Additional speedup can be achieved by translating and rotating the multipoles to the center of target clusters (called local expansion), then the local expansions are recursively passed inside the clusters down to the particles. This process is illustrated in Fig. 4.4. After P2M and M2M are performed, multipole expansions are converted to local expansions (*multipole-to-local* step, M2L) and local expansions are recursively converted to local expansions inside clusters (*local-to-local* step, or L2L). Then, local expansions are converted to particle interactions (*local-to-particle* step, or L2P), and P2P among nearby particles are the only interactions that are computed directly. This algorithm, known as the *fast multipole method* (FMM), was introduced by Greengard and Rokhlin[68] in 1987.

67 J. Barnes et al. (1986). "A hierarchical O(N log N) force-calculation algorithm".

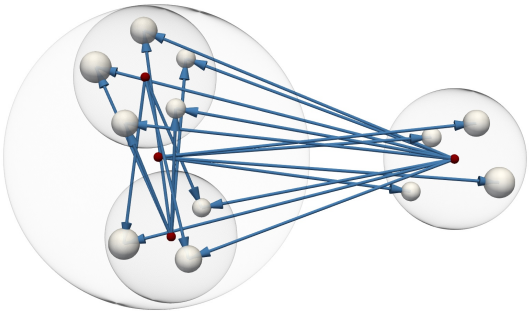
68 L. Greengard et al. (1987). "A fast algorithm for particle simulations"; L. F. Greengard (1987). "The Rapid Evaluation Of Potential Fields In Particle Systems".



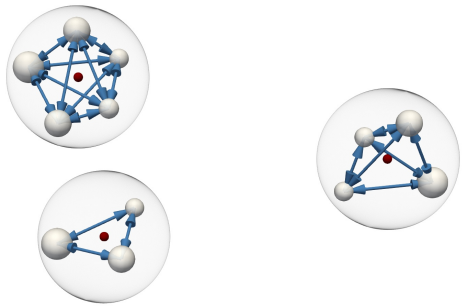
Step 1: Particle to multipole (P2M)



Step 2: Multipole to multipole (M2M)

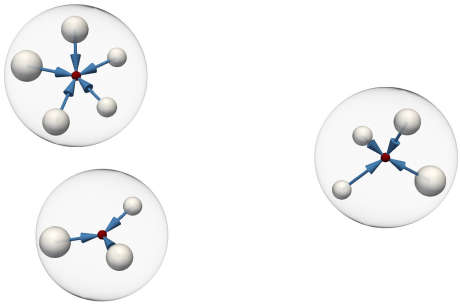


Step 3: Multipole to particle (M2P)

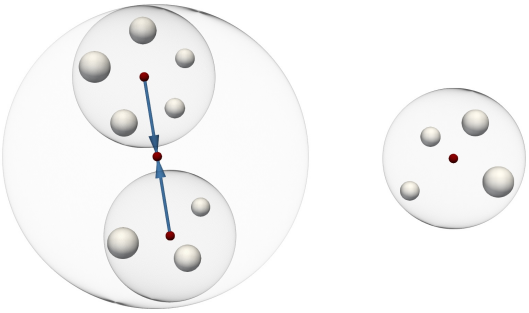


Step 4: Particle to particle (P2P)

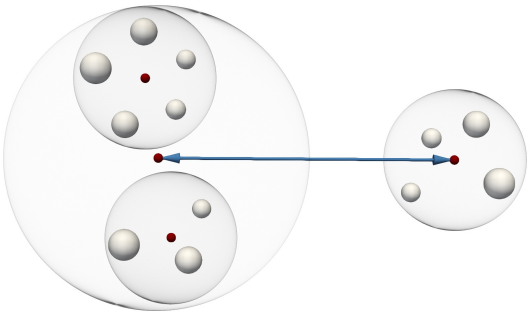
Figure 4.3: Barnes-Hut treecode approximation of particle-to-particle interactions in complexity $\mathcal{O}(N \log N)$.



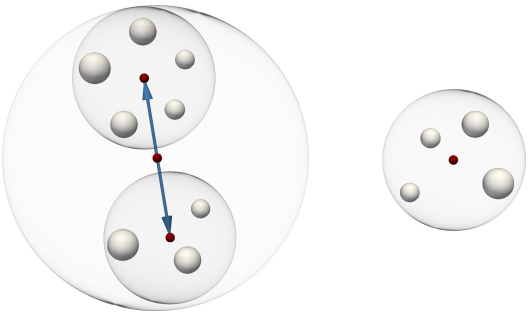
Step 1: Particle to multipole (P2M)



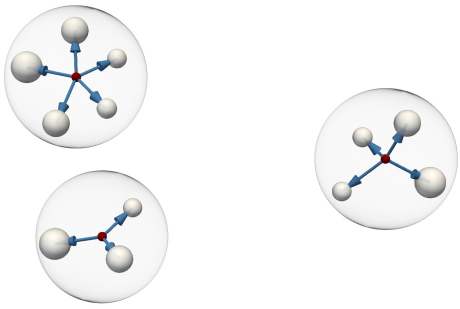
Step 2: Multipole to multipole (M2M)



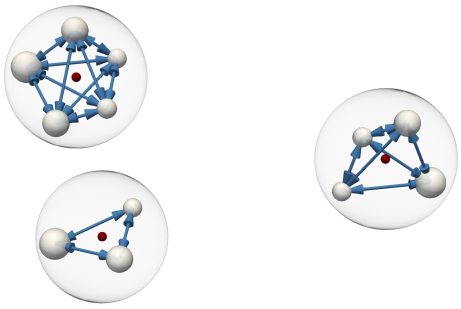
Step 3: Multipole to local (M2L)



Step 4: Local to local (L2L)



Step 5: Local to particle (L2P)



Step 6: Particle to particle (P2P)

Figure 4.4: Fast multipole method (FMM) approximation of particle-to-particle interactions in complexity $\mathcal{O}(N)$.

Selected as one of the top ten algorithms of the 20th century, FMM reduces the original $\mathcal{O}(N^2)$ N-body problem to a linear problem $\mathcal{O}(N)$, while controlling the error of the approximation down to any arbitrary precision.

While the concept behind the FMM algorithm is elegant and simple, its mathematical development is intricate and an FMM implementation usually takes thousands of lines of code. We have here described only a rough idea of the FMM algorithm, and the reader is referred to the work of Cheng, Greengard, and Rokhlin[69] for the mathematical details of the method.

In this study, FMM is applied to the computation of velocity and vortex stretching using a modified version of the open source, parallelized code ExaFMM.[70] Fig. 4.5 shows the wall-clock time of velocity and vortex stretching computed through the FMM, evidencing linear scaling from thousands to millions of particles. The FMM approximation as implemented in our modified version of ExaFMM uses the following parameters prescribed by the user.

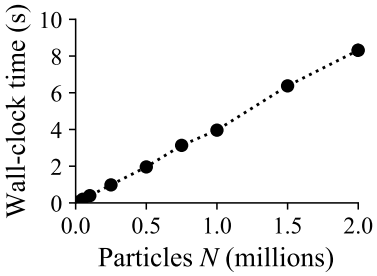


Figure 4.5: Wall-clock time of velocity and vortex stretching computation, showing linear scaling achieved through the FMM.

69 H. Cheng, L. Greengard, et al. (1999). "A Fast Adaptive Multipole Algorithm in Three Dimensions".

70 R. Yokota and L. A. Barba (2011a). "Treecode and fast multipole method for N-body simulation with CUDA"; Hu et al., "Scalable Fast Multipole Accelerated Vortex Methods"; T. Wang et al. (2021). "ExaFMM: a high-performance fast multipole method library with C++ and Python interfaces".

- **Expansion Order P:** Number of terms used in multipole and local expansions. This is analogous to the number of terms used in a Taylor expansion approximation. The FMM approximation has a computational cost $\mathcal{O}(P^4)$. We recommend using $P \geq 4$ in low and mid-fidelity simulations, and $P \geq 12$ in high-fidelity simulations.

- **Critical Cluster Size n_{crit}:** Maximum number of particles per cluster. A cluster splits up after reaching this many particles. This criterion can be used to balance the computational load across cores. All simulations in this study used either $n_{\text{crit}} = 50$ or $n_{\text{crit}} = 200$.

- **Neighborhood Criterion θ:** This criterion defines the distance where the near field (which is directly computed by P2P) ends. The interactions between two clusters i and j are resolved through direct P2P if $(R_i+R_j)/d \geq \theta$, where R_i and R_j are the radii of the clusters, and d is the distance between their centroids, shown in Fig. 4.6. Hence, with $\theta = 1$, P2P is done only between clusters that overlap; with $\theta = 0.5$, P2P is done between clusters when their distance is less than two times $R_i + R_j$; with $\theta = 0.25$, P2P is done between clusters when their distance is less than four times $R_i + R_j$; and with $\theta = 0$, P2P is done between all clusters. All simulations in this study used $\theta = 0.4$.

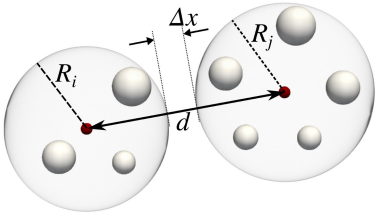


Figure 4.6: Parameters used in neighborgood criterion $(R_i+R_j)/d \geq \theta$ and regularizing criterion $\frac{\sigma_c}{\Delta x} \geq \phi$.

- **Regularizing Criterion ϕ:** The FMM approximates the singular Newtonian kernel (dashed line in Fig. 4.7), while P2P is implemented as a regularized kernel (Gaussian kernel, or solid line in Fig. 4.7). We introduce a criterion to avoid approximating interactions with the singular-FMM when particles across clusters are sufficiently close to each other, calculating them through the regularized P2P instead. P2P is performed between two clusters if $\frac{\sigma_c}{\Delta x} \geq \phi$, where σ_c is the average smoothing radius of the particles

in both clusters, and Δx is the distance between cluster boundaries defined as $\Delta x = d - (R_i + R_j)$, shown in Fig. 4.6. Hence, $\frac{1}{\phi}$ roughly corresponds to the x-axis of Fig. 4.7. With $\phi = 1$, P2P is done only between clusters with boundaries closer than σ_c, or $\frac{\Delta x}{\sigma_c} \leq 1$; with $\phi = 0.5$, P2P is done between clusters up to $\frac{\Delta x}{\sigma_c} = 2$; with $\phi = 0.25$, P2P is done between clusters up to $\frac{\Delta x}{\sigma_c} = 4$; and with $\phi = 0$, P2P is done between all clusters. As seen in Fig. 4.7, there is a large discrepancy between the singular and regularized kernel for $\|\mathbf{x}\|/\sigma < 2$, which can create a discontinuity as particles are convected and move between clusters. All simulations in this study used either $\phi = 0.3$ or $\phi = 0.5$, depending on the desired level of accuracy. $\phi = 0.5$ was used in low and mid-fidelity simulations, leading to a worst-case-scenario discontinuity of 26% in velocity jump, while $\phi = 0.3$ was used in high-fidelity simulations, corresponding to a worst-case scenario jump of less than 1%.

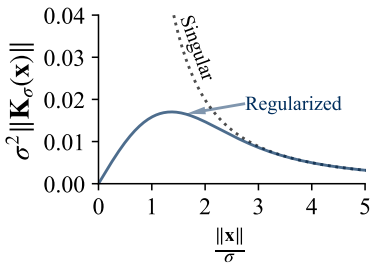


Figure 4.7: Singular kernel approximated by the FMM vs regularized kernel used in P2P.

The computation of derivates needed for vortex stretching is performed in our FMM through an efficient complex-step derivative approximation, as explained in the next section.

4.3 Vortex Stretching Through CSDA

The original FMM[71] was derived to calculate interactions of the form $1/r$, as in the case of the vector-potential $\boldsymbol{\psi}$. However, rather than $\boldsymbol{\psi}$, we are interested in the velocity $\mathbf{u}$ and vortex stretching $(\boldsymbol{\Gamma} \cdot \nabla)\mathbf{u}$. These fields can be obtained from the derivatives of $\boldsymbol{\psi}$ as

$$\mathbf{u} = \nabla \times \boldsymbol{\psi}$$

and

$$(\boldsymbol{\Gamma} \cdot \nabla)\mathbf{u} = \begin{bmatrix} \frac{\partial^2\psi_3}{\partial x_1 \partial x_2} - \frac{\partial^2\psi_2}{\partial x_1 \partial x_3} & \frac{\partial^2\psi_3}{\partial x_2{}^2} - \frac{\partial^2\psi_2}{\partial x_2 \partial x_3} & \frac{\partial^2\psi_3}{\partial x_3 \partial x_2} - \frac{\partial^2\psi_2}{\partial x_3{}^2} \\ \frac{\partial^2\psi_1}{\partial x_1 \partial x_3} - \frac{\partial^2\psi_3}{\partial x_1{}^2} & \frac{\partial^2\psi_1}{\partial x_2 \partial x_3} - \frac{\partial^2\psi_3}{\partial x_2 \partial x_1} & \frac{\partial^2\psi_1}{\partial x_3{}^2} - \frac{\partial^2\psi_3}{\partial x_3 \partial x_1} \\ \frac{\partial^2\psi_2}{\partial x_1{}^2} - \frac{\partial^2\psi_1}{\partial x_1 \partial x_2} & \frac{\partial^2\psi_2}{\partial x_2 \partial x_1} - \frac{\partial^2\psi_1}{\partial x_2{}^2} & \frac{\partial^2\psi_2}{\partial x_3 \partial x_1} - \frac{\partial^2\psi_1}{\partial x_3 \partial x_2} \end{bmatrix} \begin{bmatrix} \Gamma_1 \\ \Gamma_2 \\ \Gamma_3 \end{bmatrix}.$$

Hence, our FMM implementation must perform the computation of first and second derivatives of $\boldsymbol{\psi}$.

For first-order derivatives, most authors implement an analytical form derived from local and multipole expansions. However, for second-order derivatives, authors have differed between implementing an analytical form,[72] or computing the derivates through finite difference.[73] The former approach is carried out with exact precision, with the drawback of further complicating the FMM implementation. On the other hand, the latter approach offers simplicity while forfeiting numerical accuracy. In this study we introduce a new approach: the calculation of vortex stretching through a complex-step derivative approximation. This approach is conceptually simple and easy to implement, while achieving machine precision and without increasing the computational cost.

71 Cheng, Greengard, et al., "A Fast Adaptive Multipole Algorithm in Three Dimensions".

72 C. He et al. (2009). "Modeling Rotor Wake Dynamics with Viscous Vortex Particle Method"; Yokota and Barba, "FMM-based vortex method for simulation of isotropic turbulence on GPUs , compared with a spectral method"; Salloum and Lakkis, "Proper evaluation of spherical harmonics-based expressions for the velocity and vortex stretching vectors in three-dimensional grid-free vortex methods".

73 T. J. Berdowski (2015). "3D Lagrangian VPM-FMM for Modelling the Near-Wake of a HAWT".

Let a real-variable analytic function $f : \mathbb{R} \to \mathbb{R}$ be generalized to its complex-variable form $f^* : \mathbb{C} \to \mathbb{C}$. Let $x_0 \in \mathbb{R}$ be an arbitrary point where we desire to evaluate the first derivative of f. The function f^* can be Taylor-expanded around x_0 with an arbitrarily small step $h \in \mathbb{R}$ to obtain the derivative of f as

$$\frac{\mathrm{d}f}{\mathrm{d}x}(x_0) = \frac{\mathrm{Im}(f^*(x_0 + ih))}{h} + O(h^2).$$

Let us now choose an h sufficiently small—say $h = 10^{-30}$ for double floating point precision—and we have obtained a numerical approximation of the derivative of f that is as accurate as machine precision. This simple numerical method for calculating machine-precision derivatives is called *complex-step derivative approximation*,[74] or CSDA.

74 W. Squire et al. (1998). "Using complex variables to estimate derivatives of real functions"; J. R. R. a. Martins et al. (2003). "The complex-step derivative approximation".

CSDA can be directly applied to any real-variable real-valued function $f : \mathbb{R} \to \mathbb{R}$ by generalizing the function to the complex plane $f^* : \mathbb{C} \to \mathbb{C}$ and using this plane for carrying out the numerical differentiation. However, multipole and local expansions of the FMM are computed through complex-valued functions $Y : \mathbb{R} \to \mathbb{C}$ already using the complex plane, requiring the formulation of CSDA in a multicomplex space. Hence, we will now define the multicomplex arithmetic needed for implementing CSDA in the FMM.

Given the complex set $\mathbb{C}^1$ with imaginary unit i_1 defined as $\mathbb{C}^1 = \{x + i_1 y \,/\, x, y \in \mathbb{R}\}$, we define the multicomplex set $\mathbb{C}^2$ with imaginary unit i_2 as $\mathbb{C}^2 = \{z + i_2 w \,/\, z, w \in \mathbb{C}^1\}$. Any complex-valued analytical function $Y : \mathbb{R} \to \mathbb{C}$ will then be generalized into the multicomplex space $\mathbb{C}^2$ as $Y^* : \mathbb{C}^1 \to \mathbb{C}^2$. In this generalization we transform the set $\mathbb{R}$ into $\mathbb{C}^1$, and $\mathbb{C}$ into $\mathbb{C}^2$; meaning that a variable x originally in $\mathbb{R}$ now becomes $x^* = x + i_1 0$, and a variable $z = a + ib$ originally in $\mathbb{C}$ now becomes $z^* = a^* + i_2 b^* = (a + i_1 0) + i_2(b + i_1 0) = a + i_2 b$. With this definition, the axis i_1 will carry the CSDA computation and is not intended to interact with the axis i_2—hence, the products $i_1 i_2$ and $i_2 i_1$ will purposefully remain undefined as to avoid any unintended arithmetic. The derivative of Y is then computed as

$$\frac{\mathrm{d}Y}{\mathrm{d}x}(x) \approx \frac{\mathrm{Im}_1(Y^*(x + i_1 h))}{h}.$$

Since multipole and local expansions are computed through spherical harmonic functions $Y_n^m : \mathbb{R}^2 \to \mathbb{C}$ of the form

$$Y_n^m(\theta, \phi) = \sqrt{\frac{(n - |m|)!}{(n + |m|)!}} \cdot P_n^{|m|}(\cos\theta) e^{im\phi}, \tag{4.7}$$

where $P_n^{|m|}$ are the associated Legendre functions, we now show how this multicomplex transformation applies to the complex exponential and the complex product in Eq. (4.7). Let the complex-valued function $Y : \mathbb{R} \to \mathbb{C}$ be defined as $Y(x) = e^{ix}$, the derivative is determined analytically

as $\frac{dY}{dx}(x) = ie^{ix}$. On the other hand, its multicomplex generalization $Y^* : \mathbb{C}^1 \to \mathbb{C}^2$ is defined as

$$Y^*(z) = e^{i_2 z},$$

and the derivative $\frac{dY}{dx}(x)$ is computed through CSDA as

$$\begin{aligned}
\frac{dY}{dx}(x) &\approx \frac{\mathrm{Im}_1\left(Y^*(x+i_1h)\right)}{h} \\
&= \frac{1}{h}\mathrm{Im}_1\left(e^{i_2(x+i_1h)}\right) \\
&= \frac{1}{h}\mathrm{Im}_1\left(e^{i_2x}e^{i_1(i_2h)}\right) \\
&= \frac{1}{h}\mathrm{Im}_1\left(e^{i_2x}\left(\cos i_2h + i_1 \sin i_2h\right)\right) \\
&= \frac{1}{h}e^{i_2x}\sin i_2h \\
&= i_2e^{i_2x}\frac{\sinh h}{h} \quad \text{, where } \lim_{h\to 0}\frac{\sinh h}{h} = 1 \\
&\overset{h\to 0}{=} i_2e^{i_2x},
\end{aligned}$$

obtaining the same result as the analytical derivative in the limit $h \to 0$.

Now we show how CSDA is carried on the scalar-complex product. Let $Y : \mathbb{R} \to \mathbb{C}$ be defined as $Y(x) = cx = (c_1 + ic_2)x$ with $c_1, c_2 \in \mathbb{R}$, the derivative is trivially $\frac{dY}{dx}(x) = c$. Defining its multicomplex generalization $Y^* : \mathbb{C}^1 \to \mathbb{C}^2$ as

$$Y^*(z) = c^*z,$$

with $c^* \in \mathbb{C}^2$ / $c^* = c_1^* + i_2c_2^* = (c_1 + i_10) + i_2(c_2 + i_10)$, the derivative $\frac{dY}{dx}(x)$ is then computed through CSDA as

$$\begin{aligned}
\frac{dY}{dx}(x) &\approx \frac{\mathrm{Im}_1\left(Y^*(x+i_1h)\right)}{h} \\
&= \frac{1}{h}\mathrm{Im}_1\left(c^*(x+i_1h)\right) \\
&= \frac{1}{h}\mathrm{Im}_1\left((c_1+i_2c_2)(x+i_1h)\right) \\
&= \frac{1}{h}(c_1+i_2c_2)h \\
&= c_1 + i_2c_2 \\
&= c^*.
\end{aligned}$$

The complex exponential and the complex product are not only good example cases of multicomplex arithmetic, but they are also the only multicomplex operations needed for carrying the CSDA through multipole and local expansions of the FMM.

In order to compare accuracy between finite difference and CSDA, vortex stretching was computed through direct calculation, FMM with

finite difference, and FMM with CSDA in 1000 randomly-generated particles. The error relative to the direct calculation at each particle is shown in Fig. 4.8, evidencing the subtractive and round-off error introduced by the finite difference, meanwhile CSDA approaches floating-point precision (the remaining error is due to the FMM approximation). Furthermore, the CSDA is applied only in the L2P step of the FMM, which takes less than 2% of the computational time of the overall FMM algorithm.[75] Thus, vortex stretching calculated through CSDA has a negligible computational cost, while offering a much simpler implementation than carrying out the calculation analytically.

75 R. Yokota and L. A. Barba (2011b). *Treecode and fast multipole method for N-body simulation with CUDA.*

4.4 Relaxation of ω Divergence

Approximating the filtered vorticity field $\overline{\omega}$ as in Eq. (1.7), here repeated

$$\overline{\boldsymbol{\omega}}(\mathbf{x}) \approx \sum_p \boldsymbol{\Gamma}_p \zeta_\sigma(\mathbf{x} - \mathbf{x}_p), \tag{1.7}$$

introduces numerical instabilities by using and evolving a vorticity field that is, in general, not divergence-free. Cottet and Koumoutsakos[76] suggested that the divergence of $\overline{\omega}$ arises from unphysical small scales produced by the computation, which could be properly addressed with subfilter-scale (SFS) diffusion. Our LES formulation and SFS model remarkably increases the numerical stability of the VPM (as will be shown in Chapter 5); however, our experience is that the divergence of $\overline{\omega}$ still needs to be addressed directly to achieve complete numerical stability. A few methods have been proposed over the years for treating the divergence of $\overline{\omega}$ in a meshless approach. Although we will limit our discussion to Pedrizzetti's relaxation method, the reader is referred to Salloum's doctoral work[77] for a broader review.

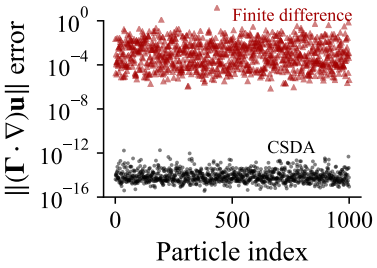


Figure 4.8: Vortex stretching computed through finite difference versus complex-step derivative approximation (CSDA). Error relative to analytic calculation.

Noticing that the velocity field $\overline{\mathbf{u}}$ induced by the particles constructs a divergence-free field by taking its curl, $\nabla \times \overline{\mathbf{u}}$, Pedrizzetti[78] proposed a divergence filtering procedure that aligns the vortex strengths $\boldsymbol{\Gamma}_p$ with this field. The divergence filter is formulated as

$$\frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{\Gamma}_p = -\eta\left(\boldsymbol{\Gamma}_p - \|\boldsymbol{\Gamma}_p\|\frac{\tilde{\boldsymbol{\omega}}(\mathbf{x}_p)}{\|\tilde{\boldsymbol{\omega}}(\mathbf{x}_p)\|}\right),$$

where $\tilde{\boldsymbol{\omega}}$ is the divergence-free vorticity field $\tilde{\boldsymbol{\omega}} \equiv \nabla \times \overline{\mathbf{u}}$, and η is the characteristic frequency of the filter with time scale $T_f = 1/\eta$. This filtering procedure is implemented without adding computational cost since the derivatives of $\overline{\mathbf{u}}$ needed to compute $\tilde{\boldsymbol{\omega}} = \nabla \times \overline{\mathbf{u}}$ are already computed in the vortex stretching calculation. Pedrizzetti suggested casting the filter equation as a relaxation process, updating the vortex strength at each time step as

$$\boldsymbol{\Gamma}_{\text{new}} = (1-\alpha)\boldsymbol{\Gamma} + \alpha\|\boldsymbol{\Gamma}\|\frac{\tilde{\boldsymbol{\omega}}}{\|\tilde{\boldsymbol{\omega}}\|},$$

which we refer to as *Pedrizzetti's relaxation*, where α is the relaxation factor $\alpha = \eta\Delta t$ and Δt is the time step.

76 Cottet and Koumoutsakos, *Vortex Methods Theory and Practice*.

77 S. O. Salloum (2021). "Towards Resurrecting Grid-Free Three Dimensional Vortex Methods".

78 Pedrizzetti, "Insight into singular vortex flows".

Even though Pedrizzetti's relaxation intends to only reorient the vortex strength, it also has the effect of decreasing its magnitude. Defining $\hat{\boldsymbol{\omega}} \equiv \tilde{\omega}/\|\tilde{\omega}\|$, $\hat{\boldsymbol{\Gamma}} \equiv \Gamma/\|\Gamma\|$, and $\Gamma \equiv \|\boldsymbol{\Gamma}\|$, it is readily shown that the magnitude of the vortex strength is not preserved:

$$\begin{aligned}\boldsymbol{\Gamma}_{\text{new}} \cdot \boldsymbol{\Gamma}_{\text{new}} &= \left[(1-\alpha)\boldsymbol{\Gamma} + \alpha\Gamma\hat{\boldsymbol{\omega}}\right] \cdot \left[(1-\alpha)\boldsymbol{\Gamma} + \alpha\Gamma\hat{\boldsymbol{\omega}}\right] \\ &= (1-\alpha)^2\Gamma^2 + 2(1-\alpha)\alpha\Gamma^2\hat{\boldsymbol{\Gamma}} \cdot \hat{\boldsymbol{\omega}} + \alpha^2\Gamma^2 \\ &= \Gamma^2\left[1 - 2(1-\alpha)\alpha(1 - \hat{\boldsymbol{\Gamma}} \cdot \hat{\boldsymbol{\omega}})\right].\end{aligned}$$

Here we see that $\|\boldsymbol{\Gamma}_{\text{new}}\| = \|\boldsymbol{\Gamma}\|$ only if $\boldsymbol{\Gamma}$ is already aligned with $\tilde{\boldsymbol{\omega}}$ (*i.e.*, $\hat{\boldsymbol{\Gamma}} \cdot \hat{\boldsymbol{\omega}} = 1$). In any other case, Pedrizzetti's relaxation decreases the strength magnitude by

$$\frac{\|\boldsymbol{\Gamma}\|^2 - \|\boldsymbol{\Gamma}_{\text{new}}\|^2}{\|\boldsymbol{\Gamma}\|^2} = 2(1-\alpha)\alpha(1 - \hat{\boldsymbol{\Gamma}} \cdot \hat{\boldsymbol{\omega}}),$$

being the largest when $\boldsymbol{\Gamma}$ and $\tilde{\boldsymbol{\omega}}$ are orthogonal to each other (*i.e.*, $\hat{\boldsymbol{\Gamma}} \cdot \hat{\boldsymbol{\omega}} = 0$) and with $\alpha = 1/2$, leading to

$$\left(\frac{\|\boldsymbol{\Gamma}\|^2 - \|\boldsymbol{\Gamma}_{\text{new}}\|^2}{\|\boldsymbol{\Gamma}\|^2}\right)_{\max} = \frac{1}{2}.$$

Thus, Pedrizzetti's relaxation unintentionally decreases the local enstrophy anywhere that vortex strength and vorticity vectors are misaligned, resembling a selective/sensor SFS model.[79] This explains its efficacy in stabilizing some simulations.

79 P. Sagaut (2001). *Large Eddy Simulation for Incompressible Flows*, Sec. 5.3, page 154.

Even though the SFS-diffusion-like behavior of Pedrizzetti's relaxation is a desirable property to increase numerical stability, it is not always effective nor physically accurate. Furthermore, in some cases the user may want to minimize all extraneous sources of numerical diffusion, like in the development and testing of a new SFS model. For this reason, we now propose a correction to Pedrizzetti's relaxation that aligns the strength with the divergence-free field while preserving its magnitude.

We rewrite the relaxation using only the direction vectors and introduce a correction factor b as

$$\hat{\boldsymbol{\Gamma}}_{\text{new}} = \frac{1}{b}\left[(1-\alpha)\hat{\boldsymbol{\Gamma}} + \alpha\hat{\boldsymbol{\omega}}\right].$$

Imposing $\hat{\boldsymbol{\Gamma}}_{\text{new}} \cdot \hat{\boldsymbol{\Gamma}}_{\text{new}} = 1$ we obtain b as

$$\begin{aligned}1 &= \hat{\boldsymbol{\Gamma}}_{\text{new}} \cdot \hat{\boldsymbol{\Gamma}}_{\text{new}} \\ &= \frac{1}{b^2}\left[(1-\alpha)^2 + 2(1-\alpha)\alpha\hat{\boldsymbol{\Gamma}} \cdot \hat{\boldsymbol{\omega}} + \alpha^2\right] \\ &= \frac{1}{b^2}\left[1 - 2\alpha + 2\alpha^2 + 2(1-\alpha)\alpha\hat{\boldsymbol{\Gamma}} \cdot \hat{\boldsymbol{\omega}}\right] \\ &= \frac{1}{b^2}\left[1 - 2(1-\alpha)\alpha + 2(1-\alpha)\alpha\hat{\boldsymbol{\Gamma}} \cdot \hat{\boldsymbol{\omega}}\right] \\ \Rightarrow b^2 &= 1 - 2(1-\alpha)\alpha\left(1 - \hat{\boldsymbol{\Gamma}} \cdot \hat{\boldsymbol{\omega}}\right).\end{aligned}$$

Then, the vortex strength is updated as

$$\boldsymbol{\Gamma}_{\text{new}} = \|\boldsymbol{\Gamma}\|\hat{\boldsymbol{\Gamma}}_{\text{new}},$$

or

$$\boldsymbol{\Gamma}_{\text{new}} = \|\boldsymbol{\Gamma}\|\frac{(1-\alpha)\hat{\boldsymbol{\Gamma}} + \alpha\hat{\boldsymbol{\omega}}}{\sqrt{1-2(1-\alpha)\alpha\left(1-\hat{\boldsymbol{\Gamma}}\cdot\hat{\boldsymbol{\omega}}\right)}}.$$

Throughout Part I, the divergence of $\overline{\omega}$ will be treated using the corrected relaxation, which, as shown in Chapter 5, will be sufficient to attain both numerical stability and physical accuracy with the LES formulation and SFS model developed in this study. However, we have observed that the actuator line and surface models used to introduce the vorticity of blades and wings in Part II can cause a numerical instability that is not damped by our corrected relaxation. This instability is triggered when the rotating blades chop through a wake, but we have observed that the original Pedrizzetti's relaxation succeeds at stabilizing it. For this reason, the rotor simulation in Section 5.4 and all simulations in Part II will use the original Pedrizzetti's relaxation. All simulation in this study will use a relaxation factor α of 0.3.

4.5 Transposed Stretching

As noticed by Rehbach,[80] the inviscid part of the unfiltered vorticity equation,

$$\frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{\omega} = (\boldsymbol{\omega}\cdot\nabla)\,\mathbf{u}$$

can be written as

$$\frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{\omega} = \left(\boldsymbol{\omega}\cdot\nabla^T\right)\mathbf{u}, \tag{4.8}$$

where ∇^T denotes the transpose of the stretching operator. Both forms are equivalent since

$$\left[\boldsymbol{\omega}\cdot\left(\nabla-\nabla^T\right)\right]\mathbf{u} = (\nabla\times\mathbf{u})\times\boldsymbol{\omega} = \boldsymbol{\omega}\times\boldsymbol{\omega} = 0.$$

The evolution equations obtained by discretizing both forms would also be equivalent if the particle-approximated vorticity field were equal to the curl of the velocity field induced by the particles, $\nabla\times\mathbf{u}$. Unfortunately, this is not the case, which is a consequence of the vorticity field not being divergence-free, as shown in Appendix C.

The transposed form offers two numerical advantages over the original form when they are both discretized with singular vortex particles: the resulting evolution equations, first, guarantee conservation of the total vorticity and, second, constitute a weak solution to the inviscid vorticity equation.[81] Thus, the reformulated VPM is able to achieve

80 C. Rehbach (1978). "Numerical calculation of three-dimensional unsteady flows with vortex sheets".

81 G. Winckelmans et al. (1988). "Weak solutions of the three-dimensional vorticity equation with vortex singularities".

numerical stability with the transposed scheme, which we believe is a consequence of having used singular vortex particles to discretize the LES-filtered vorticity equation and derive its governing equations (Sections 1.4 and 1.5).

Filtering the transposed vorticity equation given in Eq. (4.8) and discretizing it with singular particles, the evolution equations of strength and core size in the reformulated VPM, previously summarized in Section 3.6, now become

$$\frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{\Gamma}_p = \left(\boldsymbol{\Gamma}_p \cdot \nabla^T\right)\overline{\mathbf{u}}(\mathbf{x}_p) - \frac{3}{5}\left\{\left[\left(\boldsymbol{\Gamma}_p \cdot \nabla^T\right)\overline{\mathbf{u}}(\mathbf{x}_p)\right]\cdot\hat{\boldsymbol{\Gamma}}_p\right\}\hat{\boldsymbol{\Gamma}}_p - \frac{C_d}{\zeta_{\sigma_p}(\mathbf{0})}\mathbf{E}_{\mathrm{str}}(\mathbf{x}_p)$$

$$\frac{\mathrm{d}}{\mathrm{d}t}\sigma_p = -\frac{1}{5}\frac{\sigma_p}{\|\boldsymbol{\Gamma}_p\|}\left[\left(\boldsymbol{\Gamma}_p \cdot \nabla^T\right)\overline{\mathbf{u}}(\mathbf{x}_p)\right]\cdot\hat{\boldsymbol{\Gamma}}_p.$$

It also follows that the SFS vortex stretching model becomes

$$\mathbf{E}_{\mathrm{str}}(\mathbf{x}) = \sum_q \zeta_\sigma(\mathbf{x}-\mathbf{x}_q)\left(\boldsymbol{\Gamma}_q \cdot \nabla^T\right)\left(\overline{\mathbf{u}}(\mathbf{x}) - \overline{\mathbf{u}}(\mathbf{x}_q)\right),$$

and all stretching operators used in the dynamic procedure become transposed. All simulations presented in this study will use the transposed stretching scheme.

4.6 Viscous Diffusion

The viscous part of the LES-filtered vorticity equation is

$$\left(\frac{\mathrm{d}}{\mathrm{d}t}\overline{\boldsymbol{\omega}}\right)_{\mathrm{viscous}} = \nu\nabla^2\overline{\boldsymbol{\omega}}, \tag{4.9}$$

which is solved separate from the inviscid part through operator splitting. Core spreading (CS) is a numerical scheme that solves the diffusion equation by widening the core size σ (or filter width) over time, effectively diffusing the vorticity as follows.

Given a Gaussian vorticity distribution of the form

$$\overline{\boldsymbol{\omega}}(\mathbf{x}) = \frac{\boldsymbol{\Gamma}}{(2\pi\sigma^2)^{3/2}}\exp\left(-\frac{\|\mathbf{x}-\mathbf{x}_0\|^2}{2\sigma^2}\right),$$

the exact solution to Eq. (4.9) is

$$\frac{\mathrm{d}\sigma^2}{\mathrm{d}t} = 2\nu. \tag{4.10}$$

In a VPM with a Gaussian kernel, viscous diffusion can be accounted for by spreading the particle size according to Eq. (4.10). Greengard showed that this approach was inconsistent with the Navier-Stokes equations,[82] however, Rossi[83] later proved its correct convergence to the Navier-Stokes equations provided that core sizes are controlled and kept small through spatial adaptation. As shown by Rossi, the core spreading

82 C. Greengard (1985). "The core spreading vortex method approximates the wrong equation".

83 Rossi, "Resurrecting Core Spreading Vortex Methods: A New Scheme that is Both Deterministic and Convergent".

method has second-order spatial convergence, while showing linear convergence when coupled with spatial adaptation.

One approach to control core sizes is to introduce a splitting algorithm that divides overgrown particles into smaller particles. This poses the difficulty of implementing a merging algorithm that will merge particles in regions of high concentration, otherwise the number of particles will grow unboundedly.

Barba *et al.*[84] developed an alternative approach for spatial adaptation exploiting the numerical machinery of radial basis function (RBF) interpolations.[85] The method consists of resetting the spatial discretization (after particles are determined to have overgrown) with a new field of vortex particles, determining the strengths of the new particles with an RBF interpolation procedure so as to preserve the same vorticity field. The RBF interpolation can be performed through either a direct matrix inversion or an iterative method. A direct inversion readily becomes computationally infeasible with only a modest number of particles. Hence, iterative methods are usually preferred, like conjugate gradient or generalized minimal residual (GMRES). Conjugate gradient works on a system matrix that is symmetric and positive definite, which is the case when the all particles are reset to the same core size, however, this precludes the use of preconditioners. GMRES is a general method with fast convergence when the problem is correctly preconditioned, at the cost of a laborious and lengthy implementation. Both of these iterative methods are well suited to have the matrix-vector product evaluated through the FMM in complexity $\mathcal{O}(N)$.

84 Barba et al., "Advances in viscous vortex methods - Meshless spatial adaption based on radial basis function interpolation"; Barba, "Vortex Method for computing high-Reynolds number Flows: Increased accuracy with a fully mesh-less formulation".

85 Torres et al., "Fast radial basis function interpolation with Gaussians by localization and iteration".

In this study, viscous diffusion is resolved through the core spreading scheme, Eq. (4.10), coupled with Barba's spatial adaptation approach by RBF interpolation, denoted as CS-RBF. Our spatial adaptation consists of resetting all core sizes σ to a prescribed value σ_0 after a critical time t_{crit}. For simplicity, the RBF interpolation is implemented through the conjugate gradient method, with the matrix-vector product evaluated through the FMM. In order to determine the frequency of core resetting t_{crit}, we define a growth parameter β as

$$\beta(t) \equiv \frac{\sigma}{\sigma_0},$$

which, according to Eq. (4.10), becomes

$$\beta(t) = \sqrt{\frac{\sigma_0^2 + 2\nu t}{\sigma_0^2}}.$$

Then, to avoid the particles from growing beyond a critical threshold β_{crit}, the frequency of core resetting is calculated as

$$t_{\text{crit}} = (\beta_{\text{crit}}^2 - 1)\frac{\sigma_0^2}{2\nu}.$$

4.7 Implementation of the Dynamic Procedure

In Chapter 3, we developed a model of subfilter-scale vortex stretching,

$$\mathbf{E}_{\text{str}}(\mathbf{x}) = \sum_q \zeta_\sigma(\mathbf{x} - \mathbf{x}_q)\left(\boldsymbol{\Gamma}_q \cdot \nabla\right)\left(\overline{\mathbf{u}}(\mathbf{x}) - \overline{\mathbf{u}}(\mathbf{x}_q)\right),$$

with a model coefficient C_d that is calculated dynamically at the position of every particle as

$$C_d = \frac{\langle \boldsymbol{\Gamma}_p \cdot \mathbf{L} \rangle}{\langle \boldsymbol{\Gamma}_p \cdot \mathbf{m} \rangle}, \tag{4.11}$$

where $\langle \cdot \rangle$ denotes an integration along Lagrangian trajectories, and

$$\mathbf{m} \equiv \frac{\sigma^3}{\zeta(0)} \frac{\partial \mathbf{E}_{\text{str}}}{\partial \sigma}(\mathbf{x}_p)$$
$$\mathbf{L} \equiv \frac{3}{\sigma}\left(\boldsymbol{\Gamma}_p \cdot \nabla\right)\left(\mathbf{u}(\mathbf{x}_p) - \overline{\mathbf{u}}(\mathbf{x}_p)\right) + \left(\boldsymbol{\Gamma}_p \cdot \nabla\right)\frac{\partial \overline{\mathbf{u}}}{\partial \sigma}(\mathbf{x}_p).$$

Since $\mathbf{m}$ and $\mathbf{L}$ require the derivatives of both SFS model $\mathbf{E}_{\text{str}}$ and velocity field $\overline{\mathbf{u}}$ with respect to the filter width σ, we now show how these derivatives can be approximated numerically applying various filtering levels. In our proposed approach, the numerical implementation of C_d will end up resembling a multilevel approach[86] where the desired quantities are computed from variational scales obtained by applying multiple filtering levels.

86 Sagaut, *Large Eddy Simulation for Incompressible Flows*, Sec. 7.7.7.

4.7.1 Two-Level Procedure

First, assume that the primary-filter width [87] σ is small enough such that $\mathbf{u} \approx \overline{\mathbf{u}}$. The numerator term $\mathbf{L}$ then becomes

$$\mathbf{L} \approx \left(\boldsymbol{\Gamma}_p \cdot \nabla\right)\frac{\partial \overline{\mathbf{u}}}{\partial \sigma}(\mathbf{x}_p).$$

87 The primary-filter width is also the particle core size or the RBF smoothing radius. Thus, they are all denoted as σ.

We define a test filter with width $\sigma_t = \alpha_t \sigma$ and $\alpha_t > 1$, denoted as $\widehat{(\cdot)}$ while the primary filter is denoted as $\overline{(\cdot)}$. The derivatives with respect to the filter width are then approximated as

$$\frac{\partial \mathbf{E}_{\text{str}}}{\partial \sigma}(\mathbf{x}_p) \approx \frac{\mathbf{E}_{\text{str}}(\mathbf{x}_p, \sigma_t) - \mathbf{E}_{\text{str}}(\mathbf{x}_p, \sigma)}{\sigma_t - \sigma}$$
$$\frac{\partial \overline{\mathbf{u}}}{\partial \sigma}(\mathbf{x}_p) \approx \frac{\widehat{\mathbf{u}}(\mathbf{x}_p) - \overline{\mathbf{u}}(\mathbf{x}_p)}{\sigma_t - \sigma},$$

where the test-filtered fields $\mathbf{E}_{\text{str}}(\mathbf{x}_p, \sigma_t)$ and $\widehat{\mathbf{u}}$ are computed expanding each particle core by α_t. This can also be written as

$$\frac{\partial \mathbf{E}_{\text{str}}}{\partial \sigma}(\mathbf{x}_p) \approx \frac{\mathbf{E}_{\text{str}}(\mathbf{x}_p, \alpha_t\sigma) - \mathbf{E}_{\text{str}}(\mathbf{x}_p, \sigma)}{\sigma(\alpha_t - 1)}$$
$$\frac{\partial \overline{\mathbf{u}}}{\partial \sigma}(\mathbf{x}_p) \approx \frac{\widehat{\mathbf{u}}(\mathbf{x}_p) - \overline{\mathbf{u}}(\mathbf{x}_p)}{\sigma(\alpha_t - 1)}$$

Substituting this into the definition of $\mathbf{m}$ and $\mathbf{L}$, the dynamic procedure $C_d = \langle \Gamma_p \cdot \mathbf{L} \rangle / \langle \Gamma_p \cdot \mathbf{m} \rangle$ is then implemented as

$$C_d = \frac{\left\langle \boldsymbol{\Gamma}_p \cdot \left[\left(\boldsymbol{\Gamma}_p \cdot \nabla \right) \left(\widehat{\mathbf{u}}(\mathbf{x}_p) - \overline{\mathbf{u}}(\mathbf{x}_p) \right) \right] \right\rangle}{\left\langle \frac{\sigma^3}{\zeta(0)} \boldsymbol{\Gamma}_p \cdot \left(\mathbf{E}_{\text{str}}(\mathbf{x}_p, \sigma_t) - \mathbf{E}_{\text{str}}(\mathbf{x}_p, \sigma) \right) \right\rangle}. \tag{4.12}$$

This numerical scheme elucidates the variational principle underlying our dynamic procedure, which shown as follows. Using the numerical derivatives in the definition of $\mathbf{m}$ and $\mathbf{L}$, the derivative balance introduced in Section 3.3.1, here repeated

$$C_d \mathbf{m} = \mathbf{L},$$

becomes

$$C_d \frac{\sigma^3}{\zeta(0)} \left(\mathbf{E}_{\text{str}}(\mathbf{x}_p, \sigma_t) - \mathbf{E}_{\text{str}}(\mathbf{x}_p, \sigma) \right) = \underbrace{\left(\boldsymbol{\Gamma}_p \cdot \nabla \right) \widehat{\mathbf{u}}(\mathbf{x}_p)}_{\mathbf{S}_{\sigma_t}} - \underbrace{\left(\boldsymbol{\Gamma}_p \cdot \nabla \right) \overline{\mathbf{u}}(\mathbf{x}_p)}_{\mathbf{S}_{\sigma}}.$$

For simplicity, we rewrite this as

$$-C_d \left(\mathbf{E}_{\sigma_t} - \mathbf{E}_{\sigma} \right) = \mathbf{S}_{\sigma} - \mathbf{S}_{\sigma_t}, \tag{4.13}$$

where, $\mathbf{E}_{\sigma_t}$ and $\mathbf{E}_{\sigma}$ denote the subfilter-scale vortex stretching at test and primary filter levels, respectively. The two terms in the right-hand side, $\mathbf{S}_{\sigma_t}$ and $\mathbf{S}_{\sigma}$, represent the large-scale vortex stretching at test and primary levels.

Following the derivation given in Section 3.3.2, the dot-product of $\boldsymbol{\omega}$ with each $\mathbf{E}$ and $\mathbf{S}$ term in Eq. (4.13) corresponds to a contributor to enstrophy production. Fig. 4.9 depicts the range of enstrophy production that each large-scale term provides to the enstrophy spectrum, with the red hatched region corresponding to $\mathbf{S}_{\sigma} - \mathbf{S}_{\sigma_t}$. The range of each subfilter-scale term is also shown, with the blue region corresponding to $\mathbf{E}_{\sigma_t} - \mathbf{E}_{\sigma}$. The negative sign in Eq. (4.13) indicates that the enstrophy that is gained/lost in the subfilter scales must be lost/gained in the large scales, while C_d is there to enforce the equality once the SFS model is introduced.

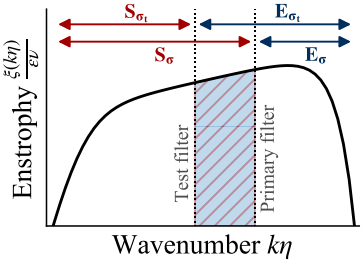


Figure 4.9: Enstrophy spectrum showing ranges of enstrophy production by large-scale vortex stretching $\mathbf{S}$ and SFS vortex stretching $\mathbf{E}_{\text{str}}$ with primary and test filters.

In short, C_d is the ratio of how sensitive the resolved enstrophy production is to the filter width, over the sensitivity of the unresolved enstrophy production: C_d increases or decreases every time and everywhere that the sensitivity of the resolved scales to σ does not match the sensitivity of the modeled scales. This comes back to the fact that the $\mathbf{E}_{\text{str}}$ model is not fully accurate, hence C_d attempts to correct its error.

4.7.2 Pseudo-Three-Level Procedure

We now repeat the same derivation but without assuming that σ is sufficiently small to justify $\mathbf{u} \approx \overline{\mathbf{u}}$. We define a test filter with width $\sigma_\tau = \alpha_\tau \sigma$ and $\alpha_\tau < 1$, denoted as $\check{(\cdot)}$. In the same manner as the two-level

procedure, the filter derivatives are approximated as

$$\frac{\partial \mathbf{E}_{\text{str}}}{\partial \sigma}(\mathbf{x}_p) \approx \frac{\mathbf{E}_{\text{str}}(\mathbf{x}_p,\, \alpha_\tau \sigma) - \mathbf{E}_{\text{str}}(\mathbf{x}_p,\, \sigma)}{\sigma(\alpha_\tau - 1)}$$
$$\frac{\partial \overline{\mathbf{u}}}{\partial \sigma}(\mathbf{x}_p) \approx \frac{\breve{\mathbf{u}}(\mathbf{x}_p) - \overline{\mathbf{u}}(\mathbf{x}_p)}{\sigma(\alpha_\tau - 1)}.$$

Now, instead of assuming that the primary filter σ is small enough to satisfy $\mathbf{u} \approx \overline{\mathbf{u}}$, we introduce a second test filter with width $\sigma_{\tau^*} = \alpha_{\tau^*}\sigma$ and $\alpha_{\tau^*} \ll 1$, denoted as $\breve{(\cdot)}$. Assuming that σ_{τ^*} is small enough such that $\mathbf{u} \approx \breve{\mathbf{u}}$, and using the numerical derivative shown above, the numerator term $\mathbf{L}$ then becomes

$$\mathbf{L} \approx \frac{3}{\sigma}\left(\mathbf{\Gamma}_p \cdot \nabla\right)\left(\breve{\mathbf{u}}(\mathbf{x}_p) - \overline{\mathbf{u}}(\mathbf{x}_p)\right) + \left(\mathbf{\Gamma}_p \cdot \nabla\right)\frac{\breve{\mathbf{u}}(\mathbf{x}_p) - \overline{\mathbf{u}}(\mathbf{x}_p)}{\sigma(\alpha_\tau - 1)}.$$

This expression calls for the velocity at three different filtering levels: $\overline{\mathbf{u}}$ at the primary filter level, and $\breve{\mathbf{u}}$ and $\breve{\breve{\mathbf{u}}}$ at the two test-filter levels, leading to a three-level procedure.

One filtering level can be avoided if we assume that the two test-filter levels are close to each other, *i.e.*, $\sigma_\tau \approx \sigma_{\tau^*}$ and $\breve{\mathbf{u}} \approx \breve{\breve{\mathbf{u}}}$ leading to $\breve{\mathbf{u}} \approx \mathbf{u}$. The numerator term then simplifies to

$$\mathbf{L} \approx \frac{3\alpha_\tau - 2}{\sigma(\alpha_\tau - 1)}\left(\mathbf{\Gamma}_p \cdot \nabla\right)\left(\breve{\mathbf{u}}(\mathbf{x}_p) - \overline{\mathbf{u}}(\mathbf{x}_p)\right),$$

and the dynamic procedure $C_d = \langle \mathbf{\Gamma}_p \cdot \mathbf{L} \rangle / \langle \mathbf{\Gamma}_p \cdot \mathbf{m} \rangle$ becomes

$$C_d = \frac{\left\langle \mathbf{\Gamma}_p \cdot \left[\frac{3\alpha_\tau - 2}{\sigma(\alpha_\tau - 1)}\left(\mathbf{\Gamma}_p \cdot \nabla\right)\left(\breve{\mathbf{u}}(\mathbf{x}_p) - \overline{\mathbf{u}}(\mathbf{x}_p)\right)\right]\right\rangle}{\left\langle \mathbf{\Gamma}_p \cdot \left(\frac{\sigma^3}{\zeta(0)} \frac{\mathbf{E}_{\text{str}}(\mathbf{x}_p, \sigma_\tau) - \mathbf{E}_{\text{str}}(\mathbf{x}_p, \sigma)}{\sigma(\alpha_\tau - 1)} \right) \right\rangle},$$

which is implemented as

$$C_d = (3\alpha_\tau - 2)\frac{\left\langle \mathbf{\Gamma}_p \cdot \left[\left(\mathbf{\Gamma}_p \cdot \nabla\right)\left(\breve{\mathbf{u}}(\mathbf{x}_p) - \overline{\mathbf{u}}(\mathbf{x}_p)\right)\right]\right\rangle}{\left\langle \frac{\sigma^3}{\zeta(0)}\mathbf{\Gamma}_p \cdot \left(\mathbf{E}_{\text{str}}(\mathbf{x}_p,\, \sigma_\tau) - \mathbf{E}_{\text{str}}(\mathbf{x}_p,\, \sigma)\right)\right\rangle}. \tag{4.14}$$

Since this was derived from a three-level procedure, but then implemented using only two levels, we call it a *pseudo-three-level procedure*.

Comparing Eq. (4.12) and Eq. (4.14), it becomes evident that the pseudo-three-level procedure is simply the two-level procedure scaled by the factor $3\alpha_\tau - 2$. Although the two-level procedure correlates directly to a balance of spectral energy transfer, the pseudo-three-level procedure lacks formalism since the scaling factor is a numerical artifact without physical significance. However, in some contexts it may be advantageous to use this procedure since it provides the user with a tunable parameter, α_τ, while converging to the two-level procedure when $\alpha_\tau \to 1$. For instance, noticing that the scaling factor $3\alpha_\tau - 2$ tends to zero when $\alpha_\tau \to 2/3$, the pseudo-three-level procedure can be used

to either arbitrarily attenuate the SFS contributions with $\alpha_\tau \to 2/3$, or let it be a trully self-regulated dynamic procedure with $\alpha_\tau \to 1$. As a reference, Table 4.1 lists the scaling factor as a function of α_τ and the resulting effect on the SFS contributions.

Since the pseudo-three-level procedure converges to the two-level procedure when $\alpha_\tau \to 1$, only the former was implemented in this study. All simulations reported in this study used $\alpha_\tau = 0.999$. Since our SFS vortex stretching model, $\mathbf{E}_{\text{str}}$, involves the radially-decaying function ζ_σ, the influence of far-away particles decays to zero. Hence, $\mathbf{E}_{\text{str}}$ is computed only over the P2P step of the FMM, rather than including interactions over the entire particle field. In this manner, both the SFS model and its dynamic procedure are computed in the same linear complexity $\mathcal{O}(N)$ of the FMM.

Table 4.1: Tunning parameter α_τ and scaling factor $3\alpha_\tau - 2$ in pseudo-three-level procedure, categorized by the effect on SFS contributions.

α_τ	$3\alpha_\tau - 2$	
4	10	} amplify
1.333	2	
1.001	1.003	
1	1	←undefined
0.999	0.997	} attenuate
0.833	0.5	
0.750	0.25	
0.700	0.1	
0.670	0.01	
0.667	0.001	
$< 2/3$	< 0	←unphysical

4.8 Implementation of the Governing Equations

As given in Section 3.6, the generalized reformulated VPM equations are

$$\frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{\Gamma}_p = \left(\boldsymbol{\Gamma}_p \cdot \nabla\right)\overline{\mathbf{u}}(\mathbf{x}_p) - \frac{g+f}{1/3+f}\left\{\left[\left(\boldsymbol{\Gamma}_p \cdot \nabla\right)\overline{\mathbf{u}}(\mathbf{x}_p)\right]\cdot\hat{\boldsymbol{\Gamma}}_p\right\}\hat{\boldsymbol{\Gamma}}_p - \frac{C_d}{\zeta_{\sigma_p}(\mathbf{0})}\left[\mathbf{E}(\mathbf{x}_p) - \frac{f}{1/3+f}\left(\mathbf{E}(\mathbf{x}_p)\cdot\hat{\boldsymbol{\Gamma}}_p\right)\hat{\boldsymbol{\Gamma}}_p\right]$$

and

$$\frac{\mathrm{d}}{\mathrm{d}t}\sigma_p = -\left(\frac{g+f}{1+3f}\right)\frac{\sigma_p}{\|\boldsymbol{\Gamma}_p\|}\left[\left(\boldsymbol{\Gamma}_p \cdot \nabla\right)\overline{\mathbf{u}}(\mathbf{x}_p)\right]\cdot\hat{\boldsymbol{\Gamma}}_p + \left(\frac{f}{1+3f}\right)\frac{\sigma_p}{\|\boldsymbol{\Gamma}_p\|}\frac{C_d}{\zeta_{\sigma_p}(\mathbf{0})}\mathbf{E}(\mathbf{x}_p)\cdot\hat{\boldsymbol{\Gamma}}_p,$$

where $\mathbf{E} = \mathbf{E}_{\text{adv}} + \mathbf{E}_{\text{str}}$. In order to ease the implementation and reduce both storage and computation, we define the following auxiliary variables:

$$\begin{aligned}
\mathbf{S}_p &= \left(\boldsymbol{\Gamma}_p \cdot \nabla\right)\overline{\mathbf{u}}(\mathbf{x}_p)\\
Z_p &= \left(\frac{g+f}{1+3f}\right)\frac{\mathbf{S}_p\cdot\boldsymbol{\Gamma}_p}{\|\boldsymbol{\Gamma}_p\|^2} - \left(\frac{f}{1+3f}\right)\frac{C_d\boldsymbol{\epsilon}_p\cdot\boldsymbol{\Gamma}_p}{\|\boldsymbol{\Gamma}_p\|^2}\\
\boldsymbol{\epsilon}_p &= \frac{1}{\zeta_{\sigma_p}(\mathbf{0})}\left(\mathbf{E}_{\text{adv}}(\mathbf{x}_p) + \mathbf{E}_{\text{str}}(\mathbf{x}_p)\right).
\end{aligned}$$

The generalized reformulated VPM equations are then implemented as

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{\Gamma}_p &= \mathbf{S}_p - 3Z_p\boldsymbol{\Gamma}_p - C_d\boldsymbol{\epsilon}_p\\
\frac{\mathrm{d}}{\mathrm{d}t}\sigma_p &= -\sigma_p Z_p.
\end{aligned}$$

The transposed stretching scheme is turned on by simply calculating $\mathbf{S}_p$ with the transposed operator.

Since the generalized reformulated VPM collapses back to the classic VPM when $f = g = 0$, a simulation using $f = g = 0$ is referred to as a classic-VPM simulation, or cVPM. A simulation using $f = 0$ and $g = 1/5$ is referred to as a reformulated-VPM simulation, or rVPM.

The equations above solve only the inviscid part of the vorticity equation, and must be modified according to the viscous scheme of preference to solve the viscous part. In the case of the core spreading scheme, the core-size equation becomes

$$\frac{\mathrm{d}}{\mathrm{d}t}\sigma_p = -\sigma_p Z_p + \frac{\nu}{\sigma_p},$$

where ν is the kinematic viscosity.

The only state variables of our meshless LES are the position of the particles $\mathbf{x}_p$, the vortex strengths $\boldsymbol{\Gamma}_p$, and the filter widths σ_p (also referred to as smoothing radius or core size). In the reformulated VPM these are all time-dependent variables. The particle field constructs a continuous vorticity field computed through radial basis functions as

$$\overline{\boldsymbol{\omega}}(\mathbf{x}) = \sum_p \boldsymbol{\Gamma}_p \zeta_{\sigma_p}\left(\mathbf{x} - \mathbf{x}_p\right)$$

and a continuous velocity field by inverting $\boldsymbol{\omega} = \nabla \times \mathbf{u}$ as explained in Section 4.1, resulting in

$$\overline{\mathbf{u}}\left(\mathbf{x}\right) = -\frac{1}{4\pi}\sum_p g_{\sigma_p}\left(\mathbf{x} - \mathbf{x}_p\right)\frac{\mathbf{x} - \mathbf{x}_p}{\|\mathbf{x} - \mathbf{x}_p\|^3} \times \boldsymbol{\Gamma}_p. \tag{4.15}$$

Hence, all fluid properties—like $\overline{\mathbf{u}}$ and any of its spatial derivatives—can be computed analytically from the state variables. When the dynamic procedure is used, both the numerator and denominator of $C_d = \langle \boldsymbol{\Gamma}_p \cdot \mathbf{L} \rangle / \langle \boldsymbol{\Gamma}_p \cdot \mathbf{m} \rangle$ also become state variables in order to perform the Lagrangian integration.

4.9 Time Integration

VPM simulations typically grow to the order of millions of particles, which poses a problem for common high-order integration schemes that require storing the value of each state variable at each subiteration step. Williamson[88] formulated alternative low-storage high-order schemes requiring only two storage locations per variable, independent of the scheme order, which we summarize as follows.

88 J. H. Williamson (1980). "Low-storage Runge-Kutta schemes".

Given the vector of state variables $\mathbf{y} \in \mathbb{R}^N$ and a set of differential equations describing the evolution of $\mathbf{y}$ as

$$\frac{\mathrm{d}\mathbf{y}}{\mathrm{d}t} = \mathbf{f}(\mathbf{y}),$$

Williamson's method integrates the equation through the algorithm

$$\begin{aligned}\mathbf{z}_i &= a_i \mathbf{z}_{i-1} + \Delta t \mathbf{f}(\mathbf{y}_{i-1})\\ \mathbf{y}_i &= \mathbf{y}_{i-1} + b_i \mathbf{z}_i.\end{aligned}$$

Here, $\mathbf{z} \in \mathbb{R}^N$ is an auxiliary storage space, a_i and b_i are constants of the integration scheme, Δt is the time step, and i is the index of the subiteration step. The values of $\mathbf{z}$ and $\mathbf{y}$ are updated in place, so only $2N$ storage locations are required, where N is the number of state variables.

In this study, the governing equations are integrated implementing Williamson's third-order low-storage integration scheme with the following three subiterations steps:

$$
\begin{aligned}
i &= 1 \\
&\qquad \mathbf{z}_1 = \Delta t \mathbf{f}(\mathbf{y}_0) \\
&\qquad \mathbf{y}_1 = \mathbf{y}_0 + \frac{1}{3}\mathbf{z}_1 \\
i &= 2 \\
&\qquad \mathbf{z}_2 = -\frac{5}{9}\mathbf{z}_1 + \Delta t \mathbf{f}(\mathbf{y}_1) \\
&\qquad \mathbf{y}_2 = \mathbf{y}_1 + \frac{15}{16}\mathbf{z}_2 \\
i &= 3 \\
&\qquad \mathbf{z}_3 = -\frac{153}{128}\mathbf{z}_2 + \Delta t \mathbf{f}(\mathbf{y}_2) \\
&\qquad \mathbf{y}_3 = \mathbf{y}_2 + \frac{8}{15}\mathbf{z}_3
\end{aligned}
$$

5 Validation of Meshless LES Scheme

In the preceding chapters we have developed a scheme for numerically solving the LES-filtered Navier-Stokes equations in their vorticity form. The proposed scheme uses a reformulation of the VPM and a novel model of SFS vortex stretching to achieve a meshless large eddy simulation.

We now proceed to test and validate both the VPM reformulation and the vortex-stretching SFS model that comprise our meshless LES. Simulations are compared to results reported in the literature from experimental work, direct numerical simulation (DNS), the lattice Boltzmann method (LBM), and unsteady Reynolds-average Navier-Stokes (URANS) simulation. These validation cases also serve as examples on how to impose initial and boundary conditions in our meshless scheme.

In Section 5.1, advection and viscous diffusion is validated on an isolated vortex ring, which is also used to test convergence of the reformulated VPM. In Section 5.2, vortex stretching is validated through the simulation of two leapfrogging vortex rings. Results with classic and reformulated VPM are first compared without SFS effects. The SFS model is then introduced to capture the subfilter energy cascade that leads the rings into turbulent breakdown. In Section 5.3, a turbulent round jet is simulated to test the accuracy and numerical stability of the LES scheme. The evolution of the jet and predicted Reynolds stress are compared to experimental measurements, validating the scheme as an LES method accurately resolving large-scale features of turbulent flow. Finally, in Section 5.4, our meshless LES is compared to conventional mesh-based CFD in a real-world engineering problem: an aircraft rotor and its wake.

5.1 Isolated Vortex Ring

A vortex ring travels with a self-induced velocity that slowly decelerates as its vorticity spreads due to viscous diffusion. This poses a good validation case for both vorticity advection and viscous diffusion. The initial vorticity inside the ring core is assumed to follow a Gaussian distribution of the form $\omega_\theta(r) = \frac{\Gamma_0}{\pi a^2} e^{-r^2/a^2}$, where Γ_0 and a are the ring's initial circulation and core size, respectively. The initial Reynolds number is defined as $\mathrm{Re} = \Gamma_0/\nu$. In all the following simulations, the rings were initially discretized by placing particles (evenly spaced by Δx) everywhere that the local vorticity was larger than 5% of the peak vorticity $\omega_\theta(0)$, and

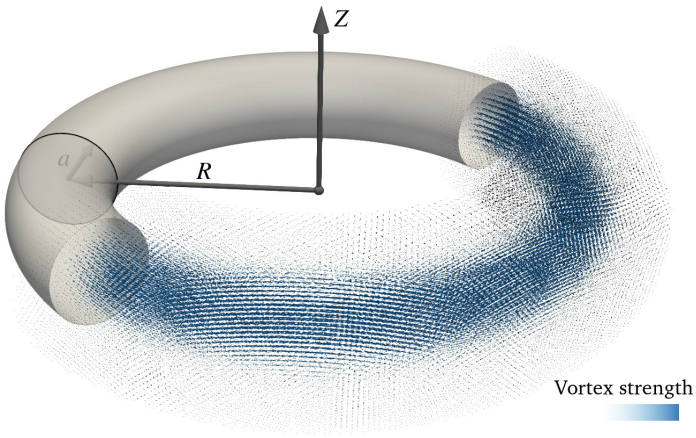


Figure 5.1: Vortex ring as discretized through particles with $\Delta x = 0.13a$.

then performing a radial-basis function (RBF) fit to the target vorticity ω_θ to determine the initial vortex strength of the particles. The initial particle size σ was set as to provide a particle overlap $\lambda = \frac{\sigma}{\Delta x}$ of 2.4. Fig. 5.1 shows a discretized vortex ring along with a depiction of the ring's geometric parameters.

A vortex ring at Re $= 7500$ and thickness $\frac{a}{R} = 0.2$ was simulated and compared to direct-numerical simulation (DNS) reported by Archer *et al.*[89] The DNS is a finite-difference method with second-order spatial accuracy solving the incompressible pressure-velocity form of the Navier Stokes equations, using over 100 million grid cells.

89 P. J. Archer et al. (2008). "Direct numerical simulation of vortex ring evolution from the laminar to the early turbulent regime".

Convergence of the reformulated VPM was tested without the SFS model by increasing the spatial resolution from $\Delta x = 0.17a$ (resulting in 60k particles) to $\Delta x = 0.08a$ (resulting in 500k particles), shown in Fig. 5.2. To minimize numerical noise, we suppressed the core-resetting step of the viscous diffusion scheme,[90] which leads the VPM simulation to become increasingly coarser as time goes by. For this reason, and without the SFS model, the VPM simulation is under-resolved and inaccurate in the turbulent regime This is observed in Fig. 5.2 as an abrupt turbulent breakdown after transition. In the laminar regime (nondimensional time between 0 and 50), the velocity shows little sensitivity to a spatial discretization finer than $\Delta x = 0.13a$, being between 4% of the DNS velocity at time 0, as shown in the zoom-in box of Fig. 5.2. The DNS shows a slow transition from laminar to turbulent starting at nondimensional time 75, while the VPM also transitions around time 75 for $\Delta x \leq 0.13a$, quickly leading the ring into turbulent breakdown. Hence, $\Delta x = 0.13a$ was used in all the simulations that follow, which results in 150k particles. These 150k vortex elements, contrasted with the 100 million grid elements used in the DNS, highlight the computational efficiency of our meshless LES, which enabled us to run our simulations in only a matter of hours on a desktop workstation.

90 Barba et al., "Advances in viscous vortex methods - Meshless spatial adaption based on radial basis function interpolation".

Fig. 5.3 compares the DNS to classic and reformulated VPM simulations, along with the analytic velocity of a viscous Gaussian vortex ring (derived in Appendix D). Archer *et al.*[91] identified that the initial Gaussian distribution goes through a transient phase until achieving a

91 Archer et al., "Direct numerical simulation of vortex ring evolution from the laminar to the early turbulent regime".

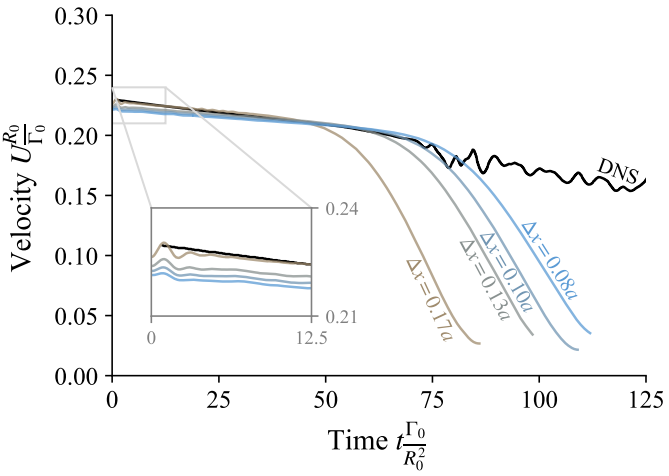


Figure 5.2: Spatial convergence of vortex ring simulated with reformulated VPM, compared to direct numerical simulation (DNS).

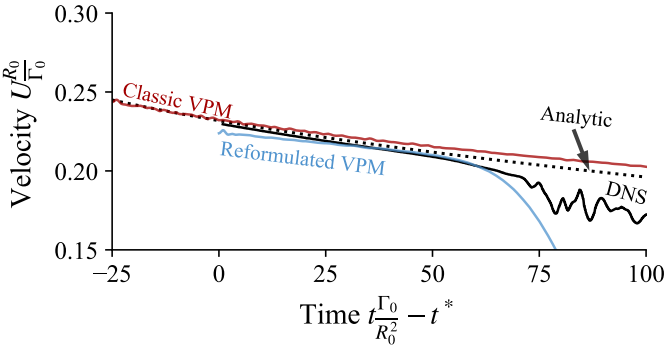


Figure 5.3: Vortex ring simulated with classic VPM ($t^* = 25$) and reformulated VPM ($t^* = 0$), compared to direct numerical simulation (DNS, $t^* = 25$) and analytic solution ($t^* = 25$).

steady skewed distribution. They denoted the time that the simulation takes to achieve this steady distribution as t^*, reporting $t^* = 25$ for their DNS. Similarly, the VPM simulations were also observed to have a phase of transient vorticity distribution. In Fig. 5.3 both the classic VPM simulation and the analytic solution were shifted by $t^* = 25$, showing good agreement with the DNS velocity at time 0. However, the classic VPM slowly drifts away from both the analytic and DNS velocity after time 0. In contrast, the reformulated VPM starts off with a velocity between 4% of the DNS velocity at time 0 and maintains good agreement up to turbulent breakdown. This confirms that the reformulated VPM accurately resolves vorticity advection and viscous diffusion in the laminar regime.

5.2 Leapfrogging Vortex Rings

Two vortex rings that travel in the same direction can repeatedly leapfrog, stretching and contracting as they pass one another. This poses a good

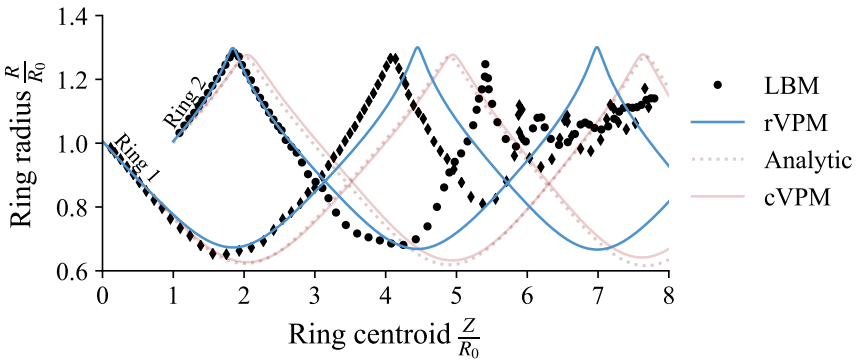


Figure 5.4: Leapfrogging vortex rings simulated with classic VPM (cVPM) and reformulated VPM (rVPM) without SFS model, compared to lattice Boltzmann method (LBM) and analytic solution.

validation case for vortex stretching. Two identical vortex rings with initial radii $R_0 = 1$, thickness $\frac{a}{R_0} = 0.1$, and separation $\Delta Z = R_0$ were simulated at $\mathrm{Re} = 3000$ and compared to the simulation through lattice Boltzmann method (LBM) reported by Cheng *et al.*[92] The LBM simulation is a finite-difference DNS with second order spatial accuracy solving the generalized lattice-Boltzmann equation, reportedly using 15 billion cells.[93] In contrast, the VPM simulations used only 600k particles, which again highlights the computational efficiency of our meshless LES.

First, the simulation was performed on both classic and reformulated VPM without the SFS model, shown in Fig. 5.4. Reformulated VPM and LBM show good agreement in the first leapfrog ($Z/R_0 \leq 3$), but they deviate in subsequent cycles ($Z/R_0 > 3$) once turbulence starts to arise in the LBM causing the rings to slow down and run into each other. This is not captured in the reformulated VPM as the simulation is under-resolved (it does not resolve all the scales of turbulent motion) and leads to a laminar behavior, which indicates the need for an SFS model. Before incorporating the SFS model, however, we conclude that the reformulated VPM accurately resolves vortex stretching in the laminar regime.

Fig. 5.4 also shows the classic VPM simulation and compares it to an analytic solution derived in Appendix D. The classic VPM leads to dynamics that are substantially different from both reformulated VPM and LBM, but it matches the analytic solution. As described in Appendix D, the analytic solution assumes that (1) the core size a is not affected by the stretching of the ring and that (2) the core size does not affect the interactions between rings. The former clearly violates conservation of mass, while the latter is suspected to violate conservation of angular momentum. Such good agreement between the classic VPM and this unphysical analytic solution supports our previous claim in Section 2.1.3 that the classic VPM may lead to unphysical results, which may also cause the method to be numerically unstable.

Finally, the leapfrog simulation was repeated with the SFS model

92 M. Cheng, J. Lou, and T. T. Lim (2015). "Leapfrogging of multiple coaxial viscous vortex rings".

93 M. Cheng, J. Lou, and L.-S. Luo (2010). "Numerical study of a vortex ring impacting a flat wall".

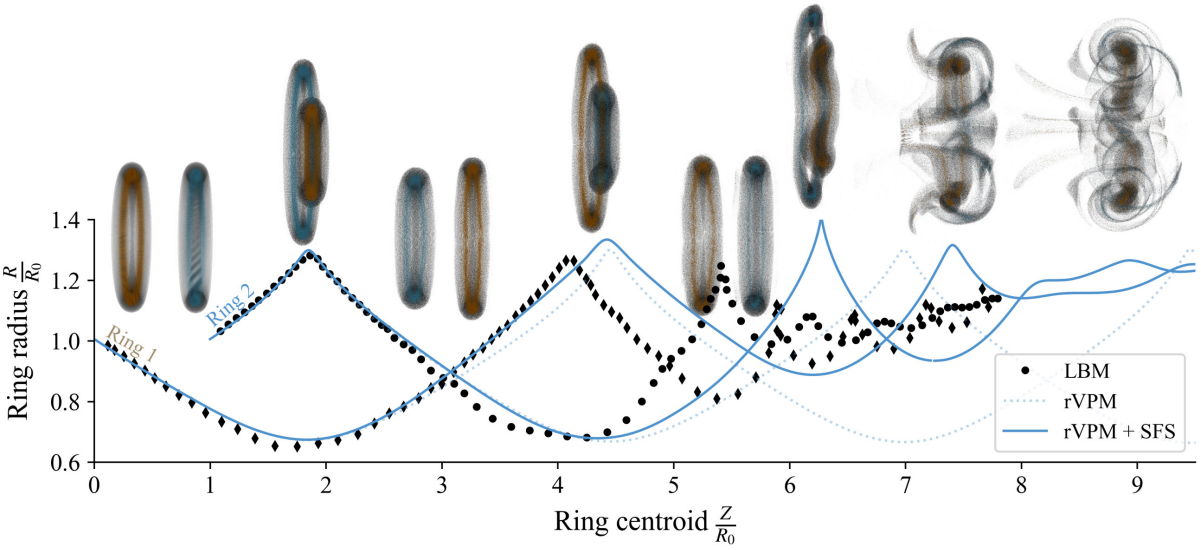


Figure 5.5: Leapfrogging of vortex rings using reformulated VPM with and without SFS model, compared to lattice Boltzmann method (LBM). Visualization of rVPM+SFS simulation overlaid on top.

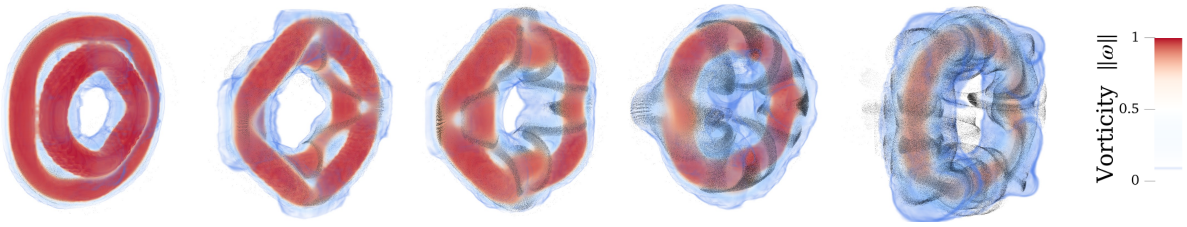


Figure 5.6: Volume rendering of vorticity field as the leapfrogging rings merge. Particles shown as black points.

added to the reformulated VPM, shown in Fig. 5.5. Notice that the SFS model only affects the dynamics after the first leapfrog ($Z/R_0 > 3$) once turbulence starts to develop. This confirms that the dynamic procedure succeeds at auto-regulating the SFS model, suppressing it in the laminar regime while activating it in the turbulent regime. The diffusivity of the SFS model slows the rings down, eventually running into each other and mixing after three leapfrogs at $Z/R_0 > 6$ (shown in more detail in Fig. 5.6). As seen in Fig. 5.5, the process of slowing down and running into each other happens at a different pace between the rVPM+SFS simulation and the LBM. It is unclear whether this discrepancy is due to our vortex-stretching SFS model lacking the full range of turbulent diffusion (requiring the addition of a vorticity-advection SFS model), or due to excessive numerical dissipation in the LBM. However, they are both consistent in that the rings mix after three leapfrog cycles, showing the ability of the SFS model to capture diffusion associated with vortex stretching in the turbulent regime.

In order to estimate the computational cost of the VPM reformulation and the SFS model, Table 5.1 compares the computational time of

Formulation	SFS Model	CPU Time	Overhead
Classic	None	t_{ref}	–
Reformulated	None	$1.01\,t_{\text{ref}}$	<1%
Reformulated	Constant C_d	$1.08\,t_{\text{ref}}$	+8%
Reformulated	Dynamic C_d	$1.43\,t_{\text{ref}}$	+43%

Table 5.1: Computational time of leapfrog VPM simulations.

each simulation to the classic VPM. The VPM reformulation runs as fast as the classic VPM, adding no significant overhead. The SFS model adds a computational overhead of 8% when a constant model coefficient is prescribed and 43% with the dynamic procedure. Comparing this to a benchmark study by Chapelier *et al.*,[94] we conclude that the overhead of our SFS model is comparable to a Smagorinsky eddy-viscosity model, while the dynamic procedure adds an overhead comparable to a Germano-identity dynamic model. The work of Chapelier *et al.* suggests that further speedup could be achieved by implementing a sensor function, but this is left for future work.

94 J.-B. Chapelier et al. (2018). "A Coherent vorticity preserving eddy-viscosity correction for Large-Eddy Simulation".

5.3 Turbulent Round Jet

A jet discharging into a quiescent environment poses a canonical case for the study of turbulence, encompassing a laminar region near the nozzle that breaks down into turbulence away from the nozzle. Experimental measurements on a round jet were used to test that the reformulated VPM is able to resolve the mean and fluctuating components of turbulent flow, while resolving Reynolds stress directly. The simulation replicated the experiment by Quinn and Militzer[95] which used a contoured nozzle with an exit diameter d of 45.4 mm, discharging air into stagnant ambient air with a centerline velocity U_c of 60 m/s. This corresponds to a Mach number of 0.18 and a diameter-based Reynolds number $\text{Re} = U_c d/\nu$ of 2×10^5.

95 W. R. Quinn et al. (1989). "Effects of nonparallel exit flow on round turbulent free jets".

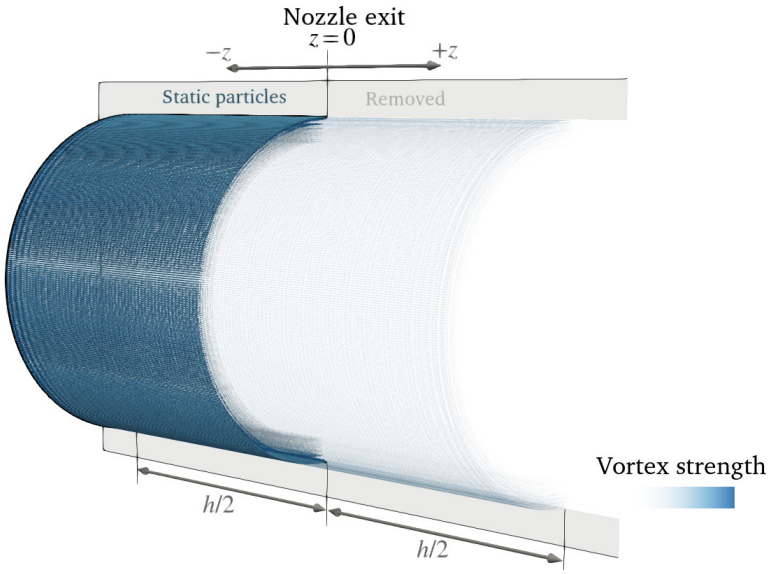


Figure 5.7: Boundary condition of round jet simulation defined with particles.

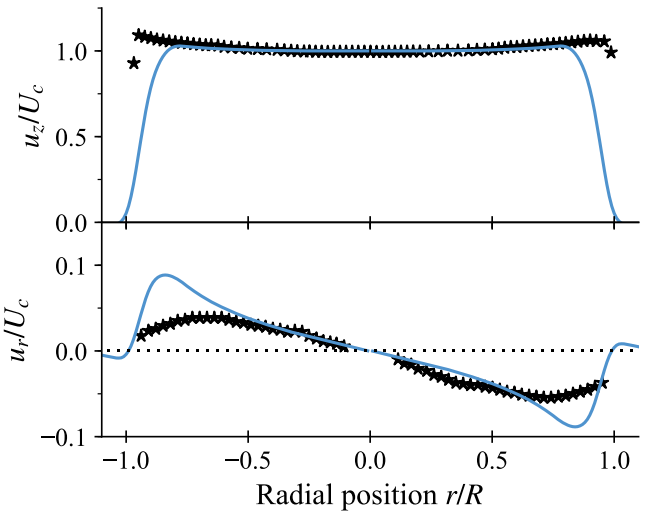


Figure 5.8: Streamwise (top) and radial (bottom) velocity close to nozzle exit in experiment by Quinn and Militzer (markers), compared to boundary condition of simulation (solid line) probed at $z/d = 0.1$.

The simulation assumed a top-hat velocity profile with smooth edges at the nozzle exit given by

$$u_e(r) = U_c \tanh\left(\frac{\frac{d}{2} - r}{\theta}\right),$$

where θ is the momentum thickness of the shear layer, assumed to be $\theta = 0.025d$. As depicted in Fig. 5.7, this velocity profile was imposed as a boundary condition at $z = 0$ by defining a volumetric cylinder of length h with particles spanning from $z = -h/2$ to $z = h/2$ and performing an RBF fit to the vorticity profile $\omega_\theta = \frac{\mathrm{d}u_e}{\mathrm{d}r}$. This computes the vortex strengths that induce the velocity profile u_e inside the cylinder. The $+z$ half of the cylinder was then removed, the $-z$ half was kept in the computational domain as static particles throughout the simulation, while the set of particles computed at $z = 0$ were injected at each time step as free particles. The resulting boundary condition showed sensitivity to the cylinder length h, leading to a straight jet when $h \geq 2d$, a contracting jet when $h < 2d$, and an actuator disk when $h \to 0$. Since Quinn and Militzer reported a jet that was slightly contracting, the length of the boundary-condition cylinder was tailored to match the streamwise and radial velocity that they measured at the nozzle exit plane shown in Fig. 5.8, finding sufficient agreement when $h = 1.7d$.

The simulation was first attempted with the classic VPM, but it quickly ended in numerical blow up at the initial stage of the jet. This is shown in Fig. 5.9 through an abrupt jump in global enstrophy. The reformulated VPM proved to be numerically stable in the initial and transition stages of the jet, however, the simulation becomes unstable in the fully-developed turbulent regime as enstrophy builds up in the absence of SFS turbulent diffusion. Introducing the SFS model, the enstrophy production of the inlet balances out with the forward-scatter of the fully-developed region and the rVPM simulation becomes indefinitely stable, as shown in Fig. 5.9. Interestingly, the SFS model did not stabilize the

classic VPM simulation, which indicates that both the reformulated VPM and the SFS model are needed to achieve numerical stability.

Once proven stable, the simulation was run for 50 ms with time step $\Delta t = 0.02\,\text{ms}$ until achieving a fully-developed region that was statistically stationary. This was run on a single node with 128 CPU cores (dual 64-core AMD EPYC 7702, 2.0 GHz) resulting on a wall-clock time of two and a half days using up to 3×10^6 particles. Fig. 5.10 shows the particle field, vortex strengths, and the SFS model coefficient C_d close to the nozzle at $t = 48\,\text{ms}$, while Fig. 5.11 shows the vorticity of the entire field. As seen in Fig. 5.11, coherent vortical structures form in the initial region ($z < 1d$), which leapfrog and mix transitioning to fully-developed turbulent flow by $z > 3d$. Notice in Fig. 5.10 that the model coefficient is negligibly small in the initial region and increases to $0.1 < C_d < 1$ in the fully-developed regime. This shows that the dynamic procedure succeeds at automatically calibrating C_d, hindering SFS turbulent diffusion in the laminar regime while facilitating it in turbulent regions.

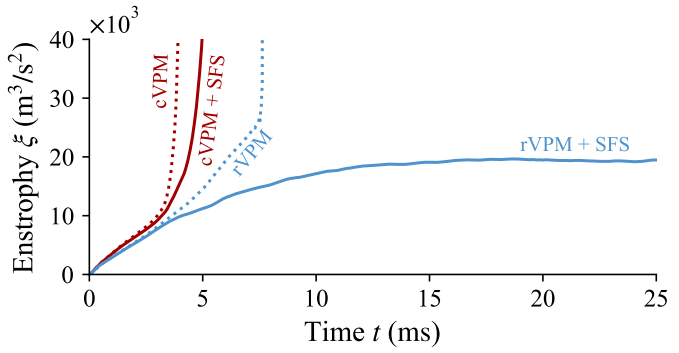


Figure 5.9: Global enstrophy of turbulent round jet with classic VPM (cVPM) and reformulated VPM (rVPM).

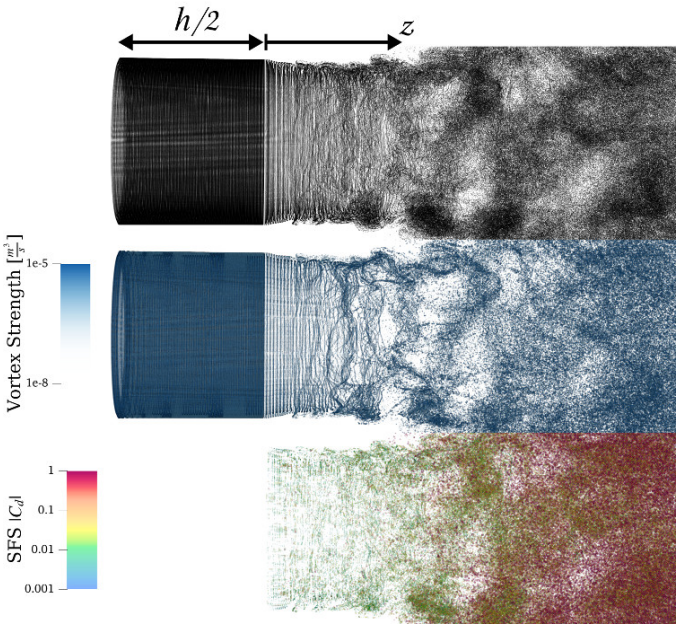


Figure 5.10: Vortex particles (top), vortex strength (middle), and SFS model coefficient (bottom) close to jet nozzle at $t = 48\,\text{ms}$.

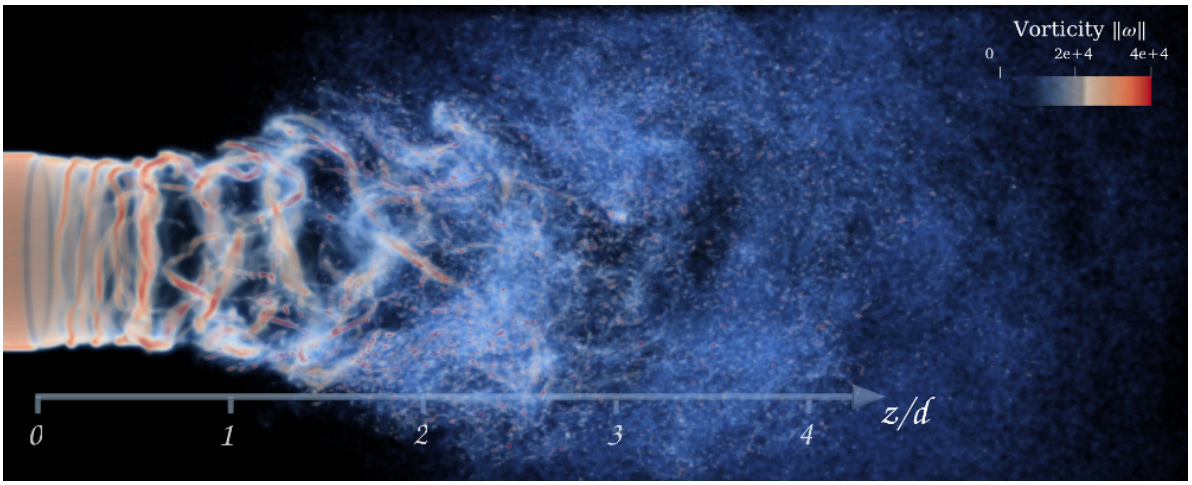


Figure 5.11: Volume rendering of vorticity field in turbulent jet at $t = 48\,\text{ms}$ showing distinct flow features: Coherent structures form in the initial region ($z < 1d$) that mix and break down by $z > 3d$.

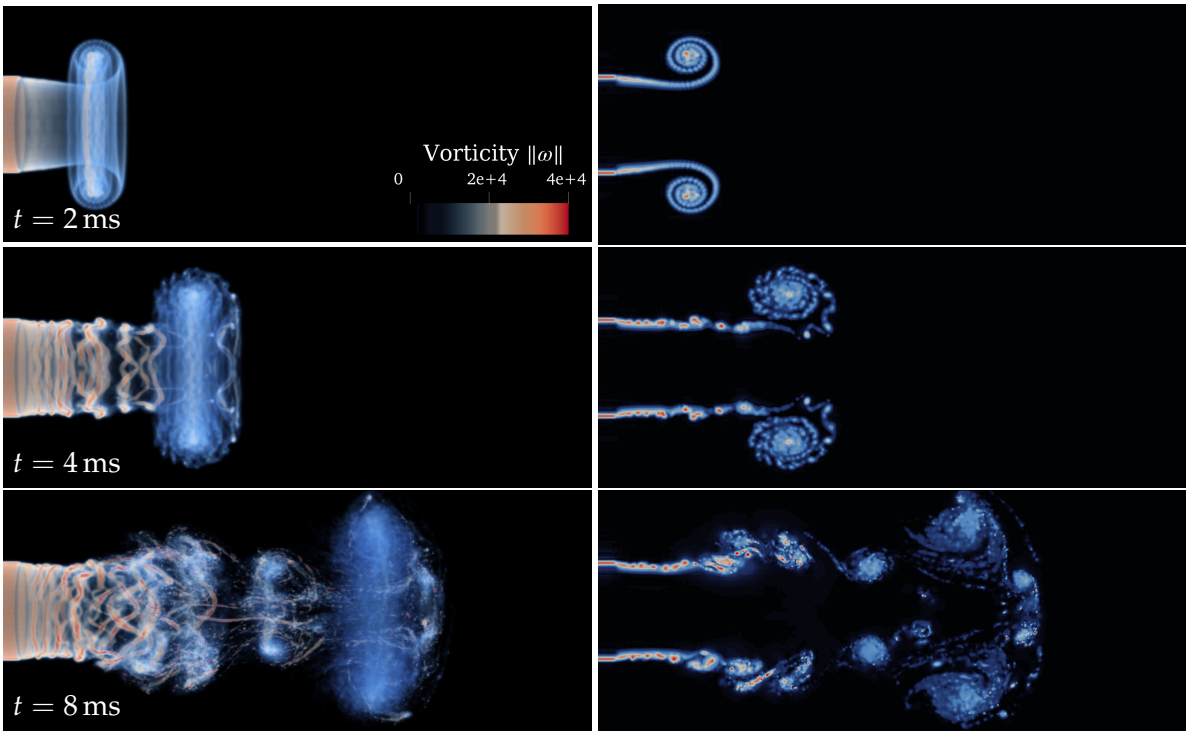


Figure 5.12: Evolution of turbulent jet simulation: (left) volume rendering and (right) slice of vorticity field. Continues in next page.

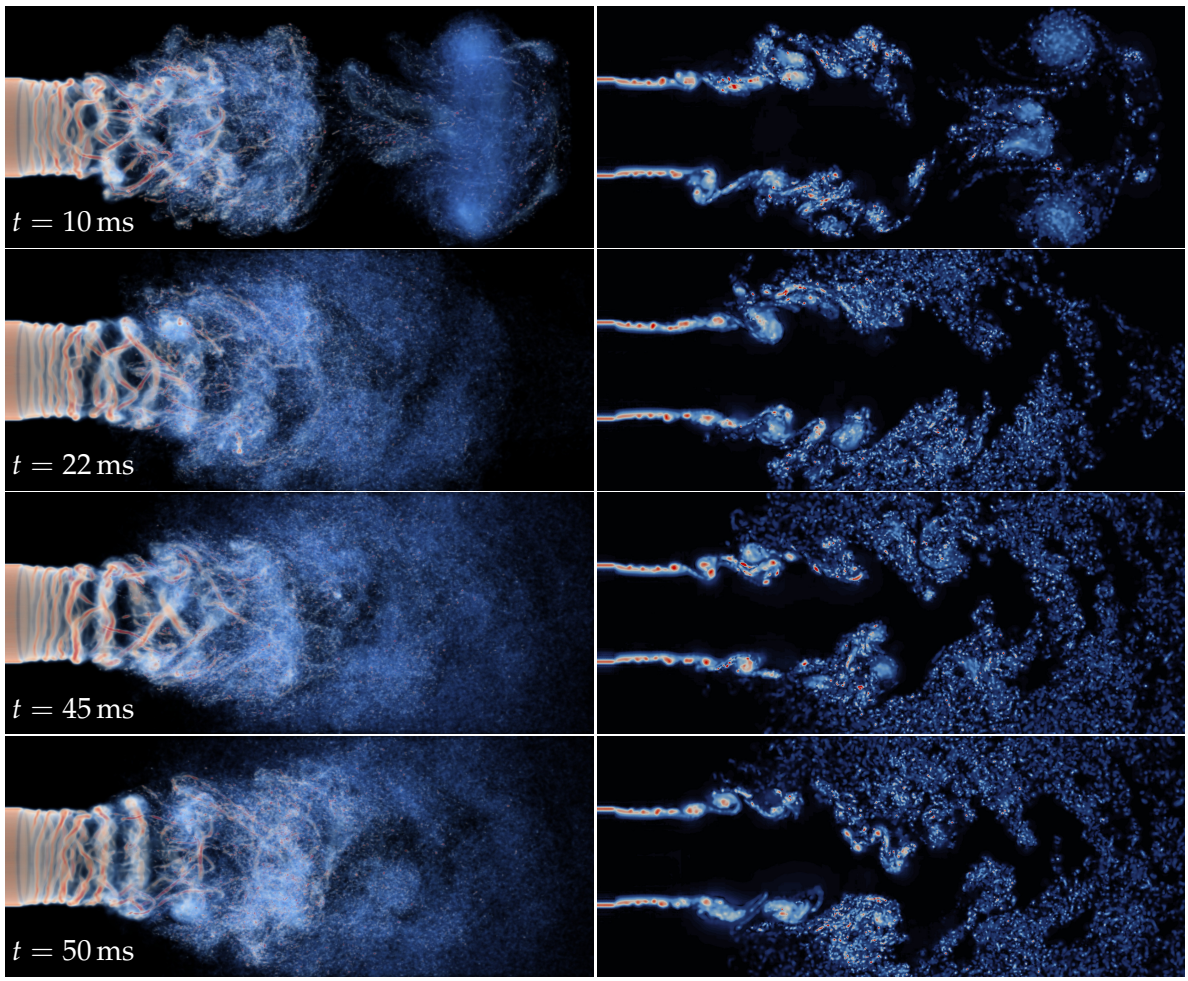


Figure 5.12: Evolution of turbulent jet simulation (continued).

The formation and time evolution of the jet are shown in Fig. 5.12. Initially, a vortex ring forms at the head of the jet. As the ring travels downstream, a vortex sheet is deployed forming the shear layer, seen at $t = 2\,\text{ms}$. The vortex sheet is stretched as the flow develops, eventually rolling up and forming filaments, seen at $t = 4\,\text{ms}$. The filaments pair up, leapfrog, merge, and breakdown (seen at $t = 8\,\text{ms}$) eventually forming the fully-developed turbulent region at $t > 20\,\text{ms}$. Fig. 5.13 shows the history of C_d throughout this process, calculated at each time step as the average $|C_d|$ over all the particles where $C_d \neq 0$. Here we see that the average C_d is negligibly small in the initial development of the jet, but it ramps up and converges to a value of 0.43 as the flow becomes fully developed, again confirming the ability of the dynamic procedure to automatically calibrate the SFS model at each flow regime.

To validate the dynamics predicted by the simulation, the velocity was probed at five stations along the jet and statistical properties were

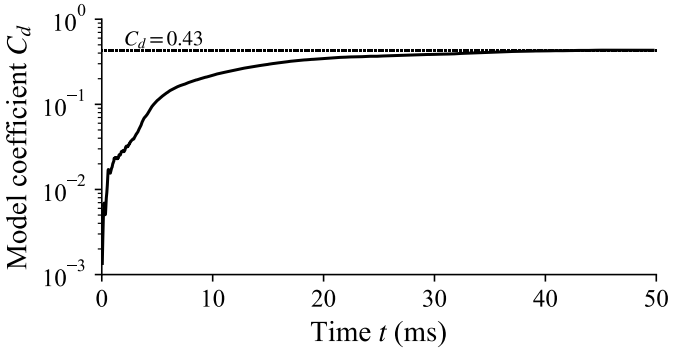


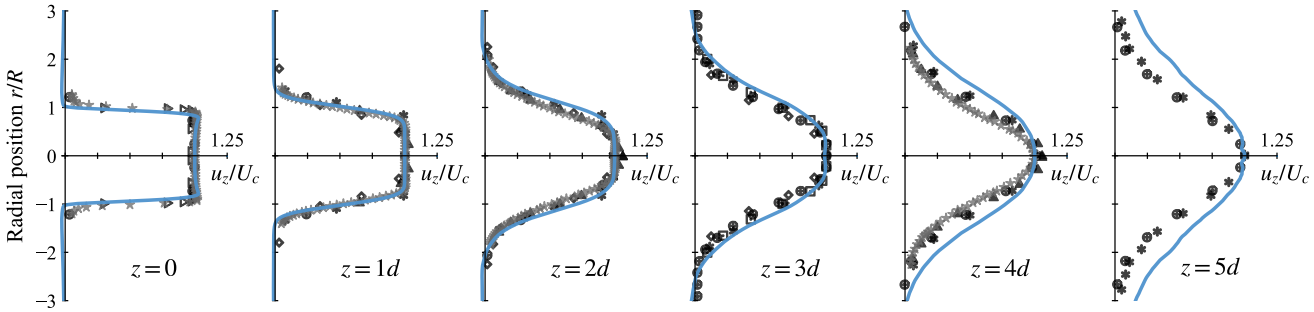
Figure 5.13: Average SFS model coefficient in turbulent jet simulation as computed through the dynamic procedure.

compared to the experimental measurements reported by Quinn and Militzer.[96] The data was also supplemented with other experiments of similar round jets compiled by Ball *et al*.[97] Statistical properties were calculated through temporal and spatial ensemble averages after the flow became statistically stationary in the region $0 \leq z \leq 5d$. Averaging was performed over the time interval $40\,\text{ms} \leq t < 50\,\text{ms}$.

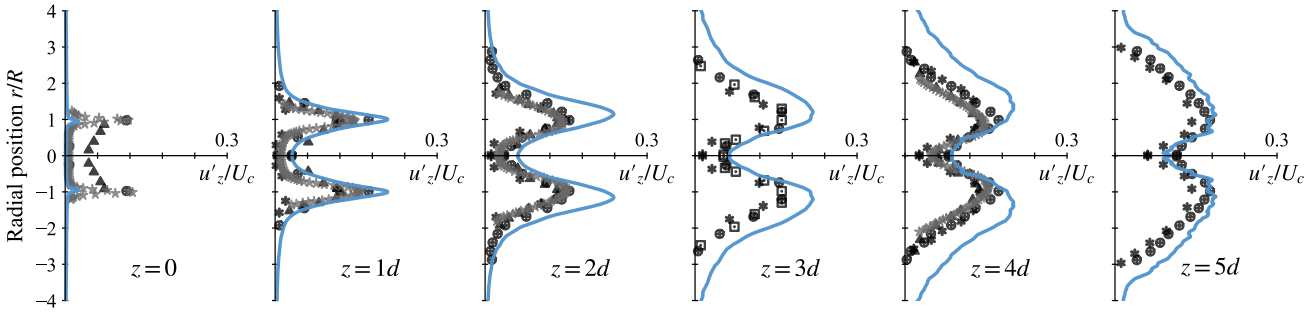
96 Quinn et al., "Effects of nonparallel exit flow on round turbulent free jets".

97 C. G. Ball et al. (2012). "The flow field in turbulent round free jets".

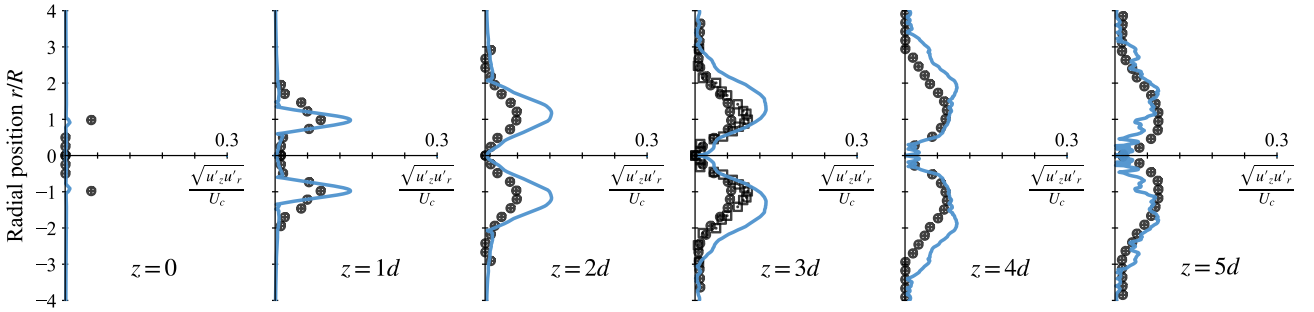
Fig. 5.14(a) shows the mean component of the streamwise velocity profile, u_z, normalized by the centerline velocity at each station. The mean component shows excellent agreement with the experiments in the initial and transition regions ($0 \leq z \leq 3d$) and reasonable agreement in the fully-developed turbulent region ($z \geq 4d$) though slightly overexpanded. Fig. 5.14(b) shows the fluctuating component of the streamwise velocity, u'_z, or standard deviation, while Fig. 5.14(c) shows the Reynolds stress between streamwise and radial velocity, defined as the square root of the covariance, $\sqrt{u'_z u'_r}$. At $z = 0$, the simulation shows only a small fluctuation and Reynolds stress as the flow is dominated by the boundary condition. Away from the nozzle exit plane ($z > 0$), fluctuations and Reynolds stress concentrate at the shear layer ($r/R = 1$) in the laminar region ($z \approx 1d$), and gradually spread as the jet breaks down in the turbulent regime ($z \geq 4d$). Fluctuations are overpredicted (and, as a consequence, also the Reynolds stress), however this is in within reasonable agreement with the experiments. These predictions can be further improved in future work with the addition of an SFS model of vorticity advection increasing the turbulent diffusion (and damping out fluctuations), or implementing a spatial adaptation strategy to better resolve small scales. However, the current agreement with the experiments suffices to confirm that our scheme is an LES able to resolve mean and fluctuating large-scale features of the flow. Also, its ability to directly resolve Reynolds stress renders the reformulated VPM a higher fidelity approach than Reynolds-average approaches like RANS and URANS (where Reynolds stresses are rather modeled), while being completely meshless.



(a) Mean streamwise velocity.



(b) Fluctuating component of the streamwise velocity.



(c) Reynolds stress between streamwise and radial fluctuating components.

Figure 5.14: Profiles along turbulent round jet. Simulation: − rVPM. Experimental: ⋆ Quinn and Militzer;[a] ⊕ Fellouah and Pollard;[b] ▷ Mi *et al.*, contraction nozzle;[c] ∗ Iqbal and Thomas;[d] ◇ Romano;[e] ⊡ Xu and Antonia, contraction nozzle[f] .

a W. R. Quinn et al. (1989). "Effects of nonparallel exit flow on round turbulent free jets".
b H. Fellouah et al. (2009). "The velocity spectra and turbulence length scale distributions in the near to intermediate regions of a round free turbulent jet".
c J. Mi et al. (2001). "Influence of jet exit conditions on the passive scalar field of an axisymmetric free jet".
d M. O. Iqbal et al. (2007). *Coherent structure in a turbulent jet via a vector implementation of the proper orthogonal decomposition*.
e G. Romano (2002). "The effect of boundary conditions by the side of the nozzle of a low Reynolds number jet".
f G. Xu et al. (2002). "Effect of different initial conditions on a turbulent round free jet".

5.4 Aircraft Rotor in Hover

The rotation of blades in static air drives a strong axial flow caused by the shedding of tip vortices. This is a challenging case to simulate since, in the absence of a freestream, the wake quickly becomes fully turbulent as tip vortices leapfrog and mix close to the rotor. Thus, a rotor in hover is a good engineering application to showcase the accuracy, numerical stability, and computational efficiency of the reformulated VPM.

In this test case, we simulated the experiment by Zawodny *et al.*[98] consisting of a DJI 9443 rotor in hover at 5400 RPM. This two-bladed rotor is 9.4 inches in diameter, resulting in a tip Mach number of 0.20 and chord and diameter-based Reynolds numbers at 70% of the blade span of 6×10^4 and 7×10^5, respectively. Our simulations were also compared to unsteady Reynolds-average Navier-Stokes (URANS) results reported by Schenk,[99] obtained with the commercial software STAR-CCM+. The URANS is an unsteady compressible solver with an SST k–ω turbulence model, resolving the blades with an all-y^+ wall treatment on a rotating mesh surrounding the rotor. It used an unstructured mesh with 14 million cells, mesh refinement down to a y^+ of only 30, and time steps equivalent to 3° of rotation on a first-order time integration scheme, which is a rather coarse simulation, but requiring very low computational resources.

The rotating blades are computed in our VPM through an actuator line model (ALM), which is a common practice in LES.[100] Our ALM discretizes the geometry into blade elements, using a two-dimensional viscous panel method to compute forces and circulation along each blade cross section as the blades move. More details on the ALM are given in Section 6.2. The vorticity of each blade is introduced in the fluid domain by embedding static particles along the surface that capture the blade's circulation distribution, while shedding free particles at the trailing edge associated with unsteady loading and trailing circulation, as shown in Fig. 5.15. The frequency of particle shedding per revolution determines the initial spacing Δx in between particles, which, along with the core size σ, determines the spatial resolution at which the wake is being resolved. The initial particle size σ was set as to provide a particle

98 N. S. Zawodny et al. (2016). "Acoustic Characterization and Prediction of Representative, Small-scale Rotary-wing Unmanned Aircraft System Components".

99 A. R. Schenk (2020). "Computational Investigation of the Effects of Rotor-on-Rotor Interactions on Thrust and Noise".

100 M. J. Churchfield et al. (2017). "An Advanced Actuator Line Method for Wind Energy Applications and Beyond".

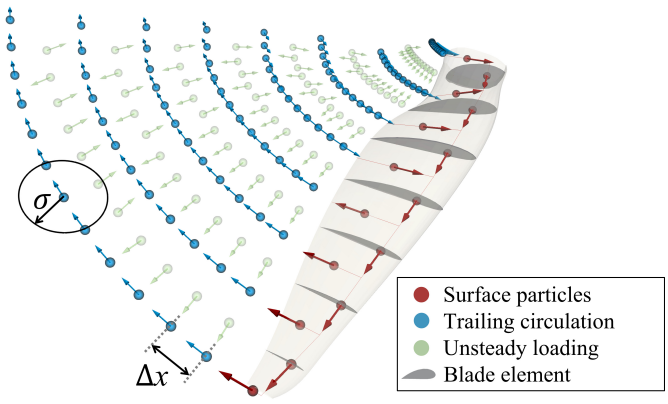


Figure 5.15: Actuator line model in rotor simulation. Particles colored by their source of vorticity; arrows indicate direction of vortex strength.

overlap $\lambda = \frac{\sigma}{\Delta x}$ of 2.125 at the blade tip. The VPM simulations used a time step equivalent to a rotation of 1° while shedding particles every 0.5° with 50 blade elements.

First, the simulation was attempted on the classic VPM without the SFS model, which quickly ended in numerical blow up after one revolution, shown in Fig. 5.16. Introducing the SFS model made the classic VPM noticeably more stable, however, the simulation still blew up before the wake became fully developed. Introducing the reformulated VPM made the simulation completely stable, using up to 1×10^6 particles after 16 revolutions. As shown in Fig. 5.16, the rate of enstrophy produced by the rotor eventually balances out with the forward scatter of the SFS model. This further asserts the numerical stability gained with the reformulated VPM and SFS model.

Next, the thrust coefficient C_T predicted with rVPM was compared to the experimental coefficient reported by Zawodny *et al.*[101] We define C_T in the propeller convention as $C_T = \frac{T}{\rho n^2 d^4}$, where T is the dimensional thrust, ρ is air density, and n is rotations per seconds. The experiment reported[102] a mean C_T of 0.072. In our simulation, the thrust is calculated integrating the force computed by the ALM at each blade element that is immersed in the fluid domain. As shown in Fig. 5.17, the VPM simulation shows excellent agreement with the experiment, predicting a mean value within 2% of the experimental mean value. This illustrates the capacity of our method to provide accurate predictions in a real engineering application.

101 Zawodny et al., "Acoustic Characterization and Prediction of Representative, Small-scale Rotary-wing Unmanned Aircraft System Components".

102 The experiment reported the thrust coefficient defined as $C_T = \frac{T}{\rho \pi R^2 (\Omega R)^2}$ which is typical in the rotorcraft community, while here we have converted their measurement to C_T as defined in the propeller convention.

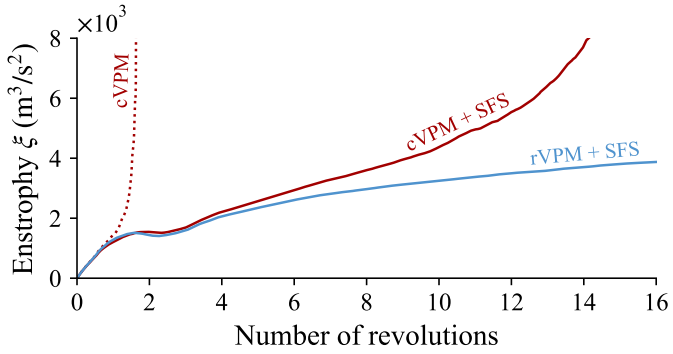


Figure 5.16: Global enstrophy in VPM rotor simulations.

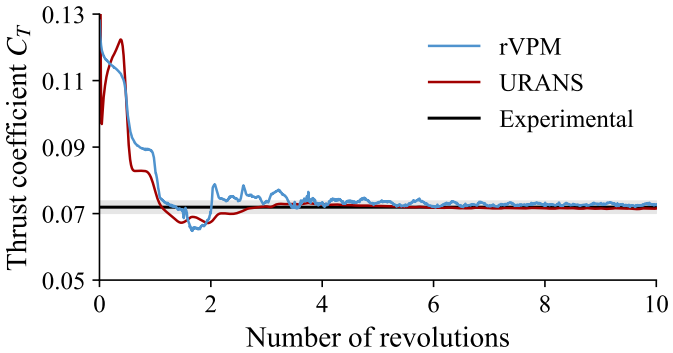


Figure 5.17: Thrust history in rotor simulations compared to experimental mean C_T. Shaded region encompasses the 95%-confidence interval of the experiment.

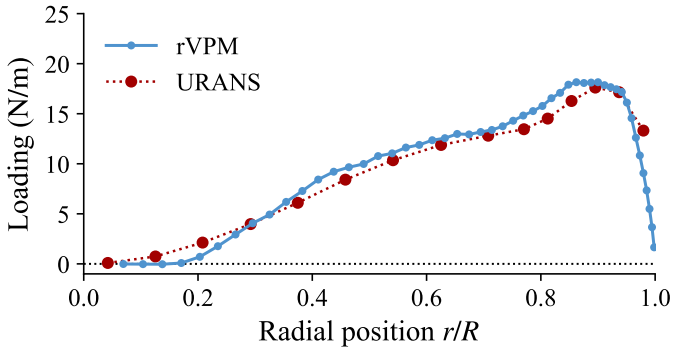


Figure 5.18: Time-average blade loading distribution in rotor simulations.

In order to illustrate the low computational cost of our meshless LES, we now compare our simulation to the URANS simulation by Schenk. As previously pointed out, for the sake of minimizing computational effort, the URANS simulation is both spatially and temporally coarse (using only 14 million cells down to a y^+ of only 30, with time steps of 3°), while using a first-order time integration. Thus, this URANS simulation represents the minimum computational effort that is plausible for a simulation of this kind while still being accurate (the C_T predicted by URANS is still within 2% of the experimental mean value, as shown in Fig. 5.17). Fig. 5.18 compares the time-average loading distribution along the blade as predicted with rVPM and URANS, showing that both methods can resolve the blade loading with similar accuracy. However, the rVPM has the advantage of being able to accurately preserve the vortical structure of the wake, shown in Fig. 5.19, with minimal computational effort. The URANS simulation reportedly took about 10 wall-clock hours to resolve 10 revolutions, using 192 CPU cores across 12 nodes on the BYU Fulton supercomputer. This is equivalent to about 1800 core-hours. On the other hand, the rVPM simulation took about 4 wall-clock hours to resolve 10 revolutions on a single node of the BYU Fulton supercomputer with 32 CPU cores, equivalent to about 140 core-hours. Hence, rVPM is 13 times faster (or one order of magnitude faster) than this coarse URANS simulation, while providing LES accuracy.

While Schenk's URANS simulation represents the low-fidelity end of mesh-based CFD, Zawodny *et al.*[103] reported a high-fidelity mesh-based simulation using OVERFLOW2. OVERFLOW2 is a dettached-eddy simulation (DES) code using a URANS solver with the Spalart-Allmaras turbulence model near solid surfaces, while switching to a subgrid scale formulation in regions fine enough for large eddy simulation. Their simulation of the DJI 9443 rotor used 260 million grid cells with time steps corresponding to 0.25° on a second-order time integration scheme, reportedly predicting a mean C_T within 2.5% of the experimental value. One rotor revolution reportedly took approximately 30 wall-clock hours using 1008 CPU cores on the NAS Pleiades supercomputer. Extrapolating this to 10 revolutions, the DES takes 300 wall-clock hours and 300k core-hours. Thus, recalling that rVPM took only 140 core-hours, the rVPM is 2200x faster (or three orders of magnitude faster) than this DES

103 Zawodny et al., "Acoustic Characterization and Prediction of Representative, Small-scale Rotary-wing Unmanned Aircraft System Components".

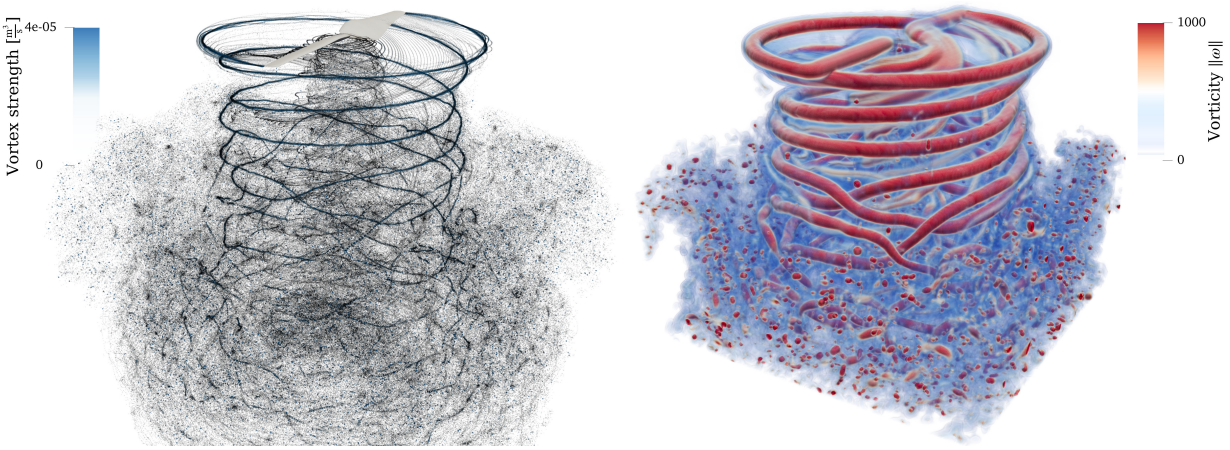


Figure 5.19: Meshless LES of rotor in hover after 15 revolutions using rVPM: (left) computational elements (vortex particles and strength), and (right) volume rendering of vorticity field.

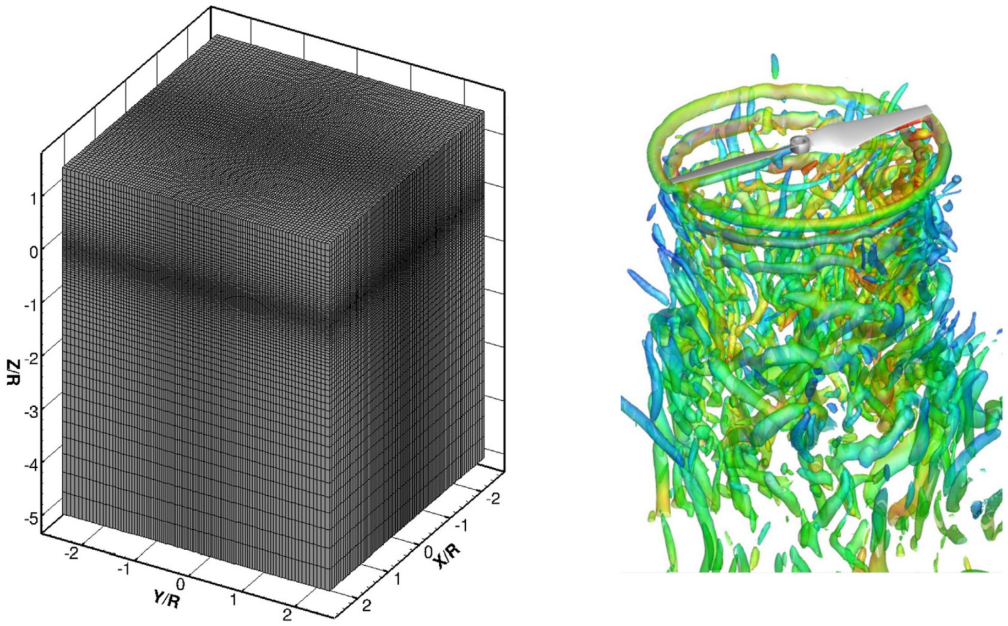


Figure 5.20: Mesh-based LES of rotor in hover reported by Delorme *et al.*: (left) computational elements, and (right) iso-surface of Q-criterion colored by velocity magnitude. Reprinted from Aerospace Science and Technology, Vol 108, Delorme *et al*, *Application of Actuator Line Model for Large Eddy Simulation of Rotor Noise Control*, Copyright 2020 Elsevier Masson SAS, with permission from Elsevier.

simulation.

While Schenk's URANS and Zawodny's DES represent the low and high end of fidelity in mesh-based CFD, respectively, Delorme *et al.*[104] reported two LES that lay somewhere in between both ends. One LES used an actuator line model (ALM) similar to our meshless LES, while the

104 Y. Delorme et al. (2021). "Application of Actuator Line Model for Large Eddy Simulation of Rotor Noise Control".

other used an immersed boundary method (IBM). All solvers previously mentioned are summarized in Table 5.2 and their computational costs are compared to our meshless LES in Table 5.3. The LES by Delorme *et al.*, shown in Fig. 5.20, is an implicit LES that relies on numerical dissipation to approximate subgrid-scale turbulent diffusion, reportedly predicting a mean C_T within 2% of the experimental value. Out of the aforementioned mesh-based simulations, their LES-ALM simulation is the most akin to our meshless LES since both use an ALM while resolving vortical structures with comparable fidelity, as shown in Fig. 5.19 (right) and Fig. 5.20 (right). The main computational advantage of our meshless LES lays in that computational elements are only placed where vorticity is originated and are automatically convected by the flow field, as shown in Fig. 5.19 (left), while mesh-based LES requires meshing the entire space, as shown in Fig. 5.20 (left). Their LES-ALM simulation of a modified DJI 9443 rotor reportedly took a wall-clock time of 24 hours per every 10 rotor revolutions using 845 CPUs, equivalent to 20k core-hours. Compared to the 140 core-hours of the rVPM, our meshless LES is 145x faster (or two orders of magnitude faster) than this mesh-based LES.

In summary, as shown in Table 5.3, our meshless LES appears to be two orders of magnitude faster than a mesh-based LES with similar fidelity, while being one order of magnitude faster than a low-fidelity URANS simulation and three orders of magnitude than high-fidelity DES. It is difficult to justify an exact comparison between the computational cost of each simulation since each solver used different order-of-convergence schemes, spatial and temporal resolution, and computing hardware. However, the comparison is still qualitatively insightful: our meshless LES can be 10x to 1000x faster than conventional mesh-based CFD approaches.

Table 5.2: Description of CFD solvers in rotor benchmark.

Simulation	Software	Blade Scheme	Turbulence Model	Computational Elements
rVPM (meshless LES)	FLOWVPM	Actuator line model	Anisotropic dynamic SFS	1M vortex elements
URANS	STAR-CCM+	Blade-resolved	SST k–ω	14M grid cells
LES-ALM	MIRACLES	Actuator line model	None (numerical dissipation)	50M grid points
LES-IBM	MIRACLES	Immersed boundary method	None (numerical dissipation)	216M grid points
DES	OVERFLOW2	Blade resolved	Spalart-Allmaras DES	260M grid points

Simulation	CPU Cores	Wall-Clock	Core-Hours	rVPM Speedup
rVPM	32	4.3 hours	140	–
URANS	192	9.6 hours	1.8k	~10x faster
LES-ALM	845	24 hours	20k	~100x faster
LES-IBM	1000	96 hours	96k	~500x faster
DES	1008	300 hours	300k	~1000x faster

Table 5.3: Rotor benchmark resolving 10 rotor revolutions. Computational cost of rVPM compared to mesh-based simulations.

Conclusions

In this study we have developed a meshless scheme for large eddy simulation that is accurate, stable, and computationally efficient. We started off by deriving a general set of VPM governing equations that solve the LES-filtered Navier-Stokes equations. The classic VPM turns out to be one of the formulations arising from these general equations, which seems to locally violate both conservation of mass and angular momentum, thus explaining its tendency to be numerically unstable. We then derived a new VPM formulation that uses the particle shape to reinforce local conservation of mass and angular momentum.

In addition to the reformulated VPM, we developed a low-dissipation anisotropic SFS model that uses vortex stretching as the physical mechanism for turbulence. We proposed a novel dynamic procedure for calculating the model coefficient and a strategy for backscatter control. The dynamic procedure is based on a simultaneous balance of enstrophy-production and derivatives between true and modeled SFS contributions. This SFS model is apt for both meshless and conventional mesh-based CFD, and is well suited for flows with coherent vortical structures where the predominant cascade mechanism is vortex stretching.

Extensive validation was presented, asserting the scheme comprised of the reformulated VPM and SFS model as a meshless LES accurately resolving large-scale turbulent dynamics. Advection, viscous diffusion, and vortex stretching were validated through the simulation of vortex rings, showing good agreement with DNS and LBM. It was shown that the reformulated VPM runs as fast as the classic VPM, adding no significant overhead, while the SFS model adds a computational overhead of 8% when a constant model coefficient is prescribed and 43% with the dynamic procedure. In the simulation of a turbulent round jet, a favorable agreement with experiments showed that our scheme is able to resolve the mean and fluctuating components of turbulent flow, while also resolving Reynolds stress directly.

Finally, computational efficiency was demonstrated in an engineering application through the simulation of an aircraft rotor in hover. Our meshless LES showed to be two orders of magnitude faster than a mesh-based LES with similar fidelity, while being one order of magnitude faster than a low-fidelity URANS simulation and three orders of magnitude than a high-fidelity DES.

Along with highlighting the strengths of the method, it is also impor-

tant to recognize its limitations. The scheme proposed in this study is especially well suited for the modeling of unbounded incompressible flow with coherent vortical structures, as in the case of a wake. However, the modeling of wall-bounded flow was not explored as this poses the challenge of defining appropriate boundary conditions for the vorticity equation. The different cases simulated in this study showed good scaling and accuracy of our LES approach for a wide range of Reynolds numbers on unbounded flow. However, none of the simulations attempted to resolve boundary layers nor flow separation. This study limited its computation to a conventional CPU paradigm, but previous work[105] shows a strong potential for massive parallelization and speed up in heterogenous GPU architectures.

This study validates the reformulated VPM as an LES that is both numerically stable and meshless, while able to accurately resolve mean and fluctuating large-scale features of turbulent flow with minimal computational effort. In future work, a wider variety of boundary conditions can be explored, especially aiming at introducing solid boundaries without a mesh. Part II will explore the introduction of boundary conditions using an actuator surface model for aerodynamic applications.

105 Yokota, Barba, et al., "Petascale turbulence simulation using a highly parallel fast multipole method on GPUs"; Hu et al., "Scalable Fast Multipole Accelerated Vortex Methods".

Part II

Meshless Large Eddy Simulation of Multirotor Aircraft

Introduction

A New Design Space

Electric vertical takeoff and landing (eVTOL) aircraft is a flourishing technology that is projected to grow into a $1.5 trillion industry of urban air mobility by the year 2040.[106] Since the first foresight[107] of this nascent technology in 2010, the list of known eVTOL aircraft concepts has grown from a handful to more than 450 today[2]. This list includes aerospace giants like Airbus and Boeing, established car makers like Toyota and Hyundai, and tech eminence like Uber and Alphabet. An example of an eVTOL vehicle is shown in Fig. II.1.

The excitement around eVTOL arises from a new design space enabled by recent and ongoing advancements in electric battery technology. An electric powertrain allows the designer to distribute the propulsion system into multiple rotors to takeoff and land vertically for intracity travel, capitalizing on the dream of a "flying car". However, these novel aircraft configurations pose technical challenges that still remain to be solved.[108] A strong noise signature[109] and a complicated transition maneuver[110] are examples of the challenges encountered in eVTOL aircraft, which are both dominated by aerodynamic interactions between the multiple rotors and lifting surfaces.[111]

The complicated aerodynamic interactions encountered in eVTOL are not well understood, are not captured through conventional design tools, and need to be addressed in the early stages of design.[112] For instance, current models used in preliminary design fail to predict and assess configurations that may lead to the wake of a rotor impinging on another rotor or a wing during the transition maneuver. These unsteady

106 *Flying Cars: Investment Implications of Autonomous Urban Air Mobility* (2018).

107 M. Moore (2010). "NASA Puffin Electric Tailsitter VTOL Concept".

2 https://evtol.news/aircraft, accessed January 26th, 2022.

108 B. J. Brelje et al. (2018). "Electric, hybrid, and turboelectric fixed-wing aircraft: A review of concepts, models, and design approaches".

109 K. Pascioni et al. (2018). "Tonal Noise Prediction of a Distributed Propulsion Unmanned Aerial Vehicle".

110 L. Persson et al. (2017). *Design and Control of an Experimental Tiltwing Aircraft*; G. Droandi et al. (2018). "Tiltwing Multi-Rotor Aerodynamic Modeling in Hover, Transition and Cruise Flight Conditions".

111 C. Silva and W. Johnson (2021). "Practical Conceptual Design of Quieter Urban VTOL Aircraft".

112 P. F. Pelz et al. (2021). "Sustainable aircraft design – A review on optimization methods for electric propulsion with derived optimal number of propulsors".

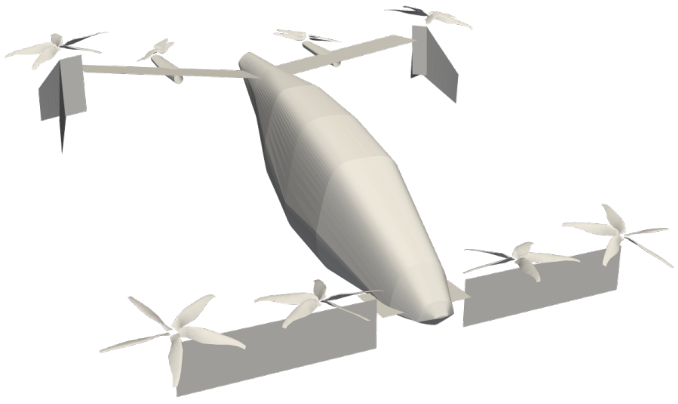


Figure II.1: Example of an eVTOL aircraft. This is a modified version of the A^3 Vahana demonstrator, which is a tandem tiltwing aircraft. Modeled in FLOWUnsteady.

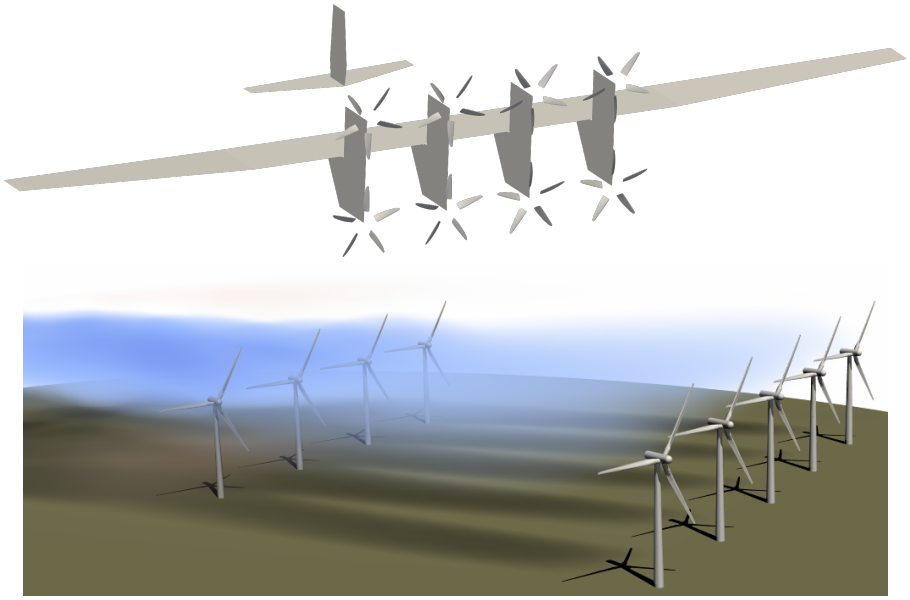


Figure II.2: Other multirotor aerospace systems that encounter unsteady aerodynamic interactions: (top) AWE wind-harvesting aircraft and (bottom) wind farm. Modeled in FLOWUnsteady.

interactions can be analyzed with computational fluid dynamics (CFD) methods. However, resolving wake dynamics in conventional CFD tools requires high-order numerical schemes and mesh resolutions with associated computational costs that make them prohibitive for design space exploration. Furthermore, these interactions are not unique to eVTOL aircraft. They are also encountered across aerospace systems that make use of multiple rotors: from recreational photography drones and fixed-wing aircraft with distributed propulsion to airborne wind energy (AWE) and conventional wind farms, the last two shown in Fig. II.2. Thus, the ability to rapidly and accurately assess unsteady interactional aerodynamics is a current shortcoming and bottleneck in the design of various next-generation aerospace systems.

The Spectrum of Analysis Tools

The spectrum of analysis tools for aircraft design, depicted in Table II.1, ranges from low-fidelity/low-computation tools with run times of seconds to high-fidelity/high-computation tools with run times of days or weeks. At the low end of fidelity, methods like blade-element momentum theory (BEMT), lifting line, the vortex lattice method (VLM), and panel methods are widely used for conceptual design, running simple analyses on a laptop computer. For instance, Ning[113] showed that gradient-based design optimization of a rotor using BEMT and 90 design variables can be done in as fast as 5 seconds. McDonnell and Ning[114] used lifting line among other low-fidelity models in a multidisciplinary design optimization (MDO) of a solar-powered, distributed electric propulsion (DEP) aircraft considering stability, flutter, solar capture, and structures, simultaneously designing powertrain, composite laminate, wing geometry, and rotor positioning.

Simple models like BEMT and VLM can be modified and corrected to include some steady interactional aerodynamics. For instance, Veld-

113 A. Ning (2021). "Using blade element momentum methods with gradient-based design optimization".

114 T. McDonnell et al. (2020). "Gradient-Based Optimization of Solar-Regenerative High-Altitude Long-Endurance Aircraft".

huis[115] used the induced velocity of BEMT as an input to VLM to capture rotor-on-wing interactions. Epema[116] later corrected the BEMT swirl velocity to capture the deformation of the streamtube due to the presence of the wing. More recently, Stokkermans *et al.*[117] showed that steady rotor-on-rotor interactions in eVTOL can be accurately captured using a modified BEMT, and van Arnhem *et al.*[118] developed a method akin to BEMT for propellers in nonuniform flow showing accurate results on a propeller encountering a wake, an over-the-wing propeller, and a wingtip-mounted propeller. Moore and Ning[119] used some of these interactional models in gradient-based MDO of a DEP aircraft considering aerodynamics, structures, and aeroacoustics, simultaneously designing blade geometry, rotor positioning, and wing geometry to minimize takeoff runway and acoustic noise.

Even though low-fidelity methods are capable of capturing some steady interactional aerodynamics during conceptual design, they fail to predict unsteady phenomena that can be critical for a minimum viable product. For instance, closely spacing the rotors can cause unsteady blade loading during takeoff,[120] which can increase the acoustic noise beyond the threshold of public acceptance.

In the low-mid fidelity range, free-wake methods like unsteady VLM (UVLM) and wake filaments are capable of predicting a limited range of unsteady interactional aerodynamics. For instance, Ho and Yeo[121] used wake filaments to predict blade-vortex interaction of a tiltrotor in

115 L. L. M. Veldhuis (2005). "Propeller Wing Aerodynamic Interference".

116 K. Epema (2017). "Wing Optimisation for Tractor Propeller Configurations".

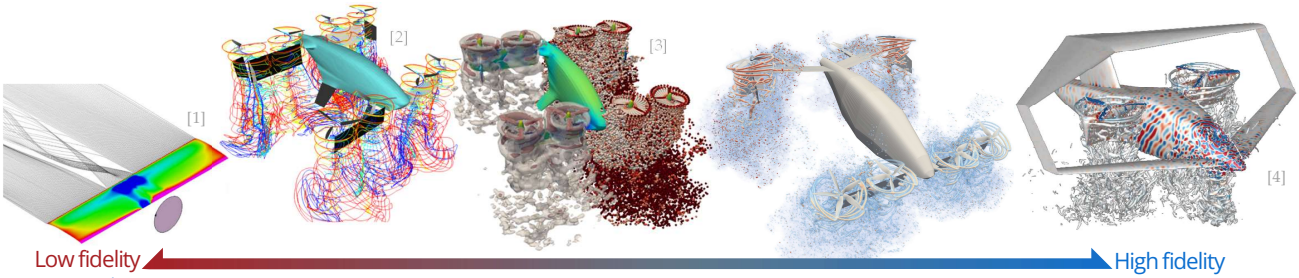
117 T. C. A. Stokkermans, D. Usai, et al. (2021). "Aerodynamic Interaction Effects Between Propellers in Typical eVTOL Vehicle Configurations".

118 N. van Arnhem, R. de Vries, et al. (2020). "Engineering Method to Estimate the Blade Loading of Propellers in Nonuniform Flow".

119 K. R. Moore and A. Ning (2019). "Takeoff and Performance Trade-Offs of Retrofit Distributed Electric Propulsion for Urban Transport".

120 W. Zhou et al. (2017). "An Experimental Investigation on Rotor-to-Rotor Interactions of Small UAV Propellers"; Alvarez, Schenk, et al., "Rotor-on-Rotor Aeroacoustic Interactions of Multirotor in Hover".

121 J. C. Ho et al. (2018). "Assessing Calculated Blade Loads of the Tilt Rotor Aeroacoustic Model".



	BEMT lifting line VLM panel methods	Free-wake Methods filament wake UVLM	Vortex Particle Method	Reformulated Vortex Particle Method meshless LES	Mesh-Based CFD URANS DES LES LBM
Unsteady interactions?	✗	✓	✓	✓	✓
Wake mixing?	✗	✗	~	✓	✓
Fast enough for design?	conceptual	conceptual / preliminary	preliminary	preliminary / detailed	detailed
Software	AVL, ASWING, XFLR5, XROTOR, CCBlade	RCAS, CAMRAD II, CHARM, VSPAERO, FlightStream	GENUVP, DUST, RCAS-VPM	FLOWVPM, FLOWUnsteady	STAR-CCM+, ANSYS Fluent, OVERFLOW, Helios, PowerFLOW

[1] Sheridan et al. (2021). *Evaluation of VSPAERO Analysis Capabilities for Conceptual Design of Aircraft with Propeller-Blown Wings*. AIAA AVIATION Forum.
[2] Droandi et al. (2018). *Tiltwing Multi-Rotor Aerodynamic Modeling in Hover, Transition and Cruise Flight Conditions*. AHS International 74th Annual Forum.
[3] Montagnani et al. (2019). *Mid-fidelity Analysis Of Unsteady Interactional Aerodynamics Of Complex VTOL Configurations*. 45th European Rotorcraft Forum.
[4] Ventura Diaz, P., & Yoon, S. (2018). *High-Fidelity Computational Aerodynamics of Multi-Rotor Unmanned Aerial Vehicles*. AIAA Aerospace Sciences Meeting.

Table II.1: Analysis tools for multirotor aircraft design.

forward flight through the code RCAS. Fei et al.[122] used UVLM to model a rotor at an incidence angle. Cole *et al.*[123] used a high-order UVLM for the simulation of a blown wing. Droandi *et al.*[124] used a wake filament approach known as constant vorticity contour through the code CHARM to model the quasi-static transition of a tiltwing eVTOL aircraft.

Free-wake methods have been widely applied in the analysis of conventional helicopters, showing run times between tens of minutes to hours on a personal computer. However, they have a limited applicability in advanced rotorcraft configurations. The requirement of preserving connectivity make free-wake methods numerically unstable when simulating the mixing of wakes and turbulent breakdown, like in the mixing of two rotor wakes.[125]

In the high-fidelity end of the spectrum, mesh-based CFD approaches like unsteady Reynolds-average Navier-Stokes (URANS), large eddy simulation (LES), detached eddy simulation (DES), and the lattice Boltzmann method (LBM) are capable of capturing the full range of interactional aerodynamics. For instance, Yoon *et al.* used DES through the solver OVERFLOW to characterize the thrust drop in a quadcopter as the separation between the rotors was decreased, while capturing the effects of blowing on the fuselage.[126] Ventura and Yoon[127] expanded on this work analyzing both a quadcopter and a tiltrotor eVTOL vehicle in forward flight. Jia and Lee[128] then integrated this scheme into the Helios simulation framework with the aeroacoustics code PSU-WOPWOP to analyze the acoustic signature of a quadcopter. Miesner *et al.*[129] used URANS to characterize rotor-rotor and rotor-fuselage interactions on the multirotor Volocopter vehicle in forward flight. De Vries[130] used URANS through the commercial software ANSYS Fluent to characterize the effects of over-the-wing DEP on wing performance accounting for boundary-layer interactions. Casalino *et al.*[131] used LBM through the commercial software PowerFLOW to resolve the acoustics signature of an eVTOL vehicle in an urban city.

Even though high-fidelity mesh-based CFD has helped researchers better understand the interactional aerodynamics of eVTOL configurations, each one of the aforementioned studies used supercomputer resources with run times of days or weeks. Its high computational cost precludes mesh-based CFD from the early stages of design, and is rather reserved for only detailed design.

In summary, simple steady interactions can be predicted with low-fidelity/low-computation methods in conceptual design, while the full range of unsteady interactions can be captured with high-fidelity/high-computation methods in detailed design. In the year 2017, when this doctoral work commenced, it became evident that there was a gap in the spectrum of analysis tools that were available to aircraft designers: there was a lack of mid-to-high fidelity methods that could predict complex unsteady aerodynamics at a computational cost fit for conceptual and preliminary design.

122 X. Fei et al. (2021). "Development of an Unsteady Vortex Lattice Method to Model Propellers at Incidence".

123 J. A. Cole et al. (2019). "Higher-Order Free-Wake Method for Propeller-Wing Systems".

124 Droandi et al., "Tiltwing Multi-Rotor Aerodynamic Modeling in Hover, Transition and Cruise Flight Conditions".

125 J. Lee, K. Yee, et al. (2009). "Aerodynamic Characteristic Analysis of Multi-Rotors Using a Modified Free-Wake Method".

126 S. Yoon et al. (2016). "Computational Analysis of Multi-Rotor Flows".

127 P. Ventura Diaz et al. (2018). "High-Fidelity Computational Aerodynamics of Multi-Rotor Unmanned Aerial Vehicles".

128 Z. Jia et al. (2019). "Acoustic Analysis of a Quadrotor eVTOL Design via High-Fidelity Simulations".

129 S. Miesner et al. (2021). "High-fidelity Simulation of the Volocopter-2X in cruise flight".

130 R. de Vries (2021). "Hybrid-Electric Aircraft with Over-the-Wing Distributed Propulsion Aerodynamic Performance and Conceptual Design".

131 D. Casalino et al. (2019). "Aeroacoustic Analysis of Urban Air Operations Using the LB/VLES Method".

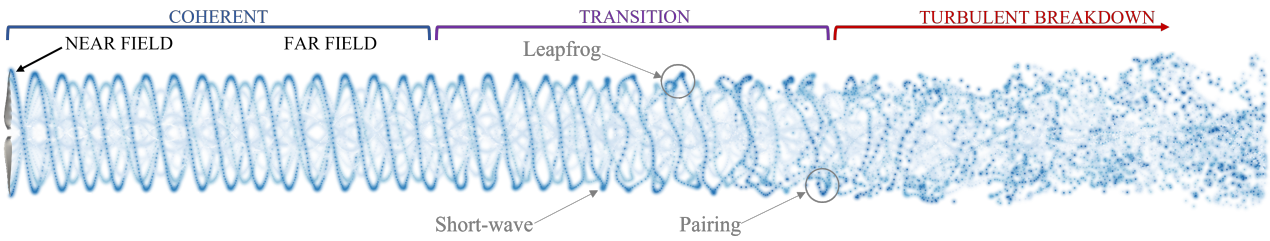


Figure II.3: Low-fidelity VPM simulation capturing the evolution of a propeller wake from a coherent vortical structure to turbulent breakdown and mixing.

The Vortex Particle Method

The vortex particle method (VPM), as introduced in Chapter 1, is a meshless CFD with several advantages over conventional mesh-based CFD. In the absence of a mesh, the VPM (1) does not suffer from the conventional Courant–Friedrichs–Lewy (CFL) condition, (2) conserves vortical structures over long distances with minimal numerical dissipation, and (3) it is 10x to 1000x faster than mesh-based CFD with comparable fidelity.

Since VPM does not suffer from the classic CFL condition, simulations retain physical accuracy even with a very coarse spatial discretization. For instance, Fig. II.3 shows a propeller wake in a coarse VPM simulation, correctly capturing each stage of the wake: from a coherent vortical structure to turbulent breakdown and mixing, predicting the mechanisms of transition in between (*i.e.*, short-wave instability, leapfrogging, and vortex pairing[132]). Fig. II.4 shows a mid-fidelity VPM simulation resolving the wake mixing and aerodynamic interactions between two propellers.[133]

VPM has gained popularity in recent years due to the growing need to predict aerodynamic interactions in eVTOL aircraft design. For example, the classic VPM has recently been used as a mid-fidelity tool for eVTOL rotor-rotor interactions,[134] flight path,[135] stacked rotors,[136] tiltrotor,[137] and multirotor tiltwing.[138] The middle column in Table II.1 shows a mid-fidelity simulation of an eVTOL aircraft using the classic VPM.

In spite of its growing popularity, VPM is known to be numerically unstable when vortical structures break down close to the turbulent regime. This has limited its range of applications in the aforementioned studies to mostly benign cases with well-behaved numerics (*e.g.*, coarse simulations of rotors with an axial inflow). Furthermore, little validation has been provided, since the lack of numerical stability typically embroils the efforts to show that simulations are convergent.

In short, the classic VPM suffers from the following shortcomings due to its poor numerical stability:

1. Applicable to only numerically well-behaved cases (*e.g.*, propellers in forward flight and wakes before breakdown/mixing).

2. Applicable to only low and mid-fidelity simulations (simulations

132 M. Felli et al. (2011). "Mechanisms of evolution of the propeller wake in the transition and far fields".

133 E. J. Alvarez and A. Ning (2018). "Development of a Vortex Particle Code for the Modeling of Wake Interaction in Distributed Propulsion".

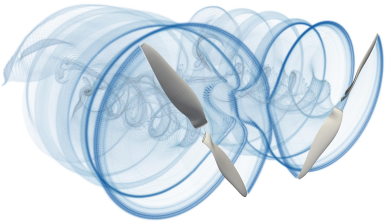


Figure II.4: Mid-fidelity VPM simulation capturing wake mixing and aerodynamic interactions between two propellers.

134 Lee and Lee, "Rotor interactional effects on aerodynamic and noise characteristics of a small multirotor unmanned aerial vehicle"; A. Zanotti and D. Algarotti (2022). "Aerodynamic interaction between tandem overlapping propellers in eVTOL airplane mode flight condition".

135 Arda Yücekayali (2020). "Noise Minimal & Green Trajectory And Flight Profile Optimization For Helicopters".

136 Jacobellis et al., "Experimental and computational investigation of stacked rotor performance in hover".

137 A. Zanotti, A. Savino, et al. (2021). "Assessment of a Mid-Fidelity Numerical Approach for the Investigation of Tiltrotor Aerodynamics".

138 Tugnoli et al., "Mid-fidelity approach to aerodynamic simulations of unconventional VTOL aircraft configurations".

become numerically unstable as spatial resolution is increased).

3. Lack of numerical convergence.
4. Scarcity of validation studies in the literature.

The Reformulated VPM: A Robust Variable-Fidelity Tool

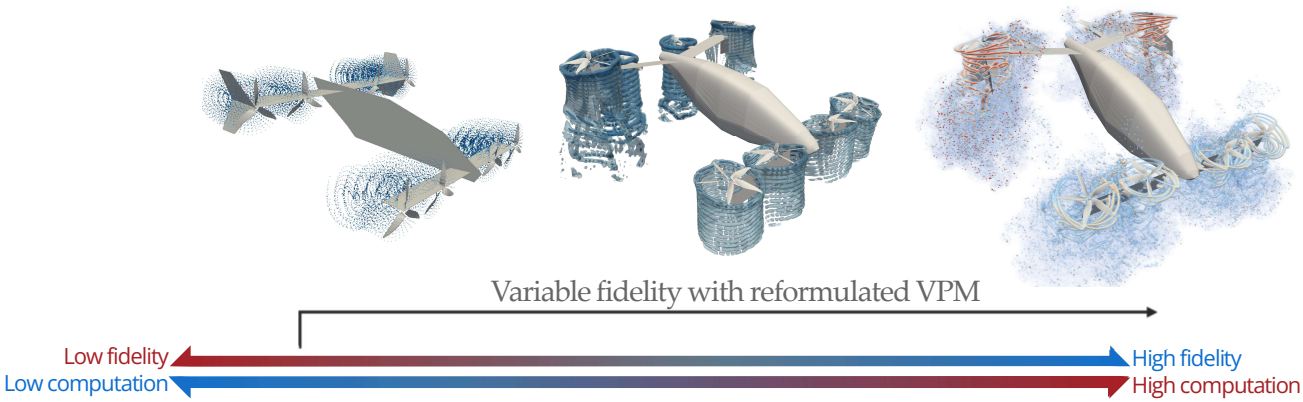
In this study, we present a CFD framework based on the reformulated VPM for simulating complex interactional aerodynamics. As shown in Part I, the rVPM is a large eddy simulation (LES) that is both numerically stable and meshless, while being 100x faster than mesh-based LES with comparable fidelity. Furthermore, since it is not limited by the classic CFL condition, rVPM can be used across all levels of fidelity, all in the same framework by simply coarsening or refining the simulation. Thus, rVPM is a high-fidelity tool that is orders of magnitude faster than mesh-based CFD (shown in Table II.1), while providing aircraft designers with a true variable-fidelity tool as depicted in Table II.2.

In Chapter 6, we will further develop our meshless LES scheme to include rotors and wings in the computational domain through actuator models. The models will be formulated by immersing their vorticity in the vorticity-velocity form of the Navier-Stokes equations. An actuator line model (ALM) based on tabulated data will be developed for rotors. An actuator surface model (ASM) based on a circulation solver will be developed for wings, which is suitable for rotor-wing interactions when a wake impinges on the surface of a wing. An aeroacoustics solver is also coupled to our meshless LES to predict tonal and broadband aeroacoustic noise radiated by rotors. This framework, called FLOWUnsteady[139], is implemented and hereby released as an open-source software.

[139] Open-source code available at github.com/byuflowlab/FLOWUnsteady

In Chapters 7 and 8, extensive validation of rotor-rotor and rotor-wing interactions predicted with our LES is presented. Verification and validation is presented on an isolated rotor, looking at cases from propeller mode in forward flight to a rotor in hover. The interactions between two side-by-side rotors in hover are then simulated and compared to experimental measurements reported in the literature. Predicting accurate rotor-wing interactions hinges on accurately resolving the rotor wake; hence, a detailed spatial resolution study characterizing the accuracy of the propeller wake is presented. A wing and rotors are then placed in a configuration resembling a tailplane with tip-mounted propellers, followed by a conventional configuration where the propellers are mounted mid-span. The predicted rotor-wing interactions are validated by comparison to experimental studies reported in the literature.

Finally, the capabilities of the framework are showcased in Chapter 9 through the simulation of a multirotor tiltwing vehicle. The vehicle is simulated mid maneuver as it transitions from powered lift to wing-borne flight. This simulation features rotors with variable RPM and variable pitch, tilting of wings and rotors, and significant rotor-rotor and rotor-wing interactions from hover to cruise. Thrust, powered lift, and wing-borne lift are characterized throughout the maneuver.



	Low Fidelity	Mid Fidelity	High Fidelity
Resolution	coarse (10k – 100k particles)	mid (100k – 1M particles)	fine (1M – 10M particles)
Wall-clock time (on laptop computer)	3 minutes – 30 minutes	1 hour – 12 hours	1 day – 4 days
Design stage	conceptual	preliminary	detailed

Table II.2: Variable fidelity achieved with the reformulated VPM.

6 Interactional Aerodynamics Solver

In this chapter we build upon the meshless LES scheme developed in Part I to create a solver for interactional aerodynamics. This poses the challenge of introducing solid boundaries without a mesh. Hence, most of the focus in this chapter, and Part II at length, will be that of developing and validating models that introduce wings and rotating blades in the computational domain.

Large eddy simulation aims at resolving the main structure of the flow, while smaller scales are modeled rather than resolved. Hence, in the proximity of a solid surface, instead of resolving the boundary layer down to the smallest y^+, solid bodies are often captured through either an immersed boundary method or an actuator model. Common immersed boundary methods introduce a penalization function in the Navier-Stokes equations that mimics the effects of both no-slip and no-flow-through boundary conditions at the surface of the body.[140] Actuator models, on the other hand, calculate and immerse the momentum imparted by the body to the fluid as a force term in the Navier-Stokes equations.[141]

In rotor simulations, the simplest actuator model is the *actuator disk model* (ADM).[142] ADM uses a spatial average of the velocity field computed by the LES at the rotor disk to calculate an effective angle of attack (AOA) along blade sections. The AOA is used with a precomputed lookup airfoil table to determine the corresponding lift and drag coefficients at each section. These blade cross sections with their respective lookup tables are referred to as *blade elements*. The force of each blade element is then distributed azimuthally along the rotor disk and introduced in the momentum Navier-Stokes equation as a momentum source. The rotor loads and momentum source terms thus computed by ADM are time averages, which result in a time-averaged rotor wake. This has the drawback that the predicted wake lacks the tip vortices and unsteady dynamics that drive many interactional aerodynamic phenomena.

An actuator model that is better suited for unsteady simulations is the *actuator line model* (ALM).[143] ALM calculates the effective AOA from a local velocity at the position of each blade element as the blade moves. Typically, the force is then placed at the aerodynamic center of the blade element (assumed to be the quarter chord) as a momentum source, thus constructing a lifting line that rotates with the blade. Tip vortices are shed as blades move and the wake is accurately resolved. Studies have

140 P. Angot et al. (1999). "A penalization method to take into account obstacles in incompressible viscous flows".

141 B. Sanderse et al. (2011). "Review of computational fluid dynamics for wind turbine wake aerodynamics".

142 I. Fejtek et al. (1992). "Navier-Stokes computation of wing/rotor interaction for a tilt rotor in hover"; R. G. Rajagopalan et al. (1993). "Three Dimensional Analysis of a Rotor in Forward Flight".

143 J. N. Sorensen et al. (2002). "Numerical Modeling of Wind Turbine Wakes"; Churchfield et al., "An Advanced Actuator Line Method for Wind Energy Applications and Beyond".

shown that wake dynamics and unsteady quantities (like sectional thrust and power) predicted with ALM can be as accurate as a blade-resolved simulation.[144]

In like manner to a rotor, ALM may also be used to simulate a fixed wing. In the case of an isolated wing, ALM can lead to an accurate lift and drag distribution as well as resolving the wake. However, when another wake is impinging directly on the surface of the wing (as in the case of a blown wing), ALM can lead to unphysical wing-on-wake interactions, which in turn lead to an inaccurate wing loading. This is because the impinging wake sees the wing only as a lifting line as opposed to a solid surface.

An actuator model that is better suited for wake impingement is the *actuator surface model* (ASM).[145] Whereas ALM places the force at the aerodynamic center of the airfoil, ASM uses precomputed pressure and skin friction distributions to distribute the force along the airfoil's mean camber line. Thus, the wing is represented as a lifting surface as opposed to a line. Studies have shown that ASM leads to accurate pressure and velocity fields in the proximity of the surface,[146] which makes ASM a promising approach to capture complex wing-wake interactions as in the case of a blown wing.

Whereas the aforementioned actuator models were developed as a momentum-source term for the Navier-Stokes equations in their pressure-velocity form, a different approach is needed for a solver based on the vorticity equation. Chatelain, Backaert, et al.[147] proposed introducing the aerodynamic force of an actuator line model into the vorticity equation by immersing the vorticity associated with its circulation. Later work applied this to both horizontal[148] and vertical[149] wind turbines. Caprace et al.[150] expounded on this by defining a "dragging line" to also immerse the vorticity associated with parasitic drag, and applied it to a wing. This lifting and dragging ALM approach was then applied to a rotor in edgewise flight, resolving the wake with unprecedented fidelity.[151] While ALM has been successfully applied to vortex methods, no precedent of an ASM application to vortex methods is found in the literature.

In this chapter we will further develop our meshless LES scheme to include rotors and wings in the computational domain through actuator models. The models will be formulated by immersing their vorticity in the vorticity-velocity form of the Navier-Stokes equations, as explained in Section 6.1. In Section 6.2, an ALM based on tabulated data will developed for rotors. In Section 6.3, an ASM based on a circulation solver will be developed for wings, which will be applied to the case of wake impingement on a blown wing. Finally, as described in Section 6.4, an aeroacoustics solver is also coupled to our meshless LES to predict tonal and broadband aeroacoustic noise radiated by rotors.

These models are implemented and integrated with the reformulated VPM code (FLOWVPM[152]) in a broader open-source solver called FLOWUnsteady.[153]

144 N. Troldborg et al. (2012). "Comparison of the wake of different types of wind turbine CFD models"; T. C. A. Stokkermans, N. van Arnhem, et al. (2019). "Validation and Comparison of RANS Propeller Modeling Methods for Tip-Mounted Applications".

145 W. Z. Shen et al. (2009). "The Actuator Surface Model: A New Navier-Stokes Based Model for Rotor Computations".

146 I. Dobrev et al. (2007). "Actuator surface hybrid model"; Shen et al., "The Actuator Surface Model: A New Navier-Stokes Based Model for Rotor Computations"; D. Linton et al. (2018). "Coupling of an Unsteady Aerodynamics Model with a Computational Fluid Dynamics Solver".

147 P. Chatelain, S. Backaert, et al. (2013). "Large eddy simulation of wind turbine wakes".

148 S. Backaert et al. (2015). "Vortex Particle-Mesh with Immersed Lifting Lines for Aerospace and Wind Engineering".

149 Chatelain, Duponcheel, et al., "Vortex particle-mesh simulations of vertical axis wind turbine flows: from the airfoil performance to the very far wake".

150 Caprace et al., "An immersed lifting and dragging line model for the vortex particle-mesh method".

151 Caprace, Chatelain, et al., "Wakes of rotorcraft in advancing flight: A large-eddy simulation study".

152 Open-source code available at github.com/byuflowlab/FLOWVPM.jl

153 Open-source code available at github.com/byuflowlab/FLOWUnsteady

6.1 Meshless LES with Immersed Vorticity

As derived in Part I, the LES-filtered vorticity equation

$$\frac{\mathrm{d}}{\mathrm{d}t}\overline{\boldsymbol{\omega}} = \left(\overline{\boldsymbol{\omega}}\cdot\nabla\right)\overline{\mathbf{u}} + \nu\nabla^2\overline{\boldsymbol{\omega}} - \mathbf{E}_{\mathrm{adv}} - \mathbf{E}_{\mathrm{str}}$$

is discretized and numerically solved in the reformulated VPM through the following governing equations

$$\begin{aligned}
&\frac{\mathrm{d}}{\mathrm{d}t}\mathbf{x}_p = \overline{\mathbf{u}}(\mathbf{x}_p)\\
&\frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{\Gamma}_p = \left(\boldsymbol{\Gamma}_p\cdot\nabla\right)\overline{\mathbf{u}}(\mathbf{x}_p) - \frac{3}{5}\left\{\left[\left(\boldsymbol{\Gamma}_p\cdot\nabla\right)\overline{\mathbf{u}}(\mathbf{x}_p)\right]\cdot\hat{\boldsymbol{\Gamma}}_p\right\}\hat{\boldsymbol{\Gamma}}_p - \frac{C_d}{\zeta_{\sigma_p}(\mathbf{0})}\mathbf{E}_{\mathrm{str}}(\mathbf{x}_p)\\
&\frac{\mathrm{d}}{\mathrm{d}t}\sigma_p = -\frac{1}{5}\frac{\sigma_p}{\|\boldsymbol{\Gamma}_p\|}\left[\left(\boldsymbol{\Gamma}_p\cdot\nabla\right)\overline{\mathbf{u}}(\mathbf{x}_p)\right]\cdot\hat{\boldsymbol{\Gamma}}_p\\
&\left(\frac{\mathrm{d}}{\mathrm{d}t}\overline{\boldsymbol{\omega}}\right)_{\mathrm{viscous}} = \nu\nabla^2\overline{\boldsymbol{\omega}}.
\end{aligned}$$

In order to immerse the vorticity of solid boundaries into these equations, the filtered vorticity field $\overline{\boldsymbol{\omega}}(\mathbf{x},t)$ is decomposed into a free-vorticity field $\overline{\boldsymbol{\omega}}_{\mathrm{free}}(\mathbf{x},t)$ and a bound-vorticity field $\overline{\boldsymbol{\omega}}_{\mathrm{bound}}(\mathbf{x},t)$ as

$$\overline{\boldsymbol{\omega}} = \overline{\boldsymbol{\omega}}_{\mathrm{free}} + \overline{\boldsymbol{\omega}}_{\mathrm{bound}}.$$

Both components can be discretized with vortex particles as

$$\overline{\boldsymbol{\omega}}(\mathbf{x}) = \underbrace{\sum_p \boldsymbol{\Gamma}_p\zeta_{\sigma_p}\left(\mathbf{x}-\mathbf{x}_p\right)}_{\overline{\boldsymbol{\omega}}_{\mathrm{free}}} + \underbrace{\sum_b \boldsymbol{\Gamma}_b\zeta_{\sigma_b}\left(\mathbf{x}-\mathbf{x}_b\right)}_{\overline{\boldsymbol{\omega}}_{\mathrm{bound}}},$$

where the particles discretizing the free-vorticity field evolve according to the governing equations shown above, while the ones discretizing the bound-vorticity are embedded on the boundary and their strength is calculated by an actuator model. As derived in Section 4.1, the velocity field is obtained by inverting the relation $\boldsymbol{\omega} = \nabla\times\mathbf{u}$, resulting in

$$\overline{\mathbf{u}}\left(\mathbf{x}\right) = \underbrace{\sum_p g_{\sigma_p}\left(\mathbf{x}-\mathbf{x}_p\right)\mathbf{K}\left(\mathbf{x}-\mathbf{x}_p\right)\times\boldsymbol{\Gamma}_p}_{\overline{\mathbf{u}}_{\mathrm{free}}} + \underbrace{\sum_b g_{\sigma_b}\left(\mathbf{x}-\mathbf{x}_b\right)\mathbf{K}\left(\mathbf{x}-\mathbf{x}_b\right)\times\boldsymbol{\Gamma}_b}_{\overline{\mathbf{u}}_{\mathrm{bound}}},$$

which includes the velocity induced by both free and bound vorticity components. Thus, the evolution of the free particles is influenced by the presence of the bound particles, affecting their convection and vortex stretching through the velocity field induced by the immersed vorticity.

The immersed vorticity not only affects the evolution of existing free vorticity, but it also creates new free vorticity at the boundary through viscous diffusion. In reality, vorticity is created in the boundary layer, it builds up as it travels along the surface, and it is eventually shed off the surface either by the Kutta condition at the trailing edge, flow separation,

or other turbulent mechanisms. In a slender body, the vorticity can be assumed to be shed at the trailing edge.

In this study, instead of creating vorticity through the viscous diffusion equation, the immersed vorticity will be shed at a prescribed trailing edge. This approach neglects the wake created by flow separation. However, the effects of flow separation on loading (like the drop in lift and increase in pressure drag on a stalled airfoil) can still be captured whenever lookup airfoil tables are used.

6.2 Rotor Model (Actuator Line Model)

Studies have shown that wake dynamics and unsteady quantities (like thrust and power) predicted with ALM can be as accurate as a blade-resolved simulation.[154] Hence, rotors will be introduced in our meshless LES through an actuator line model (ALM), as described in this section.

Actuator models typically include two schemes: one scheme for calculating blade forces from the fluid domain, and another for immersing such forces back into the fluid domain. In Section 6.2.1 we describe how the force is calculated in our ALM using blade elements with lookup airfoil tables, which is a common ALM approach. However, instead of introducing the force as a momentum source as typically done in conventional CFD, the force is introduced in our meshless LES by immersing its associated vorticity, as explained in Section 6.2.2.

154 Troldborg et al., "Comparison of the wake of different types of wind turbine CFD models"; Stokkermans, Arnhem, et al., "Validation and Comparison of RANS Propeller Modeling Methods for Tip-Mounted Applications".

6.2.1 Force Calculation

Blades are discretized through blade elements, which carry 2D airfoil data like lift and drag coefficients as a function of the angle of attack seen by the airfoil. These lift and drag curves are either automatically precomputed through the viscous panel code XFOIL or prescribed from experimental data to construct lookup tables. Hence, our ALM relies on the accuracy of the tabulated airfoil data to capture viscous effects like parasitic drag and stalled conditions, and compressible effects like wave drag, and it assumes that the data provided by the user already account for the effects of Mach and Reynolds numbers. Both lift and drag curves are then treated to capture three-dimensional drag and stall-delay effects due to centrifugal forces[155] and the Viterna method[156] is applied to obtain post-stall $\pm 180^\circ$ extrapolations of these curves.

During the simulation, the fluid domain computed by the LES is probed at the quarter-chord position of each blade element. The local velocity is used to calculate the effective angle of attack θ_{eff}, which is in turn used with the tabulated airfoil data to determine the sectional lift and drag coefficients, c_ℓ and c_d respectively. A tip correction F_{tip} is then applied to c_ℓ to account for the effects that bring the aerodynamic loading at the tip to zero, while a hub correction F_{hub} is also applied to account for the presence of the hub. F_{tip} and F_{hub} are defined as modified

155 Z. Du et al. (1998). "A 3-D stall-delay model for horizontal axis wind turbine performance prediction".

156 L. A. Viterna et al. (1982). *Theoretical and experimental power from large horizontal-axis wind turbines*.

Prandtl loss functions,

$$F_{\text{tip}} = \frac{2}{\pi}\cos^{-1}\left(\exp\left(-f_{\text{tip}}\right)\right), \qquad f_{\text{tip}} = \frac{B}{2}\frac{\left[\left(\frac{R_{\text{rotor}}}{r}\right)^{t_1} - 1\right]^{t_2}}{\left|\sin\left(\theta_{\text{eff}}\right)\right|^{t_3}}$$

and

$$F_{\text{hub}} = \frac{2}{\pi}\cos^{-1}\left(\exp\left(-f_{\text{hub}}\right)\right), \qquad f_{\text{hub}} = \frac{B}{2}\frac{\left[\left(\frac{r}{R_{\text{hub}}}\right)^{h_1} - 1\right]^{h_2}}{\left|\sin\left(\theta_{\text{eff}}\right)\right|^{h_3}},$$

where R_{rotor} and R_{hub} are the rotor and hub radii, B is the number of blades, r is the radial position of the blade element, and t_1, t_2, t_3, h_1, h_2, and h_3 are tunable parameters. Finally, the normal and tangential force coefficients, respectively c_n and c_t, are calculated as

$$c_n = F_{\text{tip}}F_{\text{hub}}c_\ell\cos\theta_{\text{eff}} + c_d\sin\theta_{\text{eff}}$$

and

$$c_t = F_{\text{tip}}F_{\text{hub}}c_\ell\sin\theta_{\text{eff}} - c_d\cos\theta_{\text{eff}}.$$

Rotor metrics like thrust, torque, and power are computed by integrating the load distribution along each blade.

6.2.2 Immersed Vorticity

The force along each blade is introduced back into the fluid domain by converting it into an equivalent immersed vorticity. The aerodynamic loading is first converted into a circulation distribution Γ using the Kutta-Joukowski theorem as

$$\Gamma = \frac{cV_{\text{local}}}{2}F_{\text{tip}}F_{\text{hub}}c_\ell,$$

where V_{local} is the local velocity seen by the blade element and c is its chord length. The vorticity is immersed by embedding particles along the surface that capture the blade's circulation distribution, while

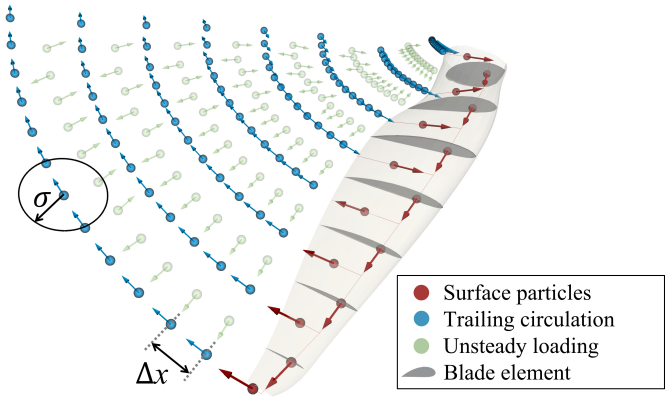


Figure 6.1: Particles used for immersed vorticity in actuator line model. Particles colored by their source of vorticity; arrows indicate direction of vortex strength.

shedding free particles at the trailing edge associated with unsteady loading and trailing circulation, as shown in Fig. 6.1.

Similarly, the vorticity associated with viscous drag can be captured by embedding a dipole of strength

$$\mu = \frac{V_{\text{local}}}{2} c_d$$

representing a dragging line, as suggested in the work of Caprace *et al.*[157] However, this showed to have a second-order effect in the applications studied in this doctoral work. Hence, this source of vorticity was neglected.

The frequency of particle shedding per revolution determines the initial spacing Δx in between particles, which, along with the core size σ, determines the spatial resolution at which the wake is being resolved. The number of elements along each blade determines the spatial resolution at which the blades are being resolved.

157 Caprace et al., "An immersed lifting and dragging line model for the vortex particle-mesh method"; D.-G. Caprace (2020). "Modeling of lifting-dragging devices for large eddy simulation of space-developing wakes: Application to wings, rotors and formation flight".

6.3 Wing Model (Actuator Surface Model)

Even though the ALM based on tabulated airfoil data is accurate for rotors, such an actuator model is only loosely-coupled with the fluid domain and imposes no boundary conditions. This makes it inadequate for cases with strong wake impingement, as in the case of a blown wing. Hence, wings will be introduced in our meshless LES through a different actuator model that is tightly coupled imposing a boundary condition at the surface of the wing. This boundary condition, called *no-flow-through condition*, consists of imposing a zero velocity normal to the surface of the wing, meaning that no flow goes through the surface. This is satisfied by solving for the circulation distribution that cancels the normal flow, as explained in Section 6.3.1. The associated vorticity is then immersed in the LES through an actuator surface model (ASM) in Section 6.3.2, while an alternative ALM is also proposed for low-fidelity simulations. Finally, the calculation of aerodynamic, viscous, and unsteady forces are described in Section 6.3.3.

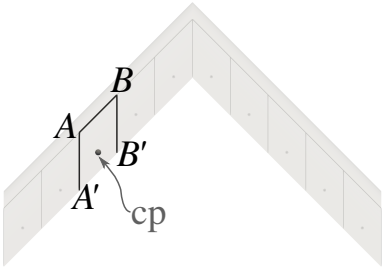


Figure 6.2: Wing element used in circulation solver.

6.3.1 Circulation Solver

The wing is discretized into wing elements in similitude to the discrete-vortex Weissinger model,[158] as shown in Fig. 6.2. Each wing element is composed of a bound vortex at the quarter-chord position (line $\overline{AB}$) and two trailing bound vortices extending to the trailing edge (lines $\overline{A'A}$ and $\overline{BB'}$). The velocity induced by the i-th wing element is approximated through vortex filaments as

$$\mathbf{u}_i(\mathbf{x}) = \Gamma_i \sum_{(a,b)\in \mathbb{H}_i} \mathbf{g}_{ab}(\mathbf{x}), \qquad \mathbb{H}_i = \{(A',A),(A,B),(B,B')\},$$

where

$$\mathbf{g}_{ab}(\mathbf{x}) = \frac{1}{4\pi} \frac{\mathbf{r}_a \times \mathbf{r}_b}{\|\mathbf{r}_a \times \mathbf{r}_b\|^2} \left(\frac{\mathbf{r}_a}{\|\mathbf{r}_a\|} - \frac{\mathbf{r}_b}{\|\mathbf{r}_b\|} \right) \cdot \mathbf{r}_{ab},$$

158 J. Weissinger (1947). *The Lift Distribution of Swept-Back Wings*.

$\mathbf{r}_{ab} = \mathbf{x}_b - \mathbf{x}_a$, $\mathbf{r}_a = \mathbf{x} - \mathbf{x}_a$, and $\mathbf{r}_b = \mathbf{x} - \mathbf{x}_b$. For ease of notation, we rewrite this as

$$\mathbf{u}_i(\mathbf{x}) = \Gamma_i \mathbf{G}_i(\mathbf{x}) \tag{6.1}$$

where $\mathbf{G}_i = \sum \mathbf{g}_{ab}$ contains the geometric information of the i-th wing element. The wing's self-induced velocity is then calculated as

$$\mathbf{u}_{\text{wing}}(\mathbf{x}) = \sum_i \mathbf{u}_i(\mathbf{x}). \tag{6.2}$$

In order to compute the circulation Γ along the wing, a control point $\mathbf{x}_{\text{cp}}$ is defined at the three-quarter-chord position of each wing element shown in Fig. 6.2, on which the no-flow-through condition is imposed. The local velocity $\mathbf{u}_{\text{local}}$ is calculated by adding the wing-induced velocity $\mathbf{u}_{\text{wing}}$, the kinematic velocity due to the motion of the wing $\mathbf{u}_{\text{kin}}$, and the velocity field calculated by the LES $\mathbf{u}_{\text{LES}}$ (computed before immersing the vorticity of the wing surface[159]). The local velocity at the i-th control point is then computed as

$$\mathbf{u}^i_{\text{local}}(t) = \mathbf{u}_{\text{wing}}(\mathbf{x}^i_{\text{cp}}, t) + \mathbf{u}^i_{\text{kin}}(t) + \mathbf{u}_{\text{LES}}(\mathbf{x}^i_{\text{cp}}, t)$$

and the no-flow-through condition is imposed as

$$\mathbf{u}^i_{\text{local}} \cdot \hat{\mathbf{n}}_i = 0,$$

leading to

$$\mathbf{u}^i_{\text{wing}} \cdot \hat{\mathbf{n}}_i = -\left(\mathbf{u}^i_{\text{kin}} + \mathbf{u}^i_{\text{LES}}\right) \cdot \hat{\mathbf{n}}_i, \tag{6.3}$$

where the superscript i denotes the corresponding function evaluated at the i-th control point, and $\hat{\mathbf{n}}_i$ is the unit vector that represents the normal to the surface of the i-th wing section. We approximate $\hat{\mathbf{n}}$ as $\mathbf{n} = (\mathbf{x}_{A'} - \mathbf{x}_A) \times (\mathbf{x}_B - \mathbf{x}_A)$ and $\hat{\mathbf{n}} = \mathbf{n}/\|\mathbf{n}\|$.

Replacing Eqs. (6.1) and (6.2) in Eq. (6.3), we arrive to

$$\sum_j \Gamma_j \mathbf{G}^i_j \cdot \hat{\mathbf{n}}_i = -\left(\mathbf{u}^i_{\text{kin}} + \mathbf{u}^i_{\text{LES}}\right) \cdot \hat{\mathbf{n}}_i.$$

Given a wing with N elements, this poses a linear system of N equations (one for each control point $\mathbf{x}^i_{\text{cp}}$) and N unknowns, Γ_j. The circulation distribution Γ_j that satisfies the boundary condition is then obtained by solving the system of equations.

When the vorticity of the wing is immersed in the fluid domain, the LES solver and the circulation solver become tightly coupled. Instead of implementing subiterations that converge this system at each time step, we let the system naturally converge as the simulation steps in time. We have observed that this can become numerically unstable when the wing experiences large velocity fluctuations, hence we introduce a relaxation procedure that updates Γ as

$$\Gamma_{\text{new}} = \alpha\Gamma + (1 - \alpha)\Gamma_{\text{old}}.$$

All simulations in this study use $\alpha = 0.3$.

159 Thus, $\mathbf{u}_{\text{LES}}$ at this point includes the velocity induced by the wing wake, but it excludes the velocity induced by the wing surface on itself.

6.3.2 Immersed Vorticity

Actuator Line for Low-Fidelity Simulations

In order to immerse the wing's vorticity in the fluid domain, each vortex segment in the wing element is converted into a static particle with strength $\mathbf{\Gamma} = \Gamma(\mathbf{x}_b - \mathbf{x}_a)$. The particle is embedded at the position $\mathbf{x} = (\mathbf{x}_a + \mathbf{x}_b)/2$, where $\mathbf{x}_a$ and $\mathbf{x}_b$ are the starting and ending points of the segment, respectively. As in the rotor ALM, free particles are shed at the trailing edge associated with unsteady loading and trailing vorticity.

Fig. 6.3 shows the vorticity field in a simulation of a planar wing of aspect ratio 5.33 at an angle of attack of 4°. Note that the vorticity is concentrated in two lines on the lifting surface. The front line corresponds to the lifting vorticity, while the back line corresponds to the trailing bound

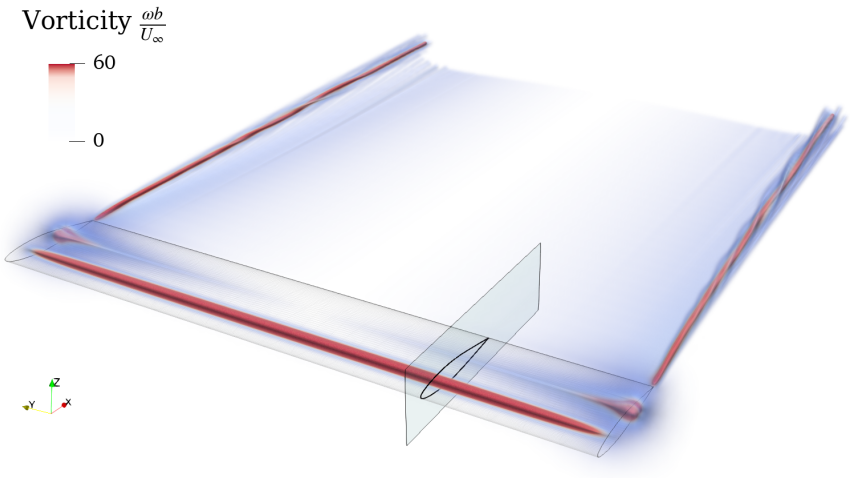


Figure 6.3: Wing simulation using ALM showing volume rendering of vorticity magnitude. Vertical plane at $2y/b = 0.5$ corresponds to slice shown in Fig. 6.4.

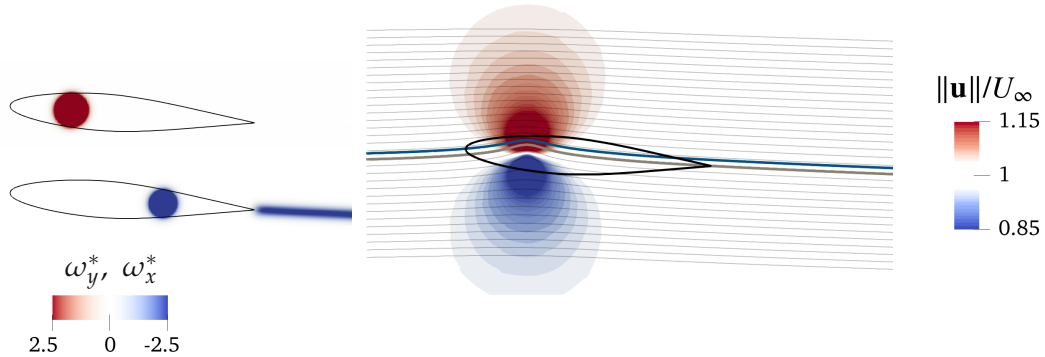


Figure 6.4: Slice at $2y/b = 0.5$ in wing simulations with ALM. (Left) lifting vorticity $\omega_y^* = \omega_y b/U_\infty$ and trailing vorticity $\omega_x^* = \omega_x b/U_\infty$, and (right) velocity magnitude and streamlines. Streamlines that pass through leading edge and trailing edge are shown in blue (━) and brown (━), respectively.

vorticity. A slice of the flow field is shown in Fig. 6.4. Even though the flow is tangent to the airfoil centerline at the three-quarter chord (satisfying the boundary condition), the streamlines cross the centerline aft of the leading edge. The streamlines passing through leading and trailing edges are shown in blue and brown, respectively, and the gap between them corresponds to the flow that crosses the centerline of the airfoil. Furthermore, notice that the velocity peaks are centered at the lifting line, but airfoils usually induce the strongest velocities by the leading edge. Because of these unphysical behaviors, the wing ALM is only recommended for low-fidelity simulations or simulations without strong aerodynamic interactions, and is not used in this study.

Actuator Surface for Higher-Fidelity Simulations

In order to achieve a more physical flow field in the vicinity of the wing, the vorticity needs to be spread chordwise rather than concentrated at the lifting line. This requires assuming a distribution $g(x^*)$ that will spread the circulation of the i-th blade element into a vortex sheet of strength $\gamma(x^*)$ as

$$\gamma(x^*) = \Gamma_i g(x^*), \qquad \text{with} \int_{-\infty}^{\infty} g(x/c)\,\mathrm{d}x = 1$$

and where $x^* = x/c$ is the chordwise position. At the same time, the trailing circulation is spread onto a vortex sheet of strength $\gamma_t(x^*)$ as

$$\gamma_t(x^*) = \int_0^{x^*} \gamma(x')\,\mathrm{d}x',$$

in order to satisfy Kelvin's theorem. Given $g(x^*)$, the center of pressure is the centroid of the distribution.

The simplest assumptions are that of singular and uniform distributions, shown in Fig. 6.5 (top and middle). The singular distribution turns out to be equivalent to a lifting line and is, in some regards, similar to the ALM discussed in the previous section, hence, it is not further discussed. The uniform distribution places the center of pressure at $x/c = 0.5$. An educated guess is that a good ASM would result, in most cases, in a center of pressure close to $x/c = 0.25$. Hence, the uniform distribution may lead to unphysical results.

Kim et al.[160] suggested using a piecewise linear distribution of the form

$$g(x^*) = \begin{cases} 0.4 + 3.04\frac{x^*}{0.25} & \text{if } 0 \le x^* < 0.25 \\ 3.44 - 3.2\left(\frac{x^*}{0.25} - 1\right) & \text{if } 0.25 \le x^* < 0.5 \\ 0.24 - 0.24\left(\frac{x^*}{0.5} - 1\right) & \text{if } 0.5 \le x^* \le 1 \\ 0 & \text{else} \end{cases}$$

which, as shown in Fig. 6.5 (bottom), leads to a center of pressure $x/c = 0.27$. The ASM distribution by Kim *et al.* seems somewhat arbitrary, but it has been used for blade-vortex interactions obtaining

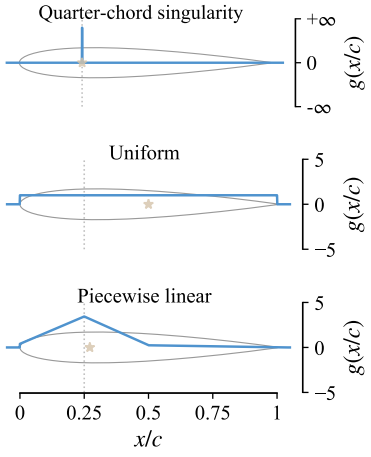


Figure 6.5: Vorticity distributions for actuator surface model. Star (⋆) indicates center of pressure.

160 T. Kim et al. (2015). "Improved actuator surface method for wind turbine application".

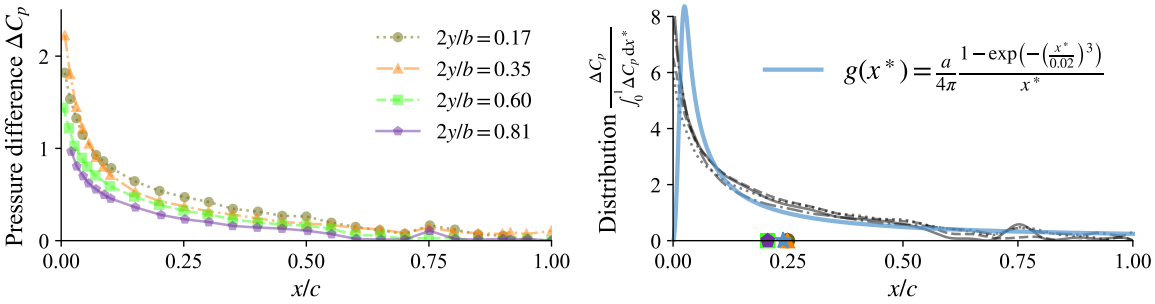


Figure 6.6: (left) Experimental chordwise pressure distribution on a planar wing reported by Veldhuis, and (right) their normalized distributions. Markers in (right) indicate corresponding centers of pressure.

favorable results.[161] However, this ASM distribution was developed for a momentum source in the pressure-velocity Navier-Stokes equation, and it is not evident that the same distribution should be applied for an immersed vorticity in the vorticity Navier-Stokes equation.

161 Linton et al., "Coupling of an Unsteady Aerodynamics Model with a Computational Fluid Dynamics Solver".

Noticing that most of the turning of the flow is usually done towards the leading edge, and that the pressure distribution typically follows that trend, we now propose a vorticity distribution akin to a pressure distribution. As a reference, Fig. 6.6 (left) shows the pressure difference between upper and lower surfaces at multiple stations measured experimentally by Veldhuis[162] on a planar wing. Even though the chordwise pressure distribution seems to vary between the different spanwise stations, normalizing each distribution evidences their similarity, as shown in Fig. 6.6 (right, black lines). Also, note that the center of pressure ranges between $x/c = 0.20$ to $x/c = 0.25$. We propose a pressure-like distribution given by

162 Veldhuis, "Propeller Wing Aerodynamic Interference".

$$g(x^*) = \begin{cases} \frac{a}{4\pi} \frac{1-\exp\left(-\left(\frac{x^*}{0.02}\right)^3\right)}{x^*} & \text{if } 0 \leq x^* \leq 1 \\ 0 & \text{else} \end{cases}$$

where a is determined numerically as $a = 3.061661$ in order to obtain a unitary distribution. This distribution is shown in Fig. 6.6 (right) and Fig. 6.7, with its center of pressure at $x/c = 0.2393$. Fig. 6.8 shows the simulation of the planar wing using the pressure-like vorticity distribution, where the vorticity is concentrated at the leading edge, while varying spanwise and chordwise.[163]

163 The spanwise distribution is calculated by the circulation solver, while the chordwise distribution is prescribed by the pressure-like g distribution.

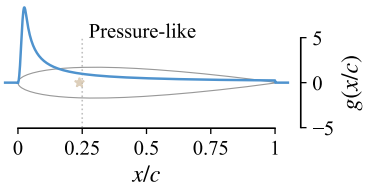


Figure 6.7: Pressure-like vorticity distribution in actuator surface model. Star ($\star$) indicates center of pressure.

In order to assess the fitness of each vorticity distribution, Fig. 6.9 shows a slice of the flow field around the planar wing (aspect ratio 5.33 and AOA 4°) obtained with each distribution. The figures in the right show the resulting velocity field and streamlines. The uniform distribution leads to somewhat of a uniform velocity field, which seems unphysical. The piecewise linear distribution leads to a velocity centered at the quarter-chord, similar to the ALM. The pressure-like distribution leads to a flow that is being turned close to the leading edge, as we hoped for. Note that they all succeed at making the flow tangent to the

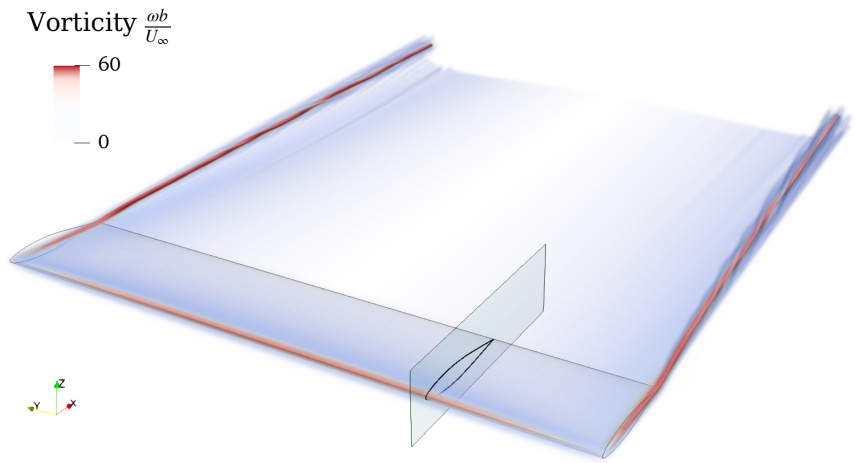
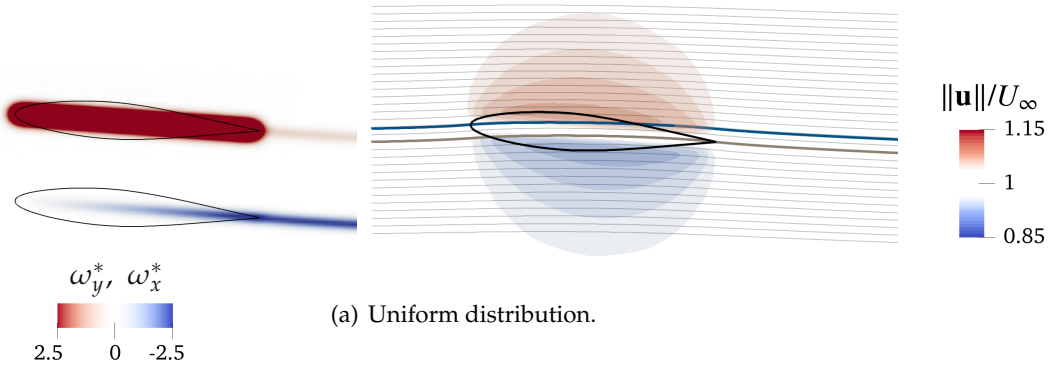


Figure 6.8: Wing simulation using ASM with pressure-like distribution. Volume rendering of vorticity magnitude. Vertical plane at $2y/b = 0.5$ corresponds to slice shown in Fig. 6.9.



(a) Uniform distribution.

(b) Piecewise linear distribution.

(c) Pressure-like distribution.

Figure 6.9: Slice at $2y/b = 0.5$ in wing simulations with various ASM distributions. (Left) lifting vorticity $\omega_y^* = \omega_y b/U_\infty$ and trailing vorticity $\omega_x^* = \omega_x b/U_\infty$, and (right) velocity magnitude and streamlines. Streamlines that pass through leading edge and trailing edge are shown in blue (━) and brown (━), respectively.

airfoil centerline at the three-quarter chord position, but there is still some amount of flow going through the centerline surface elsewhere. The streamlines passing through leading and trailing edges are shown in blue and brown, respectively, and the gap between them corresponds to the flow that crosses the centerline. Out of the three distributions, the pressure-like ASM minimizes the amount of flow permeated through the centerline surface, hence this distribution will be used for all simulations in this study.

A pressure-like ASM was previously proposed by Shen et al.;[164] however, their ASM was based on tabulated airfoil data and developed as a momentum-source term for the Navier-Stokes equations in their pressure-velocity form. The novelty of our ASM lays on that it is based on a circulation solver and developed for immersing vorticity in the vorticity form of the Navier-Stokes equations.

164 Shen et al., "The Actuator Surface Model: A New Navier-Stokes Based Model for Rotor Computations".

In this study, we will prescribe the pressure-like distribution, but future work could improve the ASM by changing the pressure-like distribution according to angle of attack, airfoil thickness, and camber as suggested by Shen et al.,[165] as well as sweep and aspect ratio of the wing. Further improvements can also include distributing the vorticity along the mean camber line as opposed to a straight line between leading and trailing edges, as well as developing a skin-friction-like distribution to immerse and shed the vorticity associated with viscous drag.

165 Ibid.

6.3.3 Force Calculation

After solving for the circulation and immersing the vorticity in the fluid domain, the force on the wing is calculated from three components: An aerodynamic force, a viscous force, and an unsteady-circulation force. The aerodynamic force is here derived from the turning of the flow around the wing, which yields a version of the Kutta-Joukowski theorem that is suitable for vortex particles. This aerodynamic force includes both lift and induced drag. The parasitic force includes form, skin friction, and wave drag. The unsteady-circulation force accounts for the fact that the circulation distribution varies in time.

Aerodynamic (Kutta-Joukowski) Force

The intensive force $\mathbf{f}$ exerted on a fluid, or force per unit volume, is defined as

$$\mathbf{f} = \frac{\mathrm{d}}{\mathrm{d}t}\left(\rho\mathbf{u}\right),$$

which in incompressible flow becomes

$$\mathbf{f} = \rho\frac{\mathrm{d}\mathbf{u}}{\mathrm{d}t}.$$

Expanding the total derivative operator,

$$\mathbf{f} = \rho\frac{\partial\mathbf{u}}{\partial t} + \rho\left(\mathbf{u}\cdot\nabla\right)\mathbf{u},$$

we notice that this is simply the left-hand side of the Navier-Stokes momentum equation. The force can then be decomposed into an unsteady component $\mathbf{f}_\mathrm{u} \equiv \rho \frac{\partial \mathbf{u}}{\partial t}$ and a quasi-steady component $\mathbf{f}_\mathrm{s} \equiv \rho \left(\mathbf{u} \cdot \nabla\right) \mathbf{u}$ as

$$\mathbf{f} = \mathbf{f}_\mathrm{u} + \mathbf{f}_\mathrm{s}.$$

Using the following identity

$$\nabla(\mathbf{A} \cdot \mathbf{B}) \;=\; (\mathbf{A} \cdot \nabla)\mathbf{B} \;+\; (\mathbf{B} \cdot \nabla)\mathbf{A} \;+\; \mathbf{A}\times(\nabla\times\mathbf{B}) \;+\; \mathbf{B}\times(\nabla\times\mathbf{A}),$$

the quasi-steady component becomes

$$\mathbf{f}_\mathrm{s} = \rho \left[\frac{1}{2}\nabla \left(\mathbf{u} \cdot \mathbf{u}\right) - \mathbf{u} \times \left(\nabla \times \mathbf{u}\right)\right].$$

For ease of notation, we write this as

$$\mathbf{f}_\mathrm{s} = \rho \frac{\nabla \mathbf{u}^2}{2} - \rho \mathbf{u} \times \boldsymbol{\omega},$$

where $\nabla \mathbf{u}^2 \equiv \nabla \left(\mathbf{u} \cdot \mathbf{u}\right)$ and $\boldsymbol{\omega} \equiv \nabla \times \mathbf{u}$. We further decompose the steady force into a kinetic component $\mathbf{f}_\mathrm{kin} \equiv \rho \frac{\nabla \mathbf{u}^2}{2}$ and an aerodynamic component $\mathbf{f}_\mathrm{aero} \equiv -\rho \mathbf{u} \times \boldsymbol{\omega}$ as

$$\mathbf{f}_\mathrm{s} = \mathbf{f}_\mathrm{kin} + \mathbf{f}_\mathrm{aero}.$$

Note that $\mathbf{f}_\mathrm{aero}$ is the Lamb vector when the density is unitary.

Consider a chunk of fluid with volume Vol, represented by a vortex particle placed at $\mathbf{x}_p$ inside the volume and vortex strength

$$\boldsymbol{\Gamma}_p \approx \int_\mathrm{Vol} \boldsymbol{\omega} \, \mathrm{d}\mathbf{x}.$$

We integrate the aerodynamic component of the intensive force, $\mathbf{f}_\mathrm{aero}$, to get an extensive force in such volume of fluid, denoted $\mathbf{F}_\mathrm{aero}$, as

$$\mathbf{F}_\mathrm{aero} = -\rho \int_\mathrm{Vol} \mathbf{u}(\mathbf{x}') \times \boldsymbol{\omega}(\mathbf{x}') \, \mathrm{d}\mathbf{x}',$$

where we have assumed a uniform density ρ and incompressible flow. Assuming that the particle is the only source of vorticity inside the volume and using the singular particle approximation, $\boldsymbol{\omega}(\mathbf{x}) \approx \boldsymbol{\Gamma}_p \delta(\mathbf{x} - \mathbf{x}_p)$, the force is then approximated as

$$\mathbf{F}_\mathrm{aero} \approx -\rho \int_\mathrm{Vol} \mathbf{u}(\mathbf{x}') \times \boldsymbol{\Gamma}_p \delta(\mathbf{x}' - \mathbf{x}_p) \, \mathrm{d}\mathbf{x}',$$

becoming

$$\mathbf{F}_\mathrm{aero} \approx -\rho \mathbf{u}(\mathbf{x}_p) \times \boldsymbol{\Gamma}_p. \tag{6.4}$$

Thus, we have arrived to a simple but general expression that approximates the aerodynamic force experienced by the fluid. We have chosen to call it "aerodynamic" force since it is caused by the vorticity in the fluid, typically associated with the presence of circulation. Furthermore, the Kutta-Joukowski theorem—the fundamental theorem of aerodynamics—can be derived directly from this expression, as follows.

Suppose that the vorticity in such volume corresponds to the immersed vorticity of a lifting line segment with length $\boldsymbol{\ell}$ and circulation Γ. The vorticity can then be represented with a bound particle of vortex strength $\boldsymbol{\Gamma}_b = \Gamma\boldsymbol{\ell}$ placed at the center of the lifting line, $\mathbf{x}_b$. The force experienced by the volume of fluid is then

$$\mathbf{F}_{\text{aero}} \approx -\rho\mathbf{u}(\mathbf{x}_b) \times \Gamma\boldsymbol{\ell}.$$

Since the lifting line corresponds to a wing section, this $\mathbf{F}_{\text{aero}}$ is an external force exerted by the wing on the fluid, and the wing feels the opposite force in response. We denote the force experienced by the wing as $\mathbf{F}_{\text{kj}}$, defined as

$$\mathbf{F}_{\text{kj}} = \rho\mathbf{u}(\mathbf{x}_b) \times \Gamma\boldsymbol{\ell}. \tag{6.5}$$

If $\mathbf{u}(\mathbf{x}_b)$ and $\boldsymbol{\ell}$ are perpendicular, the force per unit length, defined as $F'_{\text{kj}} \equiv \|\mathbf{F}_{\text{kj}}\| / \|\boldsymbol{\ell}\|$, becomes

$$F'_{\text{kj}} = \rho u(\mathbf{x}_b)\Gamma,$$

which is the Kutta-Joukowski theorem when $u(\mathbf{x}_b) = u_\infty$. Hence, the particle approximation of the aerodynamic force, as given in Eq. (6.4), is consistent with the theorem. The only assumptions we have undertaken to arrive to Eq. (6.4) is that of incompressible flow, and that the particle is the only source of vorticity inside the volume, *i.e.*, $\int_{\text{Vol}} \boldsymbol{\omega}\, \mathrm{d}\mathbf{x} \approx \boldsymbol{\Gamma}_p$.

In the case of the wing section, the velocity field $\mathbf{u}$ used in Eq. (6.5) can be the superposition of a freestream $\mathbf{u}_\infty$, a kinematic velocity $\mathbf{u}_{\text{kin}}$, the velocity induced by other lifting surfaces $\mathbf{u}_{\text{wing/blade}}$, and/or a wake velocity $\mathbf{u}_{\text{wake}}$. In simple terms, the force given by Eq. (6.5) is the reaction due to the wing section turning the local flow around it. Hence, we refer to this force as the *Kutta-Joukowski force*, $\mathbf{F}_{\text{kj}}$. Decomposing $\mathbf{u}$ as

$$\mathbf{u} = \mathbf{u}_{\text{kin}} + \underbrace{\mathbf{u}_\infty + \mathbf{u}_{\text{wing/blade}} + \mathbf{u}_{\text{wake}}}_{\mathbf{u}_{\text{LES}}},$$

the last three velocity components are calculated by the LES. Writting $\boldsymbol{\Gamma}_b = \Gamma\boldsymbol{\ell}$, the Kutta-Joukowski force is then computed as

$$\mathbf{F}_{\text{kj}} = \rho\mathbf{u}^b_{\text{kin}} \times \boldsymbol{\Gamma}_b + \rho\mathbf{u}_{\text{LES}}(\mathbf{x}_b) \times \boldsymbol{\Gamma}_b. \tag{6.6}$$

Parasitic Force

Parasitic drag along the wing is calculated using a lookup airfoil table. The drag coefficient c_d can be determined either from the local angle of

attack or the local lift coefficient, $c_\ell = \frac{2\Gamma}{U_\infty c}$. Our experience is that the most accurate results are obtained through the local lift coefficient. The parasitic drag includes both form and skin friction drag, where form drag includes both wave drag and pressure drag due to separation.

Unsteady Force

An additional force term F_{uns} is added due to the unsteady changes of circulation, which is calculated as

$$\mathbf{F}_{\text{uns}} = \rho \frac{\mathrm{d}\Gamma}{\mathrm{d}t} A \hat{\mathbf{n}},$$

where A is the area of the wing element and $\hat{\mathbf{n}}$ is its normal vector.

6.4 Aeroacoustic Solver: Ffowcs Williams-Hawkings Analogy

High-fidelity approaches for the prediction of aeroacoustic noise can be derived from the application of acoustic analogies to the Navier-Stokes equations. Ffowcs Williams and Hawkings introduced in 1969 an analogy that includes the effects of surfaces in arbitrary motion and turbulent flow.[166] Using generalized functions, they rearranged the continuity equation and the Navier-Stokes equations into a homogeneous wave equation with monopole and dipole sources on the body surface, and a quadrupole source distribution in a volume around the body.

[166] J. Ffowcs Williams et al. (1969). "Sound generation by turbulence and surfaces in arbitrary motion".

The Ffowcs Williams-Hawkings (FW-H) equation reads

$$\Box^2 p'(\mathbf{x}, t) = \underbrace{\frac{\partial}{\partial t}\left(\rho_0 u_n \delta(f)\right)}_{\text{monopole (thickness)}} - \underbrace{\frac{\partial}{\partial x_i}\left(\Delta P_{ij} \hat{n}_j \delta(f)\right)}_{\text{dipole (loading)}} + \underbrace{\frac{\partial^2}{\partial x_i \partial x_j}\left(T_{ij} H(f)\right)}_{\text{quadrupole (wake)}}, \tag{6.7}$$

where $\Box^2 \equiv \frac{1}{c^2}\frac{\partial^2}{\partial t^2} - \nabla^2$ is the wave-equation operator and p' is the acoustic pressure. The first term in the right-hand side is a monopole source representing the volume displaced by the thickness of a solid body in motion, where u_n is the local velocity normal to the body surface and $\delta(f)$ is the Dirac delta function describing the surface. The second term is a dipole source representing the force applied on the fluid by the body, where $\hat{n}_j$ is the normal vector away from the body surface. P_{ij} is the compressive stress tensor assuming $\Delta P_{ij} = (p - p_0)\delta_{ij}$, where $p - p_0$ is the gauge pressure and δ_{ij} is the Kronecker delta. The third term is a quadrupole source acting throughout the volume outside of the body, where $H(f)$ is the Heaviside function describing the exterior of the body, and T_{ij} is the Lighthill stress tensor.

The FW-H analogy has been shown to accurately predict rotorcraft noise radiated from complicated aerodynamic phenomena like blade-vortex interaction and high-speed impulsive effects.[167] When Eq. (6.7) is applied to a rotating blade, monopole sources represent the thickness of the blade and dipole sources represent the aerodynamic force distributed along the blade, or blade loading, while quadrupole sources capture the sound radiated from shear and stretching forces in the fluid as, for

[167] K. S. Brentner et al. (2003). "Modeling aerodynamically generated sound of helicopter rotors".

instance, in turbulent mixing of the wake. With the exception of high-speed impulsive noise, the aeroacoustic noise of conventional rotorcraft is typically dominated by thickness and loading of the blade, hence the quadrupole term is in many cases neglected. This considerably eases the numerical solution of the FW-H equation as the quadrupole term requires a volumetric integral, while the other two terms simply require surface integrals.

In this study we will solve the FW-H equation using the time-domain integral formulation 1A developed by Farassat.[168] This formulation neglects the quadrupole term and is valid for an arbitrary body in motion. The total acoustic pressure p' is calculated as the superposition of thickness pressure p'_T and loading pressure p'_L,

168 F. Farassat (2007). "Derivation of Formulations 1 and 1A of Farassat".

$$p'(\mathbf{x}, t) = p'_T(\mathbf{x}, t) + p'_L(\mathbf{x}, t).$$

Each component is calculated by integrating the corresponding term in Eq. (6.7) along the surface of the body, resulting in

$$4\pi p'_T(\mathbf{x}, t) = \int \left[\frac{\rho_0(\dot{u}_n) + u_{\dot{n}})}{r|1 - M_r|^2}\right] \mathrm{d}S + \int \left[\frac{\rho_0 u_n (r\dot{M}_r + cM_r - cM^2)}{r^2|1 - M_r|^3}\right] \mathrm{d}S$$

and

$$4\pi p'_L(\mathbf{x}, t) = \frac{1}{c}\int \left[\frac{\dot{\ell}_r}{r|1 - M_r|^2}\right] \mathrm{d}S + \int \left[\frac{\ell_r - \ell_M}{r|1 - M_r|^2}\right] \mathrm{d}S + \frac{1}{c}\int \left[\frac{\ell_r (r\dot{M}_r + cM_r - cM^2)}{r^2|1 - M_r|^3}\right] \mathrm{d}S,$$

where dot over a variable denotes its time derivative, $r = |\mathbf{x} - \mathbf{y}|$ is the distance between an observer at $\mathbf{x}$ and a source at $\mathbf{y}$, $\ell_r = P_{ij}\hat{r}_i n_j$ is the force intensity that acts on the fluid, M is the Mach number, $M_r = M_i \hat{r}_i$ is the Mach number of source in radiation direction, and subscripts n, r, and M denote the dot product with the unit normal vector, radiation vector, and the velocity vector divided by the speed of sound c, respectively. The derivation of this formulation can be found in the original work by Farassat and Brentner.[169] Thickness and loading noise will be computed through the code PSU-WOPWOP coupled to FLOWUnsteady, which uses a retarded-time algorithm discretizing each integral as a mid-panel quadrature over the surface as, for example,

169 Brentner et al., "Modeling aerodynamically generated sound of helicopter rotors"; Farassat, "Derivation of Formulations 1 and 1A of Farassat".

$$\int \left[\frac{Q(\mathbf{y}, \tau)}{r|1 - M_r|^2}\right] \mathrm{d}S \approx \sum_i \frac{Q(\mathbf{y}_i, t - r_i/c)}{r_i|1 - M_r|_i^2} \Delta S_i$$

where $\tau = t - r_i/c$ is the retarded time, corresponding to the time that the i-th source radiates to reach the observer by t. Thickness pressure will be calculated from the three-dimensional loft of the blade, while loading pressure will use a compact representation of two-dimensional cross sections that correspond to the blade elements used by the actuator line model discussed in Section 6.2

The different sources of aeroacoustic noise are sometimes classified as deterministic or non-deterministic sources. Deterministic sources often contribute to discrete frequencies and are perceived as tonal noise. The noise associated to blade thickness and steady loading is one example of tonal noise, which is manifested as multiples (or harmonics) of the blade-passing frequency. Non-deterministic sources are stochastic in nature (usually associated with turbulence), contribute to a wide frequency band across the spectrum, and are perceived as being atonal, referred to as broadband noise. The FW-H analogy not only captures the noise of blade thickness and loading, but if unsteady interactional aerodynamics are accurately resolved, FW-H can also predict the noise caused by unsteady loading, as in blade-vortex interaction, wake-rotor interactions, rotor-rotor interactions, and wing-rotor interactions.

The noise radiated from very small scales is difficult and computationally expensive to predict deterministically, for example, as in the case of noise radiated as vortices in the boundary layer stretch at the trailing edge. In this study, non-deterministic noise sources will be modeled through the semi-empirical methodology developed by Brooks, Pope, and Marcolini,[170] referred to as BPM. The methodology models five self-noise mechanisms due to boundary-layer phenomena: boundary-layer turbulence passing the trailing edge, separated boundary-layer and stalled-airfoil flow, vortex shedding due to laminar-boundary-layer instabilities, vortex shedding from blunt trailing edges, and turbulent flow due to vortex tip formation. The BPM equations are partially developed from theory, and supplemented with experimental measurements on a symmetric NACA 0012 airfoil. Computation will be performed through an open-source BPM code[171] developed at BYU's FLOW Lab and coupled with FLOWUnsteady. Since the method is based on a symmetric airfoil, we recognize that BPM is accurate only for simplistic blade geometries. However, the present study intends to focus on the noise caused by the aforementioned interactional aerodynamics, and the non-deterministic broadband noise captured through BPM will be mostly notional.

170 T. F. Brooks et al. (1989). *Airfoil self-noise and prediction*.

171 https://github.com/byuflowlab/BPM.jl

7 Validation of Rotor-Rotor Interactions

In this chapter we validate the actuator line model used for rotors in our LES, starting with an isolated rotor and building up to the interactional case of two side-by-side rotors in hover. First, a detailed convergence study of temporal and spatial discretizations of the isolated rotor is presented in Section 7.2. Validation on a propeller in forward flight at an incidence angle is then presented Sections 7.3 and 7.4. While a thorough validation on a rotor in hover was already presented in Section 5.4, validation of the aeroacoustic noise predicted on the same rotor in hover is now presented in Section 7.5. Finally, in Section 7.6, we characterize and validate the aerodynamic interactions between two side-by-side rotors as the distance between them is decreased.

7.1 Rotor Geometries

Throughout this dissertation, four different rotors are simulated and used in validation cases: the DJI 9443 used by Zawodny et al.[172] (9.4 inches diameter and 0.11 solidity), the DJI-like rotor designed by Ning[173] (9.4 inches diameter and 0.12 solidity), the thin-electric APC 10x7 propeller used by McCrink et al.[174] (10 inches diameter and 0.10 solidity), and the Beaver propeller described by Sinnige-de Vries *et al.*[175] (9.34 inches in diameter).

The Ning DJI rotor uses a uniform E63 airfoil shape transitioning to an E856 airfoil towards the hub, the APC 10x7 uses a NACA 4412 transitioning to a Clark Y towards the hub, while the airfoil sections of the DJI 9443 were obtained by digitizing slices of the actual rotor. Twist and chord distributions of Ning DJI, APC 10x7, and DJI 9443, along with

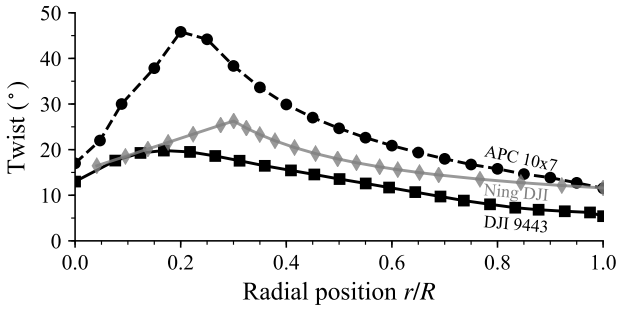


Figure 7.1: Twist distribution of each rotor.

172 Zawodny et al., "Acoustic Characterization and Prediction of Representative, Small-scale Rotary-wing Unmanned Aircraft System Components".

173 Z. Ning (2018). "Experimental investigations on the aerodynamic and aeroacoustic characteristics of small UAS propellers".

174 M. H. McCrink et al. (2017). "Blade Element Momentum Modeling of Low-Reynolds Electric Propulsion Systems".

175 T. Sinnige, R. de Vries, et al. (2018). "Unsteady Pylon Loading Caused by Propeller-Slipstream Impingement for Tip-Mounted Propellers".

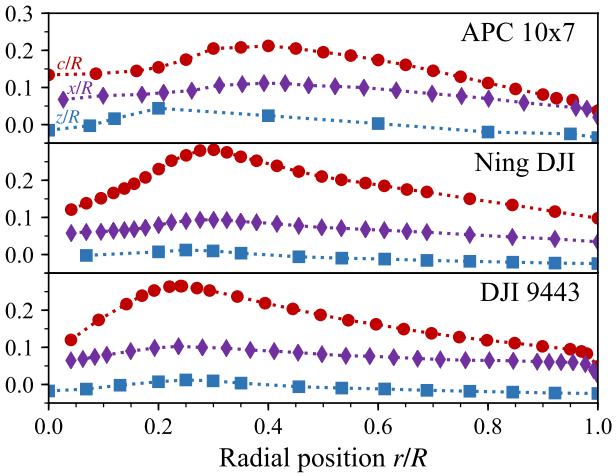


Figure 7.2: Chord distribution and leading edge curve of each rotor.

leading edge curves, are shown in Figs. 7.1 and 7.2. The reader is referred to Sinnige-de Vries *et al.* for a description of the Beaver propeller.

7.2 Convergence Study

Herein we characterize the numerical convergence of the rotor simulation with respect to spatial and temporal refinement. We focus on the convergence of predicted rotor performance, using the thrust coefficient C_T as the metric of interest. Since rotor performance is only affected by the flow field at the plane of rotation, another convergence study will be performed in Section 8.2 using metrics related to the wake away from the plane of rotation, like tip-vortex velocity profile.

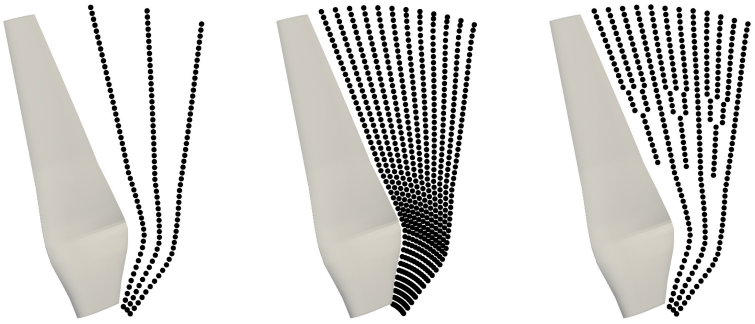
This study was performed and published[176] in the initial stages of this doctoral work before the reformulated VPM was developed, hence it used the classic VPM. We have observed that this convergence characteristics are dictated by our actuator line model used for the rotor, independent of the VPM formulation.

176 Alvarez and Ning, "High-Fidelity Modeling of Multirotor Aerodynamic Interactions for Aircraft Design".

7.2.1 Decoupling Temporal and Spatial Resolutions

In VPM-based rotor simulations, the spatial resolution is given by both the initial distance between particles Δx and the initial smoothing radius σ (or filter width). The former determines the spatial discretization, while the latter defines the smallest resolved length scale. It has been proven that numerical convergence and stability is conditional to having a spatial discretization finer than the smoothing radius,[177] meaning that decreasing σ requires increasing how often particles are shed off the blade to obtain a smaller Δx. It is common in VPM-based simulations to shed particles at every time step, thus inevitably coupling temporal and spatial resolutions: if the shedding of particles is controlled by the time stepping, a finer spatial resolution is only achieved by also increasing the temporal resolution. This spatio-temporal coupling can

177 J. T. Beale (1986). "A convergent 3-D vortex method with grid-free stretching".



(a) $N_{\text{sheds}} = N_{\text{steps}}$ (b) $N_{\text{sheds}} = 4N_{\text{steps}}$ (c) $N_{\text{sheds}} = 4N_{\text{steps}}$

Figure 7.3: Wake shedding with different N_{steps} and N_{sheds} values after three time steps: (a) case $N_{\text{steps}} \geq N_{\text{sheds}}$ where particles are shed every m steps ($m = 1$ is shown), (b) case $N_{\text{steps}} < N_{\text{sheds}}$ shedding particles m times in every time step ($m = 4$ is shown), and (c) uniform distribution of particles by varying m along the blade using Eq. (7.1).

embroil a convergence study since the numerical error will not decrease monotonically with the coupled spatio-temporal resolution.

One approach to control the spatial resolution independent of the temporal resolution is to use a background mesh onto which the particles are projected and reset after convection. This approach is called vortex-in-cell or vortex particle-mesh method.[178] However, the introduction of this mesh also forfeits some of the benefits of a purely Lagrangian (meshless) simulation.

In order to decouple temporal and spatial resolutions without a mesh, we use the following procedure for shedding particles independently of the time stepping. Let N_{steps} be the number of times steps in one rotor revolution and N_{sheds} the desired number of particle sheds in one revolution. We require N_{steps} and N_{sheds} to be multiples of each other, and denote such multiple as $m \in \mathbb{N}$. If the number of time steps is larger than or equal to the number of sheds ($N_{\text{steps}} \geq N_{\text{sheds}}$), particles are shed every m time steps as shown[179] in Fig. 7.3a. If the number of sheds is larger than the number of time steps ($N_{\text{steps}} < N_{\text{sheds}}$), particles are shed m times in every time step, as shown in Fig. 7.3b with $m = 4$. Thus, the frequency of sheds can be controlled independently of the time stepping.

Using a large N_{sheds} may lead to a large concentration of particles towards the slower-moving part of the blade, as observed in Fig. 7.3b. This does not pose a numerical difficulty, but it increases the computational cost unnecessarily. The computational cost can be reduced by varying N_{sheds} along the blade, increasing the number of sheds only towards the tip as follows. Let f be the number of particles that are shed at a given position r, f is made to vary from 1 at the root to m at the tip through the following function given as pseudo-code:

$$f(r, m) = \max\left(1, \min\left(m, \text{floor}\left(\frac{1}{1-(r/R)}\right)\right)\right). \tag{7.1}$$

Fig. 7.4 shows the number of sheds along the blade as m is increased. Fig. 7.3c shows the uniform concentration of particles achieved in this

178 Christiansen, "Numerical Simulation of Hydrodynamics by the Method of Point Vortices"; Cottet and Poncet, "Advances in direct numerical simulations of 3D wall-bounded flows by Vortex-in-Cell methods"; Koumoutsakos, "Multi-scale flow simulations using particles".

179 The case $N_{\text{steps}} > N_{\text{sheds}}$ is not shown since it is analogous to the case $N_{\text{steps}} = N_{\text{sheds}}$

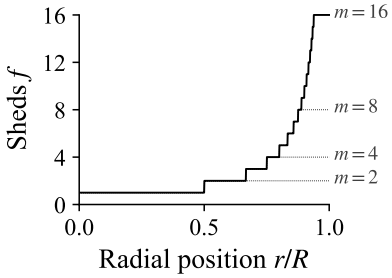


Figure 7.4: Number of particles shed along blade as a function of m.

manner.

After having decoupled the temporal and spatial resolutions, we can now explore the convergence of the rotor simulation through well-defined, independent parameters, as shown in the following sections.

7.2.2 Convergence of VPM

Before looking at the convergence of the rotor simulation, let us first verify the convergence of the vortex particle method by itself through the simulation of an inviscid vortex ring. The self-propelling vortex ring is a good VPM benchmarking case as it is unbounded, purely dominated by vorticity, and an analytic inviscid solution of its centroid velocity V exists. The analytic solution is $V = \frac{\Gamma}{4\pi R}\left(\ln\left(\frac{8R}{c}\right) - \beta\right)$, where Γ is the circulation of the ring, R is the ring's radius, c is the toroid core radius, and β is a parameter given by the vorticity distribution inside the toroid. Discretizing the ring as described by Winckelmans,[180] the algebraic kernel proposed by Winckelmans[181] leads to exactly $\beta = 0.5$. Hence, for this inviscid simulation we used Winckelmans' kernel instead of the Gaussian kernel, with $\Gamma = 1\,\mathrm{m}^2/\mathrm{s}$, $R = 2\,\mathrm{m}$, and $c = 0.002R$. This corresponds to $V = 0.310\,\mathrm{m/s}$.

Following Winckelmans' work, the vortex ring was discretized into N cross sections evenly spaced at a distance $\Delta x = 2\pi R/N$ along the toroid centerline, with one particle at the centerline and one layer of particles around the cross section, and making the smoothing radius σ equal to the toroid core radius c. The core overlap λ between contiguous particles along the centerline is defined as $\lambda \equiv \sigma/\Delta x$, resulting in

$$\lambda = \frac{\sigma}{2\pi R}N.$$

Fig. 7.5 (top) shows the centroid velocity converging to the analytic solution as the spatial resolution is increased from $N = 25$ to $N = 9425$. Fig. 7.5 (bottom) shows the numerical error in a semi-log scale as the core overlap is increased with N, evidencing that the error decreases exponentially after the threshold $\lambda > 1$. The condition $\sigma/\Delta x > 1$ is theoretically necessary to ensure convergence, and in practice constructs a smooth vorticity field. This is equivalent to requiring the spatial discretization Δx to be smaller than the filter width σ. With this practical example, it is verified that the numerical error of the VPM can be decreased down to any arbitrary precision.

180 G. S. S. Winckelmans (1989). "Topics in vortex methods for the computation of three and two dimensional incompressible unsteady flows".

181 Winckelmans et al., "Contributions to Vortex Particle Methods for the Computation of Three-Dimensional Incompressible Unsteady Flows".

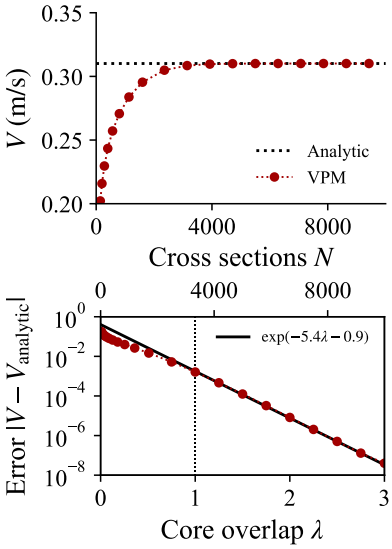


Figure 7.5: Spatial convergence of VPM in vortex ring simulation.

7.2.3 Convergence of Rotor Simulation

In order to analyze the convergence of the rotor simulation, we identified four independent discretization parameters that define the temporal resolution and spatial resolution:

- Number of time steps per revolution, N_{steps}.
- Number of particle sheds per revolution, N_{sheds}.
- Particle smoothing factor, $f_\sigma \equiv \sigma/R$.

- Number of blade elements per blade, n_{blade}.

With these parameters, N_{steps} alone determines the temporal resolution, while the spatial resolution is given by N_{sheds}, f_σ, and n_{blade}. N_{sheds} determines the discretization of vortex lines in the wake, f_σ sets the smallest length scale resolved in the wake, and n_{blade} sets the spanwise discretization of both the blade and the inner vortex sheet. To illustrate these parameters, Fig. 7.6 shows the first 12 time steps of a simulation with $N_{\text{steps}} = 90$, $N_{\text{sheds}} = 90$, $f_\sigma = 0.07$ (corresponding to $\lambda = 1.0$), and $n_{\text{blade}} = 7$.

Both f_σ and N_{sheds} determine the overlap λ between tip particles, defined as

$$\lambda \equiv \frac{\sigma}{\Delta x} = \frac{f_\sigma R}{\Delta x}.$$

In a rotor simulation, we define Δx as the spacing between particles shed from the blade tip, shown in Fig. 7.6, which is $\Delta x = 2\pi R / N_{\text{sheds}}$. Then, λ is given by

$$\lambda = \frac{f_\sigma}{2\pi} N_{\text{sheds}}.$$

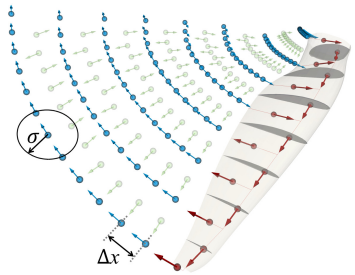


Figure 7.6: Spatial discretization of rotor simulation.

Thus, in order to maintain a constant overlap λ, the smoothing radius σ needs to decrease as the sheds per revolution N_{sheds} is increased. As previously mentioned, the stability and convergence of the VPM requires meeting the condition of sufficient core overlap, $\lambda > 1$, throughout the simulation. However, Lagrangian distortion will decrease the overlap once the wake starts to leapfrog and transition into turbulent breakdown. Our experience is that $\lambda = 2$ is the minimum overlap at the tip to achieve a stable simulation with the classic VPM up to turbulent breakdown. The reformulated VPM is needed in order to achieve numerical stability in the turbulent regime, regardless of core overlap.

The numerical error associated to each discretization parameter was characterized by varying each individual parameter from the baseline values $N_{\text{steps}} = 72$, $N_{\text{sheds}} = 144$, $\lambda = 2.125$ (or $f_\sigma = 0.093$), and $n_{\text{blade}} = 50$ in the simulation of an APC 10x7 propeller. The metric of interest was thrust coefficient C_T at an advance ratio of 0.6 (another convergence study is performed in Section 8.2.1 using the wake structure and induced velocities as metrics). Fig. 7.7 shows the convergence of C_T as N_{steps} is increased from $N_{\text{steps}} = 18$ to $N_{\text{steps}} = 360$ (corresponding to time steps of 20° and 1° of rotation, respectively) with its corresponding Richardson extrapolation, $(C_T)_{\text{extrap}}$. An approximation of the convergence order p is obtained from a curve fit of the form $(C_T)_{\text{extrap}} + a\,(\Delta t)^p$ (shown in the figure as a solid line), where $\Delta t \propto 1/N_{\text{steps}}$. This results in $p = 1.049$, indicating that the rotor simulation has first-order temporal convergence in spite of using a third-order Runge-Kutta time integration for the VPM fluid domain. This low-order convergence is believed to be due to coupling the rotor model with the VPM through a forward Euler scheme. In spite of this, the simulation shows very low numerical error associated

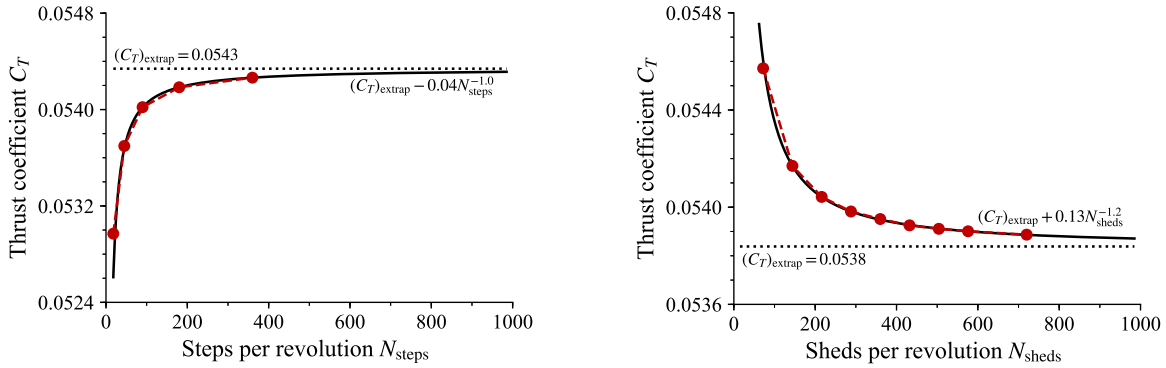


Figure 7.7: Temporal (left) and spatial (right) convergence of the rotor simulation. Dotted lines correspond to Richardson extrapolations, $(C_T)_{\text{extrap}}$.

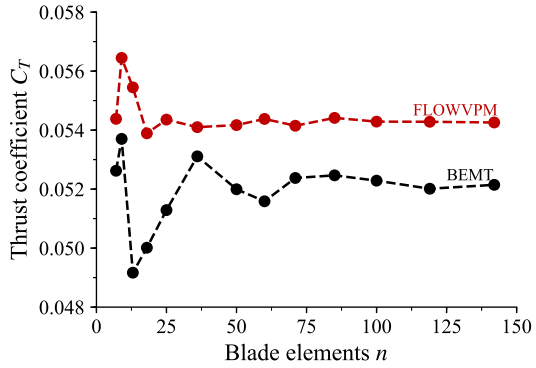


Figure 7.8: Spatial convergence of rotor simulation with blade discretization, compared to a blade-element momentum theory (BEMT) code.

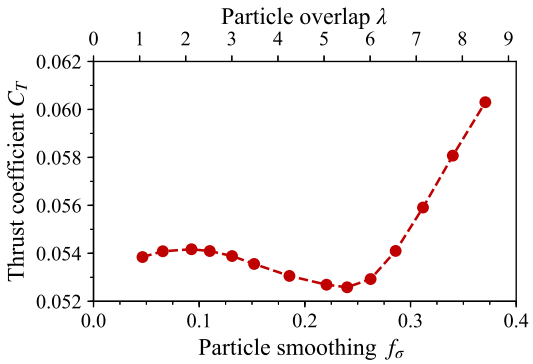


Figure 7.9: Effects of particle smoothing on C_T at a constant N_{sheds} of 144.

with temporal discretization, resulting in an error relative to $(C_T)_{\text{extrap}}$ smaller than 1% for $N_{\text{steps}} \geq 72$ (or time steps smaller than 5°).

Fig. 7.7 shows the convergence of C_T as N_{sheds} is increased from $N_{\text{sheds}} = 72$ to $N_{\text{sheds}} = 576$ (corresponding to shedding particles every 5° and 0.625° of rotation, respectively) with its corresponding Richardson extrapolation, $(C_T)_{\text{extrap}}$. A curve fit of the form $(C_T)_{\text{extrap}} + b\,(\Delta x)^p$, where $\Delta x \propto 1/N_{\text{sheds}}$, results in $p = 1.204$ indicating approximate first-order spatial convergence. For $N_{\text{sheds}} \geq 144$, the error relative to $(C_T)_{\text{extrap}}$ becomes smaller than 1%.

Fig. 7.8 shows the convergence as the number of blade elements n_{blade} is increased from 7 to 142, displaying a non-monotonic behavior. This is believed to be due to noise introduced in the spline and linear interpolation of airfoil lift and drag curves at every element along the blade. The same behavior is seen using the blade-element momentum theory (BEMT) code CCBlade [182], also shown in the figure Fig. 7.8, which supports this belief. In the VPM simulation, C_T varies by less than 0.3% with $n_{\text{blade}} \geq 25$. Thus, with $n_{\text{blade}} = 50$ and $N_{\text{sheds}} = 144$, the numerical error associated with spatial discretization is estimated to be smaller

182 Available at github.com/byuflowlab/CCBlade.jl

than 1%.

Finally, Fig. 7.9 shows the effects of particle smoothing f_σ and equivalent core overlap λ on C_T. It is observed that C_T starts to diverge as the stability threshold ($\lambda = 1$) is approached, while also diverging as $f_\sigma > 0.25$. This is because $f_\sigma > 0.25$ is equivalent to σ larger than $0.25R$, which leads to unphysical wake dynamics caused by excessive smoothing. Since the instability associated with $\lambda \to 1$ plateaus at $\lambda \approx 2$, we have chosen $\lambda = 2.125$ (or $f_\sigma = 0.093$) for all simulations in the rest of this study.

7.3 Propeller in Forward Flight

In order to validate our rotor ALM in propeller mode, we simulated the Beaver propeller originally used by Veldhuis.[183] This four-bladed propeller is 0.237 m in diameter, and, even though its design is rather outdated, it has been thoroughly tested experimentally and computationally by the Flight Performance and Propulsion research group at Delft University of Technology, producing abundant data for model validation. In this and the following section we will compare our simulation to results reported by Sinnige-van Arnhem *et al.*,[184] Sinnige-de Vries *et al.*,[185] and Sinnige-Stokkermans *et al.*[186]

The propeller was simulated across a range of advance ratio $J = {}^{V_\infty}/_{nD}$, with a diameter-based Reynolds number at 70% the blade span $\mathrm{Re}_D = {}^{0.7\pi n D^2}/_{\nu}$ of approximately 1.8×10^6, where n is the rotations per second. No collective pitch was used, resulting in a blade pitch angle of 23.9° at the radial position $r/R = 0.75$. The predicted propeller thrust, torque, and propulsive efficiently are compared in Fig. 7.10 to experimental and numerical results reported in the literature, showing satisfactory agreement across advance ratios.

183 Veldhuis, "Propeller Wing Aerodynamic Interference".

184 T. Sinnige, N. van Arnhem, et al. (2019). "Wingtip-Mounted Propellers: Aerodynamic Analysis of Interaction Effects and Comparison with Conventional Layout".

185 Sinnige, Vries, et al., "Unsteady Pylon Loading Caused by Propeller-Slipstream Impingement for Tip-Mounted Propellers".

186 T. Sinnige, T. C. Stokkermans, et al. (2019). "Aerodynamic performance of a wingtip-mounted tractor propeller configuration in windmilling and energy-harvesting conditions".

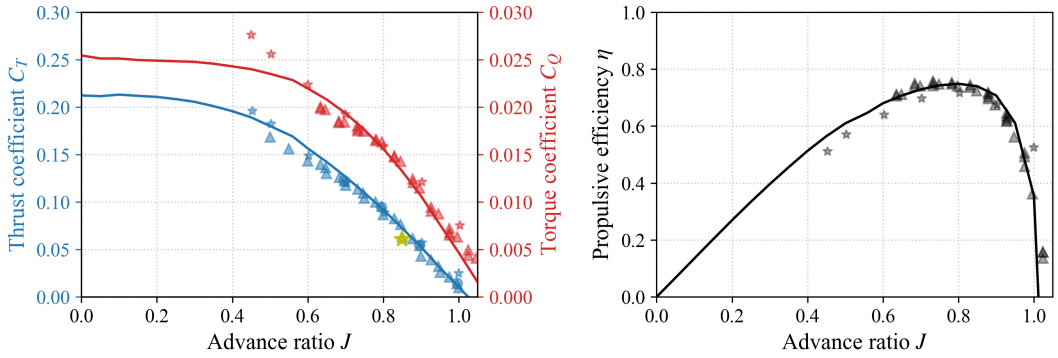


Figure 7.10: Beaver propeller in forward flight. Simulations: ━ rVPM; ★ URANS by Sinnige-Stokkermans *et al.* Experimental: △ Sinnige-van Arnhem *et al.*; ○ Sinnige-de Vries *et al.*; ⋆ Veldhuis.

7.4 Propeller at an Incidence Angle

Propellers typically operate at a mild incidence angle. This is because of the pitch of the vehicle, circulation in the vicinity of a wing, or both. In order to validate the accuracy of our simulations in these operating conditions, the isolated Beaver propeller was simulated ranging its angle of attack relative to the freestream from 0° to 20°. Fig. 7.11 compares the predicted thrust to the experimental measurements reported by Sinnige-van Arnhem *et al.* as the incidence angle is increased at a variety of advance ratios, showing reasonable agreement with the experiment.

7.5 Aeroacoustic Noise of Rotor in Hover

The ability to capture rotor aeroacoustics was validated by simulating the experiment by Zawodny *et al.*[187] on a DJI 9443 rotor in hover. The rotor was operated at 5400 RPM, resulting in a blade-passing frequency (BPF) of 180 Hz. The acoustic signature was characterized over a circular array of microphones located a distance of 7.9D from the rotor hub.

187 Zawodny et al., "Acoustic Characterization and Prediction of Representative, Small-scale Rotary-wing Unmanned Aircraft System Components".

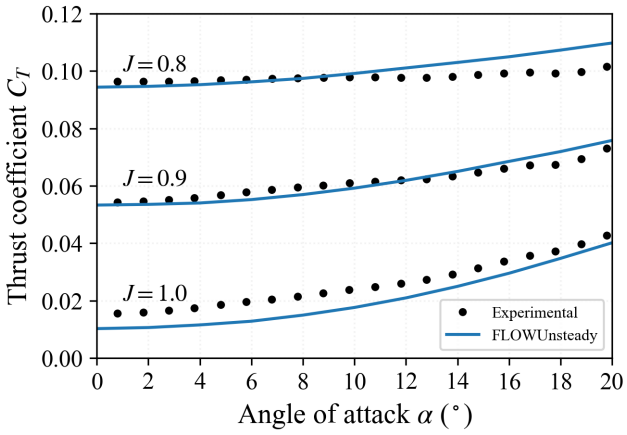


Figure 7.11: Beaver propeller at an incidence angle, compared to experimental measurements reported by Sinnige-van Arnhem *et al.*

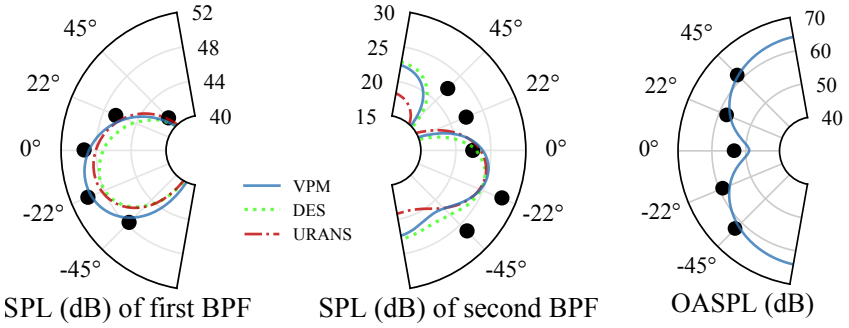


Figure 7.12: Predicted tonal directivity (left and middle) and OASPL directivity (right) of DJI 9443 rotor compared to experiment (black markers).

Zawodny *et al.* reported an experimental mean C_T of 0.072, and our simulation resulted in a mean C_T between 2% of this value, as previously shown in Fig. 5.17 in Section 5.4. Our simulation was also compared to results reported by Zawodny *et al.* using a high-fidelity detached-eddy simulation (DES) through OVERFLOW2 and a low-fidelity URANS reported by Schenk[188] using STAR-CCM+. These DES and URANS simulations were previously described in Section 5.4.

188 Schenk, "Computational Investigation of the Effects of Rotor-on-Rotor Interactions on Thrust and Noise".

Fig. 7.12 shows the directivity predicted by each simulation, compared to the experiment. The sound pressure level (SPL) of the first and second BPF is shown in left and middle figures. The first BPF is well predicted by the VPM simulation, being in all directions within 1 dB from the experiment. Note that the directivity patterns of the second BPF predicted by all simulations differ from the experiment, although the VPM is in agreement with the DES. This seems to indicate that the second BPF measured by the experiment included some other source of noise not captured in the simulations, like mechanical noise from the test stand or motor. The overall SPL (OASPL) is shown in the right, evidencing good agreement in all directions between the VPM and the experiment.

7.6 Side-by-Side Rotors in Hover

In order to validate the rotor-rotor interactions captured in our simulations, we now look at the aerodynamic interactions between two side-by-side rotors in hover. The study shown here was performed and published[189] in the initial stages of this doctoral work before the reformulated VPM was developed, hence it used the classic VPM. As explained in the journal publication, the simulation underwent significant wake treatment in order to keep the classic VPM numerically stable.

189 Alvarez and Ning, "High-Fidelity Modeling of Multirotor Aerodynamic Interactions for Aircraft Design".

The simulation replicated the experiment reported by Zhou et al.,[190] which used a two-bladed rotor resembling a DJI rotor, developed by Ning[191] and referred to as Ning DJI. The rotor has a diameter of 9.4 inches and a solidity of 0.12. The test was run at a tip Mach number of approximately 0.2, and chord and diameter-based Reynolds numbers of approximately 6×10^4 and 7×10^5, respectively. The side-by-side

190 Zhou et al., "An Experimental Investigation on Rotor-to-Rotor Interactions of Small UAV Propellers".

191 Ning, "Experimental investigations on the aerodynamic and aeroacoustic characteristics of small UAS propellers".

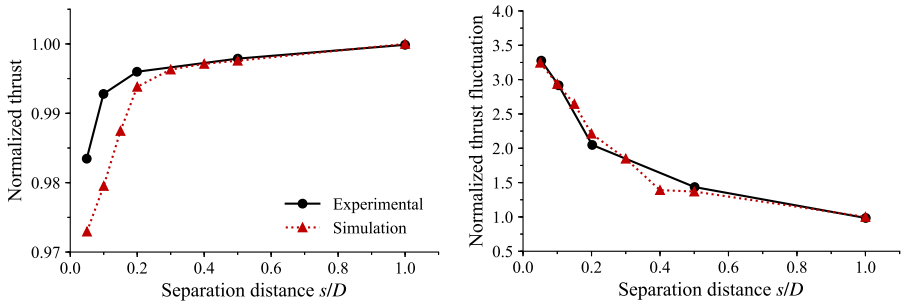


Figure 7.13: Effects of rotor-on-rotor interactions on thrust as separation between rotors is decreased, normalized by their respective values in single-rotor configuration.

configuration was in counter-rotation, meaning that one rotor rotated clockwise, meanwhile the other rotated counter-clockwise.

Every time that blades pass through the region of wake mixing in between the rotors, the loading of the blade drops, decreasing the mean thrust and increasing the fluctuation. These interactions are accentuated as the tip-to-tip separation distance s between the rotors becomes small. Fig. 7.13 compares the VPM simulation to the experimental measurements reported by Zhou *et al.* as the separation s was varied. The simulation captures both the thrust drop and the increased fluctuation, showing satisfactory agreement with the experiment.

8 Validation of Rotor-Wing Interactions

In this chapter, we incrementally validate each aspect of the interactions encountered when a rotor wake impinges on a wing. First, wing and rotors are considered in isolation. In Section 8.1, the wing loading predicted with the actuator surface model on an isolated swept-back wing (with spanwise flow) is compared to experimental measurements. Predicting accurate rotor-wing interactions hinges on accurately resolving the rotor wake. Hence, a detailed spatial resolution study characterizing the accuracy of the rotor wake is presented in Section 8.2. The predicted vortical structure and velocity in the wake is validated by comparison to experimental measurements, and also compared to conventional mesh-based CFD results reported in the literature.

Next, a wing is placed in the wake of the rotor. In Section 8.3, wing and rotor are placed in a tip-mounted configuration. The wing has a low aspect ratio and a large flap, resembling a tailplane (or horizontal stabilizer) with tip-mounted propellers. At first, the system is simulated at zero angle of attack ($\alpha = 0°$) at no elevator deflection ($\delta_e = 0°$), hence, the aerodynamic wing load is caused purely by the swirl of the wake. The wing is then aerodynamically loaded by deflecting the elevator, which causes a mild turning of the rotor wake. Stronger turning of the rotor wake is tested by pitching the propeller-wing system to an AOA of 10°. All these cases are validated by comparison to experimental measurements reported in the literature. Finally, validation on a blown wing case is presented simulating the conventional configuration of a propeller mounted mid-span on a main wing.

8.1 Isolated Wing

In order to validate the actuator surface model used for wings, a 45° swept-back wing is simulated matching the experimental conditions reported by Weber and Brebner.[192] The wing has an aspect ratio of 4.9, a span of 2.5 m, an RAE 101 airfoil section with 12% thickness, and no dihedral, twist, nor taper. This geometry is shown in Fig. 8.1. The freestream velocity V_∞ was 49.7 m/s throughout the tests, corresponding to a chord-based Reynolds number of 1.7×10^6. The high sweep of the wing causes non-negligible spanwise flow. The wing loads reported by Weber and Brebner were integrated from pressure-tap measurements, hence the drag reported in this section includes induced and form drag while excluding skin friction drag.

192 J. Weber et al. (1951). *Low-Speed Tests on 45-deg Swept-Back Wings, Part I.*

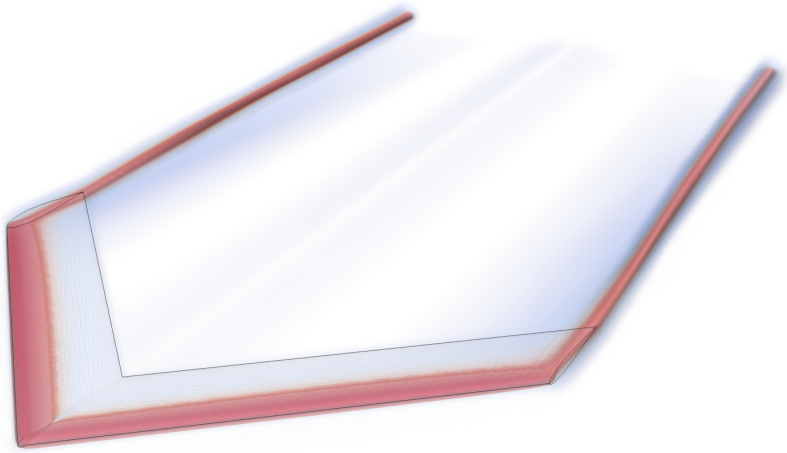


Figure 8.1: Swept-back wing simulation with $n_{\text{wing}} = 200$. Volume rendering of vorticity field.

First, spatial convergence was tested[193] by increasing the number of wing elements n_{wing} from 10 to 800. n_{wing} mostly affects how well the wake is resolved as it rolls around the tip vortices. Fig. 8.2 shows the convergence of integral quantities (lift and drag coefficients) with the wing at an angle of attack (AOA) of $\alpha = 4.2°$, resulting in both C_L and C_D changing by less than 1% with $n_{\text{wing}} \geq 100$. The respective load distributions are compared to the sectional loading reported by Weber and Brebner in Fig. 8.3, showing good agreement with the experiment even with the coarser resolutions. The drag distribution shows some variation towards the wing tip, but the trend suggests convergence to the experimental tip loading as the spatial refinement is increased. We conclude that n_{wing} between 100 and 200 should provide sufficient accuracy for most applications. From here on, $n_{\text{wing}} = 200$ is used to discretize all wings in this study.

193 Temporal and spatial discretization was further tested by decreasing the time step size Δt, the streamwise distance between shed particles Δx, and chordwise discretization of the ASM, finding little sensitivity to these parameters. Hence, these sensitivities are not hereby reported.

Fig. 8.4 shows the history of C_L as the wake is deployed in the simulation with $n_{\text{wing}} = 200$. The time has been normalized by the equivalent number of spans lengths resolved in the wake. Here we conclude that resolving the wake for about 1.5 span-distances is sufficient to obtain a

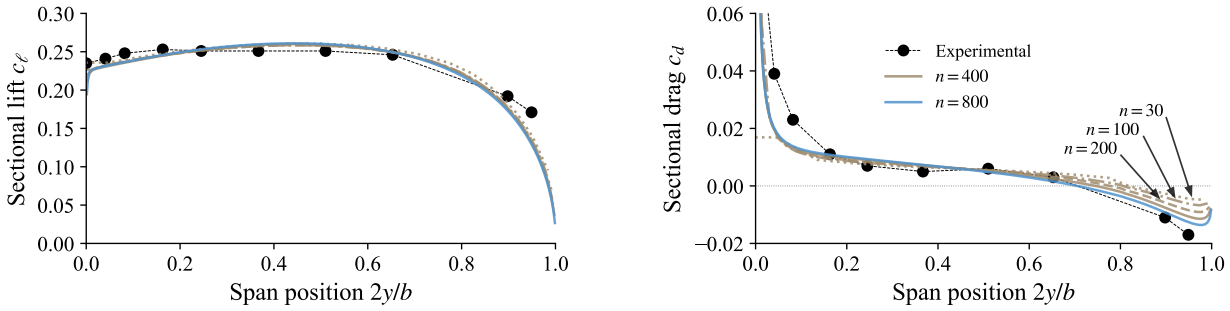


Figure 8.3: Lift and drag distribution of swept-back wing at $\alpha = 4.2°$ in rVPM simulation as spatial resolution is increased. Sectional coefficients defined as $c_\ell = \frac{\ell}{\frac{1}{2}\rho U_\infty^2 c}$ and $c_d = \frac{d}{\frac{1}{2}\rho U_\infty^2 c}$.

converged wing simulation. The predicted C_L converges to a value of 0.329 while Weber and Brebner reported a mean value of 0.328, leading to a prediction within 2% of the experiment.

Next, the case was repeated for multiple angles of attack and compared to the experiment, as seen in Fig. 8.5. The integrated lift and drag (top) show excellent agreement with the experiment from 0° to 10.5°. We expect this to be the case only for mild AOAs before approaching stall conditions as our ASM does not capture the mechanisms of flow separation. The loading distribution (bottom) also shows good agreement with the experiment across AOA. Thus, through this swept-wing case, we gain confidence that our ASM yields accurate predictions in conditions with spanwise flow as AOA is increased.

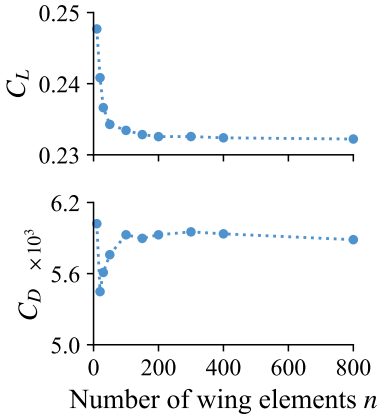


Figure 8.2: Spatial convergence of wing simulation at $\alpha = 4.2°$.

8.2 Propeller Wake

Predicting accurate rotor-wing interactions hinges on accurately resolving the rotor wake; hence, we turn our attention to the wake dynamics of the isolated propeller as predicted by our meshless LES. This test case uses the Beaver propeller (previously used in Sections 7.3 and 7.4) at an advance ratio J of 0.8 and freestream velocity V_∞ of 40 m/s at no incidence angle. This corresponds to a tip Mach number of 0.46 and a diameter-based Reynolds number of 1.8×10^6 at 70% the blade span. No collective pitch is used, resulting in a blade pitch angle of 23.9° at the radial position $r/R = 0.75$.

First, a convergence study is presented in Section 8.2.1, looking at

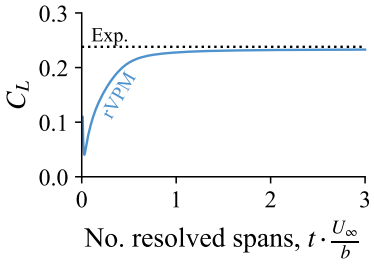


Figure 8.4: History of C_L in wing simulation at $\alpha = 4.2°$ as the wake is deployed.

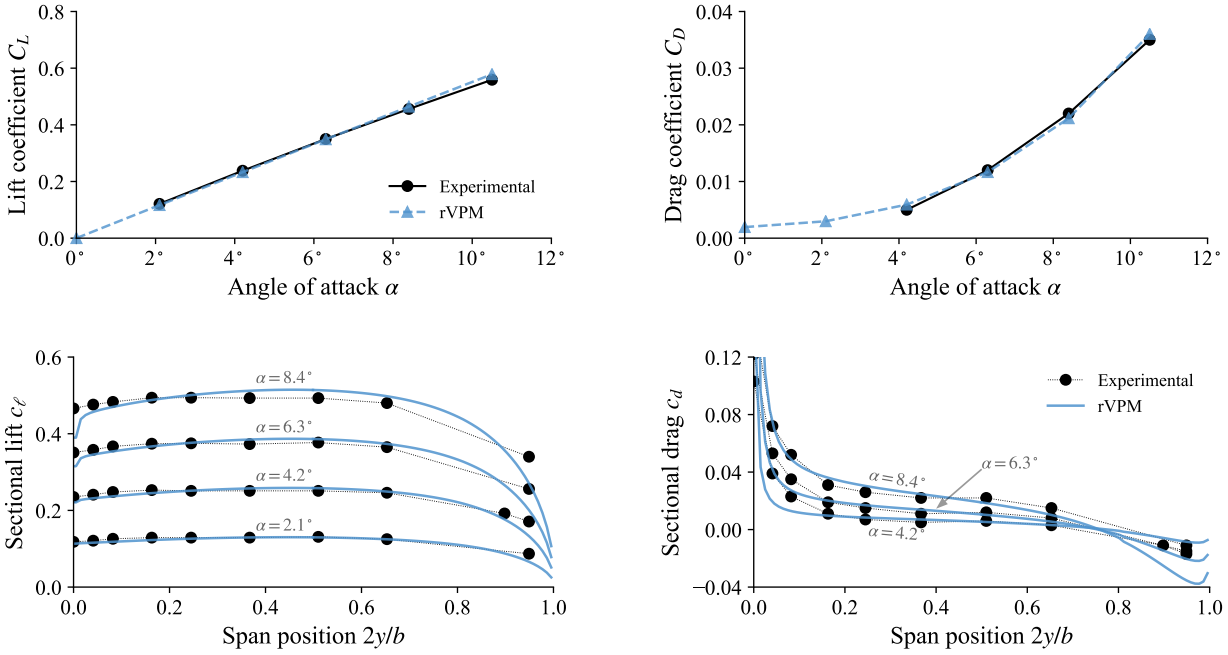


Figure 8.5: Integrated lift and drag (top), and spanwise distributions (bottom) in swept-back wing as angle of attack is increased.

the wake structure as spatial refinement is increased. This will help us determine the spatial resolution needed to accurately resolve the wake. Then, the wake structure is compared to experimental particle image velocimetry (PIV) reported by Sinnige & de Vries *et al.*[194] The predicted velocity profiles along the wake are then validated in Section 8.2.2 by comparison to experimental and mesh-based CFD results reported in the literature.

All these results will build our confidence that the propeller wake that will later be impinging on the wing is accurate and well resolved. Hence, any inconsistencies later encountered in the predicted rotor-wing interactions can be narrowed down to possible deficiencies of the wing's actuator surface model rather than the propeller wake.

194 Sinnige, Vries, et al., "Unsteady Pylon Loading Caused by Propeller-Slipstream Impingement for Tip-Mounted Propellers".

8.2.1 Spatial Resolution Study

The effects of spatial discretization were studied by varing the number of sheds per revolution N_{sheds} from 144 to 1440 (corresponding to shedding particles every 2.5° and 0.25° of rotation, respectively). Also, the number of blade elements n_{blade} was varied from 50 to 200. The initial core overlap $\lambda \equiv \frac{\Delta x}{\sigma} = \frac{2\pi R}{\sigma N_{\text{sheds}}}$ was kept at 2.125, which leads to decreasing the smoothing radius σ as N_{sheds} increases, better resolving tip vortices.

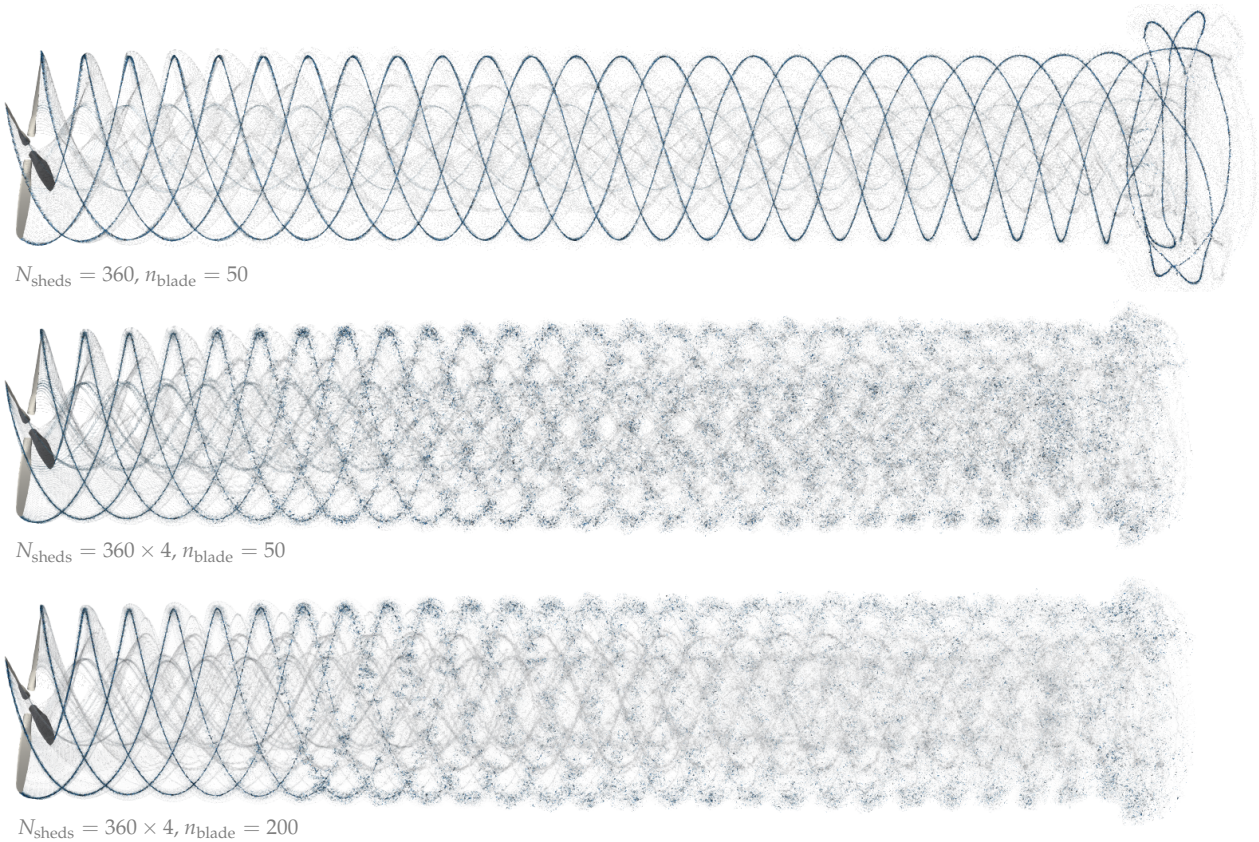


Figure 8.6: Computational elements (vortex particles) after 7 revolutions as spatial resolution is increased.

Discretization			**Mean C_T**	**Mean C_Q**
N_{sheds}	n_{blade}	Particles		
144	50	0.3M	0.0969	0.0166
360	50	0.6M	0.0939	0.0162
360×2	50	1.2M	0.0929	0.0160
360×4	50	2.3M	0.0942	0.0162
360×4	100	4.8M	0.0936	0.0161
360×4	200	9.5M	0.0935	0.0161
Experimental (Sinnige & van Arnhem *et al.*)			0.0953	0.0163

Table 8.1: Thrust and torque of isolated Beaver propeller as spatial resolution is increased. Number of particles after 8 revolutions.

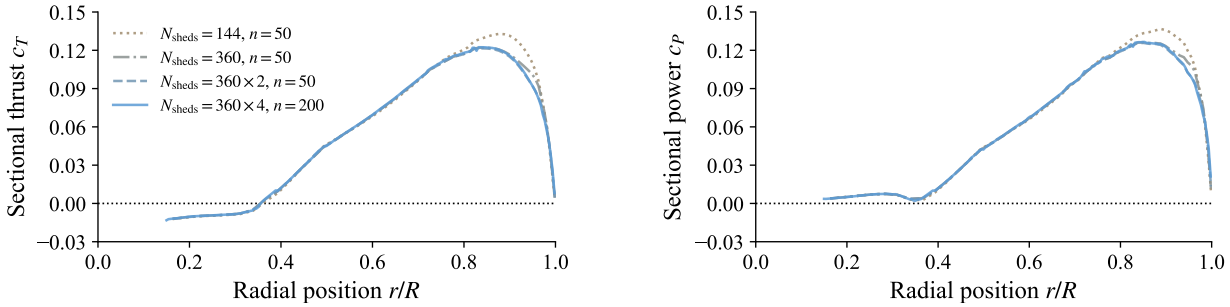


Figure 8.7: Time-average thrust and power distribution of isolated Beaver propeller in rVPM simulation as spatial resolution is increased. Sectional coefficients defined as $c_T = \frac{T'}{\rho n^2 D^3}$ and $c_P = \frac{P'}{\rho n^3 D^4}$.

Increasing n_{blade} has the effect of better resolving the inner vorticity sheet that is shed along the blade since particles are released from each blade element. The time steps per revolution N_{steps} was kept at 72 in all cases (equivalent to steps of 5°). This is a rather coarse temporal resolution, but the results presented here will show that $N_{\text{steps}} = 72$ is sufficient to fully resolve the wake in the vicinity of the plane of rotation. Further refining the temporal discretization had no effects on the results hereby reported.

The spatial refinement is visualized in Fig. 8.6, using up to 300k particles in the coarsest case after eight revolutions and up to 9.5M in the finest case. The simulations were run for eight revolutions and averaged quantities were calculated over the last three. Table 8.1 shows the change in mean thrust (C_T) and mean torque (C_Q) as the spatial resolution is increased[195]. Except for the coarsest simulation, both C_T and C_Q vary by less than 1% with spatial refinement, staying within 2% of the experimental measurements reported[196] by Sinnige & de Vries *et al.* Fig. 8.7 shows the thrust and power distribution along the blade averaged over time as the spatial resolution is increased, showing little variation with $N_{\text{sheds}} \geq 360$. Hence, we conclude that $N_{\text{sheds}} = 360$ and $n_{\text{blade}} = 50$ is sufficient to accurately resolve the performance of the isolated rotor, which is consistent with the convergence study presented in Section 7.2.

Even though the spatial resolution achieved with $N_{\text{sheds}} = 360$ and

195 Note that C_T and C_Q do not show monotonic convergence as N_{sheds} is increased. This is because imposing a constant core overlap $\lambda = \frac{2\pi R}{\sigma N_{\text{sheds}}}$ of 2.125 forces σ to change along with N_{sheds}, and in Section 7.2 we showed that these two variables need to be independent to achieve monotonic convergence.

196 Even though Sinnige & de Vries *et al.* reported an experimental C_T of 0.0953, Sinnige & van Arnhem *et al.* report that those measurements were performed at a small angle of attack of 0.2° due to an unintentional misalignment of the experimental setup. Van Arnhem later reported in his doctoral thesis a C_T of 0.0936 when the setup was correctly aligned (N. van Arnhem (2022). "Unconventional Propeller-Airframe Integration for Transport Aircraft Configurations").

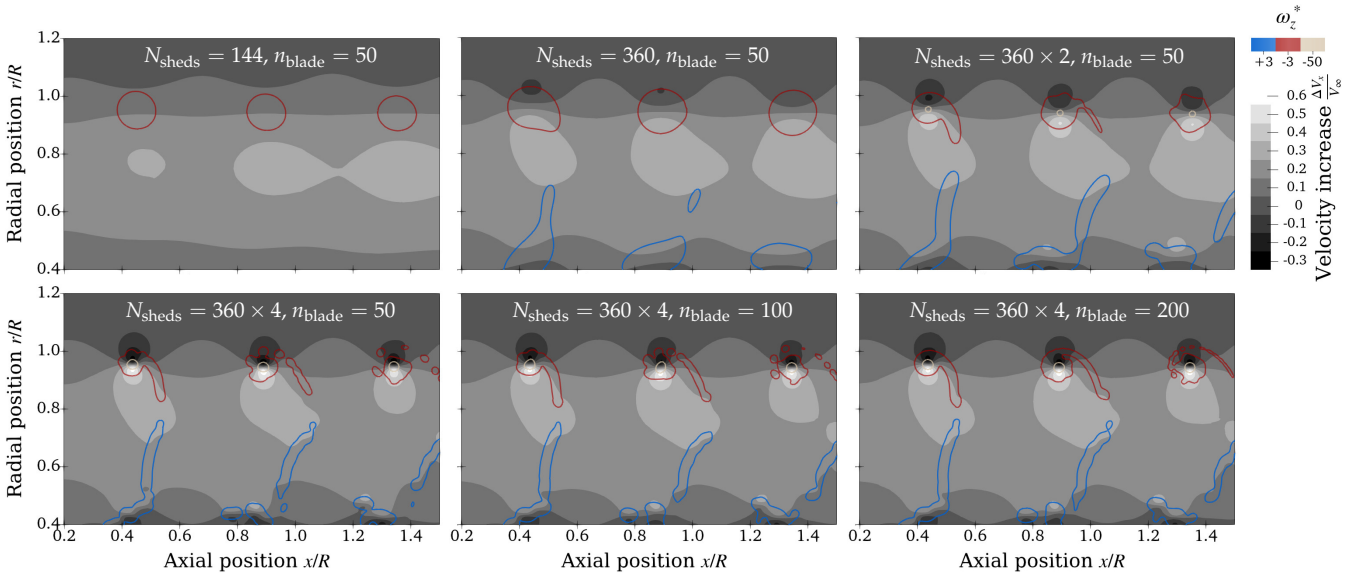


Figure 8.8: Ensemble-average flow field downstream of plane of rotation as spatial discretization is increased. Contour levels show the second, third, and fourth tip vortices and the inner vortex sheet. Ensembled as blades intersect the plane.

$n_{\text{blade}} = 50$ is enough to resolve C_T, C_Q, and blade loading, these quantities only depend on the flow field at the plane of rotation and the same discretization might not be enough to resolve the flow field in the downstream wake. Fig. 8.8 shows a slice of the ensemble-average flow field immediately downstream of the plane of rotation encompassing the second, third, and fourth tip vortices. The grayscale colormap visualizes the axial velocity induced by the wake, defined as $\Delta V_x = V_x - V_\infty$, while three contour levels of vorticity help visualize the tip vortices and the inner vortex sheet. The nondimensional vorticity ω_z^* is defined as $\omega_z^* = \omega_z D / V_{\text{disk}}$, where ω_z is the out-of-plane vorticity, D is the rotor diameter, and V_{disk} is the equivalent actuator-disk velocity $2V_{\text{disk}} = V_\infty + \sqrt{V_\infty^2 + 8T/\rho\pi D^2}$. The vortical structure is hardly recognizable with $N_{\text{sheds}} = 360$ (Fig. 8.8, top center), indicating that the smoothing radius σ (or filter width) is too large to resolve the fine features of the wake. As N_{sheds} is increased from 360 to 360×4 (bottom left), tip vortices become clearly defined and the vortical structure is evident. As n_{blade} is increased from 50 to 200 (bottom right), the inner vortex sheet becomes better resolved, capturing a smoother folding of the sheet around the third and fourth tip vortices.

Fig. 8.9 shows the axial velocity probed across the second and fourth tip vortices compared to the experimental measurements reported by Stokkermans & van Arnhem *et al.* The spikes at the tip vortices ($r/R \approx 1$) are better resolved as N_{sheds} is increased from 144 to 360×4, independent of n_{blade}. Though slightly shifted inboard, the predicted spikes converge to the experimental data, indicating that the tip vortices are fully resolved with $N_{\text{sheds}} = 360 \times 4$. Independent of the spatial resolution, the velocity profile in the inner sheet ($r/R \leq 0.8$) is under-predicted and shifted

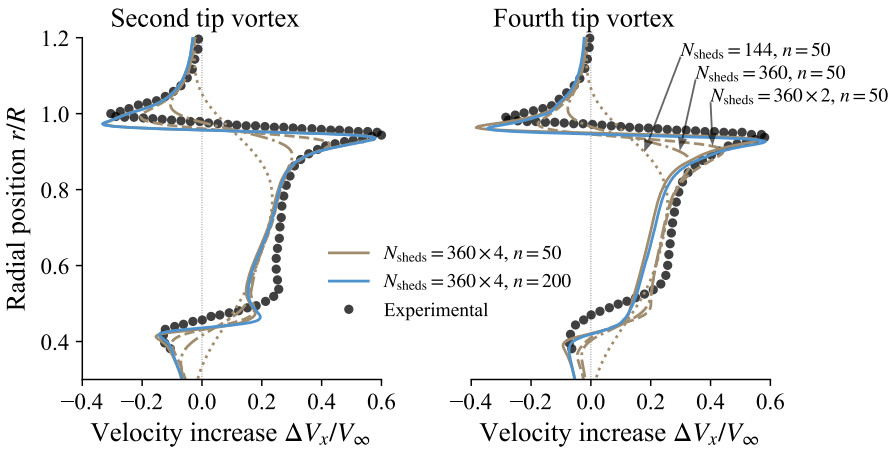


Figure 8.9: Ensemble-average induced axial velocity probed across tip vortices as spatial refinement is increased. Experimental measurements reported by Stokkermans & van Arnhem *et al.*

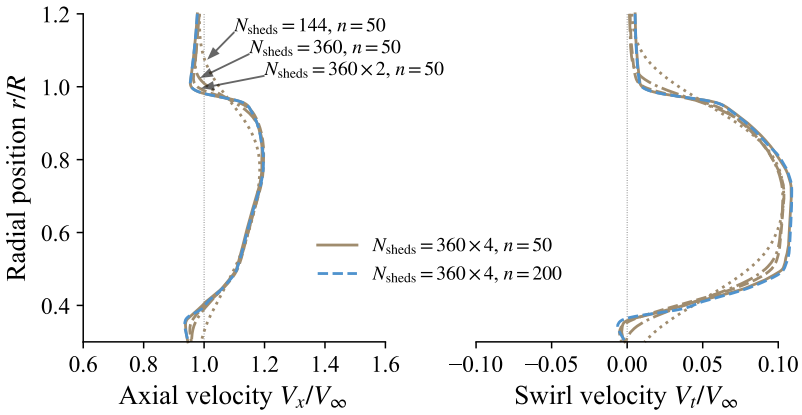


Figure 8.10: Time-average velocity profile at $x/R = 0.19$ as spatial refinement is increased.

inboard relative to the experiment. These discrepancies may be due to excluding both spinner and mounting pod from our simulations, while the experiment used a mounting pod that blocks the flow from centerline to $r/R = 0.3$. This significant blockage could both cause the wake to expand and the axial flow to speed up in the experiment. In spite of these discrepancies, and in light of the good agreement within tip vortices, we conclude that $N_{\text{sheds}} = 360 \times 4$ provides enough resolution to fully resolve the instantaneous wake velocity.

Even though we have shown that $N_{\text{sheds}} = 360 \times 4$ is necessary to fully resolve the wake structure and instantaneous velocity, one is often interested in time-average quantities rather than instantaneous ones, which might be accurately predicted with a coarser resolution. This is observed in Fig. 8.10, where time-average axial and swirl velocities are reasonably well resolved even in the coarsest simulation

using $N_{\text{sheds}} = 144$. The coarsening of the rVPM spatial resolution directly leads to using a wider LES filter, which corresponds to spatially averaging (or smearing) the instantaneous fields. In many turbulent phenomena, spatial-average and time-average processes are equivalent, which explains why in some scenarios time-average quantities are still accurate in rVPM with a coarse discretization.

8.2.2 Comparison to Experiment and Mesh-Based CFD

Our LES simulation of the Beaver propeller using $N_{\text{sheds}} = 360 \times 4$ and $n_{\text{blade}} = 200$ is shown in Fig. 8.11. The volume rendering shows the simulation capturing the fine vortical structure of the wake and the development of turbulence as the wake evolves. The flow field is shown in Fig. 8.12, with (top) a volume rendering of the instantaneous vorticity and (middle) a slice of the ensemble-average in-plane vorticity component taken as blades intersect the plane in between revolutions 4.5 and 6.5. In between the plane of rotation and $x/R = 3$, the ensemble average shows the inner vortex sheet stretching and folding around tip vortices. At $x/R > 3$, the inner sheet approaches the preceding tip vortex, causing it to deform and develop turbulence that eventually breaks the vortex down. The time-average axial velocity and streamlines are shown in Fig. 8.12 (bottom), where the streamtube is seen to contract between the plane of rotation and $x/R \approx 1$, after which it slowly expands as turbulence starts to develop.

In order to validate the flow field predicted in the wake of the propeller, we now compare our meshless LES to the experimental measurements reported by Sinnige & de Vries *et al.* and Stokkermans & van

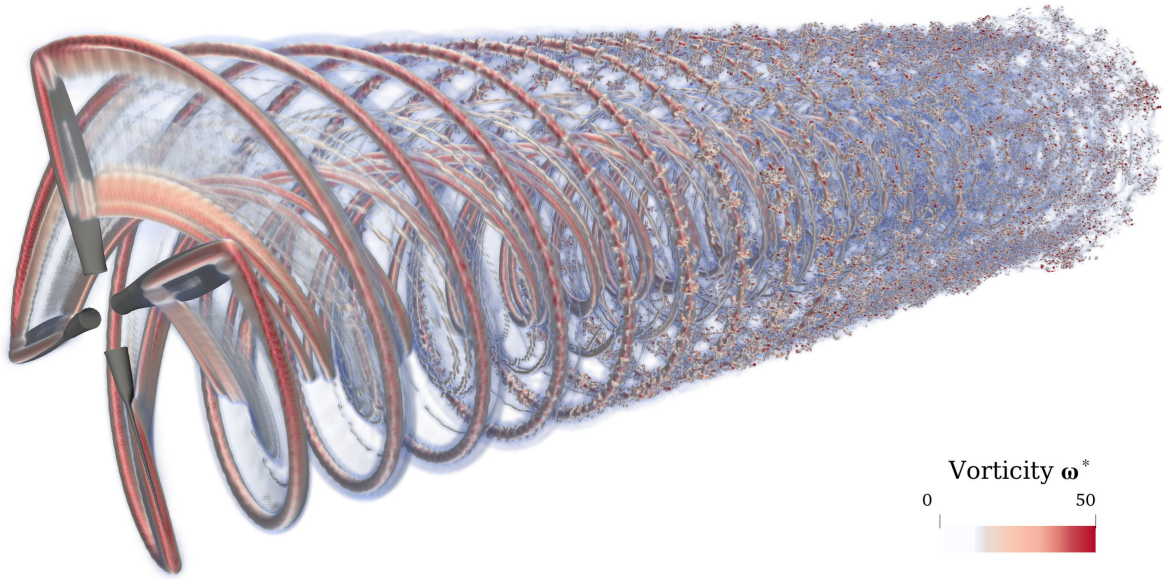


Figure 8.11: Beaver propeller simulation after five revolutions. Volume rendering of vorticity field.

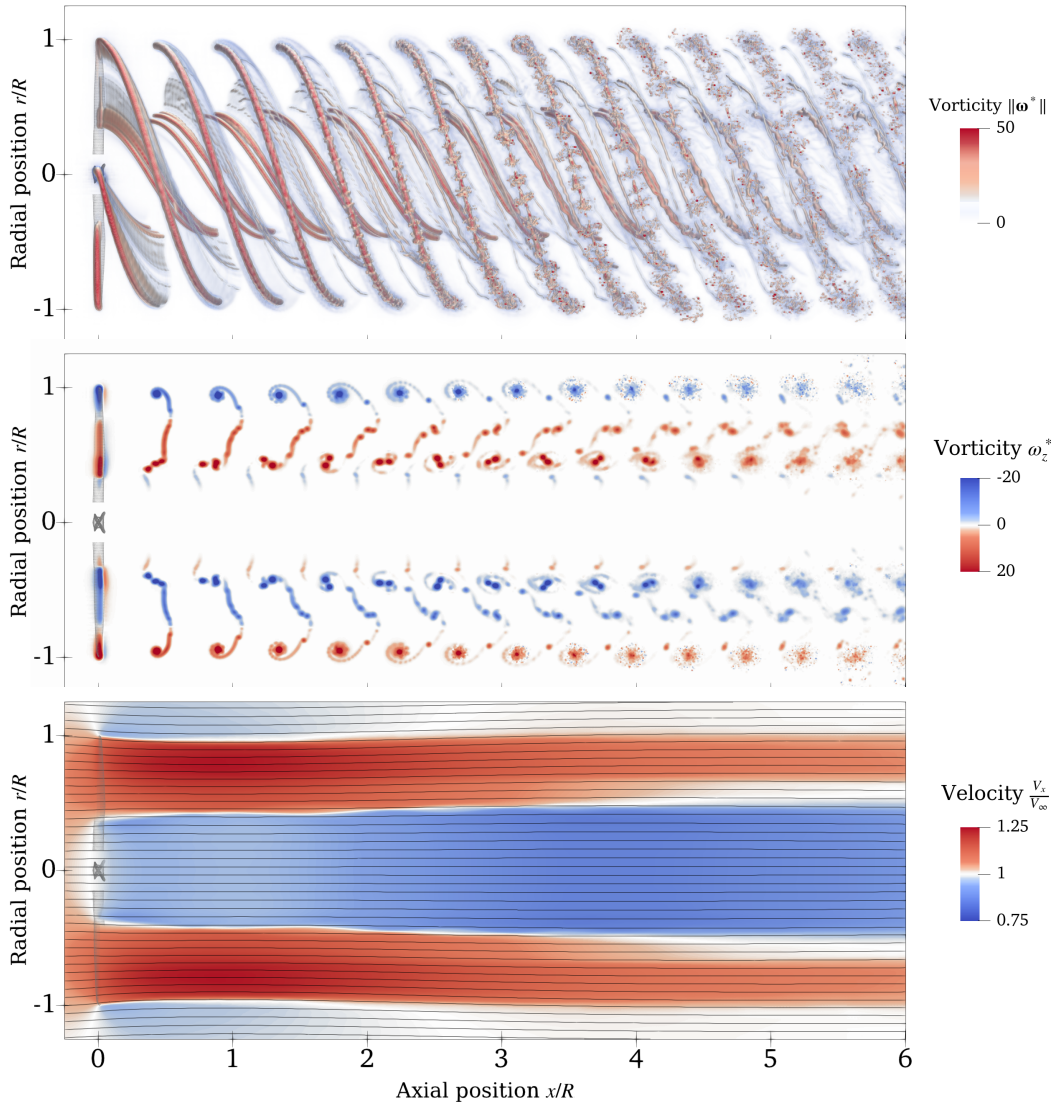


Figure 8.12: Flow field in Beaver propeller simulation: (top) instantaneous volume rendering of vorticity field after 5.5 revolutions, (middle) slice of ensemble-average in-plane vorticity, and (bottom) time-average axial velocity and streamlines.

Arnhem *et al.* Our results are also compared to the URANS simulation reported by both Sinnige & Stokkermans *et al.* and Stokkermans & van Arnhem *et al.*, and a detached-eddy simulation (DES) reported by Chu et al.[197] The different CFD solvers are summarized in Table 8.2. The URANS simulation used a compressible finite-volume solver through the commercial software ANSYS Fluent. The solver used a second-order time integration scheme, a second-order spatial scheme, the Spalart-Allmaras turbulence model resolving the blade down to a y^+ value of less than one, and periodic boundary conditions on a 90° wedge domain. The DES reported by Chu et al. resolved the mounting pod surface with the Spalart-Allmaras improved delayed DES (IDDES) turbulence model, while using an actuator surface model for the blades. It used

197 S. Chu et al. (2021). "Aerodynamic Analysis of Multi-Propeller/Wing Interaction using the Actuator Surface Model".

Table 8.2: CFD solvers compared in Beaver popeller validation.

Simulation	Software	Blade Scheme	Turbulence Model	Computational Elements
rVPM (meshless LES)	FLOWUnsteady	Actuator line model	Anisotropic dynamic SFS	9.5M vortex elements
URANS	ANSYS Fluent	Blade-resolved	Spalart-Allmaras	7.6M grid cells (Wedge-periodic domain)
DES	OpenFOAM	Actuator surface model	Spalart-Allmaras IDDES	21M grid cells

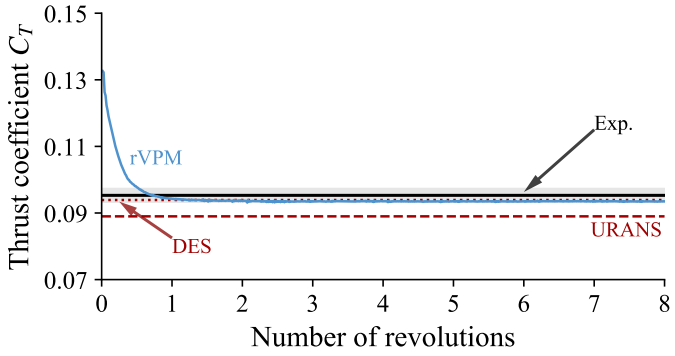


Figure 8.13: Thrust history of Beaver propeller simulation using rVPM, compared to experimental, URANS, and DES mean C_T. Shaded region encompasses the 95%-confidence interval of the experiment.

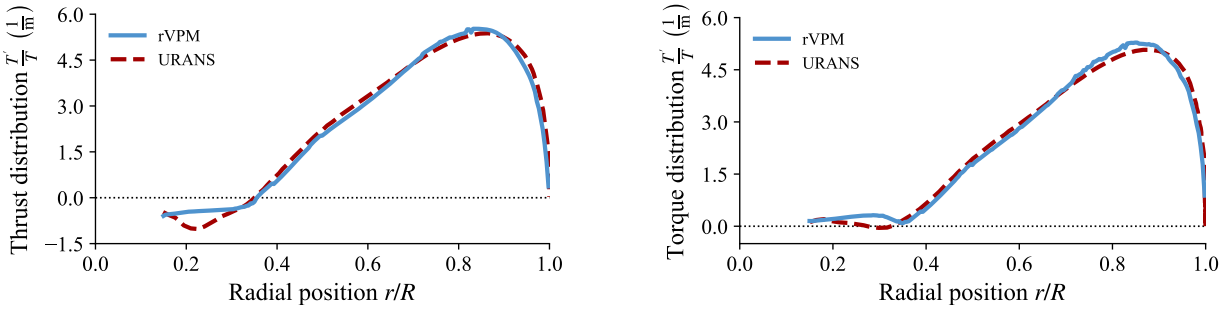


Figure 8.14: Time-average thrust and power distribution of Beaver propeller in rVPM simulation with actuator line model, compared to blade-resolved URANS results reported by Stokkermans & van Arnhem *et al.*

second-order-accurate spatial and temporal schemes implemented in an OpenFOAM incompressible solver.

Fig. 8.13 shows the thrust history of our meshless LES converging to a mean value of 0.935, which is within 1.6% of the experimental mean C_T of 0.0953 reported by Sinnige & de Vries *et al.* and within the reported experimental uncertainty. Fig. 8.13 also shows the mean C_T of the DES and URANS simulations, reported to be respectively within 1.4% and 6.6% of the experimental measurement. Fig. 8.14 shows the time-average thrust and torque distributions in our meshless LES captured with the actuator line model, compared to the blade-resolved URANS simulation reported by Stokkermans & van Arnhem *et al.* Both approaches show good agreement away from the spinner. A slight discrepancy is observed towards the spinner in the region $r/R < 0.35$, which is likely caused by both the mounting pod and the cylindrical section near the root. The mounting pod, which is included in the URANS while ignored in the

rVPM, blocks the flow from the centerline up to $r/R = 0.3$, while the cylindrical root section does not transition into a streamlined shape until about $r/R \approx 0.35$, leading to separated flow over this entire section. The good agreement over the rest of the blade (which is responsible for most of the rotor performance) and the good agreement with the experimental mean C_T confirm that our meshless LES accurately resolves the loading and performance of the propeller.

Fig. 8.15 shows a slice of the ensemble-average flow field downstream of the plane of rotation as predicted with our meshless LES, compared to the experimental PIV reported by Sinnige & de Vries *et al.* Three contour levels of vorticity help visualize the tip vortices and the inner vortex sheet. The position of tip vortices and the folding of the inner sheet predicted by rVPM show good qualitative agreement with the experiment, confirming that our meshless LES accurately resolve these wake dynamics. Fig. 8.16 shows a wider view of the flow field, comparing the rVPM simulation to the blade-resolved URANS simulation and experimental PIV, both reported by Stokkermans & van Arnhem *et al.* Both URANS and experiment show a slow velocity in the proximity of the mounting pod, while the rVPM predicts that the flow approaches the freestream velocity towards the centerline since both spinner and mounting pod are neglected. Fig. 8.17 shows the axial velocity along the second and fourth

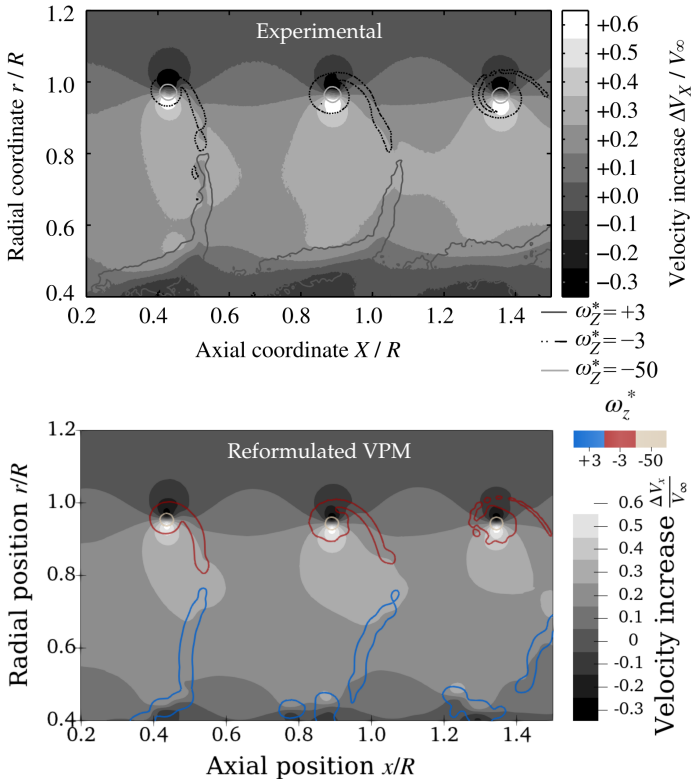


Figure 8.15: Ensemble-average flow field downstream of the plane of rotation, (top) measured experimentally and (bottom) predicted with our meshless LES. Experimental PIV (top) retrieved from Sinnige & de Vries *et al.*

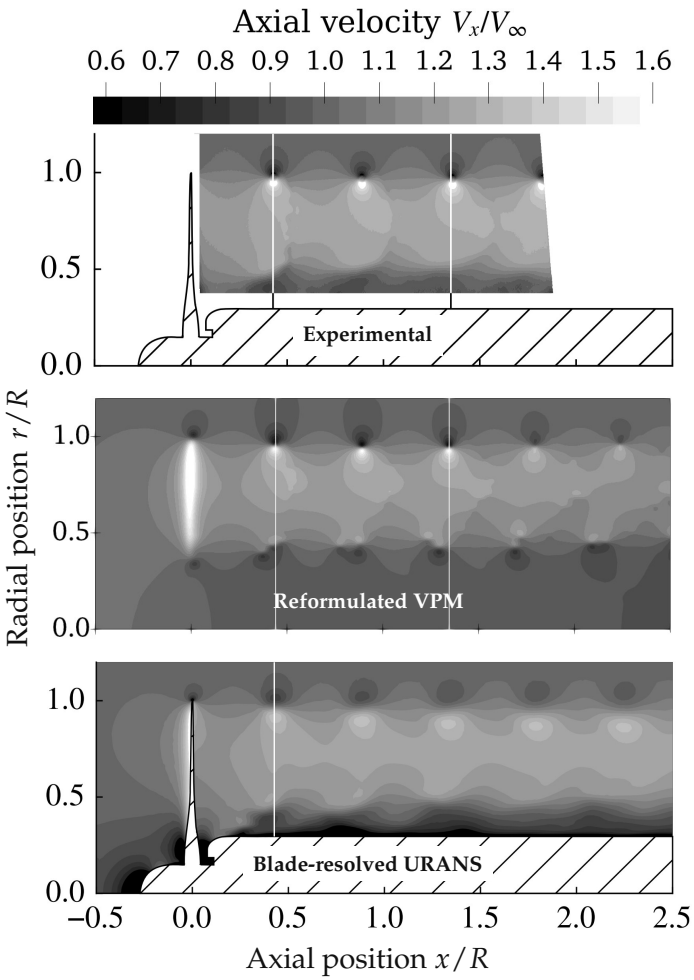


Figure 8.16: Ensemble-average flow field as measured experimentally, compared to rVPM and URANS simulations. Vertical white lines show slices plotted in Fig. 8.17. Experimental and URANS figures retrieved from Stokkermans & van Arnhem *et al.*

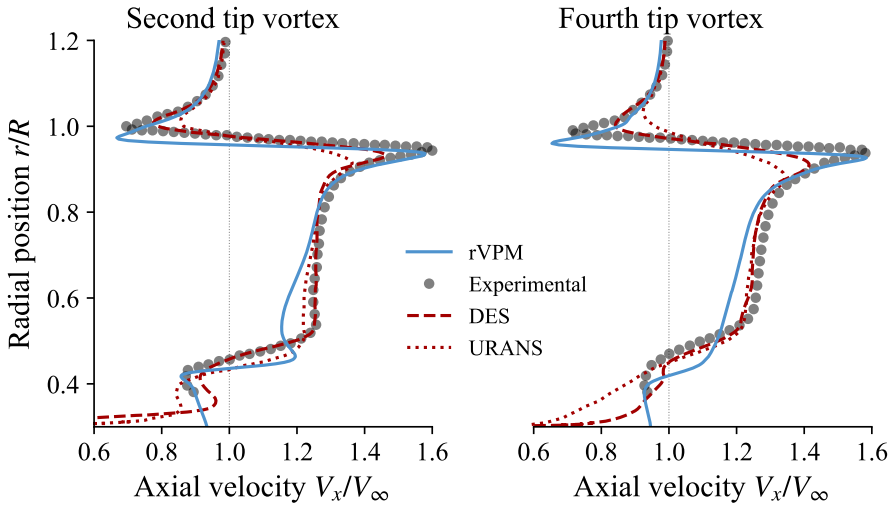


Figure 8.17: Ensemble-average axial velocity probed across tip vortices as predicted by our meshless LES, compared to mesh-based CFD simulations and experiment.

tip vortices. As discussed in Section 8.2.1, the rVPM underpredicts the

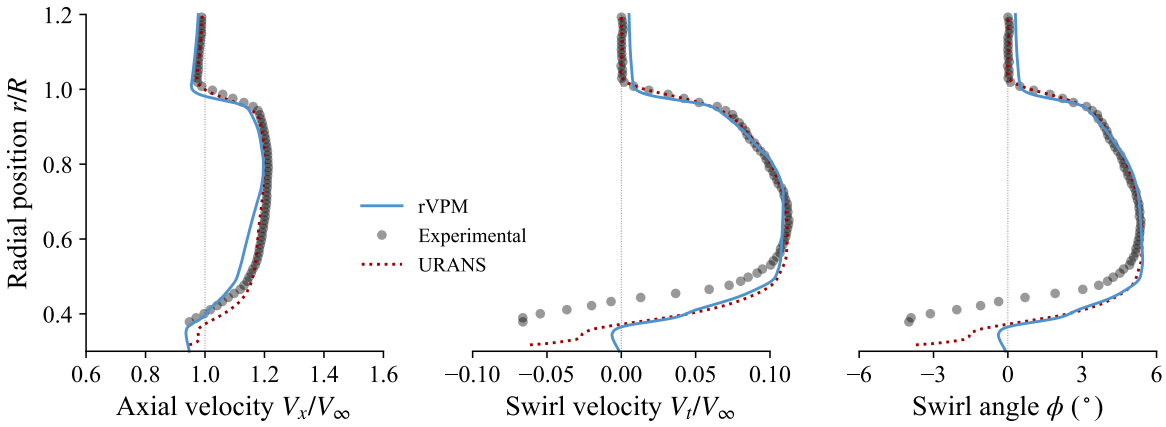


Figure 8.18: Time-average velocity profile at $r/R = 0.19$ as predicted by our meshless LES, compared to URANS and experiment.

velocity in the inboard section $r/R < 0.8$ and tip vortices are slightly shifted inboard, both effects caused by omitting the flow blockage of the mounting pod. The rVPM, however, resolves the tip vortices with remarkable accuracy as evidenced by the velocity peaks near $r/R \approx 1$, outperforming both URANS and DES. As noted by Stokkermans & van Arnhem *et al.*, the numerical dissipation associated with mesh-based CFD makes it computationally unfeasible to fully resolve the tip vortices. On the other hand, the low-numerical dissipation of our meshless LES makes it possible to preserve and resolve the vortical structure with minimal computational effort.

Fig. 8.18 shows the time-average velocity profile close to the plane of rotation as predicted with our meshless LES, compared to the experimental and blade-resolved URANS results reported by van Arnhem. Aside from the streamtube edge shifted inboard and the axial velocity underpredicted for $r/R < 0.8$ (both effects caused by omitting the flow blockage of the mounting pod), Fig. 8.18 shows reasonable agreement between rVPM and both experiment and URANS.

All these results build our confidence that the propeller wake that will later be impinging on the wing is accurate and well resolved. Hence, any inconsistencies later encountered in the predicted rotor-wing interactions can be narrowed down to possible deficiencies of the wing's actuator surface model rather than the propeller wake.

8.3 Tip-Mounted Propeller

We will now look at the interactions in a tip-mounted configuration, simulating the experiment performed by van Arnhem *et al.* known as *PROWIM-HTP*, shown in Fig. 8.19. Van Arnhem & Sinnige *et al.*[198] conducted the study of rotor-wing interactions on a tip-mounted propeller configuration using the Beaver propeller, which was further expounded in van Arnhem's doctoral thesis.[199] This dataset was later used to val-

198 N. van Arnhem, T. Sinnige, et al. (2018). "Aerodynamic Interaction Effects of Tip-Mounted Propellers Installed on the Horizontal Tailplane".

199 Arnhem, "Unconventional Propeller-Airframe Integration for Transport Aircraft Configurations".

idate a URANS study of rotor-wing interactions by Stokkermans-van Arnhem *et al.*,[200] further expounded in Stokkermans' doctoral thesis.[201] The configuration used a straight wing with low aspect ratio ($b/c = 2.7$), symmetric NACA 64_2-A015 profile, and a 25%-chord flap spanning 62% of the semi-span. This geometry resembles a tailplane (or horizontal stabilizer) with tip-mounted propellers.

The wing has a span b of 0.654 m, while the diameter D of the Beaver propeller is 0.237 m. Propeller and wing share the same angle of attack. Each test uses a freestream velocity V_∞ of 40 m/s, advance ratio J of 0.8, and inboard-up propeller rotation, unless otherwise indicated. This corresponds to a diameter-based Reynolds number of 1.8×10^6 at 70% of the blade span and a tip Mach number of 0.46 for the rotor, and a chord-based Reynolds number of 0.7×10^6 for the wing. The flow over the wing was tripped close to the leading edge in the experiment. No collective pitch is used, which, as shown in Section 8.2, leads to a thrust coefficient $C_T = {}^{T}/_{\rho n^2 D^4}$ of about 0.094 in our simulations, while van Arnhem reported an experimental C_T of 0.0936.

Rotor-wing interactions were introduced with incremental complexity. First, the prop-wing system was tested at zero angle of attack ($\alpha = 0°$) and no elevator deflection ($\delta_e = 0°$), with the wing simply acting as a flat plate. Hence, the aerodynamic wing load was caused purely by the swirl of the wake. The predicted wing loading is shown in Fig. 8.20, showing reasonable agreement with the experiment. Since this load is caused

200 Stokkermans, Arnhem, et al., "Validation and Comparison of RANS Propeller Modeling Methods for Tip-Mounted Applications".

201 T. C. A. S. Stokkermans (2020). "Aerodynamics of Propellers in Interaction Dominated Flowfields".

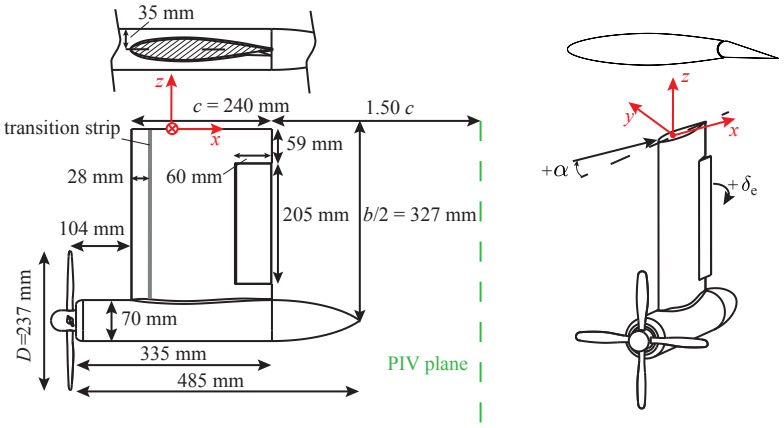


Figure 8.19: Tailplane with tip-mounted propeller case, or PROWIM-HTP. Diagram of the experiment retrieved from van Arnhem & Sinnige *et al.*

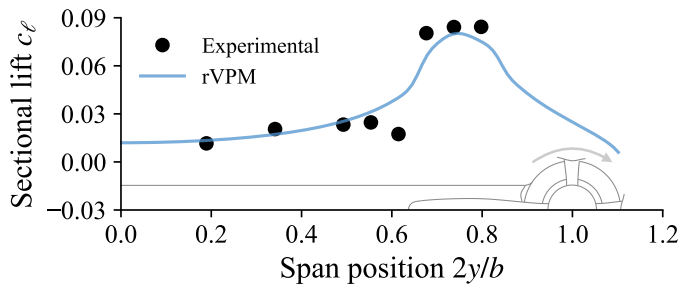


Figure 8.20: Time-average load distribution in tip-mounted case of propeller blowing on flat wing ($\alpha = 0°$, $\delta_e = 0°$, and inboard-up rotation direction).

entirely by the wake swirl, these favorable results give us confidence that both the circulation solver and force calculation in the ASM are physically accurate beyond the simple case of a uniform freestream.

Next, the wing was aerodynamically loaded by deflecting the elevator[202] by $\delta_e = +10°$. Fig. 8.21 shows the history of the lift generated by the wing with and without the propeller running. In both cases, C_L seems to converge after about seven rotor revolutions, equivalent to resolving the wake for about two span-distances. Hence, all simulation were run for 12 revolutions and all results will hereon be reported considering only the last four revolutions.

When the wing generates lift through elevator deflection, the elevator also causes a mild turning of the rotor wake, which enhances the circulation and lift of the wing. On the case $\alpha = 0°$ and $\delta_e = +10°$, Stokkermans reported an experimental mean C_L that increases from 0.189 when the prop is off to 0.257 when the prop is on, leading to a lift augmentation of 36% due to beneficial rotor-on-wing interactions. As shown in Fig. 8.21, our simulation converges to a mean C_L of 0.196 and 0.262 when the prop is off and on, respectively, leading to a lift augmentation of 34%. In order to confirm that our predicted lift augmentation is caused by the correct physical mechanisms, Fig. 8.22 shows the vorticity at a plane downstream of the wing compared to experimental particle image velocimetry (PIV) reported by van Arnhem. The vortices shed

202 Since our ASM assumes wing elements with a straight chord, the elevator deflection is modeled as an equivalent twist of the elements about the quarter-chord line, varying from 3.75° inboard to 6.5° outboard of the elevator section. This equivalent twist was determined matching the experimental wing loading in the prop-off case.

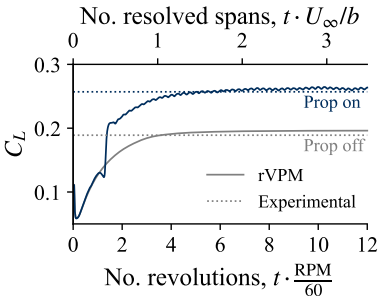


Figure 8.21: History of C_L in tip-mounted case with and without propeller. Test at $\alpha = 0°$, $\delta_e = +10°$, and inboard-up rotation direction.

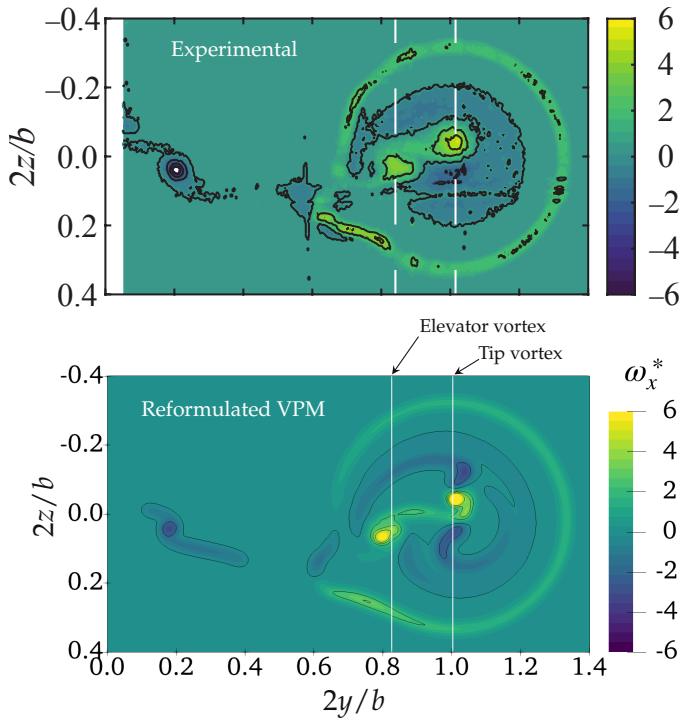


Figure 8.22: Time-average axial vorticity in tip-mounted case at plane 1.5c from trailing edge, (top) measured experimentally and (bottom) predicted with our meshless LES. Case $\alpha = 0°$, $\delta_e = +10°$, and $J = 0.8$. Experimental figure (top) retrieved from van Arnhem's doctoral thesis.

by the inboard and outboard elevator edges are seen at $2y/b \approx 0.2$ and $2y/b \approx 0.8$, respectively, along with the tip vortex at $2y/b \approx 1$. The rotor slipstream surrounds the tip vortex and is deformed by the wing surface in qualitative agreement with the experiment. Fig. 8.23 shows slices of the vorticity across the elevator vortex and tip vortex encompassed by the slipstream, evidencing quantitative agreement between simulation and experiment. This good agreement between the flow field predicted by our meshless LES and the experiment shows that the interactions between the rotor wake and the wing surface captured by our ASM, which lead to lift augmentation, are physically correct.

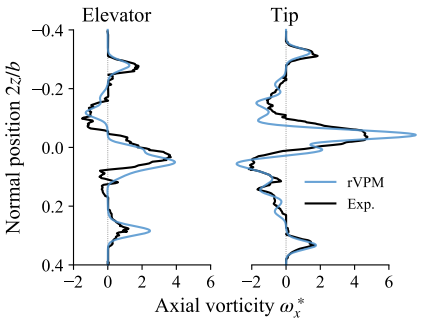


Figure 8.23: Time-average vorticity across elevator and tip vortices shown in Fig. 8.22.

The lift distribution with propeller on and off is shown in Fig. 8.24, and compared to the experiment with both positive and negative elevator deflection ($\delta_e = \pm 10^\circ$ and $\alpha = 0^\circ$). As a reference, Fig. 8.24 also includes the lift distribution reported by Van Arnhem & Sinnige *et al.* with URANS. The URANS simulation fully resolves the wing surface down to a y^+ of 1 in the boundary layer, in contrast to our LES that simply models the wing through an actuator surface model. When δ_e is positive, Fig. 8.24 (top) shows good agreement between our meshless LES and both the experiment and URANS. When δ_e is negative, Fig. 8.24 (bottom) shows some discrepancies, but overall the loading is in within reasonable agreement. This shows that the ASM is able to accurately predict the wing loading with minimal computational effort. Our predictions were also tested at different thrust settings by varying the advance ratio J. Fig. 8.25 shows that good agreement between our LES and the experiment is maintained across thrust settings.

Finally, stronger rotor-wing interactions were tested by pitching the wing system to an angle of attack of 10°. In this setting, the circulation

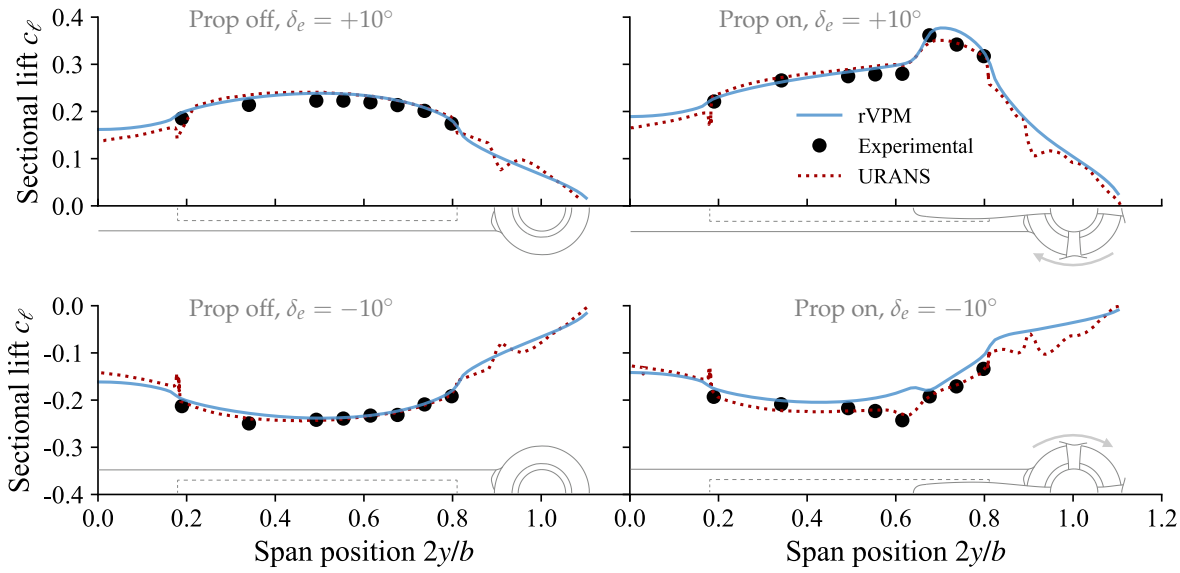


Figure 8.24: Time-average lift distribution in tip-mounted case with positive and negative elevator deflection δ_e at $\alpha = 0^\circ$. Inboard-up rotation direction.

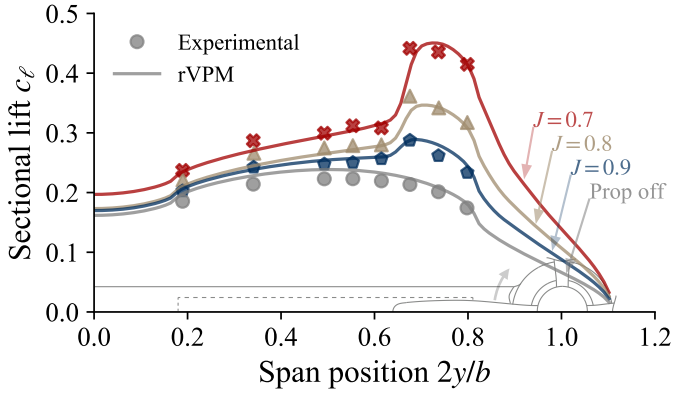


Figure 8.25: Time-average lift distribution in tip-mounted case as propeller thrust is increased. Test at $\alpha = 0°$, $\delta_e = +10°$, and inboard-up rotation direction.

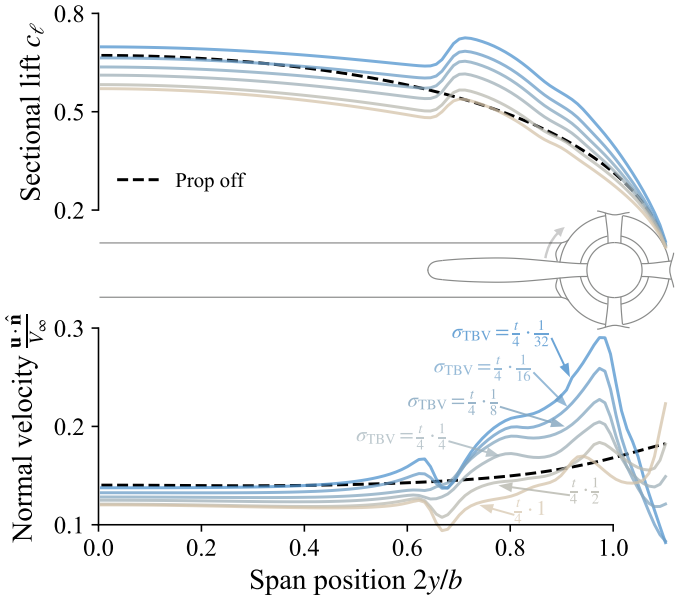


Figure 8.26: Effects of trailing bound vorticity smoothing σ_{TBV} on (top) lift distribution and (bottom) normal velocity when the system is at angle of attack $\alpha = 10°$. Time averages are shown. Test with $\delta_e = 0°$ and inboard-up rotation direction.

of the wing becomes stronger, more prominently turning the propeller slipstream. Also, since the angle of attack sets the rotor at an incidence angle relative to the freestream, the freestream pushes the wake against the wing's lower surface while also creating an asymmetric slipstream with advancing and retreating sides. Under these conditions, we first noticed that the simulation was sensitive to the smoothing radius chosen for the trailing bound vorticity (TBV) of the ASM, σ_{TBV}.

The TBV smoothing radius σ_{TBV} dictates how singular the velocity induced by the trailing bound vorticity is, which has the effect of deflecting and stretching parts of the rotor wake that come in direct contact with the immersed vorticity of the ASM. Up to this point, all simulations had defined σ_{TBV} as being equal to the filter width of the LES, which is close to $\frac{t_w}{4} \cdot \frac{1}{2}$, where t_w is the thickness of the wing. When the wing is set at an angle of attack, Fig. 8.26 (top) shows that σ_{TBV} has the effect of arbitrarily increasing the loading as σ_{TBV} is made smaller, acting as a tunable parameter. When σ_{TBV} is large, the loading drops outside of the propeller slipstream, which seems unphysical. Fig. 8.26 (bottom)

shows the velocity normal to the actuator surface as seen by each ASM control point, which is a combination of freestream, downwash of the wing tip vortices, and the rotor swirl. Note that this velocity is normal to the surface (not to the freestream), which is what is canceled by the circulation solver. Fig. 8.26 (bottom) shows that the loading drop is a consequence of reversed swirl inside the slipstream when σ_{TBV} is large, which accentuates the downwash of the tip vortices across the span. On the other hand, if σ_{TBV} is too small, the simulation can become numerically unstable as the velocity induced by the wing surface becomes singular.

The smoothing radius of the trailing bound vorticity, σ_{TBV}, was chosen to be small enough such that the extra downwash disappears outside of the rotor slipstream, which is $\sigma_{\text{TBV}} = \frac{t_w}{4} \cdot \frac{1}{32}$ as shown in Fig. 8.26 (bottom). This σ_{TBV} ends up being about two orders of magnitude smaller than the airfoil thickness t_w (which is roughly the length scale of the boundary layer along the airfoil). While $\sigma_{\text{TBV}} \approx 0.01 t_w$, we had previously chosen the smoothing of the lifting bound vorticity σ_{LBV} to be $\sigma_{\text{LBV}} = \frac{t_w}{4}\frac{3}{4} = 0.0103b$ (calculated from the thickness and aspect ratio of the wing), while the smoothing of the rotor ALM had been chosen to be $\sigma_{\text{rotor}} = \frac{R}{80} = 0.0125R$. Note that each of these parameters ended up independently and unintentionally chosen to be 0.01 times a characteristic length. This seems to point out a newly found heuristic: always choose a smoothing radius two orders of magnitude smaller than the characteristic length scale of the phenomenon at hand.

Setting $\sigma_{\text{TBV}} = \frac{t_w}{4} \cdot \frac{1}{32}$, the case with strong interactions ($\alpha = 10°$) was then compared to the experiment, finding good agreement between simulation and experiment as shown in Fig. 8.27. Furthermore, the validity of the chosen value of σ_{TBV} is further confirmed in the next section with the case of a blown wing with a higher aspect ratio, a range of angles of attack, and different rotation direction.

In summary, through this tip-mounted case we have incrementally validated each aspect of the rotor-wing interaction phenomenon. The case with the wing acting as a flat plate showed that the ASM accurately calculates the loading on the wing caused by the propeller swirl. The case with elevator deflection showed that our meshless LES captures the physical mechanisms that lead to lift augmentation when the wing

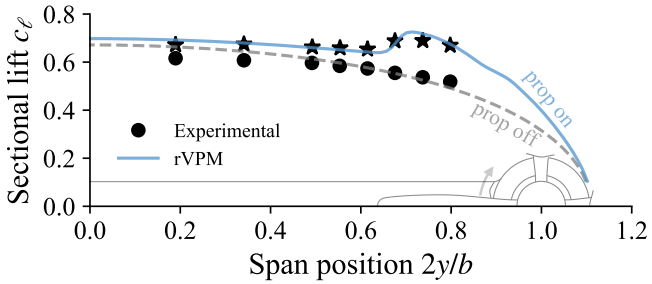


Figure 8.27: Time-average lift distribution in tip-mounted case at angle of attack $\alpha = 10°$. Test with $\delta_e = 0°$ and inboard-up rotation direction.

surface deflects the propeller slipstream. Finally, the case at an angle of attack of 10° was used to calibrate the smoothing of the ASM's trailing bound vorticity (σ_{TBV}) as to avoid unphysical results when encountering strong interactions.

8.4 Blown Wing

We will now look at the interactions in a conventional configuration where the propeller is mounted mid-span. In this configuration, the full slipstream interacts with the wing (as opposed to the tip-mounted case where only the inboard part of the slipstream does); hence, this case is refered to as the "blown wing" case. For this case, we simulated the experiment performed by Veldhuis[203] known as *PROWIM*, shown in Fig. 8.28.

[203] Veldhuis, "Propeller Wing Aerodynamic Interference".

The configuration uses a straight wing with aspect ratio $b/c = 5.33$, symmetric NACA 64_2-A015 profile, a span b of 1.28 m, and the Beaver propeller mounted at the span position $2y/b = 0.469$. Propeller and wing share the same angle of attack. Each test uses a freestream velocity V_∞ of 49.5 m/s and an advance ratio J of 0.85. This leads to similar Reynolds and Mach numbers as in the tip-mounted case. The experiment was reportedly conducted at a thrust setting $T_c = {}^{T}/_{\rho V_\infty^2 D^2}$ of 0.168 (corresponding to a thrust coefficient $C_T = {}^{T}/_{\rho n^2 D^4}$ of 0.121). A collective pitch of 2° was used in the simulation in order to match that thrust setting, leading to a blade angle of 25.9° at the radial position $r/R = 0.75$. The simulation then resulted in $T_c = 0.160$ (or $C_T = 0.117$), which is a thrust 5% lower than used in the experiment, but we deemed this difference to be negligible.

The blown wing was tested as the angle of attack varied from 0° to 10°, while determining the effects of propeller rotation direction. Fig. 8.29 shows the distribution of force normal to the wing, as predicted with our simulation and compared to the experiment. The cases with $\alpha = 0°$

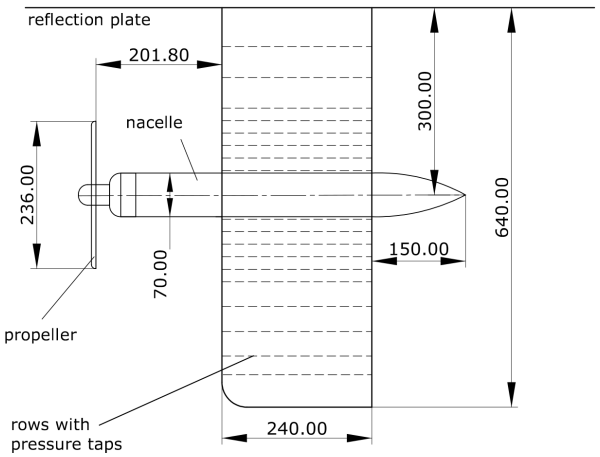


Figure 8.28: Description of blown wing case, or PROWIM, with propeller mounted mid-span. Dimensions in mm. Diagram of the experiment retrieved from Veldhuis' doctoral thesis.

(where the wing acts as a flat plate) and $\alpha = 4^\circ$ (where the wing turns the slipstream) agree reasonably well with the experiment, showing the loading increasing where the blade goes up, while decreasing where the blade goes down. These changes in the loading are caused by the swirl direction locally increasing or decreasing the AOA of the wing. In the case with $\alpha = 10^\circ$, these dynamics change due to the stronger nature of the interactions. For instance, notice in both simulation and experiment that the loading no longer drops where the blade goes down. This is because the effects of turning the slipstream become stronger than the effects of swirl. Even though the simulation shows the right trend, it overpredicts the lift augmentation in the slipstream at this AOA. Noticing that the prop off case also overpredicts the loading, it is possible that the wing in the experiment is mildly stalled at $\alpha = 10^\circ$. This could drive the discrepancy between simulation and experiment since our ASM does not capture stalled conditions. In light of this, we conclude that our LES simulation accurately predicts rotor-wing interactions up to a moderate angle of attack.

In the previous tests we have discussed only the interactional effects on normal force or lift, however, the drag force is also accurately captured. To show this, Fig. 8.30 compares our simulation to the tangential force reported by Veldhuis at $\alpha = 10^\circ$. This tangential force was measured experimentally integrating pressure taps around the wing profile, hence it includes form and induced drag, while excluding skin friction drag. Since this force is tangential to the wing (and not to the freestream), it also includes a small component of lift. In both rotation directions, the simulation agrees reasonably well with the experiment, giving confidence that the drag is accurately predicted.

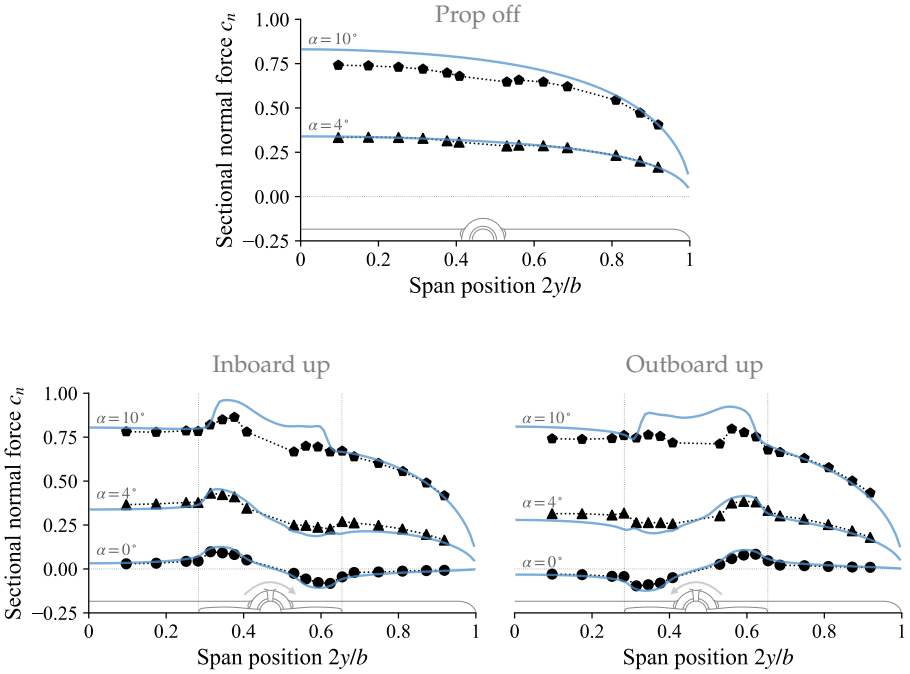


Figure 8.29: Time-average normal force distribution in blown-wing case at multiple angles of attack α (top) without the propeller, (bottom left) with the propeller rotating inboard up, and (bottom right) outboard up. rVPM simulation (solid lines) compared to experimental measurements (markers) reported by Veldhuis.

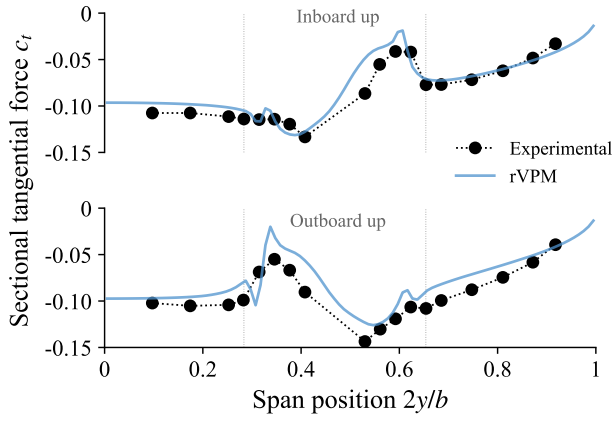
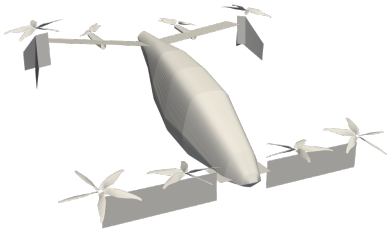


Figure 8.30: Time-average tangential force in blown-wing case at angle of attack $\alpha = 10°$.

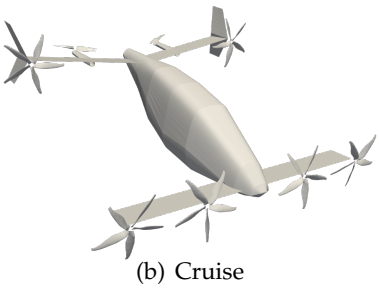
9 Maneuvering eVTOL Aircraft

In the preceding chapters, we validated a CFD framework able to predict the interactional aerodynamics that are typically encountered in multirotor aircraft. We will now exemplify the capabilities of this framework through the simulation of a full multirotor vehicle.

Since no experimental data were available on a vehicle of this kind for comparison, we used a fictional design (rather than a viable product) in order to feature as much complexity as it is relevant for eVTOL. Our fictional vehicle is a modified version of the tiltwing multirotor Vahana aircraft, shown in Fig. 9.1. The vehicle features a tilting tandem wing, tilt rotors on the main wing, and a set of stacked rotors. The vehicle was sized to an empty weight of roughly 500 kg and a cruise speed of 30 m/s with variable-pitch propellers.



(a) Takeoff and landing

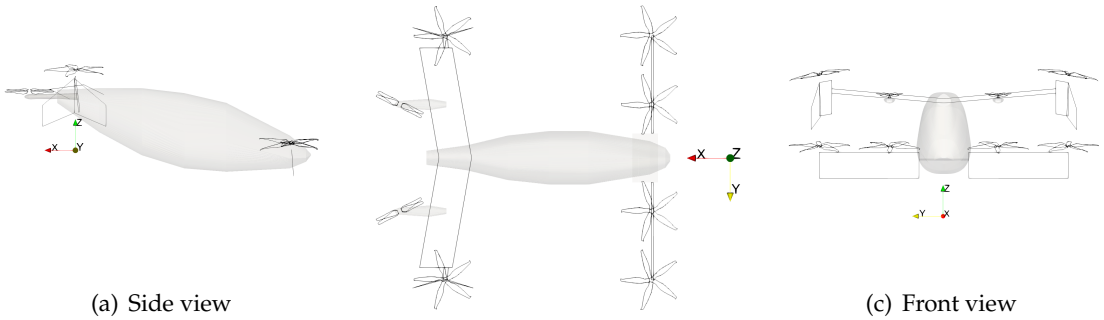


(b) Cruise

Figure 9.1: Modified Vahana vehicle.

Meshless CFD is fitting for simulations with moving boundaries. Our meshless LES allows us to effortlessly rotate and translate rotors, wings, and the whole vehicle without the hurdles of sliding/rotating/overset meshes that are typically necessary in mesh-based CFD. In order to take advantage of this feature, instead of simulating the aircraft at a quasi-static "frozen" point along a trajectory, the full trajectory of the eVTOL maneuver is simulated continuously. In this manner, the vehicle is simulated translating and pitching as it transitions from hover to cruise, while tilting rotors and the wing. Rotors also change collective blade pitch and RPM throughout the simulation. In order to simplify this example, the kinematics of the vehicle are prescribed as defined in Section 9.2. Future work could easily feed the aerodynamic forces (resulting from the control inputs) back to the vehicle dynamics to obtain a full dynamic simulation, which could we used to explore controls strategies. For simplicity, the control inputs used in this simulation are not trimmed to precisely match the prescribed kinematics; hence, the results presented here are notional and only intended to highlight the interactional aerodynamics.

A simulation with this complexity yields abundant information for analysis: from the performance of the rotors with interactions and variable RPM, to the unsteady loads that may lead to vibrations and aeroacoustic noise—not to mention the complex turbulent phenomena driving these responses. An in-depth analysis of these results can span the full scope of a PhD by itself, and is left as work for future researchers. Hence, for brevity, in Section 9.3 we will limit our discussion to only the un-



(a) Side view (b) Top view (c) Front view

Figure 9.2: Diagram of Vahana vehicle.

steady torsional loads exerted by the rotors in hover and the unsteady loading of the wings during transition.

9.1 Aircraft Geometry

Our modified Vahana aircraft features a wing span of 5.86 m for both main and front wings, as shown in Fig. 9.2. Main and front wings have an aspect ratio of 7.4 and 9.5, respectively. The main wing features tip-mounted tilt rotors and two stacked rotors. The stacked rotors can change index angle throughout the simulation, playing the same functionality as collective pitch but with potential aeroacoustic advantages; however, this feature was not used in this study. The tip-mounted rotors of the main wing were mounted at a bank angle, while the rotors on the tandem front wing were mounted at an alternating pitch angle, thus providing control authority without the need for control surfaces.

Note that the shaded components shown in Fig. 9.2 (fuselage, booms, and the front wing's middle section) are not included in the computational domain of the LES.

9.2 Controls and Prescribed Kinematics

In order to simulate a realistic eVTOL maneuver, control inputs are passed to the simulation as a function of time along with prescribing the kinematics that would roughly result from such inputs. Control inputs include: angles of tilt wing and tilt rotors, RPM of individual rotors, and collective blade pitch of each rotor. Wings and rotors with common control inputs are grouped into systems as shown in Fig. 9.3. Fig. 9.4 (left) shows the inputs used for the maneuver, tilting the wing and rotors to their cruise position as the vehicle transitions from powered lift to wing-borne lift, accompanied by an increased in RPMs to sustain flight during transition.

The kinematics of the vehicle are defined by prescribing the attitude and heading as a function of time, as well as the vehicle's velocity vector. Attitude is defined relative to the horizon given by the xy-plane of the global coordinate system, while heading is defined relative to the

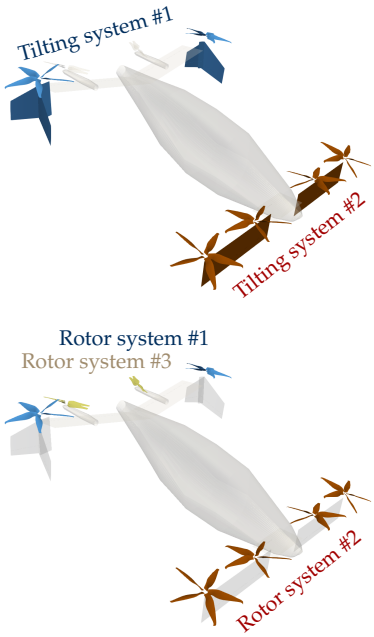
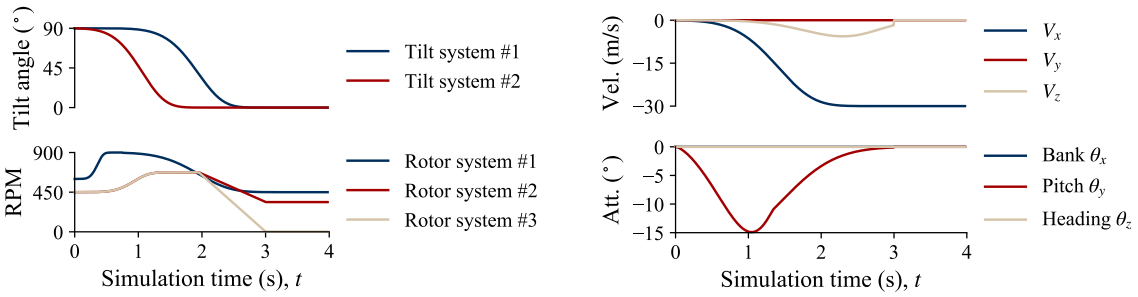
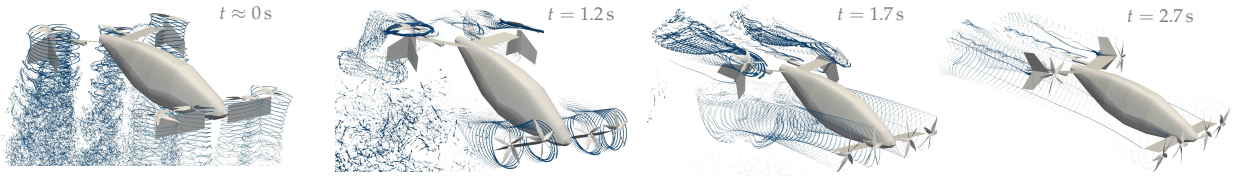


Figure 9.3: Tilting and rotors systems associated with control inputs.



(a) Control inputs (b) Vehicle kinematics

Figure 9.4: Control inputs and prescribed kinematics of the eVTOL transition maneuver.



(a) Hover (b) Start of transition (c) Mid transition (d) Cruise

Figure 9.5: eVTOL maneuver as vehicle transitions from powered-lift to wing-borne flight in mid-fidelity simulation. Arrows show the vortex strength of the particles.

zx-plane. The velocity vector is also defined in the global coordinate system (shown in Fig. 9.2). The kinematics for the simulation are shown in Fig. 9.4 (right), where we have prescribed that the vehicle initially pitches down to help transfer the powered-lift into propulsive thrust while the vehicle accelerates until sustaining wing-borne flight. The vehicle transitions in about three seconds. These prescribed kinematics are a rough estimate by the author, and suffice for the intent of this example.

The eVTOL maneuver resulting from these control inputs and kinematics is shown in Fig. 9.5 through a mid-fidelity simulation. This simulation takes a matter of hours with mid fidelity, depending on computational resources. On the other hand, the high-fidelity simulation discussed next takes approximately one day on a single node of the BYU FSL supercomputer to resolve one second of this maneuver. For this reason, the simulation was started at various points along the maneuver and resolved by about one second at a time.

9.3 Results

First, the simulation was started at $t = 0\,\mathrm{s}$ with the vehicle in hover. Fig. 9.6 shows the history of thrust and power consumption of the right tip-mounted tilt rotor, where it can be seen that the simulation takes about 0.1 s for the wake to deploy and achieve physical conditions. The RPM is ramped-up between $t = 0.2\,\mathrm{s}$ and $t = 0.4\,\mathrm{s}$, leading to an increasing in thrust and power. At $t > 0.4\,\mathrm{s}$, the RPM is relatively

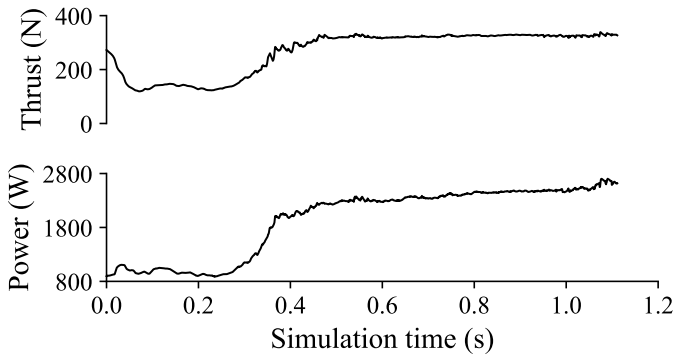


Figure 9.6: Performance of the right tip-mounted tilt rotor over time starting from hover.

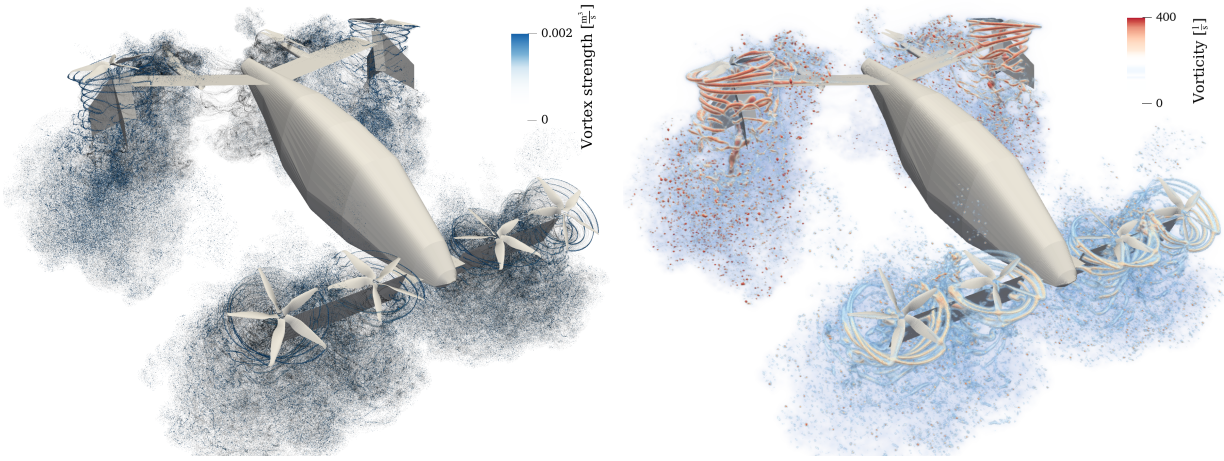


Figure 9.7: eVTOL maneuver simulation during hover at $t \approx 1\,\text{s}$: (left) computational elements (vortex particles and strength), and (right) volume rendering of vorticity field.

constant and the rotor is not yet tilting, hence any fluctuations in the rotor performance are due to aerodynamic interactions. By $t = 0.8\,\text{s}$, the performance has become relatively constant once the flow becomes fully developed.

Fig. 9.7 shows the simulation at $t \approx 1.0\,\text{s}$, visualizing the wake structure. As expected, each rotor wake breaks up into turbulence after a few diameter-distances due to the significant interactions with neighboring rotors and/or wing surfaces. In particular, we note that the wakes of the tip-mounted tilt rotors are obstructed by the presence of the horizontal wing, creating an asymmetric rotor wake. Even though Fig. 9.6 shows a somewhat steady thrust on the right rotor, the obstructed wake leads to a considerable asymmetric loading of the rotor. To illustrate this, we integrated the time-resolved loading of each blade to calculate the instantaneous moment $\mathbf{M}$ of the rotor. The moment turns out to be misaligned from the axis of rotation, as shown in Fig. 9.8, turning away from the wing surface.

The off-axis components of the moment are shown in Fig. 9.9 over time. As a reference of magnitude, the axis-aligned component is also shown. At a vehicle level, M_x cancels with the same component in the

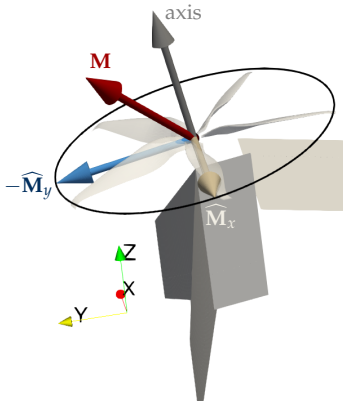


Figure 9.8: Misalignment between moment $\mathbf{M}$ of the tip-mounted rotor and its rotation axis.

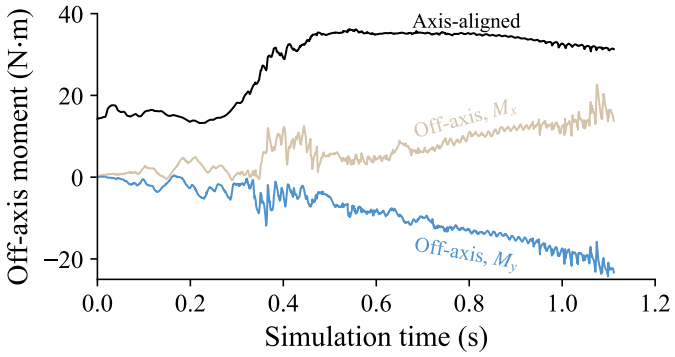


Figure 9.9: Off-axis moment of the right tip-mounted tilt rotor during hover.

rotor at the opposite wing tip, while the M_y components of both rotors get added together creating a small pitching moment. Furthermore, each off-axis component creates a torsional load on the tilting structure and its hinge, which, along with the component's fluctuations, may merit some structural and vibratory considerations. Each of these effects stem from the poor placement of the rotor, which led to the wing obstructing the wake. Our meshless LES makes it possible to identify, quantify, and address these effects early in the design process.

A second simulation was started at $t = 1.5\,\text{s}$ with the vehicle in mid transition. At this stage, the vehicle has attained a speed of $20\,\text{m/s}$ and the front wing is almost fully tilted forward, transferring the lift from the rotors to the wing. The rotors of the main wing, on the other hand, are still mostly tilted upwards carrying some of the weight until the vehicle achieves full cruise speed. Fig. 9.10 shows the loading of each wing[204] between $t = 1.9\,\text{s}$ and $t = 2.1\,\text{s}$, at which point the vehicle achieves cruise speed, $30\,\text{m/s}$. The tandem front wing carries up to a third of the weight at any time, while the other two thirds of the lift are split between the main wing and its tip-mounted tilt rotors. Fig. 9.10 shows the loading of the main wing increasing as the vehicle speeds up. When the vehicle achieves full speed by $t = 2.1\,\text{s}$, an extraneous increase in loading is observed at the spanwise positions $\pm 1\,\text{m}$. This peak also exhibits high fluctuations, observed by the shaded regions encompassing two standard deviations. Visualizing the simulation as shown in Fig. 9.11, it becomes evident that this peak in loading is caused by the wake from the advancing side of the front rotors impinging on the back wing.

204 This loading is an ensemble average over windows of 0.02 s.

In this case, it is fortuitous that the wake impingement has favorable effects on the wing loading instead of causing the lift to drop. However, that is not always the case, and the ability to predict and assess such scenarios is critical. This highlights the usefulness of our meshless LES, which is able to predict these interactions with high fidelity and modest computational resources.

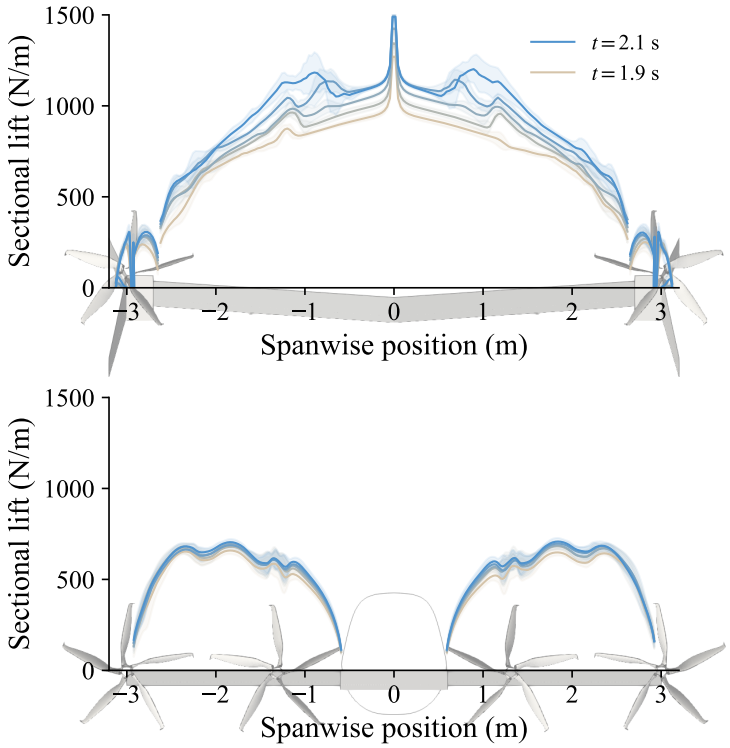


Figure 9.10: Load distribution of (top) main and (bottom) tandem wing during eVTOL transition maneuver. Solid lines show the mean distribution over an averaging window of 0.02 s, while shaded regions encompass two standard deviations.

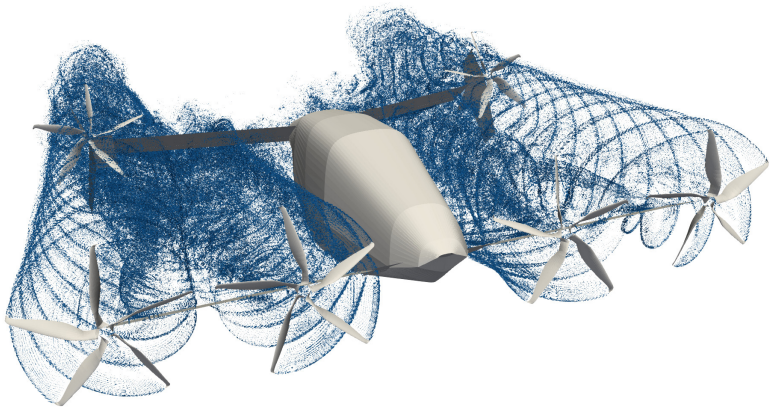


Figure 9.11: Wake of front rotors impinging on back wing during transition maneuver at $t = 2.09$.

Conclusions

In Part II, we have presented a CFD framework based on the reformulated VPM for simulating complex interactional aerodynamics. We further developed our meshless LES scheme to include rotors and wings in the computational domain through actuator models. A novel, vorticity-based, actuator surface model (ASM) was developed for wings, which is suitable for rotor-wing interactions when a wake impinges on the surface of a wing. This ASM imposes the no-flow-through condition at the airfoil centerline by calculating the circulation that meets this condition and by immersing the associated vorticity following a pressure-like distribution.

The actuator line model (ALM) used for rotors in our LES was validated as an accurate approach to capturing rotor-rotor interactions. Validation was presented starting with an isolated rotor and building up to the interactional case of two side-by-side rotors in hover. Numerical convergence of the rotor simulation was characterized through a detailed study of temporal and spatial discretization. The isolated rotor was validated by comparison to experiments in hover, forward flight, and at an incidence angle. Finally, the aerodynamic interactions between two side-by-side rotors were characterized as the distance between them was decreased, showing good agreement with experiment.

The actuator surface model (ASM) used for wings in our LES was validated as an accurate approach to capturing rotor-wing interactions. First, the numerical convergence of the propeller wake was characterized in a detailed study of spatial discretization. The predicted vortical structure and velocity in the wake showed good agreement with experimental work in the literature. Next, a wing was placed in the wake of the propeller in a tip-mounted configuration. The case with the wing acting as a flat plate showed that the ASM accurately calculates the loading on the wing caused by the propeller swirl. The case with elevator deflection showed that our meshless LES captures the physical mechanisms that lead to lift augmentation when the wing surface deflects the propeller slipstream. The case at an angle of attack (AOA) of 10° was used to calibrate the smoothing of the ASM's trailing bound vorticity, avoiding unphysical results when encountering strong interactions. Then, the interactions on a blown wing were characterized at multiple AOAs and different rotation directions, showing good agreement with experiment up to moderate AOAs.

To conclude, the capabilities of the framework were showcased through the simulation of a multirotor tiltwing vehicle. The vehicle was simulated mid maneuver as it transitions from powered lift to wing-borne flight, featuring rotors with variable RPM and variable blade pitch, tilting of wings and rotors, and significant rotor-rotor and rotor-wing interactions from hover to cruise.

This study validates our meshless LES as an accurate tool for predicting complex interactional aerodynamics. We have focused on the high fidelity achieved with rVPM; however, since it is not limited by the classic CFL condition, rVPM can be used across all levels of fidelity, all in the same framework by simply coarsening or refining the simulation. Thus, rVPM provides aircraft designers with a high-fidelity tool that is orders of magnitude faster than mesh-based CFD, while also featuring variable-fidelity capabilities.

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Appendices

A Derivations

A.1 Derivation of $\frac{\mathrm{d}}{\mathrm{d}t}\left(\zeta_{\sigma_p}(\mathbf{x}-\mathbf{x}_p)\right)$

Let us define two auxiliary functions G and g such that

$$G(\mathbf{x}, t) = g(\mathbf{x}, \mathbf{y}(t), \sigma(t))$$

and

$$g(\mathbf{x}, \mathbf{y}(t), \sigma(t)) = \zeta_\sigma(\mathbf{x}-\mathbf{y}),$$

resulting in the following properties.

Property 1

$$\frac{\mathrm{d}}{\mathrm{d}t}G(\mathbf{x}, t) = \frac{\partial G}{\partial t}(\mathbf{x}, t) + \left(\frac{\partial \mathbf{x}}{\partial t}\cdot\nabla\right)G(\mathbf{x}, t),$$

by definition.

Property 2

$$\begin{aligned}\frac{\partial G}{\partial t}(\mathbf{x}, t) &= \frac{\partial}{\partial t}(g(\mathbf{x}, \mathbf{y}(t), \sigma(t))) \\ &= \frac{\partial g}{\partial \sigma}\frac{\partial \sigma}{\partial t} + \sum_i \frac{\partial g}{\partial y_i}\frac{\partial y_i}{\partial t} \\ &= \frac{\partial g}{\partial \sigma}\frac{\partial \sigma}{\partial t} + \left(\frac{\partial \mathbf{y}}{\partial t}\cdot\nabla_y\right)g\end{aligned}$$

Property 3

$$\begin{aligned}\left(\frac{\partial \mathbf{y}}{\partial t}\cdot\nabla_y\right)g(\mathbf{x}, \mathbf{y}, \sigma) &= \left(\frac{\partial \mathbf{y}}{\partial t}\cdot\nabla_y\right)\zeta_\sigma(\mathbf{x}-\mathbf{y}) \\ &= -\left(\frac{\partial \mathbf{y}}{\partial t}\cdot\nabla\right)\zeta_\sigma(\mathbf{x}-\mathbf{y}),\end{aligned}$$

since

$$\frac{\partial}{\partial y_i}(\zeta_\sigma(\mathbf{x}-\mathbf{y})) = -\frac{\partial}{\partial x_i}(\zeta_\sigma(\mathbf{x}-\mathbf{y}))$$

Now, using $\frac{\partial \zeta_\sigma}{\partial t} = \frac{\partial \zeta_\sigma}{\partial \sigma}\frac{\partial \sigma}{\partial t}$ and replacing Property 3 in Property 2 we get

$$\frac{\partial G}{\partial t}(\mathbf{x}, t) = \frac{\partial \zeta_\sigma}{\partial t}(\mathbf{x}-\mathbf{y}) - \left(\frac{\partial \mathbf{y}}{\partial t}\cdot\nabla\right)\zeta_\sigma(\mathbf{x}-\mathbf{y}),$$

and using this in Property 1,

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}t}G(\mathbf{x}, t) &= \frac{\partial \zeta_\sigma}{\partial t}(\mathbf{x}-\mathbf{y}) - \left(\frac{\partial \mathbf{y}}{\partial t}\cdot\nabla\right)\zeta_\sigma(\mathbf{x}-\mathbf{y}) + \left(\frac{\partial \mathbf{x}}{\partial t}\cdot\nabla\right)\zeta_\sigma(\mathbf{x}-\mathbf{y}) \\ &= \frac{\partial \zeta_\sigma}{\partial t}(\mathbf{x}-\mathbf{y}) + \left[\left(\frac{\partial \mathbf{x}}{\partial t} - \frac{\partial \mathbf{y}}{\partial t}\right)\cdot\nabla\right]\zeta_\sigma(\mathbf{x}-\mathbf{y}).\end{aligned}$$

Defining $\mathbf{u}(\mathbf{x}) = \frac{\partial \mathbf{x}}{\partial t}$ and assuming $\frac{\partial \mathbf{y}}{\partial t} = \mathbf{u}(\mathbf{y})$,

$$\frac{\mathrm{d}}{\mathrm{d}t}G(\mathbf{x}, t) = \frac{\partial \zeta_\sigma}{\partial t}(\mathbf{x}-\mathbf{y}) + [(\mathbf{u}(\mathbf{x}) - \mathbf{u}(\mathbf{y}))\cdot\nabla]\zeta_\sigma(\mathbf{x}-\mathbf{y}).$$

Finally, given that

$$\frac{\mathrm{d}}{\mathrm{d}t}G(\mathbf{x}, t) = \frac{\mathrm{d}}{\mathrm{d}t}\zeta_\sigma(\mathbf{x}-\mathbf{y})$$

by definition, we conclude that

$$\frac{\mathrm{d}}{\mathrm{d}t}\zeta_\sigma(\mathbf{x}-\mathbf{y}) = \frac{\partial \zeta_\sigma}{\partial t}(\mathbf{x}-\mathbf{y}) + [(\mathbf{u}(\mathbf{x}) - \mathbf{u}(\mathbf{y}))\cdot\nabla]\zeta_\sigma(\mathbf{x}-\mathbf{y}).$$

A.2 Derivation of $\frac{\mathrm{d}\hat{\boldsymbol{\Gamma}}}{\mathrm{d}t}$

Decomposing the vortex strength $\boldsymbol{\Gamma}$ into its magnitude $\Gamma \equiv \|\boldsymbol{\Gamma}\|$ and unit vector $\hat{\boldsymbol{\Gamma}} \equiv \frac{\boldsymbol{\Gamma}}{\|\boldsymbol{\Gamma}\|}$ as $\boldsymbol{\Gamma} = \Gamma\hat{\boldsymbol{\Gamma}}$, we calculate the following

$$\begin{aligned}\frac{\mathrm{d}\boldsymbol{\Gamma}}{\mathrm{d}t} - \left(\frac{\mathrm{d}\boldsymbol{\Gamma}}{\mathrm{d}t}\cdot\hat{\boldsymbol{\Gamma}}\right)\hat{\boldsymbol{\Gamma}} &= \frac{\mathrm{d}}{\mathrm{d}t}(\Gamma\hat{\boldsymbol{\Gamma}}) - \left\{\left[\frac{\mathrm{d}}{\mathrm{d}t}(\Gamma\hat{\boldsymbol{\Gamma}})\right]\cdot\hat{\boldsymbol{\Gamma}}\right\}\hat{\boldsymbol{\Gamma}} \\ &= \frac{\mathrm{d}\Gamma}{\mathrm{d}t}\hat{\boldsymbol{\Gamma}} + \Gamma\frac{\mathrm{d}\hat{\boldsymbol{\Gamma}}}{\mathrm{d}t} - \left(\frac{\mathrm{d}\Gamma}{\mathrm{d}t}\hat{\boldsymbol{\Gamma}}\cdot\hat{\boldsymbol{\Gamma}}^{1} + \Gamma\frac{\mathrm{d}\hat{\boldsymbol{\Gamma}}}{\mathrm{d}t}\cdot\hat{\boldsymbol{\Gamma}}\right)\hat{\boldsymbol{\Gamma}} \\ &= \Gamma\frac{\mathrm{d}\hat{\boldsymbol{\Gamma}}}{\mathrm{d}t} - \left(\Gamma\frac{\mathrm{d}\hat{\boldsymbol{\Gamma}}}{\mathrm{d}t}\cdot\hat{\boldsymbol{\Gamma}}\right)\hat{\boldsymbol{\Gamma}} \\ &= \Gamma\frac{\mathrm{d}\hat{\boldsymbol{\Gamma}}}{\mathrm{d}t} - \left[\frac{\Gamma}{2}\frac{\mathrm{d}}{\mathrm{d}t}\left(\hat{\boldsymbol{\Gamma}}\cdot\hat{\boldsymbol{\Gamma}}\right)^{1}\right]\hat{\boldsymbol{\Gamma}} \\ &= \Gamma\frac{\mathrm{d}\hat{\boldsymbol{\Gamma}}}{\mathrm{d}t}\end{aligned}$$

Hence, it follows that

$$\frac{\mathrm{d}\hat{\boldsymbol{\Gamma}}}{\mathrm{d}t} = \frac{1}{\Gamma}\frac{\mathrm{d}\boldsymbol{\Gamma}}{\mathrm{d}t} - \frac{1}{\Gamma}\left(\frac{\mathrm{d}\boldsymbol{\Gamma}}{\mathrm{d}t}\cdot\hat{\boldsymbol{\Gamma}}\right)\hat{\boldsymbol{\Gamma}}.$$

B Vortex Tube Method

In the same manner that the unfiltered vorticity field $\boldsymbol{\omega}$ is discretized with singular point elements in the vortex particle method, one can also use singular lines elements defining what here we call a *vortex tube method*.

B.1 Singular Tube Discretization

Let us define a line version of the Dirac delta—denoted δ_{line}—of length ℓ, centered at the origin, and aligned with the z-axis, as

$$\delta_{\text{line}}(\mathbf{x}, \ell) = \frac{1}{2\pi\rho}\delta(\rho)H\left(\frac{\ell}{2}+z\right)H\left(\frac{\ell}{2}-z\right)$$

where δ is the Dirac delta, H is the Heaviside step function, and ρ and z are cylindrical coordinates calculated as

$$\begin{aligned}\mathbf{x} &= (x,\, y,\, z)\\ \rho &= \sqrt{x^2+y^2}\\ z &= z\end{aligned}$$

We use δ_{line} to represent a singular line arbitrarily placed and oriented in space with centroid $\mathbf{x}_c$ and centerline axis $\hat{\boldsymbol{\ell}}$ as

$$\delta_{\text{line}}(A(\mathbf{x}-\mathbf{x}_c), \ell) = \frac{1}{2\pi\rho^*}\delta(\rho^*)H\left(\frac{\ell}{2}+z^*\right)H\left(\frac{\ell}{2}-z^*\right)$$

where A is the unitary rotation matrix that transforms vectors from the global coordinate system to the coordinate system of the line, $\mathbf{x}_{\text{line}} \equiv A(\mathbf{x}-\mathbf{x}_c)$, using the centerline $\hat{\boldsymbol{\ell}}$ as one of the axes. This results in

$$\begin{aligned}\mathbf{x}_{\text{line}} &= (x^*,\, y^*,\, z^*)\\ \rho^* &= \sqrt{x^{*2}+y^{*2}} = \|(\mathbf{x}-\mathbf{x}_c) - z^*\hat{\boldsymbol{\ell}}\|\\ z^* &= (\mathbf{x}-\mathbf{x}_c)\cdot\hat{\boldsymbol{\ell}}\end{aligned}$$

The unfiltered vorticity field $\boldsymbol{\omega}$ is then discretized through singular vortex tube elements as

$$\boldsymbol{\omega}(\mathbf{x},t) \approx \sum_p \boldsymbol{\gamma}_p(t)\delta_{\text{line}}\left(A_p(t)\left(\mathbf{x}-\mathbf{x}_p(t)\right),\, \ell_p(t)\right), \tag{B.1}$$

where each tube element is aligned with the local vorticity, $\hat{\boldsymbol{\ell}}_p = \boldsymbol{\omega}(\mathbf{x}_p)/\|\boldsymbol{\omega}(\mathbf{x}_p)\|$, and likewise the coefficients, $\hat{\boldsymbol{\gamma}}_p = \boldsymbol{\gamma}_p/\|\boldsymbol{\gamma}_p\| = \hat{\boldsymbol{\ell}}_p$. Furthermore, when the $\boldsymbol{\omega}$ field encompasses a tube of vorticity, we end up with $\boldsymbol{\gamma}_p = \Gamma_p \hat{\boldsymbol{\ell}}_p$, with Γ_p the circulation of the tube.

The vortex tube discretization in Eq. (B.1) is analogous to the vortex particle discretization previously shown in Eq. (1.6), here repeated

$$\boldsymbol{\omega}(\mathbf{x}, t) \approx \sum_p \boldsymbol{\Gamma}_p(t)\delta(\mathbf{x} - \mathbf{x}_p(t)), \tag{1.6}$$

where $\boldsymbol{\Gamma}_p = \ell_p \Gamma_p \hat{\boldsymbol{\ell}}_p = \ell_p \boldsymbol{\gamma}_p$ when used to discretize tubes of vorticity. Furthermore, we will show in Appendix B.4 that the vortex tube method and the vortex particle method end up with strikingly similar governing equations.

B.2 Regularized Tube Discretization

Using the tube discretization in the filtered vorticity field $\overline{\boldsymbol{\omega}}$,

$$\begin{aligned}
\overline{\boldsymbol{\omega}}(\mathbf{x}) &= \int_{-\infty}^{\infty} \boldsymbol{\omega}(\mathbf{y})\, \zeta_\sigma(\mathbf{x} - \mathbf{y})\, \mathrm{d}\mathbf{y} \\
&\approx \int_{-\infty}^{\infty} \left(\sum_p \boldsymbol{\gamma}_p \delta_{\text{line}}\left(A_p(\mathbf{y} - \mathbf{x}_p),\, \ell_p\right) \right) \zeta_\sigma(\mathbf{x} - \mathbf{y})\, \mathrm{d}\mathbf{y} \\
&= \sum_p \boldsymbol{\gamma}_p \left(\int_{-\infty}^{\infty} \frac{1}{2\pi\rho^*} \delta(\rho^*) H\left(\frac{\ell}{2} + z^*\right) H\left(\frac{\ell}{2} - z^*\right) \zeta_\sigma(\mathbf{x} - \mathbf{y})\, \mathrm{d}\mathbf{y} \right).
\end{aligned}$$

Using the change of variable $\mathbf{y} = A_p^{-1}\mathbf{x}_{\text{line}} + \mathbf{x}_p$ while transforming the cartesian volume integral into cylindrical coordinates,

$$\begin{aligned}
\overline{\boldsymbol{\omega}}(\mathbf{x}) &\approx \sum_p \boldsymbol{\gamma}_p \left(\int_{-\infty}^{\infty} \int_0^{\infty} \int_0^{2\pi} \frac{\rho^*}{2\pi\rho^*} \delta(\rho^*) H\left(\frac{\ell}{2} + z^*\right) H\left(\frac{\ell}{2} - z^*\right) \zeta_\sigma\left(\mathbf{x} - \left(A_p^{-1}\mathbf{x}_{\text{line}} + \mathbf{x}_p\right)\right) \mathrm{d}\theta^* \mathrm{d}\rho^* \mathrm{d}z^* \right) \\
&= \sum_p \boldsymbol{\gamma}_p \left(\int_{-\infty}^{\infty} H\left(\frac{\ell}{2} + z^*\right) H\left(\frac{\ell}{2} - z^*\right) \zeta_\sigma\left(\mathbf{x} - \left[A_p^{-1}(0\hat{\boldsymbol{\rho}}^* + z^*\hat{\mathbf{z}}^*) + \mathbf{x}_p\right]\right) \mathrm{d}z^* \right) \\
&= \sum_p \boldsymbol{\gamma}_p \left(\int_{-\ell/2}^{\ell/2} \zeta_\sigma(\mathbf{x} - \mathbf{x}_p - z^* A_p^{-1}\hat{\mathbf{z}}^*)\, \mathrm{d}z^* \right).
\end{aligned}$$

Defining the auxiliary function ζ_σ^*,

$$\zeta_\sigma^*(\mathbf{x}) \equiv \int_{-\ell/2}^{\ell/2} \zeta_\sigma(\mathbf{x} - z^* A^{-1}\hat{\mathbf{z}}^*)\, \mathrm{d}z^*,$$

we obtain that the filtered field $\overline{\boldsymbol{\omega}}$ is discretized as

$$\overline{\boldsymbol{\omega}}(\mathbf{x}) \approx \sum_p \boldsymbol{\gamma}_p \zeta_{\sigma_p}^*(\mathbf{x} - \mathbf{x}_p). \tag{B.2}$$

Compare this to the particle discretization in Eq. (1.7), here repeated

$$\overline{\boldsymbol{\omega}}\left(\mathbf{x}\right) \approx \sum_p \boldsymbol{\Gamma}_p \zeta_{\sigma_p}(\mathbf{x}-\mathbf{x}_p). \tag{1.7}$$

The tube discretization differs from the particle discretization only in that it integrates the basis function ζ_{σ_p} along the line ℓ_p to obtain $\zeta^*_{\sigma_p}$, while in the particle discretization the line information is encapsulated in the coefficient as $\boldsymbol{\Gamma}_p = \ell_p \boldsymbol{\gamma}_p$.

B.3 Derivation of $\frac{\partial \zeta^*_\sigma}{\partial t}$

The temporal derivative of ζ^*_σ will be needed in order to derive the governing equation of the vortex strength $\boldsymbol{\gamma}$, hence we proceed to derive $\frac{\partial \zeta^*_\sigma}{\partial t}$. This is calculated as

$$\frac{\partial \zeta^*_\sigma}{\partial t} = \frac{\partial \zeta^*_\sigma}{\partial \sigma}\frac{\partial \sigma}{\partial t} + \frac{\partial \zeta^*_\sigma}{\partial \ell}\frac{\partial \ell}{\partial t} + \sum_i \frac{\partial \zeta^*_\sigma}{\partial b_i}\frac{\partial b_i}{\partial t}, \tag{B.3}$$

where $b_i = (A^{-1}\hat{\mathbf{z}}^*)_i$. The second term is

$$\frac{\partial \zeta^*_\sigma}{\partial \ell}\frac{\partial \ell}{\partial t} = \left(\zeta_\sigma\left(\mathbf{x}+\frac{\ell}{2}\mathbf{b}\right) + \zeta_\sigma\left(\mathbf{x}-\frac{\ell}{2}\mathbf{b}\right)\right)\frac{\partial \ell}{\partial t},$$

and the third term is

$$\sum_i \frac{\partial \zeta^*_\sigma}{\partial b_i}\frac{\partial b_i}{\partial t} = \frac{\partial \mathbf{b}}{\partial t}\cdot \int_0^{\ell/2} z^* \left(\nabla\zeta_\sigma(\mathbf{x}+z^*\mathbf{b}) - \nabla\zeta_\sigma(\mathbf{x}-z^*\mathbf{b})\right) \mathrm{d}z^*.$$

When the filter ζ_σ is defined as $\zeta_\sigma(\mathbf{x}) = \frac{1}{\sigma^3}\zeta\left(\frac{\|\mathbf{x}\|}{\sigma}\right)$, with ζ a radial basis function, evaluating Eq. (B.3) at $\mathbf{x} = \mathbf{0}$, the first term becomes

$$\frac{\partial \zeta^*_\sigma}{\partial \sigma}\frac{\partial \sigma}{\partial t}\left(\mathbf{0}\right) = -\frac{3}{\sigma}\zeta^*_\sigma(\mathbf{0})\frac{\partial \sigma}{\partial t},$$

the second term becomes

$$\frac{\partial \zeta^*_\sigma}{\partial \ell}\frac{\partial \ell}{\partial t}\left(\mathbf{0}\right) = 2\zeta_\sigma\left(\frac{\boldsymbol{\ell}}{2}\right)\frac{\partial \ell}{\partial t},$$

and the third term becomes

$$\sum_i \frac{\partial \zeta^*_\sigma}{\partial b_i}\frac{\partial b_i}{\partial t}\left(\mathbf{0}\right) = 0.$$

Hence, Eq. (B.3) evaluated at $\mathbf{x} = \mathbf{0}$ becomes

$$\frac{\partial \zeta^*_\sigma}{\partial t}(\mathbf{0}) = -\frac{3}{\sigma}\frac{\partial \sigma}{\partial t}\zeta^*_\sigma(\mathbf{0}) + 2\zeta_\sigma\left(\frac{\boldsymbol{\ell}}{2}\right)\frac{\partial \ell}{\partial t}, \tag{B.4}$$

or

$$\frac{\partial \zeta_\sigma^*}{\partial t}(\mathbf{0}) = -\frac{3}{\sigma}\frac{\partial \sigma}{\partial t}\zeta_\sigma^*(\mathbf{0}) + 2\frac{1}{\sigma^3}\zeta\left(\frac{\ell}{2\sigma}\right)\frac{\partial \ell}{\partial t}.$$

Notice that when $\frac{\ell}{\sigma} \to \infty$, $\zeta\left(\frac{\ell}{\sigma}\right) \to 0$ if the radial basis is a decaying one. Hence, for a tube with high aspect ratio ℓ/σ,

$$\frac{\partial \zeta_\sigma^*}{\partial t}(\mathbf{0}) \approx -\frac{3}{\sigma}\frac{\partial \sigma}{\partial t}\zeta_\sigma^*(\mathbf{0}).$$

B.4 Governing Equation

Starting from the LES-filtered vorticity Navier-Stokes equation and neglecting viscous and SFS effects,

$$\frac{\mathrm{d}}{\mathrm{d}t}\overline{\boldsymbol{\omega}} = (\overline{\boldsymbol{\omega}} \cdot \nabla)\,\mathbf{u},$$

we use the tube approximation of $\overline{\boldsymbol{\omega}}$ shown in Eq. (B.2) to get

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\sum_q \boldsymbol{\gamma}_q \zeta_{\sigma_q}^*\left(\mathbf{x}-\mathbf{x}_q\right)\right) = \left[\left(\sum_q \boldsymbol{\gamma}_q \zeta_{\sigma_q}^*\left(\mathbf{x}-\mathbf{x}_q\right)\right)\cdot\nabla\right]\mathbf{u}(\mathbf{x}). \tag{B.5}$$

Unfolding these equations in the same manner as done in Section 1.5, evaluating at $\mathbf{x} = \mathbf{x}_p$, and applying the localized-vorticity assumption, Eq. (B.5) becomes

$$\frac{\mathrm{d}\boldsymbol{\gamma}_p}{\mathrm{d}t}\zeta_{\sigma_p}^*(\mathbf{0}) + \boldsymbol{\gamma}_p\frac{\partial \zeta_{\sigma_p}^*}{\partial t}(\mathbf{0}) = \zeta_{\sigma_p}^*(\mathbf{0})\left(\boldsymbol{\gamma}_p\cdot\nabla\right)\mathbf{u}(\mathbf{x}_p). \tag{B.6}$$

Replacing Eq. (B.4) into Eq. (B.6) and assuming $\zeta_{\sigma_p}^*(\mathbf{0}) \neq 0$, we arrive to the equation governing the evolution of strength of the vortex tube,

$$\frac{\mathrm{d}\boldsymbol{\gamma}_p}{\mathrm{d}t} = \left(\boldsymbol{\gamma}_p\cdot\nabla\right)\mathbf{u}(\mathbf{x}_p) + 3\boldsymbol{\gamma}_p\frac{1}{\sigma_p}\frac{\partial \sigma_p}{\partial t} - 2\boldsymbol{\gamma}_p\frac{\zeta_{\sigma_p}(\boldsymbol{\ell}_p/2)}{\zeta_{\sigma_p}^*(\mathbf{0})}\frac{\partial \ell_p}{\partial t}.$$

For tubes of high aspect ratio, the last term vanishes, and the equation collapses into the same governing equation than the vortex particle method, Eq. (2.7):

$$\frac{\mathrm{d}\boldsymbol{\gamma}_p}{\mathrm{d}t} \approx \left(\boldsymbol{\gamma}_p\cdot\nabla\right)\mathbf{u}(\mathbf{x}_p) + 3\boldsymbol{\gamma}_p\frac{1}{\sigma_p}\frac{\partial \sigma_p}{\partial t}.$$

B.5 Proof that $\frac{1}{\ell}\frac{\mathrm{d}\ell}{\mathrm{d}t} = \frac{1}{\|\boldsymbol{\Gamma}\|}\frac{\mathrm{d}\boldsymbol{\Gamma}}{\mathrm{d}t}\cdot\hat{\boldsymbol{\Gamma}}$

Here we will now derive an expression that relates the rate of change of the tube length, $\frac{\mathrm{d}\ell}{\mathrm{d}t}$, to the evolution of the vortex strength of a vortex particle, $\frac{\mathrm{d}\boldsymbol{\Gamma}}{\mathrm{d}t}$. First, we will assume that each vortex tube is aligned with the local vorticity, $\hat{\boldsymbol{\ell}}_p = \boldsymbol{\omega}(\mathbf{x}_p)/\|\boldsymbol{\omega}(\mathbf{x}_p)\|$, and likewise the particle

strengths, $\hat{\mathbf{\Gamma}}_p = \mathbf{\Gamma}_p / \|\mathbf{\Gamma}_p\| = \hat{\boldsymbol{\ell}}_p$. When the $\boldsymbol{\omega}$ field encompases a tube of vorticity, we end up with

$$\mathbf{\Gamma}_p = \Gamma_p \ell \hat{\boldsymbol{\ell}}_p,$$

with Γ_p the circulation of the tube. Note that $\|\mathbf{\Gamma}_p\| = \Gamma_p \ell_p$.

Starting from

$$\Gamma^2 \ell^2 = \mathbf{\Gamma} \cdot \mathbf{\Gamma},$$

and using the fact that $\frac{\mathrm{d}\Gamma}{\mathrm{d}t} = 0$ from Kelvin's theorem, the time derivative becomes

$$2\Gamma^2 \ell \frac{\mathrm{d}\ell}{\mathrm{d}t} = \frac{\mathrm{d}}{\mathrm{d}t}\left(\mathbf{\Gamma} \cdot \mathbf{\Gamma}\right) = 2\frac{\mathrm{d}\mathbf{\Gamma}}{\mathrm{d}t} \cdot \mathbf{\Gamma}.$$

Dividing by $2\Gamma^2 \ell$,

$$\begin{aligned} \frac{\mathrm{d}\ell}{\mathrm{d}t} &= \frac{1}{\Gamma^2 \ell} \frac{\mathrm{d}\mathbf{\Gamma}}{\mathrm{d}t} \cdot \mathbf{\Gamma} \\ &= \frac{1}{\Gamma} \frac{1}{\|\mathbf{\Gamma}\|} \frac{\mathrm{d}\mathbf{\Gamma}}{\mathrm{d}t} \cdot \mathbf{\Gamma} \\ &= \frac{1}{\Gamma} \frac{\mathrm{d}\mathbf{\Gamma}}{\mathrm{d}t} \cdot \hat{\mathbf{\Gamma}}. \end{aligned}$$

Finally, dividing by ℓ we get

$$\frac{1}{\ell} \frac{\mathrm{d}\ell}{\mathrm{d}t} = \frac{1}{\|\mathbf{\Gamma}\|} \frac{\mathrm{d}\mathbf{\Gamma}}{\mathrm{d}t} \cdot \hat{\mathbf{\Gamma}}.$$

C Induced Vorticity Field $\nabla \times \overline{\mathbf{u}}$

As mentioned in Section 4.4, discretizing the filtered vorticity field $\overline{\boldsymbol{\omega}}$ with singular particles results in a radial-basis function (RBF) approximation

$$\overline{\boldsymbol{\omega}}(\mathbf{x}) \approx \sum_p \boldsymbol{\Gamma}_p \zeta_{\sigma_p}(\mathbf{x} - \mathbf{x}_p),$$

which is a field that is not divergence free. This is so because

$$\nabla \cdot \overline{\boldsymbol{\omega}}(\mathbf{x}) \approx \sum_p \boldsymbol{\Gamma}_p \cdot \nabla \left(\zeta_{\sigma_p}(\mathbf{x} - \mathbf{x}_p)\right),$$

and the gradient of the radial bases, $\nabla \zeta_{\sigma_p}$, is in general non-zero.

On the other hand, in Section 4.1 we solved the unbounded Poisson equation

$$\nabla^2 \overline{\boldsymbol{\psi}} = -\overline{\boldsymbol{\omega}}$$

to derive the velocity field $\overline{\mathbf{u}} = \nabla \times \overline{\boldsymbol{\psi}}$ induced by the particle field, resulting in Eq. (4.15) here repeated

$$\overline{\mathbf{u}}(\mathbf{x}) = \sum_p g_\sigma(\mathbf{x} - \mathbf{x}_p)\, \mathbf{K}(\mathbf{x} - \mathbf{x}_p) \times \boldsymbol{\Gamma}_p, \tag{4.15}$$

which in turns leads to an induced vorticity field, $\nabla \times \overline{\mathbf{u}}$. By construction, the induced vorticity is divergence-free since $\nabla \cdot (\nabla \times \overline{\mathbf{u}}) = 0$.

In order to gain physical and numerical insights, we will now derive the analytical form of $\nabla \times \overline{\mathbf{u}}$ and contrast it to the RBF-approximated vorticity field.

C.1 Derivation of $\nabla \times \overline{\mathbf{u}}$

Let the induced vorticity field be denoted as $\tilde{\boldsymbol{\omega}} \equiv \nabla \times \overline{\mathbf{u}}$. Replacing Eq. (4.15) in this definition,

$$\tilde{\boldsymbol{\omega}} = \nabla \times \left(\sum_p g_\sigma(\mathbf{x} - \mathbf{x}_p)\, \mathbf{K}(\mathbf{x} - \mathbf{x}_p) \times \boldsymbol{\Gamma}_p \right).$$

Using the identity $\nabla \times (a\mathbf{b}) = a(\nabla \times \mathbf{b}) + (\nabla a) \times \mathbf{b}$, this becomes

$$\tilde{\boldsymbol{\omega}} = \sum_p g_\sigma \nabla \times \left(\mathbf{K} \times \boldsymbol{\Gamma}_p\right) + \sum_p \nabla g_\sigma \times \left(\mathbf{K} \times \boldsymbol{\Gamma}_p\right), \tag{C.1}$$

and the identity $\nabla \times (\mathbf{a} \times \mathbf{b}) = \mathbf{a}(\nabla \cdot \mathbf{b}) - \mathbf{b}(\nabla \cdot \mathbf{a}) + (\mathbf{b} \cdot \nabla)\mathbf{a} - (\mathbf{a} \cdot \nabla)\mathbf{b}$ yields

$$\nabla \times \left(\mathbf{K} \times \mathbf{\Gamma}_p\right) = -\mathbf{\Gamma}_p(\nabla \cdot \mathbf{K}) + (\mathbf{\Gamma}_p \cdot \nabla)\mathbf{K}.$$

Recalling that $\mathbf{K}$ is the singular Newtonian kernel, $\mathbf{K}(\mathbf{x}) \equiv -\frac{\mathbf{x}}{4\pi\|\mathbf{x}\|^3}$, we have $\nabla \cdot \mathbf{K} = 0$, and we develop the $(\mathbf{\Gamma}_p \cdot \nabla)\mathbf{K}$ term in tensor notation as

$$\begin{aligned}
-4\pi\Gamma_j^p \frac{\partial}{\partial x_j}\left(K_i\left(\mathbf{x}-\mathbf{x}_p\right)\right) &= \Gamma_j^p \frac{\partial}{\partial x_j}\left(\frac{(x_i - x_i^p)}{\left[\sum_k (x_k - x_k^p)^2\right]^{3/2}}\right) \\
&= \frac{\Gamma_j^p \delta_{ij}}{\left[\sum_k (x_k - x_k^p)^2\right]^{3/2}} - 3\frac{\Gamma_j^p (x_j - x_j^p)(x_i - x_i^p)}{\left[\sum_k (x_k - x_k^p)^2\right]^{5/2}},
\end{aligned}$$

where the superscript p indicates the particle index rather than an exponent. Written in vector notation,

$$\begin{aligned}
-4\pi(\mathbf{\Gamma}_p \cdot \nabla)\mathbf{K}\left(\mathbf{x}-\mathbf{x}_p\right) &= \frac{\mathbf{\Gamma}_p}{\|\mathbf{x}-\mathbf{x}_p\|^3} - 3\frac{\mathbf{x}-\mathbf{x}_p}{\|\mathbf{x}-\mathbf{x}_p\|^5}\left[\mathbf{\Gamma}_p \cdot \left(\mathbf{x}-\mathbf{x}_p\right)\right] \\
&= \frac{1}{\|\mathbf{x}-\mathbf{x}_p\|^3}\left[\mathbf{\Gamma}_p - 3\frac{\mathbf{x}-\mathbf{x}_p}{\|\mathbf{x}-\mathbf{x}_p\|}\left(\mathbf{\Gamma}_p \cdot \frac{\mathbf{x}-\mathbf{x}_p}{\|\mathbf{x}-\mathbf{x}_p\|}\right)\right] \\
&= \frac{1}{r_p^3}\left[\mathbf{\Gamma}_p - 3\left(\mathbf{\Gamma}_p \cdot \hat{\mathbf{r}}_p\right)\hat{\mathbf{r}}_p\right],
\end{aligned}$$

where $\mathbf{r}_p \equiv \mathbf{x} - \mathbf{x}_p$, $r_p \equiv \|\mathbf{r}_p\|$, and $\hat{\mathbf{r}}_p \equiv \frac{\mathbf{r}_p}{r_p}$. Thus, we have

$$\nabla \times \left(\mathbf{K} \times \mathbf{\Gamma}_p\right) = -\frac{1}{4\pi r_p^3}\left[\mathbf{\Gamma}_p - 3\left(\mathbf{\Gamma}_p \cdot \hat{\mathbf{r}}_p\right)\hat{\mathbf{r}}_p\right], \tag{C.2}$$

which will be used in the first term of the right-hand side of Eq. (C.1). In the meantime, writing g_σ as

$$g_\sigma(\mathbf{x}) = 4\pi q\left(\frac{\|\mathbf{x}\|}{\sigma}\right),$$

where

$$q(r) \equiv \int_0^r \zeta(t)t^2\,\mathrm{d}t,$$

$\nabla\left(g_\sigma(\mathbf{x}-\mathbf{x}_p)\right)$ in the second term of Eq. (C.1) develops as

$$\begin{aligned}
\frac{1}{4\pi}\nabla\left(g_\sigma(\mathbf{x}-\mathbf{x}_p)\right) &= \nabla\left(q\left(\frac{\|\mathbf{x}-\mathbf{x}_p\|}{\sigma}\right)\right) \\
&= \frac{1}{\sigma}\frac{\mathbf{x}-\mathbf{x}_p}{\|\mathbf{x}-\mathbf{x}_p\|}\frac{\partial q}{\partial r}\left(\frac{\|\mathbf{x}-\mathbf{x}_p\|}{\sigma}\right) \\
&= \frac{1}{\sigma}\frac{\mathbf{x}-\mathbf{x}_p}{\|\mathbf{x}-\mathbf{x}_p\|}\zeta\left(\frac{\|\mathbf{x}-\mathbf{x}_p\|}{\sigma}\right)\left(\frac{\|\mathbf{x}-\mathbf{x}_p\|}{\sigma}\right)^2 \\
&= \|\mathbf{x}-\mathbf{x}_p\|(\mathbf{x}-\mathbf{x}_p)\frac{1}{\sigma^3}\zeta\left(\frac{\|\mathbf{x}-\mathbf{x}_p\|}{\sigma}\right) \\
&= \|\mathbf{x}-\mathbf{x}_p\|(\mathbf{x}-\mathbf{x}_p)\zeta_\sigma\left(\mathbf{x}-\mathbf{x}_p\right).
\end{aligned}$$

Thus, the $\nabla g_\sigma \times (\mathbf{K} \times \boldsymbol{\Gamma}_p)$ term in Eq. (C.1) develops as

$$\begin{aligned}
\nabla g_\sigma \times (\mathbf{K} \times \boldsymbol{\Gamma}_p) &= 4\pi \|\mathbf{x} - \mathbf{x}_p\| \zeta_\sigma (\mathbf{x} - \mathbf{x}_p) \left[(\mathbf{x} - \mathbf{x}_p) \times (\mathbf{K}(\mathbf{x} - \mathbf{x}_p) \times \boldsymbol{\Gamma}_p) \right] \\
&= 4\pi \|\mathbf{x} - \mathbf{x}_p\| \zeta_\sigma (\mathbf{x} - \mathbf{x}_p) \left[(\mathbf{x} - \mathbf{x}_p) \times \left(-\frac{\mathbf{x} - \mathbf{x}_p}{4\pi \|\mathbf{x} - \mathbf{x}_p\|^3} \times \boldsymbol{\Gamma}_p \right) \right] \\
&= -\frac{1}{\|\mathbf{x} - \mathbf{x}_p\|^2} \zeta_\sigma (\mathbf{x} - \mathbf{x}_p) \left\{ (\mathbf{x} - \mathbf{x}_p) \times \left[(\mathbf{x} - \mathbf{x}_p) \times \boldsymbol{\Gamma}_p \right] \right\} \\
&= -\frac{1}{\|\mathbf{x} - \mathbf{x}_p\|^2} \zeta_\sigma (\mathbf{x} - \mathbf{x}_p) \left\{ \left[(\mathbf{x} - \mathbf{x}_p) \cdot \boldsymbol{\Gamma}_p \right] (\mathbf{x} - \mathbf{x}_p) - \|\mathbf{x} - \mathbf{x}_p\|^2 \boldsymbol{\Gamma}_p \right\} \\
&= -\zeta_\sigma (\mathbf{x} - \mathbf{x}_p) \left[\left(\frac{\mathbf{x} - \mathbf{x}_p}{\|\mathbf{x} - \mathbf{x}_p\|} \cdot \boldsymbol{\Gamma}_p \right) \frac{\mathbf{x} - \mathbf{x}_p}{\|\mathbf{x} - \mathbf{x}_p\|} - \boldsymbol{\Gamma}_p \right] \\
&= \zeta_\sigma (\mathbf{r}_p) \left[\boldsymbol{\Gamma}_p - (\boldsymbol{\Gamma}_p \cdot \hat{\mathbf{r}}_p) \hat{\mathbf{r}}_p \right],
\end{aligned}$$

where in the fourth line we have used the triple-product identity $\mathbf{a} \times (\mathbf{a} \times \mathbf{c}) = (\mathbf{a} \cdot \mathbf{c})\mathbf{a} - (\mathbf{a} \cdot \mathbf{a})\mathbf{c}$. Finally, replacing this and Eq. (C.2) in Eq. (C.1), we arrive to

$$\tilde{\boldsymbol{\omega}}(\mathbf{x}) = \sum_p \left\{ \zeta_\sigma (\mathbf{r}_p) \left[\boldsymbol{\Gamma}_p - (\boldsymbol{\Gamma}_p \cdot \hat{\mathbf{r}}_p) \hat{\mathbf{r}}_p \right] - \frac{g_\sigma (\mathbf{r}_p)}{4\pi \|\mathbf{r}_p\|^3} \left[\boldsymbol{\Gamma}_p - 3 (\boldsymbol{\Gamma}_p \cdot \hat{\mathbf{r}}_p) \hat{\mathbf{r}}_p \right] \right\}, \tag{C.3}$$

where $\mathbf{r}_p = \mathbf{x} - \mathbf{x}_p$.

C.2 Comparing $\sum_p \boldsymbol{\Gamma}_p \zeta_{\sigma_p}(\mathbf{x} - \mathbf{x}_p)$ and $\nabla \times \bar{\mathbf{u}}$

In order to attain a rough intuition on how the induced vorticity field

$$\tilde{\boldsymbol{\omega}} = \nabla \times \bar{\mathbf{u}}$$

compares to the RBF vorticity field

$$\boldsymbol{\omega}_\sigma = \sum_p \boldsymbol{\Gamma}_p \zeta_{\sigma_p}(\mathbf{x} - \mathbf{x}_p),$$

we now assume that the fields have only one particle. If we let $\mathbf{x}$ only take the values of a line perpendicular to the vortex strength passing through $\mathbf{x}_p$, we have $\boldsymbol{\Gamma}_p \cdot \hat{\mathbf{r}}_p = 0$, and $\tilde{\boldsymbol{\omega}}$ given by Eq. (C.3) becomes

$$\tilde{\boldsymbol{\omega}}(\mathbf{x}) = \boldsymbol{\Gamma}_p \zeta_\sigma (\mathbf{x} - \mathbf{x}_p) - \boldsymbol{\Gamma}_p \frac{g_\sigma (\mathbf{x} - \mathbf{x}_p)}{4\pi \|\mathbf{x} - \mathbf{x}_p\|^3}.$$

On the other hand, the RBF-approximated field is

$$\boldsymbol{\omega}_\sigma(\mathbf{x}) = \boldsymbol{\Gamma}_p \zeta_\sigma (\mathbf{x} - \mathbf{x}_p).$$

Both fields are shown in Fig. C.1 obtained with a Gaussian radial basis, evidencing that the induced vorticity is smaller than the RBF-approximated vorticity and that it attains a negative value is it decays.

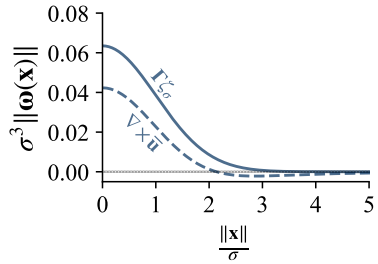

Figure C.1: Vorticity fields generated by a particle of unitary strength and Gaussian basis, probed along line $\boldsymbol{\Gamma}_p \cdot (\mathbf{x} - \mathbf{x}_p) = 0$.

Let us rewrite the induced vorticity as

$$\tilde{\boldsymbol{\omega}}(\mathbf{x}) = \boldsymbol{\Gamma}_p \left(\zeta_\sigma \left(\mathbf{x} - \mathbf{x}_p \right) - J_\sigma \left(\mathbf{x} - \mathbf{x}_p \right) \right),$$

with

$$J_\sigma(\mathbf{x}) \equiv \frac{g_\sigma \left(\mathbf{x} - \mathbf{x}_p \right)}{4\pi \|\mathbf{x} - \mathbf{x}_p\|^3}.$$

Fig. C.2 (top) shows the distribution of induced vorticity normalized by the peak value of the RBF-vorticity, $\zeta_\sigma(\mathbf{0})$, as calculated with three different basis functions. All three bases reach the same peak, which seems to indicate that the peak of induced vorticity is about 2/3 the value of the RBF-approximated peak. Furthermore, numerical results confirm that $J_\sigma\left(\mathbf{0}\right)/\zeta_\sigma\left(\mathbf{0}\right) = 1/3$ for all three basis functions, as shown in Fig. C.2 (bottom).

While we have been evaluating the fields only along the line $\boldsymbol{\Gamma}_p \cdot (\mathbf{x} - \mathbf{x}_p) = 0$, now we evaluate the entire field as given by Eq. (C.3). The magnitude of the vorticity fields is shown in Fig. C.3, scaled by a factor of 2/3. We see that the RBF-approximated field is radially symmetric, as expected. Surprisingly, the induced field is not symmetric, but is rather elongated in the direction of the vortex strength. A slice of both fields is shown in Fig. C.4, superimposed with vortex lines. The RBF-approximated field results in straight vortex lines all throughout, while the induced field results in vortex lines that close on themselves with two loci of zero vorticity near $y/\sigma \approx 2$. This zero-vorticity point had been observed in Fig. C.1, beyond which point the vorticity changes direction in Fig. C.4 while becoming negative in Fig. C.1.

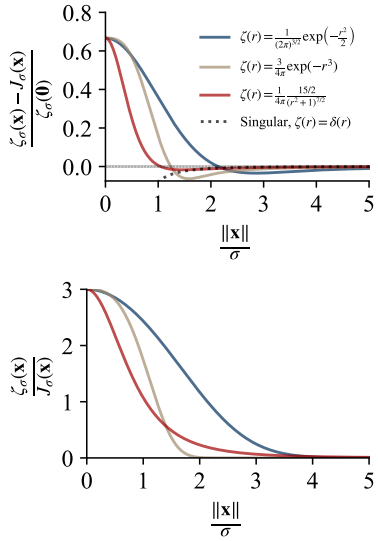

Figure C.2: (top) distribution of induced vorticity and (bottom) ratio of factor J_σ to the radial basis, probed along line $\boldsymbol{\Gamma}_p \cdot (\mathbf{x} - \mathbf{x}_p) = 0$.

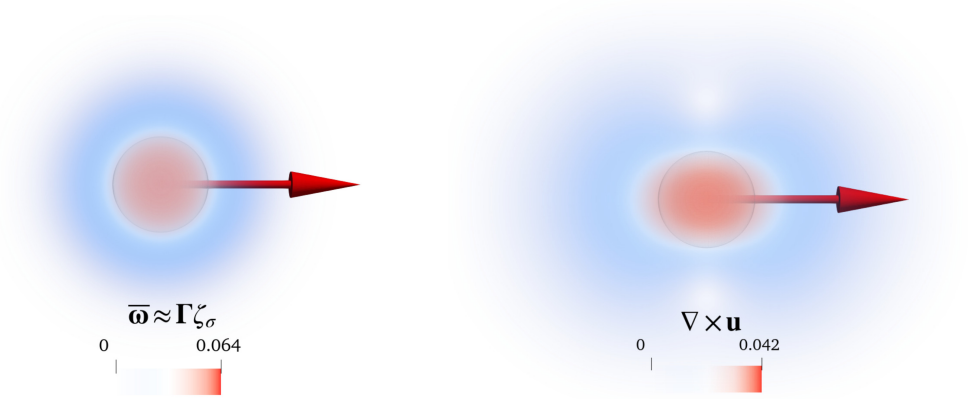

Figure C.3: Volume rendering of (left) RBF-approximated vorticity field and (right) induced vorticity field, both generated by a particle with unitary strength and Gaussian basis. Vortex strength $\boldsymbol{\Gamma}$ shown as arrow, while the smoothing radius σ is shown as a sphere.

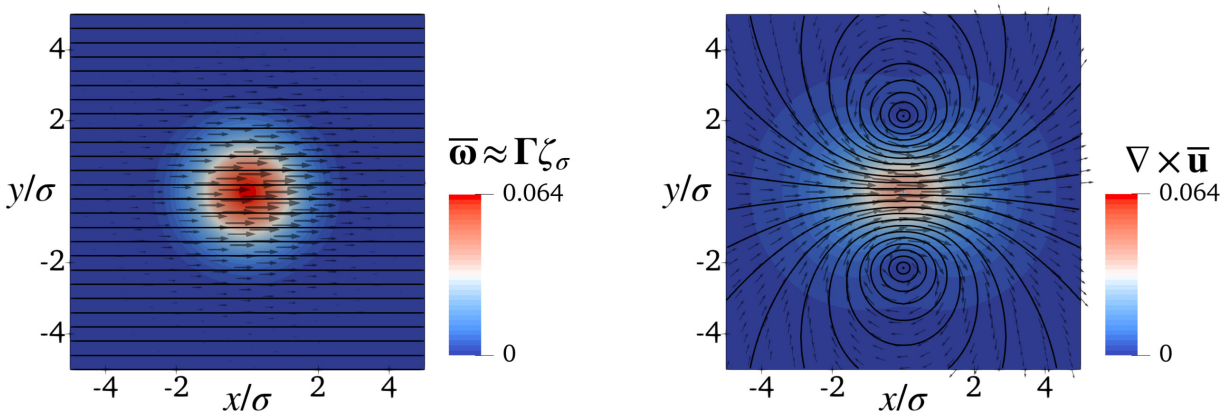

Figure C.4: Slice of (left) RBF-approximated vorticity field and (right) induced vorticity field, both generated by a particle with unitary strength and Gaussian basis. Vortex lines shown in black. Direction of vorticity shown as arrows.

Analytical Dynamics of Vortex Rings

D

Given a system of N coaxial vortex rings with circulation Γ_i, centroid Z_i, and radius R_i for $i = 1, ..., N$, let us assume a singular vorticity along each ring's centerline,

$$\omega(r, z, t) = \sum_{i=1}^{N} \Gamma_i \delta\left(z - Z_i(t)\right) \delta\left(r - R_i(t)\right)$$

where δ is the Dirac delta, and r and z are the radial and axial coordinates of a cylindrical system. As described by Borisov *et al.*,[205] the interactions between singular rings lead to dynamics dictated by

[205] A. V. Borisov et al. (2013). "The dynamics of vortex rings: Leapfrogging, choreographies and the stability problem".

$$\frac{\mathrm{d}R_i}{\mathrm{d}t} = -\frac{1}{R_i}\frac{\partial}{\partial Z_i}\sum_{j\neq i}\Gamma_j G\left(R_i, Z_i, R_j, Z_j\right) \tag{D.1}$$

and

$$\frac{\mathrm{d}Z_i}{\mathrm{d}t} = V_i + \frac{1}{R_i}\frac{\partial}{\partial R_i}\sum_{j\neq i}\Gamma_j G\left(R_i, Z_i, R_j, Z_j\right) \tag{D.2}$$

where V_i is the self-induced velocity of the ring and G is the streamfunction of an infinitely thin circular vortex filament of unit intensity. G is given by

$$G(r, z, R, Z) = \frac{\sqrt{rR}}{2\pi}\left[\left(\frac{2}{k} - k\right)K(k^2) - \frac{2}{k}E(k^2)\right],$$

where

$$k = \sqrt{\frac{4rR}{(z-Z)^2 + (r+R)^2}},$$

and $K(m)$ and $E(m)$ are the complete elliptic integrals of the first and second kind, respectively:

$$K(m) = \int_0^{\frac{\pi}{2}} \frac{1}{\sqrt{1 - m\sin^2\theta}}\,\mathrm{d}\theta$$

and

$$E(m) = \int_0^{\frac{\pi}{2}} \sqrt{1 - m\sin^2\theta}\,\mathrm{d}\theta.$$

Since the velocity induced by a singular filament on itself is infinite, a regularized vorticity distribution must be assumed when calculating V_i. Assuming a thin toroidal vortex ring, Saffman[206] obtained

$$V = \frac{\Gamma}{4\pi R}\left(\ln\frac{8}{a/R} + C\left(a/R\right)\right), \tag{D.3}$$

where the function C is determined from the vorticity distribution inside the ring core. For thin Gaussian rings with $a \ll R$, $C(\epsilon)$ becomes $C = -0.558$. For thick Gaussian rings, Archer *et al.*[207] observed through direct numerical simulation that the initial Gaussian distribution quickly becomes skewed and leads to

$$C(\epsilon) \approx -0.558 - 1.12\epsilon^2 - 5.0\epsilon^4, \tag{D.4}$$

where $\epsilon \equiv a/R$. Thus, the self-induced velocity depends on the current core size a of the ring. At the same time, viscous diffusion constantly spreads the vorticity distribution according to $\frac{\mathrm{d}\boldsymbol{\omega}}{\mathrm{d}t} = \nu\nabla^2\boldsymbol{\omega}$. In a vortex filament with a Gaussian vorticity distribution, $\omega(r) = \frac{\Gamma}{\pi a^2}e^{-r^2/a^2}$, the viscous diffusion equation leads to

$$\frac{\mathrm{d}a^2}{\mathrm{d}t} = 4\nu. \tag{D.5}$$

Finally, when viscous effects are ignored through operator splitting, the ring must conserve mass due to Helmholtz' circulation theorem in incompressible flow. This is expressed as

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(Ra^2\right) = 0,$$

which leads to

$$\frac{\mathrm{d}a}{\mathrm{d}t} = -\frac{a}{2R}\frac{\mathrm{d}R}{\mathrm{d}t}, \tag{D.6}$$

meaning that the core size must shrink as the ring is stretched by the interaction with neighboring rings. We have observed that incorporating this last equation introduces dynamics that are incongruent with the numerical results from DNS, LBM, and VPM discussed in Sections 5.1 and 5.2. This could be due to the fact that Eqs. (D.1) and (D.2) ignore the effects of core size in the mutual interactions between rings, thus Eq. (D.6) is introducing an unphysical one-way coupling between $\frac{\mathrm{d}a}{\mathrm{d}t}$ and $\frac{\mathrm{d}R}{\mathrm{d}t}$. Hence, Eq. (D.6) was not used in the analytical solutions shown in Sections 5.1 and 5.2.

In summary, Eqs. (D.1) to (D.5) form a system of analytical ordinary differential equations that describe the dynamics of an arbitrary number of coaxial, viscous, incompressible vortex rings. In this derivation we have assumed that the core thickness only affects the self-induced velocity of each ring, while its effects are ignored in the interactions between the rings and the core size is not affected by vortex stretching.

206 P. G. Saffman (1970). "The Velocity of Viscous Vortex Rings"; P. G. Saffman (1985). "Vortex Dynamics".

207 Archer et al., "Direct numerical simulation of vortex ring evolution from the laminar to the early turbulent regime".